

The Rise and Potential of Large Language Model Based Agents: A Survey

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Abstract

For a long time, humanity has pursued artificial intelligence (AI) equivalent to or surpassing the human level, with AI agents considered a promising vehicle for this pursuit. AI agents are artificial entities that sense their environment, make decisions, and take actions. Many efforts have been made to develop intelligent agents, but they mainly focus on advancement in algorithms or training strategies to enhance specific capabilities or performance on particular tasks. Actually, what the community lacks is a general and powerful model to serve as a starting point for designing AI agents that can adapt to diverse scenarios. Due to the versatile capabilities they demonstrate, large language models (LLMs) are regarded as potential sparks for Artificial General Intelligence (AGI), offering hope for building general AI agents. Many researchers have leveraged LLMs as the foundation to build AI agents and have achieved significant progress. In this paper, we perform a comprehensive survey on LLM-based agents. We start by tracing the concept of agents from its philosophical origins to its development in AI, and explain why LLMs are suitable foundations for agents. Building upon this, we present a general framework for LLM-based agents, comprising three main components: brain, perception, and action, and the framework can be tailored for different applications. Subsequently, we explore the extensive applications of LLM-based agents in three aspects: single-agent scenarios, multi-agent scenarios, and human-agent cooperation. Following this, we delve into agent societies, exploring the behavior and personality of LLM-based agents, the social phenomena that emerge from an agent society, and the insights they offer for human society. Finally, we discuss several key topics and open problems within the field. A repository for the related papers at <https://github.com/WooooDyy/LLM-Agent-Paper-List>.

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大型语言模型基础代理的崛起与潜力：综述

《基于大型语言模型的智能体的崛起与潜力：综述》

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摘要

《大型语言模型基础的智能体的崛起与潜力：综述》长久以来，人类一直追求具有等同或超越人类水平的人工智能（AI），而AI智能体被视为实现这一目标的有前景的载体。AI智能体是感知其环境、做出决策并采取行动的人工实体。为了开发智能体，已进行了大量努力，但这些工作主要集中在算法或训练策略的改进，以增强特定能力或在特定任务上的性能。实际上，社区缺乏一个通用且强大的模型，作为设计能够适应多样场景的AI智能体的起点。由于其展现出多功能能力，大型语言模型（LLMs）被认为是人工通用智能（AGI）的潜在火花，为构建通用AI智能体带来了希望。许多研究者利用LLMs作为基础，构建AI智能体，并取得了显著进展。本文对基于LLMs的智能体进行了全面的综述。我们首先追溯智能体的概念，从其哲学起源到在AI中的发展，并阐述为何LLMs适合作为智能体的基础。在此基础上，我们提出一个通用的基于LLMs的智能体框架，包括三个主要组成部分：大脑、感知和行动，该框架可根据不同应用进行定制。随后，我们探讨了基于LLMs的智能体在三个方面的广泛应用：单智能体场景、多智能体场景以及人-智能体合作。接着，我们深入研究智能体社会，探讨基于LLMs的智能体的行为与个性、智能体社会中出现的社会现象，以及它们对人类社会的启示。最后，我们讨论了该领域的若干关键主题和未解难题。相关论文的存储库在

<https://github.com/WooooDyy/LLM-Agent-Paper-List>.

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1 Introduction

“If they find a parrot who could answer to everything, I would claim it to be an intelligent being without hesitation.”

—Denis Diderot, 1875

Artificial Intelligence (AI) is a field dedicated to designing and developing systems that can replicate human-like intelligence and abilities [1]. As early as the 18th century, philosopher Denis Diderot introduced the idea that if a parrot could respond to every question, it could be considered intelligent [2]. While Diderot was referring to living beings, like the parrot, his notion highlights the profound concept that a highly intelligent organism could resemble human intelligence. In the 1950s, Alan Turing expanded this notion to artificial entities and proposed the renowned Turing Test [3]. This test is a cornerstone in AI and aims to explore whether machines can display intelligent behavior comparable to humans. These AI entities are often termed “agents”, forming the essential building blocks of AI systems. Typically in AI, an agent refers to an artificial entity capable of perceiving its surroundings using sensors, making decisions, and then taking actions in response using actuators [1; 4].

The concept of agents originated in Philosophy, with roots tracing back to thinkers like Aristotle and Hume [5]. It describes entities possessing desires, beliefs, intentions, and the ability to take actions [5]. This idea transitioned into computer science, intending to enable computers to understand users’ interests and autonomously perform actions on their behalf [6; 7; 8]. As AI advanced, the term “agent” found its place in AI research to depict entities showcasing intelligent behavior and possessing qualities like autonomy, reactivity, pro-activeness, and social ability [4; 9]. Since then, the exploration and technical advancement of agents have become focal points within the AI community [1; 10]. AI agents are now acknowledged as a pivotal stride towards achieving Artificial General Intelligence (AGI)¹, as they encompass the potential for a wide range of intelligent activities [4; 11; 12].

From the mid-20th century, significant strides were made in developing smart AI agents as research delved deep into their design and advancement [13; 14; 15; 16; 17; 18]. However, these efforts have predominantly focused on enhancing specific capabilities, such as symbolic reasoning, or mastering particular tasks like Go or Chess [19; 20; 21]. Achieving a broad adaptability across varied scenarios remained elusive. Moreover, previous studies have placed more emphasis on the design of algorithms and training strategies, overlooking the development of the model’s inherent general abilities like knowledge memorization, long-term planning, effective generalization, and efficient interaction [22; 23]. Actually, enhancing the inherent capabilities of the model is the pivotal factor for advancing the agent further, and the domain is in need of a powerful foundational model endowed with a variety of key attributes mentioned above to serve as a starting point for agent systems.

The development of large language models (LLMs) has brought a glimmer of hope for the further development of agents [24; 25; 26], and significant progress has been made by the community [22; 27; 28; 29]. According to the notion of World Scope (WS) [30] which encompasses five levels that depict the research progress from NLP to general AI (i.e., Corpus, Internet, Perception, Embodiment, and Social), the pure LLMs are built on the second level with internet-scale textual inputs and outputs. Despite this, LLMs have demonstrated powerful capabilities in knowledge acquisition, instruction comprehension, generalization, planning, and reasoning, while displaying effective natural language interactions with humans. These advantages have earned LLMs the designation of sparks for AGI [31], making them highly desirable for building intelligent agents to foster a world where humans and agents coexist harmoniously [22]. Starting from this, if we elevate LLMs to the status of agents and equip them with an expanded perception space and action space, they have the potential to reach the third and fourth levels of WS. Furthermore, these LLMs-based agents can tackle more complex tasks through cooperation or competition, and emergent social phenomena can be observed when placing them together, potentially achieving the fifth WS level. As shown in Figure 1, we envision a harmonious society composed of AI agents where human can also participate.

In this paper, we present a comprehensive and systematic survey focusing on LLM-based agents, attempting to investigate the existing studies and prospective avenues in this burgeoning field. To this end, we begin by delving into crucial **background** information (§ 2). In particular, we commence by tracing the origin of AI agents from philosophy to the AI domain, along with a brief overview of the

¹Also known as Strong AI.

1 引言

“如果他们找到一只能够回答所有问题的鹦鹉，我会毫不犹豫地认为它是一个具有智能的存在。”

——德尼·狄德罗，1875年

人工智能（AI）是一个致力于设计和开发能够模拟人类智能和能力的系统的领域 [1]。早在18世纪，哲学家丹尼斯·狄德罗提出了这样一个观点：如果一只鹦鹉能够回答每一个问题，那么它可以被认为具有智能[2]。虽然狄德罗指的是活生生的生物，比如鹦鹉，但他的观点强调了一个深刻的概念：高度智能的生物体可能类似于人类的智能。在20世纪50年代，艾伦·图灵将这一概念扩展到人工实体，并提出了著名的图灵测试 [3]。该测试是人工智能的基石，旨在探索机器是否能够表现出与人类相当的智能行为。这些人工智能实体通常被称为“代理”，构成了人工智能系统的基本组成部分。在人工智能中，代理通常指能够利用传感器感知环境、做出决策，并通过执行器采取行动的人工实体[1; 4]。

代理的概念起源于哲学，其根源可以追溯到亚里士多德和休谟等思想家。它描述具有欲望、信念、意图以及采取行动能力的实体 [5]。这一思想逐渐转入计算机科学，旨在使计算机能够理解用户的兴趣，并自主代表用户执行行动 [5]。随着人工智能的发展，“代理”一词在人工智能研究中逐渐占据一席之地，用以描述展现出智能行为、具有自主性、反应性、主动性和社会能力的实体 [6; 7; 8]。随着人工智能的不断进步, "代理"一词在人工智能领域的探索与技术发展成为研究的重点 [4; 9]。如今，人工智能代理被认为是实现通用人工智能（AGI）的关键一步，因为它们具有执行广泛智能活动的潜力 [4; 11; 12]。

自20世纪中叶起，随着对智能AI代理的设计与发展深入研究，取得了显著的进展From the mid-20th century, significant strides were made in developing smart AI agents as research delved deep into their design and advancement [13; 14; 15; 16; 17; 18]. 然而，这些努力主要集中在提升特定能力，如符号推理，或掌握特定任务，如围棋或国际象棋 [19; 20; 21]. 实现跨多场景的广泛适应性仍然具有挑战性。此外，先前的研究更侧重于算法设计和训练策略，忽视了模型固有的通用能力的发展，如知识记忆、长期规划、有效泛化和高效交互[22; 23]. 实际上，提升模型的固有能力和推动代理进一步发展的关键因素，领域内亟需一个具备上述多项关键属性的强大基础模型，作为代理系统的起点。

大型语言模型（LLMs）的发展为代理的进一步发展带来了希望的曙光 [24; 25; 26], 社区取得了显著的进展[22; 27; 28; 29]。根据世界范围（WS）的概念 [30], 它涵盖五个层级，描述从自然语言处理（NLP）到通用人工智能（即语料库、互联网、感知、具身和社会）的研究进展，纯粹的LLMs建立在第二层级，利用互联网规模的文本输入和输出。尽管如此，LLMs在知识获取、指令理解、泛化、规划和推理方面表现出强大的能力，同时展现出与人类进行自然语言交互的有效性。这些优势使得LLMs被誉为通向AGI的火花 [31], 使其在构建智能代理、促进人类与代理和谐共存的世界中具有高度的吸引力 [22]。基于此，如果我们将LLMs提升到代理的地位，并赋予它们扩展的感知空间和行动空间，它们有潜力达到WS的第三和第四层级。此外，这些基于LLMs的代理可以通过合作或竞争应对更复杂的任务，当它们共同存在时，还可以观察到新兴的社会现象，可能实现第五层级的WS。如图1所示，我们设想一个由AI代理组成的和谐社会，人类也可以参与其中。

在本文中，我们对基于大规模语言模型的智能体进行了全面而系统的综述，旨在探讨该新兴领域中的现有研究和未来潜在方向。为此，我们首先深入探讨关键的背景信息（§ 2）。特别是，我们从哲学角度追溯AI智能体的起源，直到AI domain，并对其进行了简要概述。

¹也被称为强人工智能。



Figure 1: Scenario of an envisioned society composed of AI agents, in which humans can also participate. The above image depicts some specific scenes within society. In the kitchen, one agent orders dishes, while another agent is responsible for planning and solving the cooking task. At the concert, three agents are collaborating to perform in a band. Outdoors, two agents are discussing lantern-making, planning the required materials, and finances by selecting and using tools. Users can participate in any of these stages of this social activity.

debate surrounding the existence of artificial agents (§ 2.1). Next, we take the lens of technological trends to provide a concise historical review of the development of AI agents (§ 2.2). Finally, we delve into an in-depth introduction of the essential characteristics of agents and elucidate why large language models are well-suited to serve as the main component of brains or controllers for AI agents (§ 2.3).

Inspired by the definition of the agent, we present a general conceptual **framework** for the LLM-based agents with three key parts: **brain, perception, and action** (§ 3), and the framework can be tailored to suit different applications. We first introduce the brain, which is primarily composed of a large language model (§ 3.1). Similar to humans, the brain is the core of an AI agent because it not only stores crucial memories, information, and knowledge but also undertakes essential tasks of information processing, decision-making, reasoning, and planning. It is the key determinant of whether the agent can exhibit intelligent behaviors. Next, we introduce the perception module (§ 3.2). For an agent, this module serves a role similar to that of sensory organs for humans. Its primary function is to expand the agent’s perceptual space from text-only to a multimodal space that includes diverse sensory modalities like text, sound, visuals, touch, smell, and more. This expansion enables the agent to better perceive information from the external environment. Finally, we present the action module for expanding the action space of an agent (§ 3.3). Specifically, we expect the agent to be able to possess textual output, take embodied actions, and use tools so that it can better respond to environmental changes and provide feedback, and even alter and shape the environment.

After that, we provide a detailed and thorough introduction to the **practical applications** of LLM-based agents and elucidate the foundational design pursuit—“Harnessing AI for good” (§ 4). To start, we delve into the current applications of a single agent and discuss their performance in text-based tasks and simulated exploration environments, with a highlight on their capabilities in handling specific tasks, driving innovation, and exhibiting human-like survival skills and adaptability (§ 4.1). Following that, we take a retrospective look at the development history of multi-agents. We introduce the interactions between agents in LLM-based multi-agent system applications, where they engage in



图1: 设想中的由AI代理组成的社会场景， humans也可以参与。上图展示了社会中的一些具体场景。在厨房中，一个代理点餐，另一个代理负责规划和解决烹饪任务。在音乐会中，三个代理合作组成乐队演出。在户外，两个代理讨论灯笼制作，规划所需材料和财务，通过选择和使用工具。用户可以参与这些社会活动的任何阶段。

关于人工智能代理存在的辩论 (§ 2.1)。接下来，我们将以技术趋势为视角，简要回顾人工智能代理的发展历程 (§ 2.2)。最后，我们将深入介绍代理的基本特性，并阐明为何大型语言模型非常适合作为人工智能代理的“脑”或“控制器”的主要组成部分 (§ 2.3)。

受到代理定义的启发，我们提出了一个面向LLM的通用概念框架，包含三个关键部分：大脑、感知和行动 (§ 3)，该框架可以根据不同应用进行定制。我们首先介绍大脑，主要由大型语言模型组成 (§ 3.1)。与人类类似，大脑是AI代理的核心，因为它不仅存储关键的记忆、信息和知识，还承担信息处理、决策、推理和规划等基本任务。它是决定代理是否能表现出智能行为的关键因素。接下来，我们介绍感知模块 (§ 3.2)。对于代理而言，该模块的作用类似于人类的感官器官，其主要功能是将代理的感知空间从仅文本扩展到包括多种感官模态的多模态空间，如文本、声音、视觉、触觉、嗅觉等。这一扩展使代理能够更好地感知外部环境中的信息。最后，我们介绍行动模块 (§ 3.3)，旨在扩展代理的行动空间。具体而言，我们希望代理能够进行文本输出、采取具象行动、使用工具，从而更好地应对环境变化、提供反馈，甚至改变和塑造环境。

之后，我们将对基于大语言模型 (LLM) 代理的实际应用进行详细而全面的介绍，并阐明其基础设计追求——“善用人工智能” (§ 4.7)。首先，我们探讨单一代理的当前应用，分析其在文本任务和模拟探索环境中的表现，重点关注其在处理特定任务、推动创新以及展现类人生存技能和适应能力方面的能力 (§ 4.1)。随后，我们回顾多智能体的发展历史，介绍在基于大语言模型的多智能体系统应用中，代理之间的交互方式。

collaboration, negotiation, or competition. Regardless of the mode of interaction, agents collectively strive toward a shared objective (§ 4.2). Lastly, considering the potential limitations of LLM-based agents in aspects such as privacy security, ethical constraints, and data deficiencies, we discuss the human-agent collaboration. We summarize the paradigms of collaboration between agents and humans: the instructor-executor paradigm and the equal partnership paradigm, along with specific applications in practice (§ 4.3).

Building upon the exploration of practical applications of LLM-based agents, we now shift our focus to the concept of the “**Agent Society**”, examining the intricate interactions between agents and their surrounding environments (§ 5). This section begins with an investigation into whether these agents exhibit human-like behavior and possess corresponding personality (§5.1). Furthermore, we introduce the social environments within which the agents operate, including text-based environment, virtual sandbox, and the physical world (§5.2). Unlike the previous section (§ 3.2), here we will focus on diverse types of the environment rather than how the agents perceive it. Having established the foundation of agents and their environments, we proceed to unveil the simulated societies that they form (§5.3). We will discuss the construction of a simulated society, and go on to examine the social phenomena that emerge from it. Specifically, we will emphasize the lessons and potential risks inherent in simulated societies.

Finally, we discuss a range of key **topics** (§ 6) and open problems within the field of LLM-based agents: (1) the mutual benefits and inspirations of the LLM research and the agent research, where we demonstrate that the development of LLM-based agents has provided many opportunities for both agent and LLM communities (§ 6.1); (2) existing evaluation efforts and some prospects for LLM-based agents from four dimensions, including utility, sociability, values and the ability to continually evolve (§ 6.2); (3) potential risks of LLM-based agents, where we discuss adversarial robustness and trustworthiness of LLM-based agents. We also include the discussion of some other risks like misuse, unemployment and the threat to the well-being of the human race (§ 6.3); (4) scaling up the number of agents, where we discuss the potential advantages and challenges of scaling up agent counts, along with the approaches of pre-determined and dynamic scaling (§ 6.4); (5) several open problems, such as the debate over whether LLM-based agents represent a potential path to AGI, challenges from virtual simulated environment to physical environment, collective Intelligence in AI agents, and Agent as a Service (§ 6.5). After all, we hope this paper could provide inspiration to the researchers and practitioners from relevant fields.

2 Background

In this section, we provide crucial background information to lay the groundwork for the subsequent content (§ 2.1). We first discuss the origin of AI agents, from philosophy to the realm of AI, coupled with a discussion of the discourse regarding the existence of artificial agents (§ 2.2). Subsequently, we summarize the development of AI agents through the lens of technological trends. Finally, we introduce the key characteristics of agents and demonstrate why LLMs are suitable to serve as the main part of the brains of AI agents (§ 2.3).

2.1 Origin of AI Agent

“Agent” is a concept with a long history that has been explored and interpreted in many fields. Here, we first explore its origins in philosophy, discuss whether artificial products can possess agency in a philosophical sense, and examine how related concepts have been introduced into the field of AI.

Agent in philosophy. The core idea of an agent has a historical background in philosophical discussions, with its roots traceable to influential thinkers such as Aristotle and Hume, among others [5]. In a general sense, an “agent” is an entity with the capacity to act, and the term “agency” denotes the exercise or manifestation of this capacity [5]. While in a narrow sense, “agency” is usually used to refer to the performance of intentional actions; and correspondingly, the term “agent” denotes entities that possess desires, beliefs, intentions, and the ability to act [32; 33; 34; 35]. Note that agents can encompass not only individual human beings but also other entities in both the physical and virtual world. Importantly, the concept of an agent involves individual autonomy, granting them the ability to exercise volition, make choices, and take actions, rather than passively reacting to external stimuli.

合作、协商或竞争。无论交互方式如何，代理人都共同朝着共享目标努力 (§ 4.2)。最后，考虑到基于大规模语言模型（LLM）代理在隐私安全、伦理约束和数据不足等方面的潜在限制，我们讨论了人机协作。我们总结了代理人与人类合作的范式：指导-执行范式和平等伙伴关系范式，以及在实际中的具体应用 (§ 4.3)。

在基于大规模语言模型的代理的实际应用探索基础上，我们现在将焦点转向“**代理社会**”的概念，探讨代理与其周围环境之间的复杂互动 (§ 5)。本节首先调查这些代理是否表现出类人行为并具有相应的个性 (§ 5.1)。此外，我们还介绍了代理所处的社会环境，包括基于文本的环境、虚拟沙箱和物理世界 (§ 5.2)。与前一节 (§ 3.2) 不同，这里我们将关注环境的多样类型，而非代理对环境的感知。在建立了代理及其环境的基础之后，我们将揭示它们形成的模拟社会 (§ 5.3)。我们将讨论模拟社会的构建，并进一步考察由此产生的社会现象，特别强调模拟社会中蕴含的教训和潜在风险。

最后，我们讨论了LLM基础智能体领域中的一系列关键问题和未解难题：(1) LLM研究与智能体研究的相互促进与启示，本文展示了基于LLM的智能体的发展为智能体和LLM社区都提供了许多机遇 (§ 6.1)；(2) 现有的评估工作及从实用性、社交性、价值观和持续演化能力四个维度对基于LLM的智能体的展望 (§ 6.2)；(3) 基于LLM的智能体的潜在风险，本文讨论了对抗鲁棒性与可信度问题，同时也涉及滥用、失业以及对人类福祉的威胁等其他风险 (§ 6.3)；(4) 扩大智能体数量的潜在优势与挑战，探讨了预设规模与动态扩展的方法 (§ 6.4)；(5) 若干未解难题，如关于基于LLM的智能体是否是通向AGI的潜在路径的争论，从虚拟仿真环境到实体环境的挑战，AI智能体中的集体智能，以及智能体即服务（Agent as a Service）的问题 (§ 6.5)。总之，我们希望本文能为相关领域的研究人员和实践者提供启发。

2 背景

在本节中，我们提供关键的背景信息，为后续内容奠定基础 (§ 2.1)。我们首先讨论人工智能代理的起源，从哲学到人工智能领域，并结合关于人工代理存在的讨论 (§ 2.2)。随后，我们从技术发展趋势的角度总结人工智能代理的演变。最后，我们介绍代理的关键特性，并阐述为何大型语言模型（LLMs）适合作为人工智能代理“脑”的主要部分 (§ 2.3)。

2.1 人工智能代理的起源

“Agent”是一个具有悠久历史的概念，在许多领域中都曾被探索和诠释。在此，我们首先探讨其在哲学中的起源，讨论人工产物是否可以在哲学意义上具有“agency”，以及相关概念是如何引入人工智能领域的。

哲学中的代理人。代理人的核心思想具有哲学讨论的历史背景，其根源可以追溯到亚里士多德和休谟等有影响力的思想家。[5]一般而言，“代理人”是具有行动能力的实体，而“代理”一词则表示这种能力的行使或体现。[5]狭义上，“代理”通常用来指有意图的行为的执行；相应地，“代理人”则指具有欲望、信念、意图以及行动能力的实体。[32; 33; 34; 35]。需要注意的是，代理人不仅包括个体人类，还涵盖物理世界和虚拟世界中的其他实体。重要的是，代理人的概念涉及个体自主性，赋予其行使意志、做出选择和采取行动的能力，而非被动地对外部刺激作出反应。

From the perspective of philosophy, is artificial entities capable of agency? In a general sense, if we define agents as entities with the capacity to act, AI systems do exhibit a form of agency [5]. However, the term agent is more usually used to refer to entities or subjects that possess consciousness, intentionality, and the ability to act [32; 33; 34]. Within this framework, it’s not immediately clear whether artificial systems can possess agency, as it remains uncertain whether they possess internal states that form the basis for attributing desires, beliefs, and intentions. Some people argue that attributing psychological states like intention to artificial agents is a form of anthropomorphism and lacks scientific rigor [5; 36]. As Barandiaran et al. [36] stated, “Being specific about the requirements for agency has told us a lot about how much is still needed for the development of artificial forms of agency.” In contrast, there are also researchers who believe that, in certain circumstances, employing the intentional stance (that is, interpreting agent behavior in terms of intentions) can provide a better description, explanation and abstraction of the actions of artificial agents, much like it is done for humans [11; 37; 38].

With the advancement of language models, the potential emergence of artificial intentional agents appears more promising [24; 25; 39; 40; 41]. In a rigorous sense, language models merely function as conditional probability models, using input to predict the next token [42]. Different from this, humans incorporate social and perceptual context, and speak according to their mental states [43; 44]. Consequently, some researchers argue that the current paradigm of language modeling is not compatible with the intentional actions of an agent [30; 45]. However, there are also researchers who propose that language models can, in a narrow sense, serve as models of agents [46; 47]. They argue that during the process of context-based next-word prediction, current language models can sometimes infer approximate, partial representations of the beliefs, desires, and intentions held by the agent who generated the context. With these representations, the language models can then generate utterances like humans. To support their viewpoint, they conduct experiments to provide some empirical evidence [46; 48; 49].

Introduction of agents into AI. It might come as a surprise that researchers within the mainstream AI community devoted relatively minimal attention to concepts related to agents until the mid to late 1980s. Nevertheless, there has been a significant surge of interest in this topic within the realms of computer science and artificial intelligence communities since then [50; 51; 52; 53]. As Wooldridge et al. [4] stated, we can define AI by saying that it is a subfield of computer science that aims to design and build computer-based agents that exhibit aspects of intelligent behavior. So we can treat “agent” as a central concept in AI. When the concept of agent is introduced into the field of AI, its meaning undergoes some changes. In the realm of Philosophy, an agent can be a human, an animal, or even a concept or entity with autonomy [5]. However, in the field of artificial intelligence, an agent is a computational entity [4; 7]. Due to the seemingly metaphysical nature of concepts like consciousness and desires for computational entities [11], and given that we can only observe the behavior of the machine, many AI researchers, including Alan Turing, suggest temporarily setting aside the question of whether an agent is “actually” thinking or literally possesses a “mind” [3]. Instead, researchers employ other attributes to help describe an agent, such as properties of autonomy, reactivity, pro-activeness and social ability [4; 9]. There are also researchers who held that intelligence is “in the eye of the beholder”; it is not an innate, isolated property [15; 16; 54; 55]. In essence, an AI agent is not equivalent to a philosophical agent; rather, it is a concretization of the philosophical concept of an agent in the context of AI. In this paper, we treat AI agents as artificial entities that are capable of perceiving their surroundings using sensors, making decisions, and then taking actions in response using actuators [1; 4].

2.2 Technological Trends in Agent Research

The evolution of AI agents has undergone several stages, and here we take the lens of technological trends to review its development briefly.

Symbolic Agents. In the early stages of artificial intelligence research, the predominant approach utilized was symbolic AI, characterized by its reliance on symbolic logic [56; 57]. This approach employed logical rules and symbolic representations to encapsulate knowledge and facilitate reasoning processes. Early AI agents were built based on this approach [58], and they primarily focused on two problems: the transduction problem and the representation/reasoning problem [59]. These agents are aimed to emulate human thinking patterns. They possess explicit and interpretable reasoning

从哲学角度来看，人工实体是否具有能动性？一般而言，如果我们将能动体定义为具有行动能力的实体，人工智能系统确实表现出某种形式的能动性 [5]。然而，能动体一词更常用来指具有意识、意向性和行动能力的实体或主体 [32; 33; 34]。在此框架下，尚不清楚人工系统是否能够拥有能动性，因为尚不确定它们是否具有构成赋予欲望、信念和意图基础的内部状态。一些人认为，将意图等心理状态归于人工代理是一种拟人化，缺乏科学严谨性 [5; 36]。正如Barandiaran等人所述，“明确能动性的要求为我们提供了许多关于人工智能形式发展仍需努力的线索。”相比之下，也有研究者认为，在某些情况下，采用意向立场（即以意图解释代理行为）可以更好地描述、解释和抽象人工代理的行为，就像对人类所做的那样 [11; 37; 38]。

随着语言模型的不断发展，人工意向代理可能出现的潜力变得更加有希望 [24; 25; 39; 40; 41]。从严格意义上讲，语言模型仅仅作为条件概率模型，利用输入预测下一个标记 [42]。与此不同，人类结合社会和感知背景，根据他们的心理状态进行交流 [43;44]。因此，一些研究人员认为，当前的语言建模范式与代理的意向行为不兼容 [30; 45]。但也有研究人员提出，狭义上，语言模型可以作为代理的模型 [46; 47]。他们认为，在基于上下文的下一个词预测过程中，当前的语言模型有时可以推断出代理在生成上下文时所持有的信念、欲望和意图的近似、部分表示。借助这些表示，语言模型可以像人类一样生成话语。为了支持他们的观点，他们进行了实验以提供一些实证证据 [46; 48; 49]。

引入AI中的代理。或许令人惊讶的是，主流AI社区的研究人员直到20世纪80年代中后期才相对较少关注与代理相关的概念。然而，自那时起，在计算机科学和人工智能领域对此主题的兴趣显著增加 [50; 51; 52; 53]。正如 Wooldridge 等人所述， [4] 我们可以将AI定义为：它是计算机科学的一个子领域，旨在设计和构建表现出智能行为特征的基于计算机的代理。因此，我们可以将“代理”视为AI中的核心概念。当“代理”这一概念引入AI领域时，其含义发生了一些变化。在哲学领域，代理可以是人类、动物，甚至是具有自主性的概念或实体 [5]。然而，在人工智能领域，代理是一个计算实体 [4; 7]。由于诸如意识和欲望等概念在计算实体中具有似乎是形而上学的性质， [11]，而且鉴于我们只能观察机器的行为，许多AI研究人员，包括Alan Turing，建议暂时搁置代理是否“真正”思考或字面上拥有“心灵”的问题 [3]。相反，研究人员采用其他属性来帮助描述代理，例如自主性、反应性、主动性和社会能力 [4; 9]。也有研究者认为，智能是“在观察者眼中”的；它不是一种天生的、孤立的属性 [15; 16; 54; 55]。本质上，AI代理并不等同于哲学中的代理；它是在AI背景下对哲学代理概念的具体化。在本文中，我们将AI代理视为具有感知周围环境能力、做出决策并通过执行器采取行动的人工实体 [1; 4]。

2.2 代理研究的技术趋势

人工智能代理的演变经历了多个阶段，本文将以技术趋势为视角，简要回顾其发展历程。

符号代理。在人工智能研究的早期阶段，主要采用的方法是符号人工智能，其特点是依赖符号逻辑 [56; 57]。该方法利用逻辑规则和符号表示来封装知识并促进推理过程。早期的人工智能代理基于此方法构建 [58]，，主要关注两个问题：转换问题和表示/推理问题 [59]。这些代理旨在模拟人类的思维模式，具有明确且可解释的推理能力。

frameworks, and due to their symbolic nature, they exhibit a high degree of expressive capability [13; 14; 60]. A classic example of this approach is knowledge-based expert systems. However, symbolic agents faced limitations in handling uncertainty and large-scale real-world problems [19; 20]. Additionally, due to the intricacies of symbolic reasoning algorithms, it was challenging to find an efficient algorithm capable of producing meaningful results within a finite timeframe [20; 61].

Reactive agents. Different from symbolic agents, reactive agents do not use complex symbolic reasoning. Instead, they primarily focus on the interaction between the agent and its environment, emphasizing quick and real-time responses [15; 16; 20; 62; 63]. These agents are mainly based on a sense-act loop, efficiently perceiving and reacting to the environment. The design of such agents prioritizes direct input-output mappings rather than intricate reasoning and symbolic operations [52]. However, Reactive agents also have limitations. They typically require fewer computational resources, enabling quicker responses, but they might lack complex higher-level decision-making and planning capabilities.

Reinforcement learning-based agents. With the improvement of computational capabilities and data availability, along with a growing interest in simulating interactions between intelligent agents and their environments, researchers have begun to utilize reinforcement learning methods to train agents for tackling more challenging and complex tasks [17; 18; 64; 65]. The primary concern in this field is how to enable agents to learn through interactions with their environments, enabling them to achieve maximum cumulative rewards in specific tasks [21]. Initially, reinforcement learning (RL) agents were primarily based on fundamental techniques such as policy search and value function optimization, exemplified by Q-learning [66] and SARSA [67]. With the rise of deep learning, the integration of deep neural networks and reinforcement learning, known as Deep Reinforcement Learning (DRL), has emerged [68; 69]. This allows agents to learn intricate policies from high-dimensional inputs, leading to numerous significant accomplishments like AlphaGo [70] and DQN [71]. The advantage of this approach lies in its capacity to enable agents to autonomously learn in unknown environments, without explicit human intervention. This allows for its wide application in an array of domains, from gaming to robot control and beyond. Nonetheless, reinforcement learning faces challenges including long training times, low sample efficiency, and stability concerns, particularly when applied in complex real-world environments [21].

Agents with transfer learning and meta learning. Traditionally, training a reinforcement learning agent requires huge sample sizes and long training time, and lacks generalization capability [72; 73; 74; 75; 76]. Consequently, researchers have introduced transfer learning to expedite an agent’s learning on new tasks [77; 78; 79]. Transfer learning reduces the burden of training on new tasks and facilitates the sharing and migration of knowledge across different tasks, thereby enhancing learning efficiency, performance, and generalization capabilities. Furthermore, meta-learning has also been introduced to AI agents [80; 81; 82; 83; 84]. Meta-learning focuses on learning how to learn, enabling an agent to swiftly infer optimal policies for new tasks from a small number of samples [85]. Such an agent, when confronted with a new task, can rapidly adjust its learning approach by leveraging acquired general knowledge and policies, consequently reducing the reliance on a large volume of samples. However, when there exist significant disparities between source and target tasks, the effectiveness of transfer learning might fall short of expectations and there may exist negative transfer [86; 87]. Additionally, the substantial amount of pre-training and large sample sizes required by meta learning make it hard to establish a universal learning policy [81; 88].

Large language model-based agents. As large language models have demonstrated impressive emergent capabilities and have gained immense popularity [24; 25; 26; 41], researchers have started to leverage these models to construct AI agents [22; 27; 28; 89]. Specifically, they employ LLMs as the primary component of brain or controller of these agents and expand their perceptual and action space through strategies such as multimodal perception and tool utilization [90; 91; 92; 93; 94]. These LLM-based agents can exhibit reasoning and planning abilities comparable to symbolic agents through techniques like Chain-of-Thought (CoT) and problem decomposition [95; 96; 97; 98; 99; 100; 101]. They can also acquire interactive capabilities with the environment, akin to reactive agents, by learning from feedback and performing new actions [102; 103; 104]. Similarly, large language models undergo pre-training on large-scale corpora and demonstrate the capacity for few-shot and zero-shot generalization, allowing for seamless transfer between tasks without the need to update parameters [41; 105; 106; 107]. LLM-based agents have been applied to various real-world scenarios,

框架, 由于其符号性质, 表现出较高的表达能力[13; 14; 60]。这种方法的经典例子是基于知识的专家系统。然而, 符号代理在处理不确定性和大规模实际问题方面面临限制[19; 20]。此外, 由于符号推理算法的复杂性, 难以找到一种在有限时间内能够产生有意义结果的高效算法 [20; 61]。

反应型代理。不同于符号型代理, 反应型代理不使用复杂的符号推理, 而主要关注代理与环境之间的交互, 强调快速和实时的响应。这些代理主要基于感知-行动循环, 能够高效地感知和反应环境。这类代理的设计优先考虑直接的输入-输出映射, 而非复杂的推理和符号操作。然而, 反应型代理也存在局限性。它们通常需要较少的计算资源, 从而实现更快的响应, 但可能缺乏复杂的高级决策和规划能力。

基于强化学习的智能体。随着计算能力和数据可用性的提升, 以及对模拟智能体与环境交互的兴趣日益增长, 研究人员开始利用强化学习方法训练智能体, 以应对更具挑战性和复杂性的任务。该领域的主要关注点在于如何使智能体通过与环境的交互进行学习, 从而在特定任务中实现最大化的累积奖励。最初, 强化学习 (RL) 智能体主要基于策略搜索和价值函数优化等基础技术, 例如Q-learning和SARSA。随着深度学习的兴起, 深度神经网络与强化学习的结合, 即深度强化学习 (Deep Reinforcement Learning, DRL) 逐渐出现。这使得智能体能够从高维输入中学习复杂的策略, 取得了诸如AlphaGo和DQN等诸多重要成就。这一方法的优势在于其能够使智能体在未知环境中自主学习, 无需明确的人类干预, 从而在游戏、机器人控制等多个领域得到广泛应用。然而, 强化学习也面临诸如训练时间长、样本效率低以及在复杂实际环境中稳定性不足等挑战。

具有迁移学习和元学习的Agent。 传统上, 训练一个强化学习Agent需要大量的样本和长时间的训练, 且缺乏泛化能力 [72; 73; 74; 75; 76]。因此, 研究人员引入了迁移学习以加快Agent在新任务上的学习 [77; 78; 79]。迁移学习减轻了新任务训练的负担, 促进了知识在不同任务之间的共享和迁移, 从而提升了学习效率、性能和泛化能力。此外, 元学习也被引入到AI Agent中 [80; 81; 82; 83; 84]。元学习专注于学习如何学习, 使Agent能够从少量样本中迅速推断出最优策略[85]。当面对新任务时, 这样的Agent可以利用已获得的通用知识和策略, 快速调整其学习方法, 从而减少对大量样本的依赖。然而, 当源任务与目标任务存在显著差异时, 迁移学习的效果可能不尽如人意, 甚至可能出现负迁移 [86; 87]。此外, 元学习所需的大量预训练和大量样本使得建立通用学习策略变得困难 [81; 88]。

基于大规模语言模型的智能体。由于大规模语言模型展现出令人印象深刻的突现能力, 并获得了广泛的关注 [24; 25; 26; 41], 研究人员开始利用这些模型构建人工智能智能体 [22; 27; 28; 89]。具体而言, 他们将大规模语言模型作为这些智能体的核心组件或控制器, 并通过多模态感知和工具利用等策略扩展其感知和行动空间 [90; 91; 92; 93; 94]。这些基于大规模语言模型的智能体能够通过链式思维 (CoT) 和问题分解等技术展现出与符号智能体相当的推理和规划能力 [95; 96; 97; 98; 99; 100; 101]。它们还可以通过学习反馈并执行新动作, 获得类似反应型智能体的交互能力 [102; 103; 104]。类似地, 大规模语言模型经过大规模语料库的预训练, 展现出少样本和零样本泛化能力, 能够在任务之间实现无缝迁移, 无需更新参数 [41; 105; 106; 107]。基于大规模语言模型的智能体已被应用于各种实际场景,

such as software development [108; 109] and scientific research [110]. Due to their natural language comprehension and generation capabilities, they can interact with each other seamlessly, giving rise to collaboration and competition among multiple agents [108; 109; 111; 112]. Furthermore, research suggests that allowing multiple agents to coexist can lead to the emergence of social phenomena [22].

2.3 Why is LLM suitable as the primary component of an Agent’s brain?

As mentioned before, researchers have introduced several properties to help describe and define agents in the field of AI. Here, we will delve into some key properties, elucidate their relevance to LLMs, and thereby expound on why LLMs are highly suited to serve as the main part of brains of AI agents.

Autonomy. Autonomy means that an agent operates without direct intervention from humans or others and possesses a degree of control over its actions and internal states [4; 113]. This implies that an agent should not only possess the capability to follow explicit human instructions for task completion but also exhibit the capacity to initiate and execute actions independently. LLMs can demonstrate a form of autonomy through their ability to generate human-like text, engage in conversations, and perform various tasks without detailed step-by-step instructions [114; 115]. Moreover, they can dynamically adjust their outputs based on environmental input, reflecting a degree of adaptive autonomy [23; 27; 104]. Furthermore, they can showcase autonomy through exhibiting creativity like coming up with novel ideas, stories, or solutions that haven’t been explicitly programmed into them [116; 117]. This implies a certain level of self-directed exploration and decision-making. Applications like Auto-GPT [114] exemplify the significant potential of LLMs in constructing autonomous agents. Simply by providing them with a task and a set of available tools, they can autonomously formulate plans and execute them to achieve the ultimate goal.

Reactivity. Reactivity in an agent refers to its ability to respond rapidly to immediate changes and stimuli in its environment [9]. This implies that the agent can perceive alterations in its surroundings and promptly take appropriate actions. Traditionally, the perceptual space of language models has been confined to textual inputs, while the action space has been limited to textual outputs. However, researchers have demonstrated the potential to expand the perceptual space of LLMs using multimodal fusion techniques, enabling them to rapidly process visual and auditory information from the environment [25; 118; 119]. Similarly, it’s also feasible to expand the action space of LLMs through embodiment techniques [120; 121] and tool usage [92; 94]. These advancements enable LLMs to effectively interact with the real-world physical environment and carry out tasks within it. One major challenge is that LLM-based agents, when performing non-textual actions, require an intermediate step of generating thoughts or formulating tool usage in textual form before eventually translating them into concrete actions. This intermediary process consumes time and reduces the response speed. However, this aligns closely with human behavioral patterns, where the principle of “think before you act” is observed [122; 123].

Pro-activeness. Pro-activeness denotes that agents don’t merely react to their environments; they possess the capacity to display goal-oriented actions by proactively taking the initiative [9]. This property emphasizes that agents can reason, make plans, and take proactive measures in their actions to achieve specific goals or adapt to environmental changes. Although intuitively the paradigm of next token prediction in LLMs may not possess intention or desire, research has shown that they can implicitly generate representations of these states and guide the model’s inference process [46; 48; 49]. LLMs have demonstrated a strong capacity for generalized reasoning and planning. By prompting large language models with instructions like “let’s think step by step”, we can elicit their reasoning abilities, such as logical and mathematical reasoning [95; 96; 97]. Similarly, large language models have shown the emergent ability of planning in forms of goal reformulation [99; 124], task decomposition [98; 125], and adjusting plans in response to environmental changes [100; 126].

Social ability. Social ability refers to an agent’s capacity to interact with other agents, including humans, through some kind of agent-communication language [8]. Large language models exhibit strong natural language interaction abilities like understanding and generation [23; 127; 128]. Compared to structured languages or other communication protocols, such capability enables them to interact with other models or humans in an interpretable manner. This forms the cornerstone of social ability for LLM-based agents [22; 108]. Many researchers have demonstrated that LLM-based

《基于大型语言模型的代理的崛起与潜力：一项综述》例如软件开发 [108; 109] 以及科学研究 [110]。由于其自然语言理解与生成能力，它们可以实现相互无缝交互，促使多个代理之间出现合作与竞争 [108; 109; 111; 112]。此外，研究表明，允许多个代理共存可能导致社会现象的出现 [22]。

2.3 为什么大型语言模型（LLM）适合作为代理大脑的主要组成部分？

正如前文所述，研究人员在人工智能领域引入了若干属性，以帮助描述和定义智能体。在此，我们将深入探讨一些关键属性，阐明它们与大型语言模型（LLMs）的相关性，从而说明为何LLMs非常适合作为人工智能智能体大脑的主要组成部分。

自主性。 自主性意味着代理在没有人类或其他人直接干预的情况下运行，并且对其行为和内部状态具有一定的控制能力 [4; 113]这意味着代理不仅应具备按照明确的人类指令完成任务的能力，还应展现出自主发起和执行行动的能力。大型语言模型（LLMs）通过其生成类人文本、进行对话以及执行各种任务的能力，展示出一种自主性，而无需详细的逐步指令 [114; 115]此外，它们还能根据环境输入动态调整输出，反映出一定程度的适应性自主性 [23; 27; 104]此外，它们还能通过展现创造力，例如提出新颖的想法、故事或解决方案（这些未被明确编程进去），来体现自主性 [116; 117]这意味着一定程度的自主探索和决策能力。像Auto-GPT这样的应用 [114]，充分展示了LLMs在构建自主代理方面的巨大潜力。只需为其提供任务和一套可用工具，它们便能自主制定计划并执行，以实现最终目标。

Reactivity. 代理中的Reactivity指的是其对环境中即时变化和刺激做出快速反应的能力 [9]. 这意味着代理能够感知其周围环境的变化，并迅速采取适当的行动。传统上，语言模型的感知空间局限于文本输入，而行动空间则限制在文本输出。然而，研究人员已证明通过多模态融合技术可以扩展LLMs的感知空间，使其能够快速处理来自环境的视觉和听觉信息 [25; 118; 119]. 类似地，通过具身技术也可以扩展 LLMs 的行动空间 [120; 121] 以及工具使用 [92; 94]。这些进展使得LLMs能够有效地与现实世界的物理环境互动并执行任务。一个主要的挑战是，基于LLM的代理在执行非文本行为时，需要一个中间步骤，即生成思想或用文本形式制定工具使用计划，随后再将其转化为具体行动。这一中间过程耗时较长，降低了反应速度。然而，这与人类的行为模式密切相关，其中“先思考再行动”的原则被广泛观察到 [122; 123].

主动性。 主动性指的是代理不仅仅对其环境做出反应；它们具有通过主动采取行动展示目标导向行为的能力 [9]。该特性强调代理能够推理、制定计划，并采取主动措施以实现特定目标或适应环境变化。虽然直观上，LLMs中的下一个词预测范式可能不具备意图或欲望，但研究表明，它们可以隐式生成这些状态的表示，并引导模型的推理过程[46; 48; 49]。LLMs已展现出强大的泛化推理和规划能力。通过向大型语言模型提示诸如“让我们一步步思考”的指令，我们可以激发它们的推理能力，例如逻辑和数学推理 [95; 96; 97]。类似地，大型语言模型还展现出在目标重述 [99; 124], 任务分解 [98; 125], 以及根据环境变化调整计划 [100; 126]的涌现能力。

社会能力。Large language models 展示出强大的自然语言交互能力，如理解与生成；与结构化语言或其他通信协议相比，这种能力使它们能够以可解释的方式与其他模型或人类进行交互。这构成了基于大语言模型的代理的社会能力的基础。许多研究人员已经证明，基于大语言模型的

agents can enhance task performance through social behaviors such as collaboration and competition [108; 111; 129; 130]. By inputting specific prompts, LLMs can also play different roles, thereby simulating the social division of labor in the real world [109]. Furthermore, when we place multiple agents with distinct identities into a society, emergent social phenomena can be observed [22].

3 The Birth of An Agent: Construction of LLM-based Agents

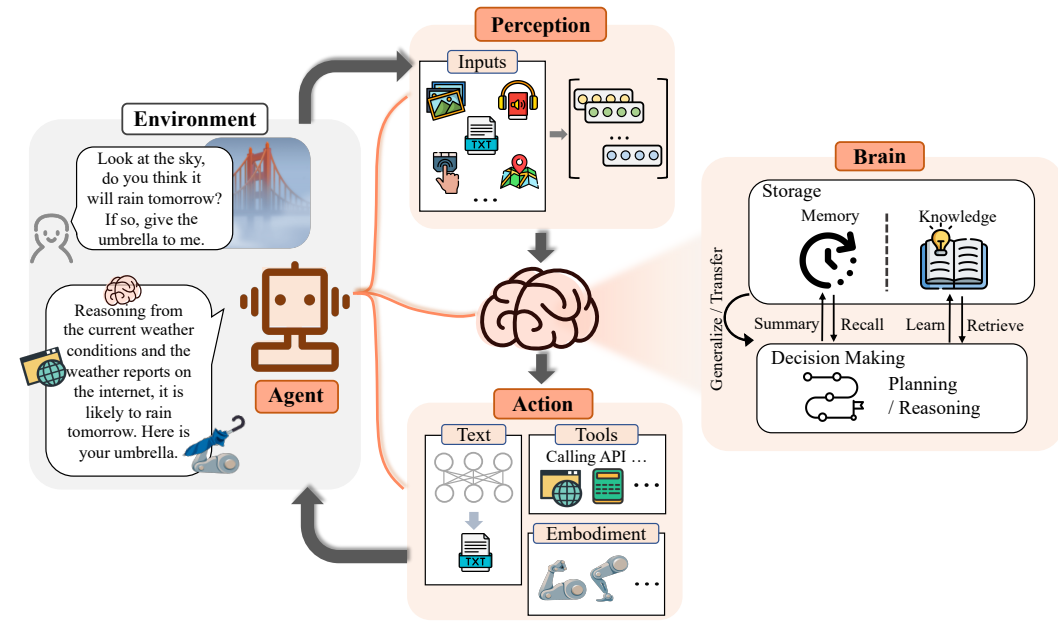


Figure 2: Conceptual framework of LLM-based agent with three components: brain, perception, and action. Serving as the controller, the brain module undertakes basic tasks like memorizing, thinking, and decision-making. The perception module perceives and processes multimodal information from the external environment, and the action module carries out the execution using tools and influences the surroundings. Here we give an example to illustrate the workflow: When a human asks whether it will rain, the perception module converts the instruction into an understandable representation for LLMs. Then the brain module begins to reason according to the current weather and the weather reports on the internet. Finally, the action module responds and hands the umbrella to the human. By repeating the above process, an agent can continuously get feedback and interact with the environment.

“Survival of the Fittest” [131] shows that if an individual wants to survive in the external environment, he must adapt to the surroundings efficiently. This requires him to be cognitive, able to perceive and respond to changes in the outside world, which is consistent with the definition of “agent” mentioned in §2.1. Inspired by this, we present a general conceptual framework of an LLM-based agent composed of three key parts: brain, perception, and action (see Figure 2). We first describe the structure and working mechanism of the brain, which is primarily composed of a large language model (§ 3.1). The brain is the core of an AI agent because it not only stores knowledge and memories but also undertakes indispensable functions like information processing and decision-making. It can present the process of reasoning and planning, and cope well with unseen tasks, exhibiting the intelligence of an agent. Next, we introduce the perception module (§ 3.2). Its core purpose is to broaden the agent’s perception space from a text-only domain to a multimodal sphere that includes textual, auditory, and visual modalities. This extension equips the agent to grasp and utilize information from its surroundings more effectively. Finally, we present the action module designed to expand the action space of an agent (§ 3.3). Specifically, we empower the agent with embodied action ability and tool-handling skills, enabling it to adeptly adapt to environmental changes, provide feedback, and even influence and mold the environment.

The framework can be tailored for different application scenarios, i.e. not every specific component will be used in all studies. In general, agents operate in the following workflow: First, the **perception**

代理可以通过合作与竞争等社会行为提升任务表现[108; 111; 129;]。通过输入特定提示，大型语言模型（LLMs）还可以扮演不同角色，从而模拟现实世界中的社会分工 [130]。此外，当我们将具有不同身份的多个代理置入一个社会中时，可以观察到涌现的社会现象 [109]。

3 代理的诞生：基于大语言模型的代理构建

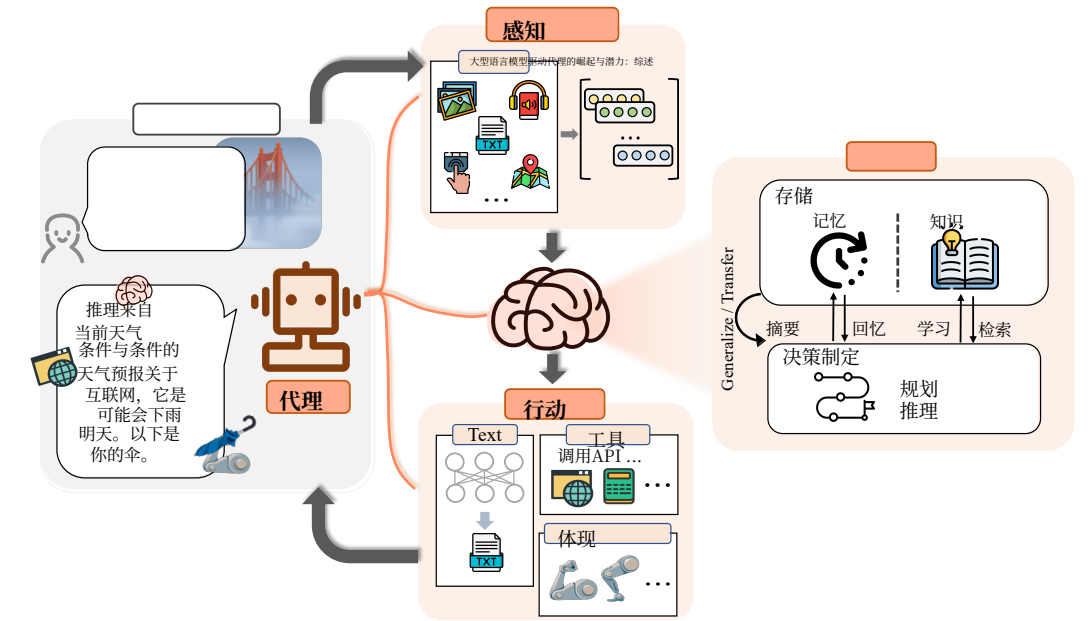


图2：基于大语言模型的代理的概念框架，包含脑、感知和行动三个组成部分。作为控制器，脑模块承担记忆、思考和决策等基本任务。感知模块感知并处理来自外部环境的多模态信息，行动模块则通过工具执行操作并影响环境。这里举例说明工作流程：当人类询问是否会下雨时，感知模块将指令转换为大语言模型可以理解的表示。然后，脑模块根据当前天气和网络上的天气预报进行推理。最后，行动模块做出回应，将伞递给人类。通过重复上述过程，代理可以不断获得反馈并与环境互动。

“适者生存” [131] 显示，如果个体想在外部环境中生存，就必须高效地适应周围环境。这要求其具备认知能力，能够感知并应对外部世界的变化，这与 § 2.1中提到的“智能体”定义一致。受到此启发，我们提出了一个由三个关键部分组成的基于大规模语言模型（LLM）的智能体的通用概念框架：大脑、感知和行动（见图2）。我们首先描述大脑的结构和工作机制，大脑主要由大规模语言模型构成（§ 3.1）。大脑是AI智能体的核心，因为它不仅存储知识和记忆，还承担信息处理和决策等不可或缺的功能。它能够展现推理和规划的过程，并能应对未见过的任务，体现出智能体的智能。接下来，我们介绍感知模块（§ 3.2）。其核心目标是将智能体的感知空间从纯文本领域扩展到包括文本、听觉和视觉多模态的空间。这一扩展使智能体能够更有效地理解和利用其周围环境中的信息。最后，我们提出行动模块（§ 3.3），旨在扩展智能体的行动空间。具体而言，我们赋予智能体具象化的行动能力和工具操作技能，使其能够灵活适应环境变化、提供反馈，甚至影响和塑造环境。

该框架可以根据不同的应用场景进行定制，即并非所有研究中都会使用每个特定组件。一般而言，代理的工作流程如下：首先，**感知**

module, corresponding to human sensory systems such as the eyes and ears, perceives changes in the external environment and then converts multimodal information into an understandable representation for the agent. Subsequently, the **brain** module, serving as the control center, engages in information processing activities such as thinking, decision-making, and operations with storage including memory and knowledge. Finally, the **action** module, corresponding to human limbs, carries out the execution with the assistance of tools and leaves an impact on the surroundings. By repeating the above process, an agent can continuously get feedback and interact with the environment.

3.1 Brain

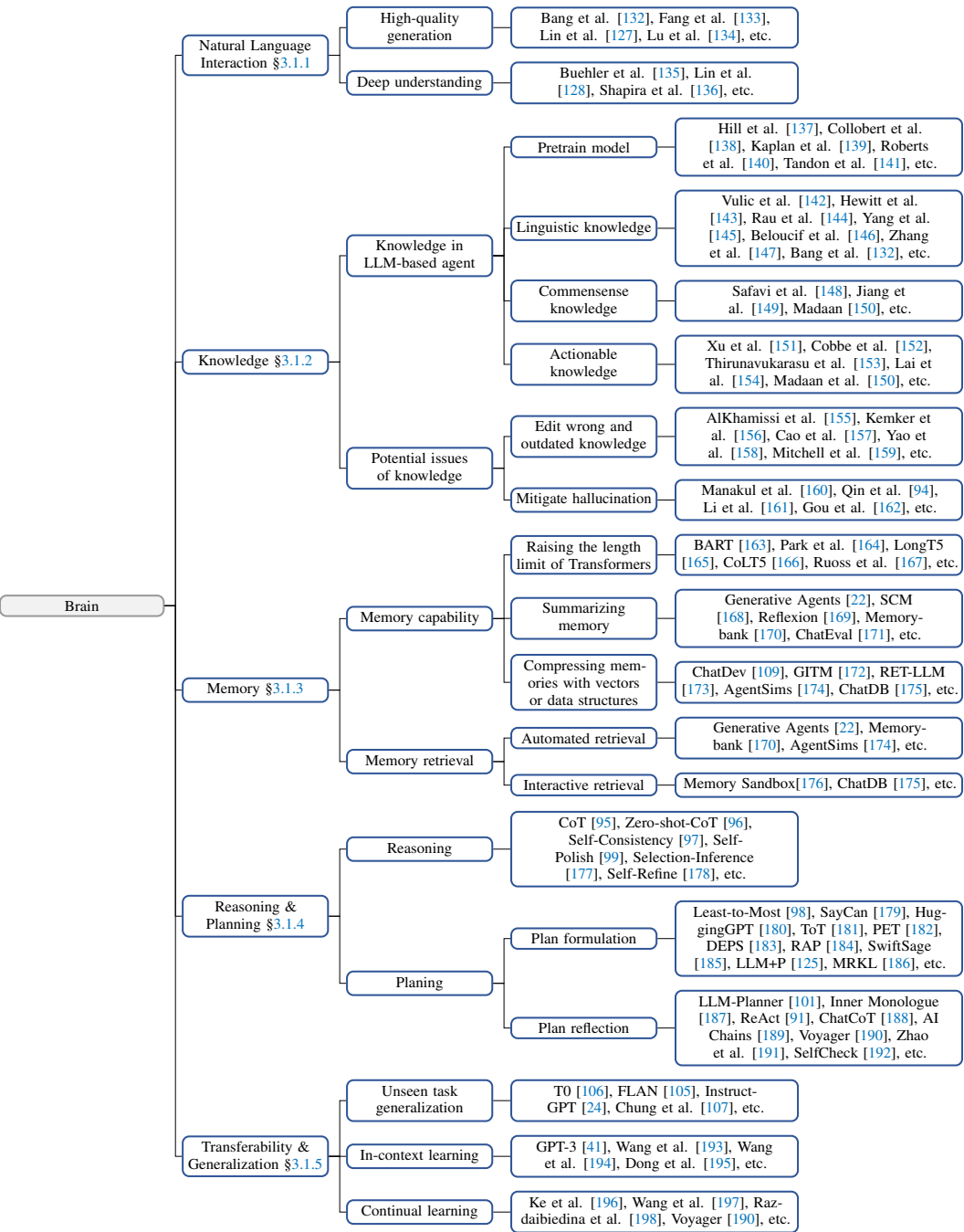


Figure 3: Typology of the brain module.

模块，对应于人类的感官系统如眼睛和耳朵，感知外部环境的变化，然后将多模态信息转换为对代理人而言可理解的表征。随后，**大脑**模块，作为控制中心，进行信息处理活动，如思考、决策和操作，包含存储的记忆和知识。最后，**行动**模块，对应于人类的肢体，在工具的协助下执行任务，并对环境产生影响。通过重复上述过程，代理人可以不断获得反馈并与环境进行交互。

3.1 大脑

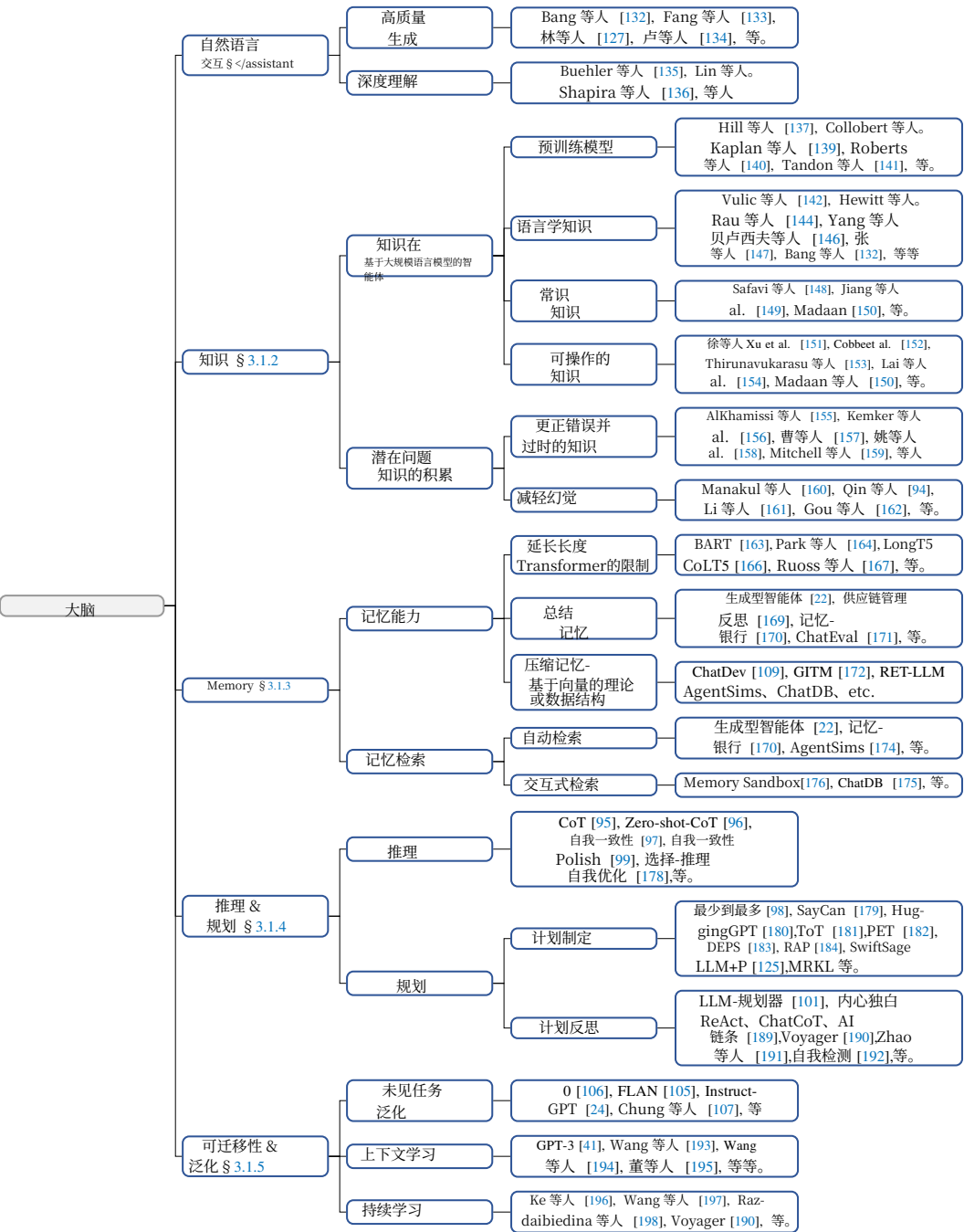


图3: 大脑模块的类型学。

The human brain is a sophisticated structure comprised of a vast number of interconnected neurons, capable of processing various information, generating diverse thoughts, controlling different behaviors, and even creating art and culture [199]. Much like humans, the brain serves as the central nucleus of an AI agent, primarily composed of a large language model.

Operating mechanism. To ensure effective communication, the ability to engage in natural language interaction (§3.1.1) is paramount. After receiving the information processed by the perception module, the brain module first turns to storage, retrieving in knowledge (§3.1.2) and recalling from memory (§3.1.3). These outcomes aid the agent in devising plans, reasoning, and making informed decisions (§3.1.4). Additionally, the brain module may memorize the agent’s past observations, thoughts, and actions in the form of summaries, vectors, or other data structures. Meanwhile, it can also update the knowledge such as common sense and domain knowledge for future use. The LLM-based agent may also adapt to unfamiliar scenarios with its inherent generalization and transfer ability (§3.1.5). In the subsequent sections, we delve into a detailed exploration of these extraordinary facets of the brain module as depicted in Figure 3.

3.1.1 Natural Language Interaction

As a medium for communication, language contains a wealth of information. In addition to the intuitively expressed content, there may also be the speaker’s beliefs, desires, and intentions hidden behind it [200]. Thanks to the powerful natural language understanding and generation capabilities inherent in LLMs [25; 201; 202; 203], agents can proficiently engage in not only basic interactive conversations [204; 205; 206] in multiple languages [132; 202] but also exhibit in-depth comprehension abilities, which allow humans to easily understand and interact with agents [207; 208]. Besides, LLM-based agents that communicate in natural language can earn more trust and cooperate more effectively with humans [130].

Multi-turn interactive conversation. The capability of multi-turn conversation is the foundation of effective and consistent communication. As the core of the brain module, LLMs, such as GPT series [40; 41; 201], LLaMA series [201; 209] and T5 series [107; 210], can understand natural language and generate coherent and contextually relevant responses, which helps agents to comprehend better and handle various problems [211]. However, even humans find it hard to communicate without confusion in one sitting, so multiple rounds of dialogue are necessary. Compared with traditional text-only reading comprehension tasks like SQuAD [212], multi-turn conversations (1) are interactive, involving multiple speakers, and lack continuity; (2) may involve multiple topics, and the information of the dialogue may also be redundant, making the text structure more complex [147]. In general, the multi-turn conversation is mainly divided into three steps: (1) Understanding the history of natural language dialogue, (2) Deciding what action to take, and (3) Generating natural language responses. LLM-based agents are capable of continuously refining outputs using existing information to conduct multi-turn conversations and effectively achieve the ultimate goal [132; 147].

High-quality natural language generation. Recent LLMs show exceptional natural language generation capabilities, consistently producing high-quality text in multiple languages [132; 213]. The coherency [214] and grammatical accuracy [133] of LLM-generated content have shown steady enhancement, evolving progressively from GPT-3 [41] to InstructGPT [24], and culminating in GPT-4 [25]. See et al. [214] empirically affirm that these language models can “adapt to the style and content of the conditioning text” [215]. And the results of Fang et al. [133] suggest that ChatGPT excels in grammar error detection, underscoring its powerful language capabilities. In conversational contexts, LLMs also perform well in key metrics of dialogue quality, including content, relevance, and appropriateness [127]. Importantly, they do not merely copy training data but display a certain degree of creativity, generating diverse texts that are equally novel or even more novel than the benchmarks crafted by humans [216]. Meanwhile, human oversight remains effective through the use of controllable prompts, ensuring precise control over the content generated by these language models [134].

Intention and implication understanding. Although models trained on the large-scale corpus are already intelligent enough to understand instructions, most are still incapable of emulating human dialogues or fully leveraging the information conveyed in language [217]. Understanding the implied meanings is essential for effective communication and cooperation with other intelligent agents [135].

人脑是一种复杂的结构，由大量相互连接的神经元组成，能够处理各种信息，产生多样的思想，控制不同的行为，甚至创造艺术和文化。，与人类类似，大脑作为人工智能代理的核心部分，主要由大型语言模型构成。

操作机制。 为了确保有效的沟通，具备自然语言交互能力（§3.1.1）至关重要。在感知模块处理信息后，脑模块首先进行存储，检索知识（§3.1.2）并从记忆中回忆（§3.1.3）。这些结果帮助代理制定计划、推理和做出明智的决策（§3.1.4）。此外，脑模块还可能以摘要、向量或其他数据结构的形式记忆代理的过去观察、思考和行为。同时，它还能更新常识和领域知识等知识，以备将来使用。基于大型语言模型的代理还具有固有的泛化和迁移能力（§3.1.5），能够适应不熟悉的场景。在后续章节中，我们将对图3所示的脑模块的这些卓越特性进行详细探讨。

3.1.1 自然语言交互

作为一种交流媒介，语言蕴含着丰富的信息。除了直观表达的内容外，背后还可能隐藏着说话者的信念、欲望和意图。得益于大型语言模型（LLMs）固有的强大自然语言理解与生成能力，代理不仅能够熟练进行多语言的基础互动对话，还展现出深层次的理解能力，使人类能够轻松理解与代理进行交互。此外，基于LLMs的自然语言交流代理能够赢得更多信任，并与人类更高效地合作。

多轮交互对话。 多轮对话能力是实现有效且连贯沟通的基础。作为大脑模块的核心，LLMs（如GPT系列）<v5>、LLaMA系列</v13>以及T5系列<v21>，能够理解自然语言并生成连贯且具有上下文相关性的响应，这有助于代理更好地理解 and 处理各种问题< v27>。然而，即使是人类也难以在一次对话中完全避免混淆，因此需要多轮对话。与传统的纯文本阅读理解任务（如SQuAD）相比，<v30>多轮对话（1）具有交互性，涉及多个发言者，且缺乏连续性；（2）可能涉及多个主题，且对话中的信息可能存在冗余，使文本结构更为复杂< v33>。一般而言，多轮对话主要分为三个步骤：（1）理解自然语言对话的历史；（2）决定采取的行动；（3）生成自然语言响应。基于LLM的代理能够利用现有信息不断优化输出，以进行多轮对话，并有效实现最终目标< v36>。

高质量的自然语言生成。近期的LLMs展现出卓越的自然语言生成能力，持续在多种语言 [132; 213]中生成高质量文本。LLM生成内容的连贯性 [214] 和语法准确性 [133] 已显示出稳步提升，逐步从GPT-3 [41]演变到InstructGPT [24], , 最终达到GPT-4[25]。见et al. [214] 实证确认这些语言模型能够“适应条件文本的风格和内容” [215]。Fang等人 [133] 的结果表明，ChatGPT在语法错误检测方面表现出色，凸显其强大的语言能力。在对话场景中，LLMs在对话质量的关键指标上也表现良好，包括内容、相关性和适当性 [127]。重要的是，它们不仅仅复制训练数据，还展现出一定程度的创造力，生成的多样化文本同样具有新颖性，甚至比人类设计的基准更具新颖性 [216]。同时，通过可控提示的使用，人类监督仍然有效，确保对这些语言模型生成内容的精确控制 [134]。

意图与含义理解。虽然在大规模语料库上训练的模型已经具有足够的智能来理解指令，但大多数仍无法模拟人类对话或充分利用语言中传达的信息 [217]。理解隐含的意义对于与其他智能体进行有效沟通与合作至关重要 [135]。

and enables one to interpret others’ feedback. The emergence of LLMs highlights the potential of foundation models to understand human intentions, but when it comes to vague instructions or other implications, it poses a significant challenge for agents [94; 136]. For humans, grasping the implied meanings from a conversation comes naturally, whereas for agents, they should formalize implied meanings into a reward function that allows them to choose the option in line with the speaker’s preferences in unseen contexts [128]. One of the main ways for reward modeling is inferring rewards based on feedback, which is primarily presented in the form of comparisons [218] (possibly supplemented with reasons [219]) and unconstrained natural language [220]. Another way involves recovering rewards from descriptions, using the action space as a bridge [128]. Jeon et al. [221] suggests that human behavior can be mapped to a choice from an implicit set of options, which helps to interpret all the information in a single unifying formalism. By utilizing their understanding of context, agents can take highly personalized and accurate action, tailored to specific requirements.

3.1.2 Knowledge

Due to the diversity of the real world, many NLP researchers attempt to utilize data that has a larger scale. This data usually is unstructured and unlabeled [137; 138], yet it contains enormous knowledge that language models could learn. In theory, language models can learn more knowledge as they have more parameters [139], and it is possible for language models to learn and comprehend everything in natural language. Research [140] shows that language models trained on a large-scale dataset can encode a wide range of knowledge into their parameters and respond correctly to various types of queries. Furthermore, the knowledge can assist LLM-based agents in making informed decisions [222]. All of this knowledge can be roughly categorized into the following types:

- **Linguistic knowledge.** Linguistic knowledge [142; 143; 144] is represented as a system of constraints, a grammar, which defines all and only the possible sentences of the language. It includes morphology, syntax, semantics [145; 146], and pragmatics. Only the agents that acquire linguistic knowledge can comprehend sentences and engage in multi-turn conversations [147]. Moreover, these agents can acquire multilingual knowledge [132] by training on datasets that contain multiple languages, eliminating the need for extra translation models.
- **Commonsense knowledge.** Commonsense knowledge [148; 149; 150] refers to general world facts that are typically taught to most individuals at an early age. For example, people commonly know that medicine is used for curing diseases, and umbrellas are used to protect against rain. Such information is usually not explicitly mentioned in the context. Therefore, the models lacking the corresponding commonsense knowledge may fail to grasp or misinterpret the intended meaning [141]. Similarly, agents without commonsense knowledge may make incorrect decisions, such as not bringing an umbrella when it rains heavily.
- **Professional domain knowledge.** Professional domain knowledge refers to the knowledge associated with a specific domain like programming [151; 154; 150], mathematics [152], medicine [153], etc. It is essential for models to effectively solve problems within a particular domain [223]. For example, models designed to perform programming tasks need to possess programming knowledge, such as code format. Similarly, models intended for diagnostic purposes should possess medical knowledge like the names of specific diseases and prescription drugs.

Although LLMs demonstrate excellent performance in acquiring, storing, and utilizing knowledge [155], there remain potential issues and unresolved problems. For example, the knowledge acquired by models during training could become outdated or even be incorrect from the start. A simple way to address this is retraining. However, it requires advanced data, extensive time, and computing resources. Even worse, it can lead to catastrophic forgetting [156]. Therefore, some researchers[157; 158; 159] try editing LLMs to locate and modify specific knowledge stored within the models. This involved unloading incorrect knowledge while simultaneously acquiring new knowledge. Their experiments show that this method can partially edit factual knowledge, but its underlying mechanism still requires further research. Besides, LLMs may generate content that conflicts with the source or factual information [224], a phenomenon often referred to as hallucinations [225]. It is one of the critical reasons why LLMs can not be widely used in factually rigorous tasks. To tackle this issue, some researchers [160] proposed a metric to measure the level of hallucinations and provide developers with an effective reference to evaluate the trustworthiness of LLM outputs. Moreover, some researchers[161; 162] enable LLMs to utilize external tools[94; 226; 227] to avoid incorrect

并使其能够解释他人的反馈。LLMs的出现凸显了基础模型理解人类意图的潜力，但在面对模糊指令或其他隐含意义时，给代理带来了重大挑战；对于人类而言，从对话中把握隐含意义是自然而然的，而对于代理，则应将隐含意义形式化为奖励函数，以便在未见过的情境中选择符合说话者偏好的选项。一种主要的奖励建模方式是基于反馈推断奖励，反馈主要以比较的形式呈现（可能辅以理由）和非限制性的自然语言。另一种方式是通过描述恢复奖励，利用动作空间作为桥梁。Jeon等人提出，人类行为可以映射为从一组隐含选项中做出的选择，这有助于用统一的形式主义解释所有信息。通过利用对上下文的理解，代理可以采取高度个性化和精准的行动，以满足特定需求。

3.1.2 知识

由于现实世界的多样性，许多自然语言处理研究者试图利用规模更大的数据。这些数据通常是非结构化且未标注的 [137; 138], 但它包含了大量语言模型可以学习的知识。从理论上讲，随着参数数量的增加，语言模型可以学习更多的知识 [139], 并且有可能学习和理解自然语言中的所有内容。研究 [140] 表明，在大规模数据集上训练的语言模型能够将广泛的知识编码到其参数中，并对各种类型的查询做出正确的响应。此外，这些知识可以帮助基于大型语言模型的代理做出明智的决策[222]。所有这些知识大致可以归类为以下几类：

语言学知识。语言学知识被表示为一套约束系统，即语法，定义了该语言所有且仅有的可能句子。它包括形态学、句法学、语义学和语用学。只有获得语言学知识的智能体才能理解句子并进行多轮对话。此外，这些智能体还可以通过在包含多种语言的数据集上训练，获得多语言知识，从而无需额外的翻译模型。

常识知识。常识知识指的是通常在早期教育中教授给大多数人的关于世界的普遍事实。例如，人们普遍知道药物用于治愈疾病，雨伞用于遮挡雨水。这类信息通常不会在上下文中明确提及。因此，缺乏相应常识知识的模型可能无法理解或误解其意图。同样，没有常识知识的智能体可能会做出错误的决策，例如在大雨时不带伞。

专业领域知识。专业领域知识是指与特定领域相关的知识，例如编程、数学、医学等。它对于模型在特定领域内有效解决问题至关重要。例如，设计用于执行编程任务的模型需要具备编程知识，如代码格式。同样，用于诊断的模型应具备医学知识，如特定疾病和处方药的名称。

虽然LLMs在获取、存储和利用知识方面表现出色，但仍存在潜在问题和未解决的难题。例如，模型在训练过程中获得的知识可能会变得过时，甚至从一开始就存在错误。一种简单的解决方法是重新训练，但这需要先进的数据、耗费大量时间和计算资源。更糟糕的是，这可能导致灾难性遗忘。因此，一些研究人员尝试编辑LLMs，以定位并修改模型中存储的特定知识。这涉及在卸载错误知识的同时获取新知识。他们的实验表明，这种方法可以部分编辑事实知识，但其底层机制仍需进一步研究。此外，LLMs可能会生成与源信息或事实相冲突的内容，这种现象通常被称为“幻觉”。这是LLMs无法在事实严谨任务中广泛应用的关键原因之一。为解决这一问题，一些研究人员提出了一种衡量幻觉程度的指标，为开发者提供了评估LLM输出可信度的有效参考。此外，还有研究者致力于使LLMs能够利用外部工具，以避免生成错误信息。

knowledge. Both of these methods can alleviate the impact of hallucinations, but further exploration of more effective approaches is still needed.

3.1.3 Memory

In our framework, “memory” stores sequences of the agent’s past observations, thoughts and actions, which is akin to the definition presented by Nuxoll et al. [228]. Just as the human brain relies on memory systems to retrospectively harness prior experiences for strategy formulation and decision-making, agents necessitate specific memory mechanisms to ensure their proficient handling of a sequence of consecutive tasks [229; 230; 231]. When faced with complex problems, memory mechanisms help the agent to revisit and apply antecedent strategies effectively. Furthermore, these memory mechanisms enable individuals to adjust to unfamiliar environments by drawing on past experiences.

With the expansion of interaction cycles in LLM-based agents, two primary challenges arise. The first pertains to the sheer length of historical records. LLM-based agents process prior interactions in natural language format, appending historical records to each subsequent input. As these records expand, they might surpass the constraints of the Transformer architecture that most LLM-based agents rely on. When this occurs, the system might truncate some content. The second challenge is the difficulty in extracting relevant memories. As agents amass a vast array of historical observations and action sequences, they grapple with an escalating memory burden. This makes establishing connections between related topics increasingly challenging, potentially causing the agent to misalign its responses with the ongoing context.

Methods for better memory capability. Here we introduce several methods to enhance the memory of LLM-based agents.

- **Raising the length limit of Transformers.** The first method tries to address or mitigate the inherent sequence length constraints. The Transformer architecture struggles with long sequences due to these intrinsic limits. As sequence length expands, computational demand grows exponentially due to the pairwise token calculations in the self-attention mechanism. Strategies to mitigate these length restrictions encompass text truncation [163; 164; 232], segmenting inputs [233; 234], and emphasizing key portions of text [235; 236; 237]. Some other works modify the attention mechanism to reduce complexity, thereby accommodating longer sequences [238; 165; 166; 167].
- **Summarizing memory.** The second strategy for amplifying memory efficiency hinges on the concept of memory summarization. This ensures agents effortlessly extract pivotal details from historical interactions. Various techniques have been proposed for summarizing memory. Using prompts, some methods succinctly integrate memories [168], while others emphasize reflective processes to create condensed memory representations [22; 239]. Hierarchical methods streamline dialogues into both daily snapshots and overarching summaries [170]. Notably, specific strategies translate environmental feedback into textual encapsulations, bolstering agents’ contextual grasp for future engagements [169]. Moreover, in multi-agent environments, vital elements of agent communication are captured and retained [171].
- **Compressing memories with vectors or data structures.** By employing suitable data structures, intelligent agents boost memory retrieval efficiency, facilitating prompt responses to interactions. Notably, several methodologies lean on embedding vectors for memory sections, plans, or dialogue histories [109; 170; 172; 174]. Another approach translates sentences into triplet configurations [173], while some perceive memory as a unique data object, fostering varied interactions [176]. Furthermore, ChatDB [175] and DB-GPT [240] integrate the LLMrollers with SQL databases, enabling data manipulation through SQL commands.

Methods for memory retrieval. When an agent interacts with its environment or users, it is imperative to retrieve the most appropriate content from its memory. This ensures that the agent accesses relevant and accurate information to execute specific actions. An important question arises: How can an agent select the most suitable memory? Typically, agents retrieve memories in an automated manner [170; 174]. A significant approach in automated retrieval considers three metrics: Recency, Relevance, and Importance. The memory score is determined as a weighted combination of these metrics, with memories having the highest scores being prioritized in the model’s context [22].

知识。这两种方法都可以减轻幻觉的影响，但仍需进一步探索更有效的方法。

3.1.3 记忆

在我们的框架中，“记忆”存储代理过去的观察、思考和行动的序列，这类似于Nuxoll等人提出的定义。正如人类大脑依赖记忆系统以回顾性地利用先前的经验进行策略制定和决策，代理也需要特定的记忆机制以确保其高效处理一系列连续任务。当面对复杂问题时，记忆机制帮助代理有效地回顾和应用先前的策略。此外，这些记忆机制使个体能够通过借鉴过去的经验，适应不熟悉的环境。

随着基于大规模语言模型（LLM）代理的交互周期扩展，出现了两个主要挑战。第一项涉及历史记录的长度问题。基于LLM的代理以自然语言格式处理先前的交互，将历史记录附加到每次后续输入中。随着这些记录的增长，可能会超出大多数LLM代理所依赖的Transformer架构的限制。当发生这种情况时，系统可能会截断部分内容。第二项挑战是提取相关记忆的困难。随着代理积累大量的历史观察和行动序列，它们面临着日益增加的记忆负担。这使得建立相关主题之间的联系变得愈发困难，可能导致代理在响应中与当前上下文出现偏差。

Methods for better memory capability. 这里我们介绍几种增强记忆能力的方法。
of L 基于大语言模型的人工智能体

提高Transformer的长度限制。 第一种方法试图解决或缓解固有的序列长度限制。由于这些内在限制，Transformer架构在处理长序列时面临挑战。随着序列长度的增加，由于自注意力机制中成对的标记计算，计算需求呈指数增长。缓解这些长度限制的策略包括文本截断、输入分段以及强调文本的关键部分；一些其他工作则通过修改注意力机制以降低复杂度，从而适应更长的序列。

• **总结记忆。** 增强记忆效率的第二种策略依赖于记忆总结的概念。这确保了代理能够轻松提取历史交互中的关键细节。已有多项记忆总结技术被提出。利用提示，一些方法简洁地整合记忆 [168]，而另一些则强调反思过程以创建浓缩的记忆表示 [22; 239]。层级方法将对话简化为日常快照和总体摘要 [170]。值得注意的是，特定策略将环境反馈转化为文本封装，从而增强代理对未来交互的上下文理解 [169]。此外，在多代理环境中，关键的代理通信要素被捕捉并保留 [171]。

• **使用向量或数据结构压缩记忆。** 通过采用合适的数据结构，智能体提升了记忆检索效率，便于对交互做出迅速响应。值得注意的是，几种方法依赖于嵌入向量，用于记忆部分、计划或对话历史 [109; 170; 172; 174]。另一种方法将句子转化为三元组配置[173]，而一些人将记忆视为一个独特的数据对象，从而促进多样化的交互 [176]。此外，ChatDB [175] 和 DB-GPT [240] 将LLM控制器与SQL数据库集成，通过SQL命令实现数据操作。

记忆检索方法。 当智能体与环境或用户交互时，检索其记忆中最合适的内容至关重要。这确保智能体能够访问相关且准确的信息，以执行特定操作。一个重要的问题是：智能体如何选择最合适的记忆？通常，智能体以自动化方式检索记忆 [170; 174]。在自动检索中，一个重要的方法考虑三个指标：新颖性、相关性和重要性。记忆得分通过这些指标的加权组合确定，得分最高的记忆在模型的上下文中被优先考虑 [22]。

Some research introduces the concept of interactive memory objects, which are representations of dialogue history that can be moved, edited, deleted, or combined through summarization. Users can view and manipulate these objects, influencing how the agent perceives the dialogue [176]. Similarly, other studies allow for memory operations like deletion based on specific commands provided by users [175]. Such methods ensure that the memory content aligns closely with user expectations.

3.1.4 Reasoning and Planning

Reasoning. Reasoning, underpinned by evidence and logic, is fundamental to human intellectual endeavors, serving as the cornerstone for problem-solving, decision-making, and critical analysis [241; 242; 243]. Deductive, inductive, and abductive are the primary forms of reasoning commonly recognized in intellectual endeavor [244]. For LLM-based agents, like humans, reasoning capacity is crucial for solving complex tasks [25].

Differing academic views exist regarding the reasoning capabilities of large language models. Some argue language models possess reasoning during pre-training or fine-tuning [244], while others believe it emerges after reaching a certain scale in size [26; 245]. Specifically, the representative Chain-of-Thought (CoT) method [95; 96] has been demonstrated to elicit the reasoning capacities of large language models by guiding LLMs to generate rationales before outputting the answer. Some other strategies have also been presented to enhance the performance of LLMs like self-consistency [97], self-polish [99], self-refine [178] and selection-inference [177], among others. Some studies suggest that the effectiveness of step-by-step reasoning can be attributed to the local statistical structure of training data, with locally structured dependencies between variables yielding higher data efficiency than training on all variables [246].

Planning. Planning is a key strategy humans employ when facing complex challenges. For humans, planning helps organize thoughts, set objectives, and determine the steps to achieve those objectives [247; 248; 249]. Just as with humans, the ability to plan is crucial for agents, and central to this planning module is the capacity for reasoning [250; 251; 252]. This offers a structured thought process for agents based on LLMs. Through reasoning, agents deconstruct complex tasks into more manageable sub-tasks, devising appropriate plans for each [253; 254]. Moreover, as tasks progress, agents can employ introspection to modify their plans, ensuring they align better with real-world circumstances, leading to adaptive and successful task execution.

Typically, planning comprises two stages: plan formulation and plan reflection.

- **Plan formulation.** During the process of plan formulation, agents generally decompose an overarching task into numerous sub-tasks, and various approaches have been proposed in this phase. Notably, some works advocate for LLM-based agents to decompose problems comprehensively in one go, formulating a complete plan at once and then executing it sequentially [98; 179; 255; 256]. In contrast, other studies like the CoT-series employ an adaptive strategy, where they plan and address sub-tasks one at a time, allowing for more fluidity in handling intricate tasks in their entirety [95; 96; 257]. Additionally, some methods emphasize hierarchical planning [182; 185], while others underscore a strategy in which final plans are derived from reasoning steps structured in a tree-like format. The latter approach argues that agents should assess all possible paths before finalizing a plan [97; 181; 184; 258; 184]. While LLM-based agents demonstrate a broad scope of general knowledge, they can occasionally face challenges when tasked with situations that require expertise knowledge. Enhancing these agents by integrating them with planners of specific domains has shown to yield better performance [125; 130; 186; 259].
- **Plan reflection.** Upon formulating a plan, it’s imperative to reflect upon and evaluate its merits. LLM-based agents leverage internal feedback mechanisms, often drawing insights from pre-existing models, to hone and enhance their strategies and planning approaches [169; 178; 188; 192]. To better align with human values and preferences, agents actively engage with humans, allowing them to rectify some misunderstandings and assimilate this tailored feedback into their planning methodology [108; 189; 190]. Furthermore, they could draw feedback from tangible or virtual surroundings, such as cues from task accomplishments or post-action observations, aiding them in revising and refining their plans [91; 101; 187; 191; 260].

一些研究引入了交互式记忆对象的概念，这些对象是对对话历史的表示，可以通过总结进行移动、编辑、删除或组合。用户可以查看并操作这些对象，从而影响代理对对话的感知 [176]。类似地，其他研究也允许根据用户提供的特定指令进行记忆操作，如删除 [175]。这些方法确保记忆内容与用户预期高度一致。

3.1.4 推理与规划

推理。推理以证据和逻辑为基础，是人类智力活动的核心，作为解决问题、决策和批判性分析的基石；242；推理的主要形式包括演绎、归纳和溯因，通常在智力活动中被广泛认可。对于基于大规模语言模型的代理而言，像人类一样，推理能力对于解决复杂任务至关重要。

关于大型语言模型的推理能力，学术界存在不同的观点。一些观点认为语言模型在预训练或微调过程中具备推理能力 [244]，而另一些则认为推理能力是在达到一定规模后自然出现的 [26; 245]。具体而言，代表性的方法“链式思维” (Chain-of-Thought, CoT) [95; 96]已被证明能够激发大型语言模型的推理能力，通过引导模型在输出答案前生成推理过程。一些其他策略也被提出以提升大型语言模型的性能，如自一致性[97]、自我润色 [99]、自我优化 [178] 以及选择推理 [177], 等。部分研究表明，逐步推理的有效性可以归因于训练数据的局部统计结构，变量之间的局部结构依赖关系比在所有变量上训练具有更高的数据效率 [246]。

规划。规划是人类在面对复杂挑战时采用的关键策略。对于人类而言，规划有助于组织思维、设定目标，并确定实现这些目标的步骤；这与人类的认知过程密切相关。对于智能体而言，规划能力同样至关重要，而这一规划模块的核心在于推理能力。这为基于大规模语言模型（LLMs）的智能体提供了一种结构化的思考过程。通过推理，智能体能够将复杂任务拆解为更易管理的子任务，并为每个子任务制定合适的计划。此外，随着任务的推进，智能体还可以利用内省能力调整其计划，以更好地适应实际环境，从而实现具有适应性和成功性的任务执行。

通常，规划包括两个阶段：计划制定和计划反思。

计划制定。在计划制定过程中，代理通常将总体任务分解为多个子任务，并在此阶段提出了多种方法。值得注意的是，一些研究倡导基于大规模语言模型（LLM）的代理一次性全面分解问题，形成完整的计划，然后按顺序执行；而另一些研究，如连锁推理（CoT）系列，则采用自适应策略，逐个规划和处理子任务，从而在处理复杂任务时具有更大的灵活性。此外，一些方法强调层次化规划，而另一些则强调通过树状结构的推理步骤导出最终计划。后者认为，代理在最终确定计划之前，应评估所有可能的路径。虽然基于大规模语言模型的代理展现出广泛的通用知识，但在面对需要专业知识的情境时，偶尔会遇到挑战。通过将这些代理与特定领域的规划器结合，已显示出性能的提升。

计划反思。在制定计划时，反思并评估其优劣势至关重要。基于大型语言模型的代理利用内部反馈机制，通常借助预先存在的模型，优化和提升其策略与规划方法。为了更好地与人类价值观和偏好保持一致，代理主动与人类互动，允许他们纠正一些误解，并将这些定制化的反馈融入其规划方法中。此外，它们还可以从有形或虚拟环境中获取反馈，例如任务完成的线索或行动后的观察，帮助它们修正和完善计划。

3.1.5 Transferability and Generalization

Intelligence shouldn’t be limited to a specific domain or task, but rather encompass a broad range of cognitive skills and abilities [31]. The remarkable nature of the human brain is largely attributed to its high degree of plasticity and adaptability. It can continuously adjust its structure and function in response to external stimuli and internal needs, thereby adapting to different environments and tasks. These years, plenty of research indicates that pre-trained models on large-scale corpora can learn universal language representations [36; 261; 262]. Leveraging the power of pre-trained models, with only a small amount of data for fine-tuning, LLMs can demonstrate excellent performance in downstream tasks [263]. There is no need to train new models from scratch, which saves a lot of computation resources. However, through this task-specific fine-tuning, the models lack versatility and struggle to be generalized to other tasks. Instead of merely functioning as a static knowledge repository, LLM-based agents exhibit dynamic learning ability which enables them to adapt to novel tasks swiftly and robustly [24; 105; 106].

Unseen task generalization. Studies show that instruction-tuned LLMs exhibit zero-shot generalization without the need for task-specific fine-tuning [24; 25; 105; 106; 107]. With the expansion of model size and corpus size, LLMs gradually exhibit remarkable emergent abilities in unfamiliar tasks [132]. Specifically, LLMs can complete new tasks they do not encounter in the training stage by following the instructions based on their own understanding. One of the implementations is multi-task learning, for example, FLAN [105] finetunes language models on a collection of tasks described via instructions, and T0 [106] introduces a unified framework that converts every language problem into a text-to-text format. Despite being purely a language model, GPT-4 [25] demonstrates remarkable capabilities in a variety of domains and tasks, including abstraction, comprehension, vision, coding, mathematics, medicine, law, understanding of human motives and emotions, and others [31]. It is noticed that the choices in prompting are critical for appropriate predictions, and training directly on the prompts can improve the models’ robustness in generalizing to unseen tasks [264]. Promisingly, such generalization capability can further be enhanced by scaling up both the model size and the quantity or diversity of training instructions [94; 265].

In-context learning. Numerous studies indicate that LLMs can perform a variety of complex tasks through in-context learning (ICL), which refers to the models’ ability to learn from a few examples in the context [195]. Few-shot in-context learning enhances the predictive performance of language models by concatenating the original input with several complete examples as prompts to enrich the context [41]. The key idea of ICL is learning from analogy, which is similar to the learning process of humans [266]. Furthermore, since the prompts are written in natural language, the interaction is interpretable and changeable, making it easier to incorporate human knowledge into LLMs [95; 267]. Unlike the supervised learning process, ICL doesn’t involve fine-tuning or parameter updates, which could greatly reduce the computation costs for adapting the models to new tasks. Beyond text, researchers also explore the potential ICL capabilities in different multimodal tasks [193; 194; 268; 269; 270; 271], making it possible for agents to be applied to large-scale real-world tasks.

Continual learning. Recent studies [190; 272] have highlighted the potential of LLMs’ planning capabilities in facilitating continuous learning [196; 197] for agents, which involves continuous acquisition and update of skills. A core challenge in continual learning is catastrophic forgetting [273]: as a model learns new tasks, it tends to lose knowledge from previous tasks. Numerous efforts have been devoted to addressing the above challenge, which can be broadly separated into three groups, introducing regularly used terms in reference to the previous model [274; 275; 276; 277], approximating prior data distributions [278; 279; 280], and designing architectures with task-adaptive parameters [281; 198]. LLM-based agents have emerged as a novel paradigm, leveraging the planning capabilities of LLMs to combine existing skills and address more intricate challenges. Voyager [190] attempts to solve progressively harder tasks proposed by the automatic curriculum devised by GPT-4 [25]. By synthesizing complex skills from simpler programs, the agent not only rapidly enhances its capabilities but also effectively counters catastrophic forgetting.

3.1.5 转移能力与泛化性

智能不应局限于特定的领域或任务，而应涵盖广泛的认知技能和能力 [31]。人类大脑的卓越特性在很大程度上归因于其高度的可塑性和适应性。它能够不断调整其结构和功能，以应对外部刺激和内部需求，从而适应不同的环境和任务。这些年来，许多研究表明，在大规模语料库上预训练的模型能够学习到通用的语言表征 [36]。利用预训练模型的强大能力，只需少量数据进行微调，LLMs（大型语言模型）就能在下游任务中展现出出色的性能 [263]。无需从零训练新模型，节省了大量的计算资源。然而，通过这种任务特定的微调，模型缺乏通用性，难以推广到其他任务。与仅作为静态知识库的角色不同，基于LLMs的代理展现出动态学习能力，使其能够迅速且稳健地适应新任务 [24]。

未见任务泛化。研究表明，经过指令调优的大型语言模型（LLMs）在无需任务特定微调的情况下表现出零样本泛化能力。随着模型规模和语料库规模的扩大，LLMs在不熟悉的任务中逐渐展现出显著的突现能力。具体而言，LLMs能够根据自身理解，遵循指令完成在训练阶段未曾遇到的新任务。其中一种实现方式是多任务学习，例如，FLAN通过在一组由指令描述的任务上微调语言模型，T0则引入了一个统一框架，将每个语言问题转换为文本到文本的格式。尽管仅是纯粹的语言模型，GPT-4则在抽象、理解、视觉、编码、数学、医学、法律、理解人类动机与情感等多个领域和任务中展现出卓越的能力。值得注意的是，提示的选择对预测的准确性至关重要，直接在提示上进行训练可以提升模型在泛化到未见任务时的鲁棒性。令人鼓舞的是，这种泛化能力还可以通过扩大模型规模以及增加训练指令的数量或多样性进一步增强。

In-context learning. 大量研究表明，LLMs可以通过in-context learning（ICL）执行各种复杂任务，ICL指的是模型从上下文中的少量示例中学习的能力 [195]。少样本in-context learning通过将原始输入与多个完整示例作为提示连接起来，丰富上下文，从而提升语言模型的预测性能 [41]。ICL的核心思想是通过类比学习，这类似于人类的学习过程 [266]。此外，由于提示是用自然语言编写的，交互具有可解释性和可变性，更容易将人类知识融入LLMs中 [95; 267]。与监督学习过程不同，ICL不涉及微调或参数更新，这可以大大降低模型适应新任务的计算成本。除了文本之外，研究人员还探索了ICL在不同多模态任务中的潜在能力[193; 194; 268; 269; 270; 271],使得代理能够应用于大规模的实际任务中。

连续学习。近期研究 [190; 272] 已凸显大规模语言模型（LLMs）在促进连续学习方面的规划能力潜力 [196; 197]对于智能体而言，连续学习涉及技能的持续获取与更新。连续学习中的核心挑战是灾难性遗忘[273]: 随着模型学习新任务，往往会丧失先前任务的知识。为应对这一挑战，已有大量研究，主要可分为三类，涉及引入与先前模型相关的常用术语 [274; 275; 276; 277],近似先验数据分布 [278; 279; 280], 以及设计具有任务自适应参数的架构 [281; 198]。基于大规模语言模型的智能体已成为一种新颖的范式，利用LLMs的规划能力，将现有技能结合起来，应对更复杂的挑战。Voyager [190]试图解决由GPT-4设计的自动课程提出的逐步增加难度的任务[25]。通过从简单程序合成复杂技能，智能体不仅能迅速提升其能力，还能有效应对灾难性遗忘。

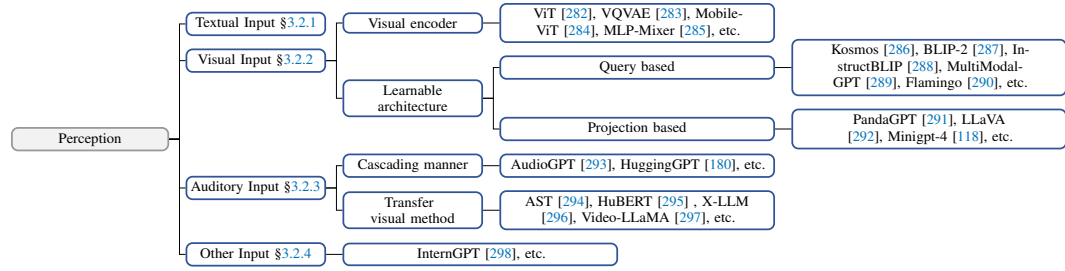


Figure 4: Typology of the perception module.

3.2 Perception

Both humans and animals rely on sensory organs like eyes and ears to gather information from their surroundings. These perceptual inputs are converted into neural signals and sent to the brain for processing [299; 300], allowing us to perceive and interact with the world. Similarly, it’s crucial for LLM-based agents to receive information from various sources and modalities. This expanded perceptual space helps agents better understand their environment, make informed decisions, and excel in a broader range of tasks, making it an essential development direction. Agent handles this information to the Brain module for processing through the perception module.

In this section, we introduce how to enable LLM-based agents to acquire multimodal perception capabilities, encompassing textual (§ 3.2.1), visual (§ 3.2.2), and auditory inputs (§ 3.2.3). We also consider other potential input forms (§ 3.2.4) such as tactile feedback, gestures, and 3D maps to enrich the agent’s perception domain and enhance its versatility.3). The typology diagram for the LLM-based agent perception is depicted in Figure 4.

3.2.1 Textual Input

Text is a way to carry data, information, and knowledge, making text communication one of the most important ways humans interact with the world. An LLM-based agent already has the fundamental ability to communicate with humans through textual input and output [114]. In a user’s textual input, aside from the explicit content, there are also beliefs, desires, and intentions hidden behind it. Understanding implied meanings is crucial for the agent to grasp the potential and underlying intentions of human users, thereby enhancing its communication efficiency and quality with users. However, as discussed in § 3.1.1, understanding implied meanings within textual input remains challenging for the current LLM-based agent. For example, some works [128; 218; 219; 220] employ reinforcement learning to perceive implied meanings and models feedback to derive rewards. This helps deduce the speaker’s preferences, leading to more personalized and accurate responses from the agent. Additionally, as the agent is designed for use in complex real-world situations, it will inevitably encounter many entirely new tasks. Understanding text instructions for unknown tasks places higher demands on the agent’s text perception abilities. As described in § 3.1.5, an LLM that has undergone instruction tuning [105] can exhibit remarkable zero-shot instruction understanding and generalization abilities, eliminating the need for task-specific fine-tuning.

3.2.2 Visual Input

Although LLMs exhibit outstanding performance in language comprehension [25; 301] and multi-turn conversations [302], they inherently lack visual perception and can only understand discrete textual content. Visual input usually contains a wealth of information about the world, including properties of objects, spatial relationships, scene layouts, and more in the agent’s surroundings. Therefore, integrating visual information with data from other modalities can offer the agent a broader context and a more precise understanding [120], deepening the agent’s perception of the environment.

To help the agent understand the information contained within images, a straightforward approach is to generate corresponding text descriptions for image inputs, known as image captioning [303; 304; 305; 306; 307]. Captions can be directly linked with standard text instructions and fed into the agent. This approach is highly interpretable and doesn’t require additional training for caption generation, which can save a significant number of computational resources. However, caption

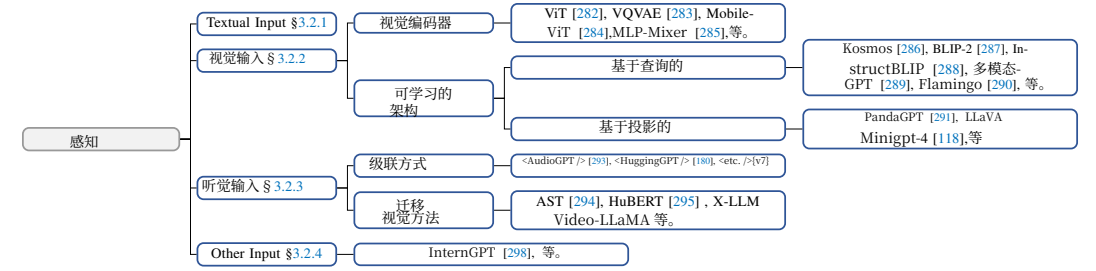


图4：感知模块的类型学。

3.2 感知

人类和动物都依赖于眼睛、耳朵等感官器官从环境中获取信息。这些感知输入被转换为神经信号并传送到大脑进行处理，从而使我们能够感知和与世界互动。同样，基于大型语言模型的智能体也需要从各种来源和模式接收信息。这种扩展的感知空间有助于智能体更好地理解环境，做出明智的决策，并在更广泛的任务中表现出色，这使其成为一个关键的发展方向。智能体通过感知模块将这些信息传递给大脑模块进行处理。

在本节中，我们介绍如何使基于大型语言模型（LLM）的代理获得多模态感知能力，包括文本输入（§ 3.2.1）、视觉输入（§ 3.2.2）和听觉输入（§ 3.2.3）。我们还考虑其他潜在的输入形式（§ 3.2.4），如触觉反馈、手势和三维地图，以丰富代理的感知领域并提升其多样性。基于LLM的代理感知的类型学图示如图4所示。

3.2.1 文本输入

文本是一种携带数据、信息和知识的方式，使得文本交流成为人类与世界互动的最重要方式之一。基于大型语言模型（LLM）的智能体已经具备通过文本输入和输出与人类进行交流的基本能力。在用户的文本输入中，除了明确的内容外，还隐藏着信念、欲望和意图。理解隐含的意义对于智能体把握人类用户的潜在意图和深层意图至关重要，从而提升其与用户的交流效率和质量。然而，正如在 § 3.1.1中所讨论的，目前的基于大型语言模型的智能体在理解文本输入中的隐含意义方面仍然具有挑战性。例如，一些研究工作采用强化学习感知隐含意义并对反馈进行建模以导出奖励。这有助于推断发言者的偏好，从而使智能体的回应更加个性化和准确。此外，由于智能体设计用于应对复杂的现实场景，它不可避免地会遇到许多全新的任务。理解未知任务的文本指令对智能体的文本感知能力提出了更高的要求。正如在 § 3.1.5中所描述的，经过指令调优的LLM能够展现出卓越的零样本指令理解和泛化能力，无需进行任务特定的微调。

3.2.2 视觉输入

尽管大型语言模型在语言理解 [25、301] 和多轮对话 [302] 方面表现出色，但它们本质上缺乏视觉感知能力，只能理解离散的文本内容。视觉输入通常包含关于世界的丰富信息，包括物体的属性、空间关系、场景布局等，涉及代理人周围的环境。因此，将视觉信息与其他模态的数据结合，可以为代理人提供更广泛的上下文和更为精准的理解 [120]，，深化代理人对环境的感知。

为了帮助代理理解图像中包含的信息，一种简单的方法是为图像输入生成相应的文本描述，称为图像描述生成。描述可以直接与标准文本指令关联，并输入到代理中。这种方法具有高度的可解释性，并且不需要额外的训练来生成描述，从而可以节省大量的计算资源。然而，描述

generation is a low-bandwidth method [120; 308], and it may lose a lot of potential information during the conversion process. Furthermore, the agent’s focus on images may introduce biases.

Inspired by the excellent performance of transformers [309] in natural language processing, researchers have extended their use to the field of computer vision. Representative works like ViT/VQVAE [282; 283; 284; 285; 310] have successfully encoded visual information using transformers. Researchers first divide an image into fixed-size patches and then treat these patches, after linear projection, as input tokens for Transformers [292]. In the end, by calculating self-attention between tokens, they are able to integrate information across the entire image, resulting in a highly effective way to perceive visual content. Therefore, some works [311] try to combine the image encoder and LLM directly to train the entire model in an end-to-end way. While the agent can achieve remarkable visual perception abilities, it comes at the cost of substantial computational resources.

Extensively pre-trained visual encoders and LLMs can greatly enhance the agent’s visual perception and language expression abilities [286; 312]. Freezing one or both of them during training is a widely adopted paradigm that achieves a balance between training resources and model performance [287]. However, LLMs cannot directly understand the output of a visual encoder, so it’s necessary to convert the image encoding into embeddings that LLMs can comprehend. In other words, it involves aligning the visual encoder with the LLM. This usually requires adding an extra learnable interface layer between them. For example, BLIP-2 [287] and InstructBLIP [288] use the Querying Transformer(Q-Former) module as an intermediate layer between the visual encoder and the LLM [288]. Q-Former is a transformer that employs learnable query vectors [289], giving it the capability to extract language-informative visual representations. It can provide the most valuable information to the LLM, reducing the agent’s burden of learning visual-language alignment and thereby mitigating the issue of catastrophic forgetting. At the same time, some researchers adopt a computationally efficient method by using a single projection layer to achieve visual-text alignment, reducing the need for training additional parameters [118; 291; 312]. Moreover, the projection layer can effectively integrate with the learnable interface to adapt the dimensions of its outputs, making them compatible with LLMs [296; 297; 313; 314].

Video input consists of a series of continuous image frames. As a result, the methods used by agents to perceive images [287] may be applicable to the realm of videos, allowing the agent to have good perception of video inputs as well. Compared to image information, video information adds a temporal dimension. Therefore, the agent’s understanding of the relationships between different frames in time is crucial for perceiving video information. Some works like Flamingo [290; 315] ensure temporal order when understanding videos using a mask mechanism. The mask mechanism restricts the agent’s view to only access visual information from frames that occurred earlier in time when it perceives a specific frame in the video.

3.2.3 Auditory Input

Undoubtedly, auditory information is a crucial component of world information. When an agent possesses auditory capabilities, it can improve its awareness of interactive content, the surrounding environment, and even potential dangers. Indeed, there are numerous well-established models and approaches [293; 316; 317] for processing audio as a standalone modality. However, these models often excel at specific tasks. Given the excellent tool-using capabilities of LLMs (which will be discussed in detail in §3.3), a very intuitive idea is that the agent can use LLMs as control hubs, invoking existing toolsets or model repositories in a cascading manner to perceive audio information. For instance, AudioGPT [293], makes full use of the capabilities of models like FastSpeech [317], GenerSpeech [316], and others [318; 319; 320; 321; 322] which have achieved excellent results in tasks such as Text-to-Speech, Style Transfer, and Speech Recognition.

An audio spectrogram provides an intuitive representation of the frequency spectrum of an audio signal as it changes over time [323]. For a segment of audio data over a period of time, it can be abstracted into a finite-length audio spectrogram. An audio spectrogram has a 2D representation, which can be visualized as a flat image. Hence, some research [294; 295] efforts aim to migrate perceptual methods from the visual domain to audio. AST (Audio Spectrogram Transformer) [294] employs a Transformer architecture similar to ViT to process audio spectrogram images. By segmenting the audio spectrogram into patches, it achieves effective encoding of audio information. Moreover, some researchers [296; 297] have drawn inspiration from the idea of freezing encoders to reduce training

生成是一种低带宽的方法 [120; 308], 在转换过程中可能会丢失大量潜在信息。此外, 代理对图像的关注可能引入偏差。

受到变换器在自然语言处理中的出色表现的启发, 研究人员已将其应用扩展到计算机视觉领域。代表性工作如ViT/VQVAE成功利用变换器对视觉信息进行编码。研究者首先将图像划分为固定大小的块, 然后对这些块进行线性投影, 作为变换器的输入标记。最终, 通过计算标记之间的自注意力, 能够整合整个图像中的信息, 从而实现一种高效的视觉感知方式。因此, 一些研究尝试将图像编码器与大型语言模型 (LLM) 直接结合, 进行端到端的模型训练。虽然该方法使智能体具备了卓越的视觉感知能力, 但也伴随着大量计算资源的消耗。

广泛预训练的视觉编码器和大型语言模型 (LLMs) 可以极大地增强代理的视觉感知能力和语言表达能力; [312]。在训练过程中冻结其中一个或两个模型是一种被广泛采用的范式, 能够在训练资源和模型性能之间取得平衡[287]。然而, LLMs无法直接理解视觉编码器的输出, 因此有必要将图像编码转换为LLMs能够理解的嵌入向量。换句话说, 这涉及到将视觉编码器与LLM对齐, 通常需要在它们之间添加一个额外的可学习接口层。例如, BLIP-2 [287] 和InstructBLIP [288] 使用查询变换器 (Q-Former) 模块作为视觉编码器与LLM之间的中间层[288]。Q-Former是一种采用可学习查询向量的变换器 [289], 赋予其提取具有语言信息的视觉表征的能力。它可以向LLM提供最有价值的信息, 减轻代理在学习视觉-语言对齐方面的负担, 从而缓解灾难性遗忘的问题。同时, 一些研究者采用一种计算效率较高的方法, 通过使用单一投影层实现视觉-文本对齐, 减少了训练额外参数的需求 [118; 291; 312]。此外, 投影层可以有效地与可学习接口结合, 调整其输出的维度, 使其与LLMs兼容 [296; 297; 313; 314]。

视频输入由一系列连续的图像帧组成。因此, 代理在感知图像时采用的方法 [287] 可能也适用于视频领域, 使代理能够对视频输入具有良好的感知能力。与图像信息相比, 视频信息增加了时间维度。因此, 代理对不同帧之间时间关系的理解对于感知视频信息至关重要。一些工作如 Flamingo [290; 315]在理解视频时通过掩码机制确保时间顺序。该掩码机制限制代理在感知视频中的特定帧时, 仅访问发生在该帧之前的视觉信息。

3.2.3 听觉输入

毫无疑问, 听觉信息是世界信息的重要组成部分。当智能体具备听觉能力时, 能够提升其对交互内容、周围环境甚至潜在危险的感知能力。实际上, 已有许多成熟的模型和方法用于将音频作为单一模态进行处理。然而, 这些模型通常在特定任务中表现出色。鉴于大规模语言模型 (LLMs) 具有出色的工具使用能力 (将在 § 3.3中详细讨论), 一个非常直观的想法是, 智能体可以将LLMs作为控制枢纽, 级联调用现有的工具集或模型仓库, 以感知音频信息。例如, AudioGPT充分利用了FastSpeech、GenerSpeech、Whisper等模型的能力, 这些模型在文本转语音、风格迁移和语音识别等任务中取得了优异的成果。

声谱图提供了音频信号频谱随时间变化的直观表现 [323]。对于一段时间内的音频数据, 可以抽象为有限长度的声谱图。声谱图具有二维表示, 可以直观地作为一张平面图像进行可视化。因此, 一些研究 [294; 295] 的努力旨在将感知方法从视觉领域迁移到音频领域。AST (音频声谱变换器) [294]采用类似于ViT的Transformer架构来处理声谱图图像。通过将声谱图划分为块, 它实现了对音频信息的有效编码。此外, 一些研究人员 [296; 297] 受到冻结编码器思想的启发, 以减少训练成本。

time and computational costs. They align audio encoding with data encoding from other modalities by adding the same learnable interface layer.

3.2.4 Other Input

As mentioned earlier, many studies have looked into perception units for text, visual, and audio. However, LLM-based agents might be equipped with richer perception modules. In the future, they could perceive and understand diverse modalities in the real world, much like humans. For example, agents could have unique touch and smell organs, allowing them to gather more detailed information when interacting with objects. At the same time, agents can also have a clear sense of the temperature, humidity, and brightness in their surroundings, enabling them to take environment-aware actions. Moreover, by efficiently integrating basic perceptual abilities like vision, text, and light sensitivity, agents can develop various user-friendly perception modules for humans. InternGPT [298] introduces pointing instructions. Users can interact with specific, hard-to-describe portions of an image by using gestures or moving the cursor to select, drag, or draw. The addition of pointing instructions helps provide more precise specifications for individual text instructions. Building upon this, agents have the potential to perceive more complex user inputs. For example, technologies such as eye-tracking in AR/VR devices, body motion capture, and even brainwave signals in brain-computer interaction.

Finally, a human-like LLM-based agent should possess awareness of a broader overall environment. At present, numerous mature and widely adopted hardware devices can assist agents in accomplishing this. Lidar [324] can create 3D point cloud maps to help agents detect and identify objects in their surroundings. GPS [325] can provide accurate location coordinates and can be integrated with map data. Inertial Measurement Units (IMUs) can measure and record the three-dimensional motion of objects, offering details about an object’s speed and direction. However, these sensory data are complex and cannot be directly understood by LLM-based agents. Exploring how agents can perceive more comprehensive input is a promising direction for the future.

3.3 Action

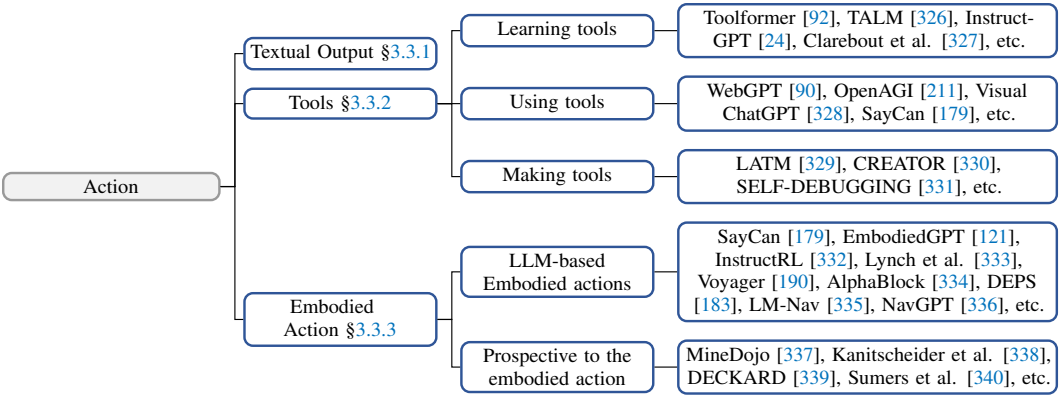


Figure 5: Typology of the action module.

After humans perceive their environment, their brains integrate, analyze, and reason with the perceived information and make decisions. Subsequently, they employ their nervous systems to control their bodies, enabling adaptive or creative actions in response to the environment, such as engaging in conversation, evading obstacles, or starting a fire. When an agent possesses a brain-like structure with capabilities of knowledge, memory, reasoning, planning, and generalization, as well as multimodal perception, it is also expected to possess a diverse range of actions akin to humans to respond to its surrounding environment. In the construction of the agent, the action module receives action sequences sent by the brain module and carries out actions to interact with the environment. As Figure 5 shows, this section begins with textual output (§ 3.3.1), which is the inherent capability of LLM-based agents. Next we talk about the tool-using capability of LLM-based agents (§ 3.3.2), which has proved effective in enhancing their versatility and expertise. Finally, we discuss equipping the LLM-based agent with embodied action to facilitate its grounding in the physical world (§ 3.3.3).

时间和计算成本。它们通过添加相同的可学习接口层，将音频编码与来自其他模态的数据编码对齐。

3.2.4 其他输入

正如前文所述，许多研究已探讨了文本、视觉和音频的感知单元。然而，基于大型语言模型的代理可能配备更丰富的感知模块。未来，它们或许能够像人类一样感知和理解多模态信息。例如，代理可以拥有独特的触觉和嗅觉器官，使其在与物体交互时获取更详细的信息。同时，代理还能清楚感知周围环境的温度、湿度和亮度，从而实现环境感知驱动的行动。此外，通过高效整合视觉、文本和光敏等基本感知能力，代理可以开发出多种面向人类的友好型感知模块。InternGPT引入了指示指令。用户可以通过手势或移动光标来选择、拖拽或绘制图像中的特定、难以描述的部分，从而与之交互。指示指令的加入有助于为个别文本指令提供更为精确的规范。在此基础上，代理有潜力感知更复杂的用户输入，例如AR/VR设备中的眼动追踪技术、身体运动捕捉，甚至脑电波信号在脑机接口中的应用。<style>

最后，类人化的基于大语言模型（LLM）代理应具备对更广泛整体环境的感知能力。目前，许多成熟且广泛应用的硬件设备可以协助代理实现这一目标。激光雷达（Lidar） [324] 可以创建三维点云地图，帮助代理检测和识别周围的物体。全球定位系统（GPS） [325] 能够提供精确的位置信息，并可与地图数据集成。惯性测量单元（IMU）可以测量和记录物体的三维运动，提供关于速度和方向的详细信息。然而，这些感知数据复杂，不能被基于大语言模型的代理直接理解。探索代理如何感知更全面的输入，是未来的一个有前景的研究方向。

3.3 行动

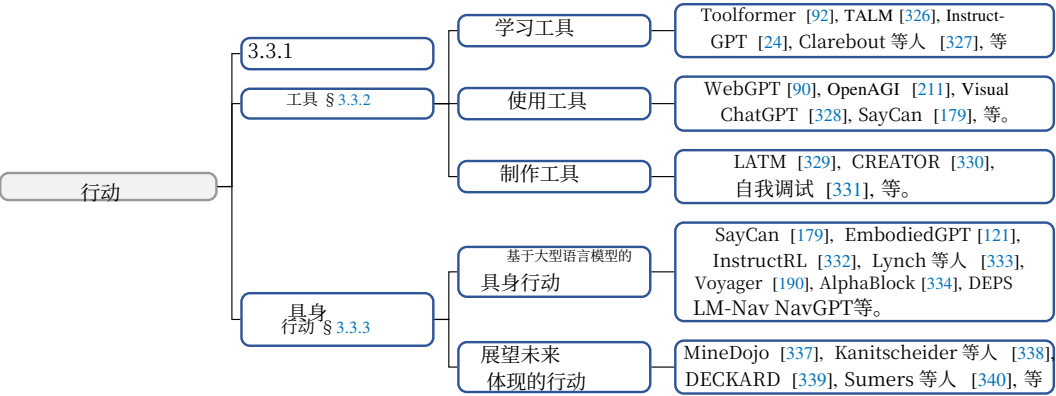


图5：行动模块的类型学。

在人类感知其环境后，大脑会整合、分析并推理所感知的信息，从而做出决策。随后，他们利用神经系统控制身体，执行适应性或创造性的行动，以应对环境，例如进行对话、躲避障碍或点燃火源。当一个智能体具有类似大脑的结构，具备知识、记忆、推理、规划和泛化能力，以及多模态感知时，也应具备类似人类的多样化行动，以应对其周围环境。在智能体的构建中，行动模块接收由大脑模块传递的行动序列，并执行相应动作以与环境互动。如图所示，本节首先介绍基于大语言模型（LLM）智能体的固有能力——文本输出 (§ 3.3.1)，接着讨论其工具使用能力 (§ 3.3.2)，该能力已被证明能有效提升其多功能性和专业性。最后，我们探讨赋予基于大语言模型的智能体具身行动的可能性，以促进其在物理世界中的落地 (§ 3.3.3)。

3.3.1 Textual Output

As discussed in § 3.1.1, the rise and development of Transformer-based generative large language models have endowed LLM-based agents with inherent language generation capabilities [132; 213]. The text quality they generate excels in various aspects such as fluency, relevance, diversity, controllability [127; 214; 134; 216]. Consequently, LLM-based agents can be exceptionally strong language generators.

3.3.2 Tool Using

Tools are extensions of the capabilities of tool users. When faced with complex tasks, humans employ tools to simplify task-solving and enhance efficiency, freeing time and resources. Similarly, agents have the potential to accomplish complex tasks more efficiently and with higher quality if they also learn to use and utilize tools [94].

LLM-based agents have *limitations* in some aspects, and the use of tools can *strengthen the agents’ capabilities*. First, although LLM-based agents have a strong knowledge base and expertise, they don’t have the ability to memorize every piece of training data [341; 342]. They may also fail to steer to correct knowledge due to the influence of contextual prompts [226], or even generate hallucinate knowledge [208]. Coupled with the lack of corpus, training data, and tuning for specific fields and scenarios, agents’ expertise is also limited when specializing in specific domains [343]. Specialized tools enable LLMs to enhance their expertise, adapt domain knowledge, and be more suitable for domain-specific needs in a pluggable form. Furthermore, the decision-making process of LLM-based agents lacks transparency, making them less trustworthy in high-risk domains such as healthcare and finance [344]. Additionally, LLMs are susceptible to adversarial attacks [345], and their robustness against slight input modifications is inadequate. In contrast, agents that accomplish tasks with the assistance of tools exhibit stronger interpretability and robustness. The execution process of tools can reflect the agents’ approach to addressing complex requirements and enhance the credibility of their decisions. Moreover, for the reason that tools are specifically designed for their respective usage scenarios, agents utilizing such tools are better equipped to handle slight input modifications and are more resilient against adversarial attacks [94].

LLM-based agents not only require the use of tools, but are also *well-suited* for tool integration. Leveraging the rich world knowledge accumulated through the pre-training process and CoT prompting, LLMs have demonstrated remarkable reasoning and decision-making abilities in complex interactive environments [97], which help agents break down and address tasks specified by users in an appropriate way. What’s more, LLMs show significant potential in intent understanding and other aspects [25; 201; 202; 203]. When agents are combined with tools, the threshold for tool utilization can be lowered, thereby fully unleashing the creative potential of human users [94].

Understanding tools. A prerequisite for an agent to use tools effectively is a comprehensive understanding of the tools’ application scenarios and invocation methods. Without this understanding, the process of the agent using tools will become untrustworthy and fail to genuinely enhance the agent’s capabilities. Leveraging the powerful zero-shot and few-shot learning abilities of LLMs [40; 41], agents can acquire knowledge about tools by utilizing *zero-shot prompts* that describe tool functionalities and parameters, or *few-shot prompts* that provide demonstrations of specific tool usage scenarios and corresponding methods [92; 326]. These learning approaches parallel human methods of learning by consulting tool manuals or observing others using tools [94]. A single tool is often insufficient when facing complex tasks. Therefore, the agents should first decompose the complex task into subtasks in an appropriate manner, and their understanding of tools play a significant role in task decomposition.

Learning to use tools. The methods for agents to learn to utilize tools primarily consist of *learning from demonstrations* and *learning from feedback*. This involves mimicking the behavior of human experts [346; 347; 348], as well as understanding the consequences of their actions and making adjustments based on feedback received from both the environment and humans [24; 349; 350]. Environmental feedback encompasses result feedback on whether actions have successfully completed the task and intermediate feedback that captures changes in the environmental state caused by actions; human feedback comprises explicit evaluations and implicit behaviors, such as clicking on links [94].

3.3.1 文本输出

正如在 § 3.1.1中所讨论的，基于Transformer的生成型大规模语言模型的崛起与发展赋予了LLM（大规模语言模型）代理固有的语言生成能力 [132; 213]。它们生成的文本在流畅性、相关性、多样性和可控性等多个方面表现出色 [127; {v14}; {v16}; {v18}; {v20}; {v22}; 216]。因此，基于LLM的代理在语言生成方面具有极强的能力。

3.3.2 工具使用

工具是工具使用者能力的扩展。在面对复杂任务时，人类利用工具简化任务解决过程并提高效率，从而节省时间和资源。同样，代理如果也能学习使用和利用工具 [94]，则有潜力以更高的效率和更高的质量完成复杂任务。

基于大型语言模型（LLM）的代理在某些方面存在局限性，而工具的使用可以增强代理的能力。首先，尽管基于LLM的代理具有强大的知识基础和专业能力，但它们并不具备记忆所有训练数据的能力 [341; 342]。它们也可能由于上下文提示的影响而未能引导到正确的知识，甚至会生成幻觉知识 [208]。加之缺乏语料库、训练数据以及针对特定领域和场景的调优，代理在专业化特定领域时的专业能力也受到限制 [343]。专业工具使得LLMs能够增强其专业知识、适应领域知识，并以可插拔的形式更好地满足领域特定需求。此外，基于LLM的代理的决策过程缺乏透明性，导致其在医疗、金融等高风险领域的可信度较低 [344]。此外，LLMs容易受到对抗性攻击 [345]，其对微小输入修改的鲁棒性不足。相比之下，借助工具完成任务的代理表现出更强的可解释性和鲁棒性。工具的执行过程可以反映代理应对复杂需求的方法，并增强其决策的可信度。此外，由于工具是为其各自的使用场景专门设计的，利用此类工具的代理更能应对微小输入变化，并在对抗性攻击中表现出更强的韧性 [94]。

LLM基础的代理不仅需要使用工具，而且非常适合工具集成。利用预训练过程中积累的丰富世界知识和链式推理（CoT）提示，LLMs在复杂交互环境中展现出卓越的推理和决策能力 [97]，这有助于代理以适当的方式拆解和应对用户指定的任务。更重要的是，LLMs在意图理解等方面显示出显著潜力[25; 201; 202; 203]。当代理与工具结合时，工具利用的门槛可以降低，从而充分释放人类用户的创造潜能 [94]。

理解工具。 使代理能够有效使用工具的前提是对工具的应用场景和调用方法有全面的理解。没有这种理解，代理使用工具的过程将变得不可靠，无法真正提升代理的能力。利用大规模语言模型（LLMs）[40; 41], 的强大零样本和少样本学习能力，代理可以通过使用描述工具功能和参数的零样本提示，或提供特定工具使用场景及相应方法示范的少样本提示，来获取关于工具的知识 [92; 326]。这些学习方法类似于人类通过查阅工具手册或观察他人使用工具来学习 [94]。在面对复杂任务时，单一工具往往不足以应对。因此，代理应首先以适当的方式将复杂任务分解为子任务，而它们对工具的理解在任务分解中起着重要作用。

学习使用工具。 代理学习利用工具的方法主要包括 模仿人类专家的行为 [346; 以及理解其行为的后果，并根据从环境和人类获得的反馈进行调整 [24]。 环境反馈包括对行动是否成功完成任务的结果反馈以及反映行动引起的环境状态变化的中间反馈；人类反馈则包括明确的评价和隐性行为，例如点击链接 [94]。

If an agent rigidly applies tools without *adaptability*, it cannot achieve acceptable performance in all scenarios. Agents need to generalize their tool usage skills learned in specific contexts to more general situations, such as transferring a model trained on Yahoo search to Google search. To accomplish this, it’s necessary for agents to grasp the common principles or patterns in tool usage strategies, which can potentially be achieved through meta-tool learning [327]. Enhancing the agent’s understanding of relationships between simple and complex tools, such as how complex tools are built on simpler ones, can contribute to the agents’ capacity to generalize tool usage. This allows agents to effectively discern nuances across various application scenarios and transfer previously learned knowledge to new tools [94]. Curriculum learning [351], which allows an agent to start from simple tools and progressively learn complex ones, aligns with the requirements. Moreover, benefiting from the understanding of user intent reasoning and planning abilities, agents can better design methods of tool utilization and collaboration and then provide higher-quality outcomes.

Making tools for self-sufficiency. Existing tools are often designed for human convenience, which might not be optimal for agents. To make agents use tools better, there’s a need for tools specifically designed for agents. These tools should be more modular and have input-output formats that are more suitable for agents. If instructions and demonstrations are provided, LLM-based agents also possess the ability to create tools by generating executable programs, or integrating existing tools into more powerful ones [94; 330; 352]. and they can learn to perform self-debugging [331]. Moreover, if the agent that serves as a tool maker successfully creates a tool, it can produce packages containing the tool’s code and demonstrations for other agents in a multi-agent system, in addition to using the tool itself [329]. Speculatively, in the future, agents might become self-sufficient and exhibit a high degree of autonomy in terms of tools.

Tools can expand the action space of LLM-based agents. With the help of tools, agents can utilize various external *resources* such as web applications and other LMs during the reasoning and planning phase [92]. This process can provide information with high expertise, reliability, diversity, and quality for LLM-based agents, facilitating their decision-making and action. For example, search-based tools can improve the scope and quality of the knowledge accessible to the agents with the aid of external databases, knowledge graphs, and web pages, while domain-specific tools can enhance an agent’s expertise in the corresponding field [211; 353]. Some researchers have already developed LLM-based controllers that generate SQL statements to query databases, or to convert user queries into search requests and use search engines to obtain the desired results [90; 175]. What’s more, LLM-based agents can use scientific tools to execute tasks like organic synthesis in chemistry, or interface with Python interpreters to enhance their performance on intricate mathematical computation tasks [354; 355]. For multi-agent systems, communication tools (e.g., emails) may serve as a means for agents to interact with each other under strict security constraints, facilitating their *collaboration*, and showing autonomy and flexibility [94].

Although the tools mentioned before enhance the capabilities of agents, the medium of interaction with the environment remains text-based. However, tools are designed to expand the functionality of language models, and their outputs are not limited to text. Tools for non-textual output can diversify the *modalities* of agent actions, thereby expanding the application scenarios of LLM-based agents. For example, image processing and generation can be accomplished by an agent that draws on a visual model [328]. In aerospace engineering, agents are being explored for modeling physics and solving complex differential equations [356]; in the field of robotics, agents are required to plan physical operations and control the robot execution [179]; and so on. Agents that are capable of dynamically interacting with the environment or the world through tools, or in a multimodal manner, can be referred to as digitally embodied [94]. The *embodiment* of agents has been a central focus of embodied learning research. We will make a deep discussion on agents’ embodied action in §3.3.3.

3.3.3 Embodied Action

In the pursuit of Artificial General Intelligence (AGI), the embodied agent is considered a pivotal paradigm while it strives to integrate model intelligence with the physical world. *The Embodiment hypothesis* [357] draws inspiration from the human intelligence development process, posing that an agent’s intelligence arises from continuous interaction and feedback with the environment rather than relying solely on well-curated textbooks. Similarly, unlike traditional deep learning models that learn explicit capabilities from the internet datasets to solve domain problems, people anticipate that LLM-based agents’ behaviors will no longer be limited to pure text output or calling exact tools to perform

如果一个代理僵硬地应用工具而缺乏适应性，它无法在所有场景中实现令人满意的性能。代理需要将其在特定环境中学习到的工具使用技能推广到更一般的情境，例如将在Yahoo搜索中训练的模型迁移到Google搜索。为此，代理必须掌握工具使用策略中的共通原则或模式，这可以通过元工具学习实现。增强代理对简单工具与复杂工具之间关系的理解，例如复杂工具是如何建立在简单工具之上的，有助于提升其工具使用的泛化能力。这使得代理能够有效识别各种应用场景中的细微差别，并将先前学到的知识迁移到新工具<style id='10'>中。课程学习，即让代理从简单工具开始，逐步学习复杂工具，符合这一需求。此外，借助对用户意图推理与规划能力的理解，代理可以更好地设计工具的利用与协作方法，从而提供更高质量的结果。

为自给自足而制作工具。现有工具通常是为人类便利而设计，可能并不最适合代理。为了让代理更好地使用工具，需要专门为代理设计的工具。这些工具应具有更高的模块化，并采用更适合代理的输入输出格式。如果提供指令和示范，基于大型语言模型的代理还具备通过生成可执行程序或将现有工具整合成更强大工具来创建工具的能力。此外，如果作为工具制造者的代理成功创建了工具，它不仅可以使用该工具，还能为多代理系统中的其他代理生成包含工具代码和示范的包。推测未来，代理可能变得自给自足，在工具方面表现出高度的自主性。

Tools can expand the action space of LLM-based agents. 通过工具，代理可以在推理和规划阶段利用各种外部资源，如网络应用和其他语言模型（LMs）[92]。这一过程可以为基于大语言模型的代理提供高专业性、可靠性、多样性和高质量的信息，促进其决策和行动。例如，基于搜索的工具可以借助外部数据库、知识图谱和网页，扩大代理可访问的知识范围和提升其质量，而领域特定工具则可以增强代理在相应领域的专业能力 [211; 353]。一些研究人员已经开发出基于大语言模型的控制​​器，能够生成SQL语句以查询数据库，或将用户查询转换为搜索请求，并利用搜索引擎获取所需结果 [90; 175]。此外，基于大语言模型的代理还可以使用科学工具执行有机合成等化学任务，或与Python解释器接口，提升其在复杂数学计算任务中的表现 [354; 355]。对于多智能体系统，通信工具（如电子邮件）可以作为代理之间在严格安全约束下交互的手段，促进它们的协作，并展现出自主性和灵活性 [94]。

虽然前面提到的工具增强了代理的能力，但与环境交互的媒介仍然是基于文本的。然而，工具旨在扩展语言模型的功能，其输出不仅限于文本。非文本输出的工具可以丰富模态，从而拓展基于LLM的代理的应用场景。例如，图像处理和生成可以由依赖视觉模型的代理完成 [328]。在航空航天工程中，正在探索利用代理建模物理和解决复杂的微分方程 [356]；在机器人领域，代理需要规划物理操作并控制机器人执行 [179]；等等。能够通过工具或多模态方式与环境或世界进行动态交互的代理，可以被称为数字化具身 [94]。代理的具身一直是具身学习研究的核心焦点。我们将在 § 3.3.3中对代理的具身行动进行深入讨论。

3.3.3 具身行动

在追求人工通用智能（AGI）的过程中，具身代理被认为是一种关键范式，它旨在将模型智能与物理世界相结合。具身假设 [357]的灵感来自于人类智能的发展过程，提出代理的智能源于与环境的持续交互和反馈，而不仅仅依赖于精心整理的教科书。类似地，与传统的深度学习模型通过互联网数据集学习明确能力以解决领域问题不同，人们预期基于大规模语言模型（LLM）的代理行为将不再局限于纯文本输出或调用特定工具来执行任务。

particular domain tasks [358]. Instead, they should be capable of actively perceiving, comprehending, and interacting with physical environments, making decisions, and generating specific behaviors to modify the environment based on LLM’s extensive internal knowledge. We collectively term these as **embodied actions**, which enable agents’ ability to interact with and comprehend the world in a manner closely resembling human behavior.

The potential of LLM-based agents for embodied actions. Before the widespread rise of LLMs, researchers tended to use methods like reinforcement learning to explore the embodied actions of agents. Despite the extensive success of RL-based embodiment [359; 360; 361], it does have certain limitations in some aspects. In brief, RL algorithms face limitations in terms of data efficiency, generalization, and complex problem reasoning due to challenges in modeling the dynamic and often ambiguous real environment, or their heavy reliance on precise reward signal representations [362]. Recent studies have indicated that leveraging the rich internal knowledge acquired during the pre-training of LLMs can effectively alleviate these issues [120; 187; 258; 363].

- **Cost efficiency.** Some on-policy algorithms struggle with sample efficiency as they require fresh data for policy updates while gathering enough embodied data for high-performance training is costly and noisy. The constraint is also found in some end-to-end models [364; 365; 366]. By leveraging the intrinsic knowledge from LLMs, agents like PaLM-E [120] jointly train robotic data with general visual-language data to achieve significant transfer ability in embodied tasks while also showcasing that geometric input representations can improve training data efficiency.
- **Embodied action generalization.** As discussed in section §3.1.5, an agent’s competence should extend beyond specific tasks. When faced with intricate, uncharted real-world environments, it’s imperative that the agent exhibits dynamic learning and generalization capabilities. However, the majority of RL algorithms are designed to train and evaluate relevant skills for specific tasks [101; 367; 368; 369]. In contrast, fine-tuned by diverse forms and rich task types, LLMs have showcased remarkable cross-task generalization capabilities [370; 371]. For instance, PaLM-E exhibits surprising zero-shot or one-shot generalization capabilities to new objects or novel combinations of existing objects [120]. Further, language proficiency represents a distinctive advantage of LLM-based agents, serving both as a means to interact with the environment and as a medium for transferring foundational skills to new tasks [372]. SayCan [179] decomposes task instructions presented in prompts using LLMs into corresponding skill commands, but in partially observable environments, limited prior skills often do not achieve satisfactory performance [101]. To address this, Voyager [190] introduces the skill library component to continuously collect novel self-verified skills, which allows for the agent’s lifelong learning capabilities.
- **Embodied action planning.** Planning constitutes a pivotal strategy employed by humans in response to complex problems as well as LLM-based agents. Before LLMs exhibited remarkable reasoning abilities, researchers introduced Hierarchical Reinforcement Learning (HRL) methods while the high-level policy constraints sub-goals for the low-level policy and the low-level policy produces appropriate action signals [373; 374; 375]. Similar to the role of high-level policies, LLMs with emerging reasoning abilities [26] can be seamlessly applied to complex tasks in a zero-shot or few-shot manner [95; 97; 98; 99]. In addition, external feedback from the environment can further enhance LLM-based agents’ planning performance. Based on the current environmental feedback, some work [101; 91; 100; 376] dynamically generate, maintain, and adjust high-level action plans in order to minimize dependency on prior knowledge in partially observable environments, thereby grounding the plan. Feedback can also come from models or humans, which can usually be referred to as the critics, assessing task completion based on the current state and task prompts [25; 190].

Embodied actions for LLM-based agents. Depending on the agents’ level of autonomy in a task or the complexity of actions, there are several fundamental LLM-based embodied actions, primarily including observation, manipulation, and navigation.

- **Observation.** Observation constitutes the primary ways by which the agent acquires environmental information and updates states, playing a crucial role in enhancing the efficiency of subsequent embodied actions. As mentioned in §3.2, observation by embodied agents primarily occurs in environments with various inputs, which are ultimately converged into a multimodal signal. A common approach entails a pre-trained Vision Transformer (ViT) used as the alignment module for text and visual information and special tokens are marked to denote the positions of multimodal data [120; 332; 121]. Soundspaces [377] proposes the identification of physical spatial geometric

particular domain tasks [358]. 相反，它们应具备主动感知、理解和与物理环境交互的能力，能够基于LLM丰富的内部知识做出决策并生成特定行为以改变环境。我们统称这些为**体现行动**，使得代理能够以类似人类行为的方式与世界互动和理解。

LLM基础代理在具身行动方面的潜力。在LLM广泛崛起之前，研究人员倾向于采用强化学习等方法探索代理的具身行动。尽管基于RL的具身表现取得了广泛成功，但在某些方面仍存在一定的局限性。简而言之，由于在建模动态且常常模糊的真实环境方面面临挑战，或过于依赖精确的奖励信号表示，RL算法在数据效率、泛化能力和复杂问题推理方面受到限制。最近的研究表明，利用LLM在预训练过程中获得的丰富内部知识，可以有效缓解这些问题；187；258。

成本效率。一些基于策略的算法在样本效率方面存在困难，因为它们需要新鲜数据进行策略更新，而收集足够的体现数据以实现高性能训练既昂贵又噪声较大。这一限制也存在于一些端到端模型中。通过利用大型语言模型（LLMs）内在的知识，像 PaLM-E 这样的代理可以将机器人数据与通用视觉-语言数据联合训练，以在体现任务中实现显著的迁移能力，同时也展示了几何输入表示可以提高训练数据的效率。

- **具身行动的泛化。**如第 § 3.1.5 节所述，代理的能力应超越特定任务。当面对复杂的、未知的现实世界环境时，代理展现出动态学习和泛化能力至关重要。然而，大多数RL算法旨在训练和评估与特定任务相关的技能[101; 367; 368; 369]。相比之下，通过多样的形式和丰富的任务类型进行微调的LLMs展现出卓越的跨任务泛化能力 [370; 371]。例如，PaLM-E 在面对新对象或现有对象的组合时，表现出令人惊讶的零-shot或一-shot泛化能力 [120]。此外，语言能力是基于LLM的代理的一个显著优势，既作为与环境交互的手段，也作为将基础技能转移到新任务的媒介 [372]。SayCan [179] 将提示中呈现的任务指令分解为相应的技能命令，但在部分可观察环境中，有限的先前技能往往难以取得令人满意的表现 [101]。为了解决这一问题，Voyager [190] 引入技能库组件，持续收集新颖的自验证技能，从而实现代理的终身学习能力。

- **体现行动规划。**规划是人类应对复杂问题以及基于LLM的代理所采用的关键策略。在LLMs展现出卓越推理能力之前，研究人员引入了层级强化学习（HRL）方法，该方法在高层策略中约束子目标，而低层策略则产生适当的行动信号 [373; 374; 375]。与高层策略的作用类似，具有新兴推理能力的LLMs [26] 可以无缝应用于复杂任务的零样本或少样本方式 [95; 97; 98; 99]。此外，来自环境的外部反馈还可以进一步提升基于LLM的代理的规划性能。根据当前环境反馈，一些工作 [101; 91; 100; 376] 能够动态生成、维护和调整高层行动计划，以最小化对部分可观察环境中先验知识的依赖，从而使计划具有基础性。反馈还可以来自模型或人类，通常被称为评论者，根据当前状态和任务提示评估任务完成情况 [25; 190]。

基于LLM的代理的具身动作。根据代理在任务中的自主性水平或动作的复杂性，存在几种基本的基于LLM的具身动作，主要包括观察、操作和导航。

观察。观察构成了代理获取环境信息和更新状态的主要方式，在提升后续具身行动效率方面起着关键作用。如 § 3.2 所述，具身代理的观察主要发生在具有多种输入的环境中，最终汇聚成多模态信号。一种常用的方法是采用预训练的视觉变换器（Vision Transformer, ViT）作为文本与视觉信息的对齐模块，并用特殊标记标示多模态数据的位置；声空间（Soundspaces）提出了物理空间几何的识别。

elements guided by reverberant audio input, enhancing the agent’s observations with a more comprehensive perspective [375]. In recent times, even more research takes audio as a modality for embedded observation. Apart from the widely employed cascading paradigm [293; 378; 316], audio information encoding similar to ViT further enhances the seamless integration of audio with other modalities of inputs [294]. The agent’s observation of the environment can also be derived from real-time linguistic instructions from humans, while human feedback helps the agent in acquiring detail information that may not be readily obtained or parsed [333; 190].

- **Manipulation.** In general, manipulation tasks for embodied agents include object rearrangements, tabletop manipulation, and mobile manipulation [23; 120]. The typical case entails the agent executing a sequence of tasks in the kitchen, which includes retrieving items from drawers and handing them to the user, as well as cleaning the tabletop [179]. Besides precise observation, this involves combining a series of subgoals by leveraging LLM. Consequently, maintaining synchronization between the agent’s state and the subgoals is of significance. DEPS [183] utilizes an LLM-based interactive planning approach to maintain this consistency and help error correction from agent’s feedback throughout the multi-step, long-haul reasoning process. In contrast to these, AlphaBlock [334] focuses on more challenging manipulation tasks (e.g. making a smiley face using building blocks), which requires the agent to have a more grounded understanding of the instructions. Unlike the existing open-loop paradigm, AlphaBlock constructs a dataset comprising 35 complex high-level tasks, along with corresponding multi-step planning and observation pairs, and then fine-tunes a multimodal model to enhance its comprehension of high-level cognitive instructions.
- **Navigation.** Navigation permits agents to dynamically alter their positions within the environment, which often involves multi-angle and multi-object observations, as well as long-horizon manipulations based on current exploration [23]. Before navigation, it is essential for embodied agents to establish prior internal maps about the external environment, which are typically in the form of a topological map, semantic map or occupancy map [358]. For example, LM-Nav [335] utilizes the VNM [379] to create an internal topological map. It further leverages the LLM and VLM for decomposing input commands and analyzing the environment to find the optimal path. Furthermore, some [380; 381] highlight the importance of spatial representation to achieve the precise localization of spatial targets rather than conventional point or object-centric navigation actions by leveraging the pre-trained VLM model to combine visual features from images with 3D reconstructions of the physical world [358]. Navigation is usually a long-horizon task, where the upcoming states of the agent are influenced by its past actions. A memory buffer and summary mechanism are needed to serve as a reference for historical information [336], which is also employed in Smallville and Voyager [22; 190; 382; 383]. Additionally, as mentioned in §3.2, some works have proposed the audio input is also of great significance, but integrating audio information presents challenges in associating it with the visual environment. A basic framework includes a dynamic path planner that uses visual and auditory observations along with spatial memories to plan a series of actions for navigation [375; 384].

By integrating these, the agent can accomplish more complex tasks, such as embodied question answering, whose primary objective is autonomous exploration of the environment, and responding to pre-defined multimodal questions, such as *Is the watermelon in the kitchen larger than the pot? Which one is harder?* To address these questions, the agent needs to navigate to the kitchen, observe the sizes of both objects and then answer the questions through comparison [358].

In terms of control strategies, as previously mentioned, LLM-based agents trained on particular embodied datasets typically generate high-level policy commands to control low-level policies for achieving specific sub-goals. The low-level policy can be a robotic transformer [120; 385; 386], which takes images and instructions as inputs and produces control commands for the end effector as well as robotic arms in particular embodied tasks [179]. Recently, in virtual embodied environments, the high-level strategies are utilized to control agents in gaming [172; 183; 190; 337] or simulated worlds [22; 108; 109]. For instance, Voyager [190] calls the Mineflayer [387] API interface to continuously acquire various skills and explore the world.

Prospective future of the embodied action. LLM-based embodied actions are seen as the bridge between virtual intelligence and the physical world, enabling agents to perceive and modify the environment much like humans. However, there remain several constraints such as high costs of physical-world robotic operators and the scarcity of embodied datasets, which foster a growing

由回声音频输入引导的元素，增强代理的观察能力，提供更全面的视角 [375]。近年来，越来越多的研究将音频作为嵌入观察的模式。除了广泛采用的级联范式 [293; {v9}378{v11}; 316]，类似于ViT的音频信息编码进一步促进了音频与其他输入模式的无缝融合 [294]。代理对环境的观察还可以来自人类的实时语言指令，而人类反馈则帮助代理获取可能难以直接获得或解析的细节信息 [333; 190]。

- **操控。** 一般而言，具身代理的操控任务包括物体重排、桌面操控和移动操控 [23; 120]。典型的情况是在厨房中执行一系列任务，包括从抽屉中取出物品并交给用户，以及清理桌面 [179]。除了精确的观察外，这还涉及通过利用LLM将一系列子目标结合起来。因此，保持代理的状态与子目标之间的同步具有重要意义。DEPS [183] 采用基于LLM的交互式规划方法来保持这种一致性，并在多步骤、长距离推理过程中帮助错误校正，利用代理的反馈。与此相对，AlphaBlock [334] 专注于更具挑战性的操控任务（例如用积木搭建笑脸），这要求代理对指令有更为扎实的理解。不同于现有的开环范式，AlphaBlock构建了一个包含35个复杂高层次任务的数据集，以及相应的多步骤规划和观察对，然后对多模态模型进行微调，以增强其对高层次认知指令的理解。

- **导航。** 导航使代理能够动态改变其在环境中的位置，这通常涉及多角度和多目标的观察，以及基于当前探索的长远操控 [23]。在导航之前，具有体现能力的代理必须建立关于外部环境的先验内部地图，通常以拓扑地图、语义地图或占用地图的形式存在 [358]。例如，LM-Nav [335]利用 VNM [379] 创建内部拓扑地图。它进一步利用 LLM 和 VLM 来分解输入指令和分析环境，以找到最优路径。此外，一些 [380; 381] 强调空间表示的重要性，以实现空间目标的精确定位，而不是传统的点或以对象为中心的导航动作，利用预训练的 VLM 模型将图像的视觉特征与物理世界的 3D 重建相结合 [358]。导航通常是一个长远任务，代理的未来状态受到其过去动作的影响。需要一个记忆缓冲区和总结机制，作为历史信息的参考 [336]，这在 Smallville 和 Voyager 中也有应用 [22; 190; 382; 383]。此外，正如 § 3.2中提到的，一些研究提出音频输入 also 具有重要意义，但整合音频信息在将其与视觉环境关联方面存在挑战。一个基本框架包括一个动态路径规划器，利用视觉和听觉观察以及空间记忆来规划一系列导航动作 [375; 384]。

通过整合这些能力，代理可以完成更复杂的任务，例如具身问答，其主要目标是自主探索环境，并对预定义的多模态问题做出回应，例如厨房里的西瓜比锅大吗？哪个更硬？为了回答这些问题，代理需要导航到厨房，观察两个物体的大小，然后通过比较 [358]来回答问题。

在控制策略方面，如前所述，基于大型语言模型（LLM）的代理在特定的具身数据集上训练，通常会生成高层次的策略指令，以控制低层次的策略，从而实现特定的子目标。低层策略可以是机器人变换器 [120，它以图像和指令作为输入，生成末端执行器以及机器人手臂在特定具身任务中的控制指令 [179]。近年来，在虚拟具身环境中，高层策略被用来控制代理进行游戏 [172; [22或模拟世界 109]。例如，Voyager [190] 调用Mineflayer [387] API接口，持续获取各种技能并探索世界。

《基于大型语言模型的代理的崛起与潜力：一项综述》**具身行动的未来展望。** 基于大型语言模型的具身行动被视为虚拟智能与物理世界之间的桥梁，使代理能够像人类一样感知和改变环境。然而，仍存在一些限制，例如物理世界机器人操作的高成本以及具身数据集的稀缺，这促使着一个不断增长的

interest in investigating agents’ embodied actions within simulated environments like Minecraft [183; 338; 337; 190; 339]. By utilizing the Mineflayer [387] API, these investigations enable cost-effective examination of a wide range of embodied agents’ operations including exploration, planning, self-improvement, and even lifelong learning [190]. Despite notable progress, achieving optimal embodied actions remains a challenge due to the significant disparity between simulated platforms and the physical world. To enable the effective deployment of embodied agents in real-world scenarios, an increasing demand exists for embodied task paradigms and evaluation criteria that closely mirror real-world conditions [358]. On the other hand, learning to ground language for agents is also an obstacle. For example, expressions like “jump down like a cat” primarily convey a sense of lightness and tranquility, but this linguistic metaphor requires adequate world knowledge [30]. [340] endeavors to amalgamate text distillation with Hindsight Experience Replay (HER) to construct a dataset as the supervised signal for the training process. Nevertheless, additional investigation on grounding embodied datasets still remains necessary while embodied action plays an increasingly pivotal role across various domains in human life.

4 Agents in Practice: Harnessing AI for Good

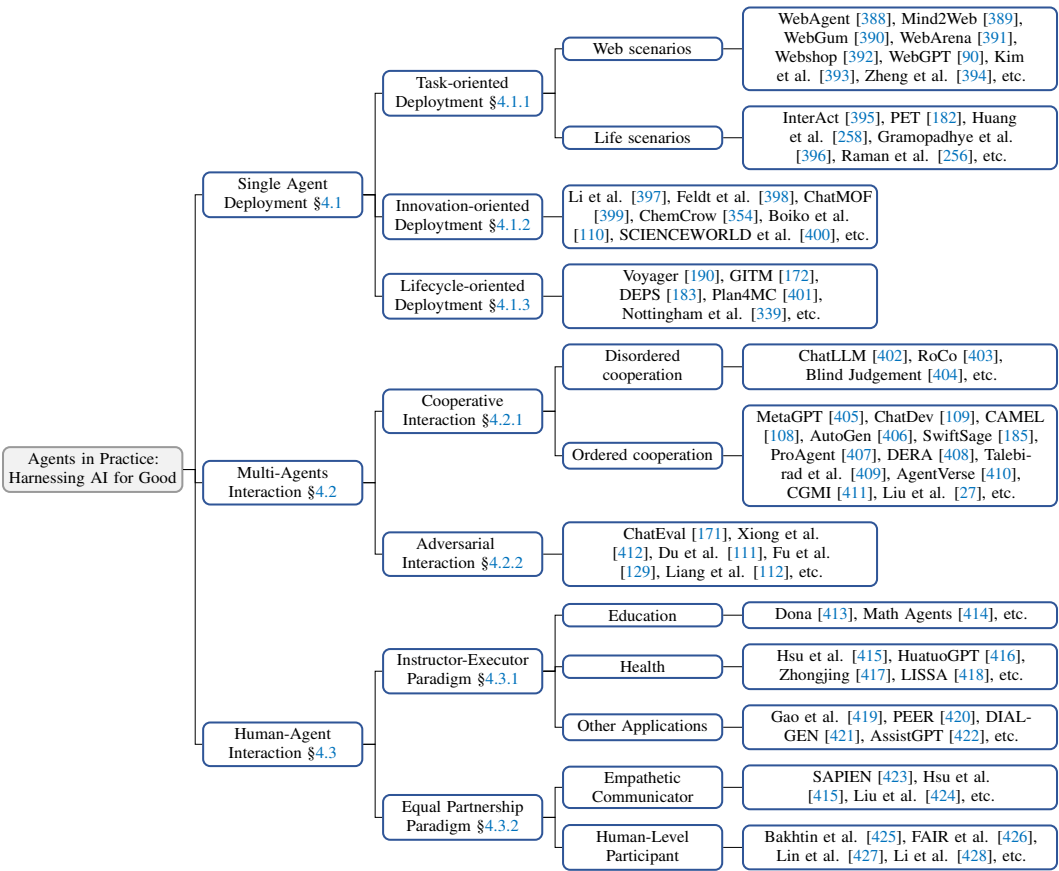


Figure 6: Typology of applications of LLM-based agents.

The LLM-based agent, as an emerging direction, has gained increasing attention from researchers. Many applications in specific domains and tasks have already been developed, showcasing the powerful and versatile capabilities of agents. We can state with great confidence that, the possibility of having a personal agent capable of assisting users with typical daily tasks is larger than ever before [398]. As an LLM-based agent, its design objective should always be beneficial to humans, i.e., humans can *harness AI for good*. Specifically, we expect the agent to achieve the following objectives:

对在像Minecraft这样的模拟环境中研究代理人具身行为的兴趣日益增加；338；337；190； 339。通过利用Mineflayer API，这些研究能够以成本效益的方式检验各种具身代理人的操作，包括探索、规划、自我提升甚至终身学习；尽管取得了显著进展，但由于模拟平台与物理世界之间存在巨大差异，实现最优的具身行为仍然具有挑战性。为了在实际场景中有效部署具身代理人，越来越多的需求出现，用于模拟真实条件的具身任务范式和评估标准；另一方面，学习为代理人建立语言基础也是一大难题。例如，“像猫一样跳下来”这样的表达主要传达轻盈和宁静的感觉，但这种语言隐喻需要充分的世界知识。[340]。努力将文本提炼与后见之明经验回放（HER）相结合，构建作为训练过程监督信号的数据集。然而，随着具身行为在各个领域中的日益重要，关于具身数据集的基础研究仍然是必要的。

4 实践中的智能体：利用人工智能造福社会

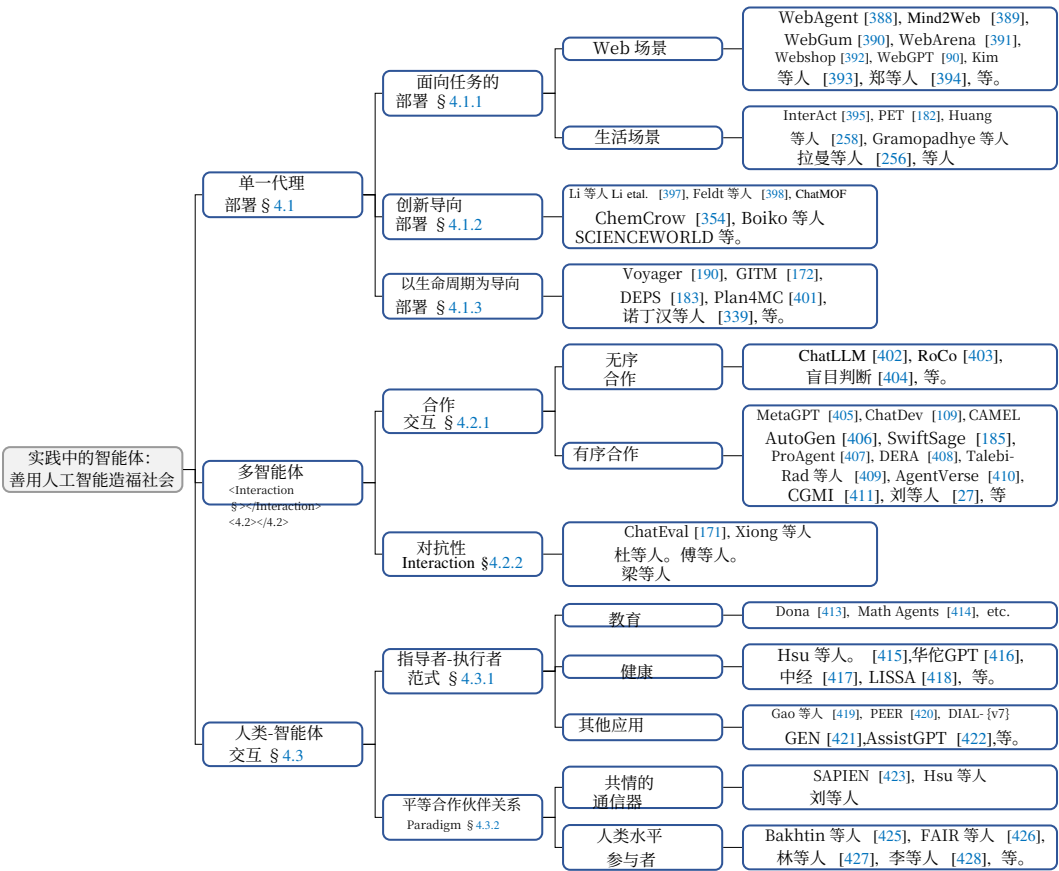


图6：基于大型语言模型的代理应用类型学。

基于大型语言模型的代理，作为一个新兴方向，已引起研究人员日益关注。许多在特定领域和任务中的应用已经被开发出来，展示了代理的强大和多功能能力。我们可以非常自信地断言，拥有一个能够协助用户完成日常典型任务的个人代理的可能性比以往任何时候都要大 [398]。作为一个基于大型语言模型的代理，其设计目标应始终以造福人类为宗旨，即人类可以善用人工智能。具体而言，我们期望该代理实现以下目标：

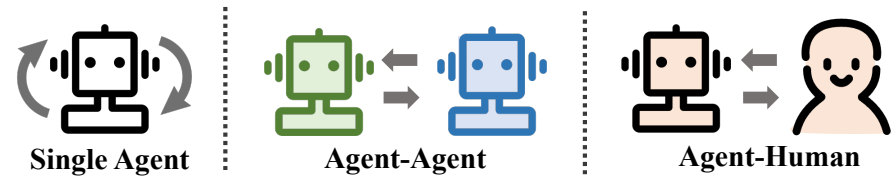


Figure 7: Scenarios of LLM-based agent applications. We mainly introduce three scenarios: single-agent deployment, multi-agent interaction, and human-agent interaction. A **single agent** possesses diverse capabilities and can demonstrate outstanding task-solving performance in various application orientations. When **multiple agents** interact, they can achieve advancement through cooperative or adversarial interactions. Furthermore, in **human-agent** interactions, human feedback can enable agents to perform tasks more efficiently and safely, while agents can also provide better service to humans.

1. Assist users in breaking free from daily tasks and repetitive labor, thereby Alleviating human work pressure and enhancing task-solving efficiency.
2. No longer necessitates users to provide explicit low-level instructions. Instead, the agent can independently analyze, plan, and solve problems.
3. After freeing users' hands, the agent also liberates their minds to engage in exploratory and innovative work, realizing their full potential in cutting-edge scientific fields.

In this section, we provide an in-depth overview of current applications of LLM-based agents, aiming to offer a broad perspective for the practical deployment scenarios (see Figure 7). First, we elucidate the diverse application scenarios of Single Agent, including task-oriented, innovation-oriented, and lifecycle-oriented scenarios (§ 4.1). Then, we present the significant coordinating potential of Multiple Agents. Whether through cooperative interaction for complementarity or adversarial interaction for advancement, both approaches can lead to higher task efficiency and response quality (§ 4.2). Finally, we categorize the interactive collaboration between humans and agents into two paradigms and introduce the main forms and specific applications respectively (§ 4.3). The topological diagram for LLM-based agent applications is depicted in Figure 6.

4.1 General Ability of Single Agent

Currently, there is a vibrant development of application instances of LLM-based agents [429; 430; 431]. AutoGPT [114] is one of the ongoing popular open-source projects aiming to achieve a fully autonomous system. Apart from the basic functions of large language models like GPT-4, the AutoGPT framework also incorporates various practical external tools and long/short-term memory management. After users input their customized objectives, they can free their hands and wait for AutoGPT to automatically generate thoughts and perform specific tasks, all without requiring additional user prompts.

As shown in Figure 8, we introduce the astonishingly diverse capabilities that the agent exhibits in scenarios where only one single agent is present.

4.1.1 Task-oriented Deployment

The LLM-based agents, which can understand human natural language commands and perform everyday tasks [391], are currently among the most favored and practically valuable agents by users. This is because they have the potential to enhance task efficiency, alleviate user workload, and promote access for a broader user base. In **task-oriented deployment**, the agent follows high-level instructions from users, undertaking tasks such as goal decomposition [182; 258; 388; 394], sequence planning of sub-goals [182; 395], interactive exploration of the environment [256; 391; 390; 392], until the final objective is achieved.

To explore whether agents can perform basic tasks, they are first deployed in text-based game scenarios. In this type of game, agents interact with the world purely using natural language [432]. By reading textual descriptions of their surroundings and utilizing skills like memory, planning,

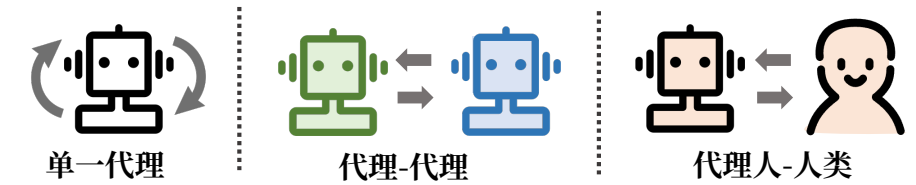


图7：基于大语言模型的智能体应用场景。我们主要介绍三种场景：单一智能体部署、多智能体交互和人机交互。单一智能体具有多样的能力，能够在各种应用方向上展现出出色的任务解决性能。当多智能体进行交互时，它们可以通过合作或对抗的方式实现共同进步。此外，在人机交互中，人类反馈可以使智能体更高效、更安全地完成任务，同时智能体也能为人类提供更优质的服务。

1. 协助用户摆脱日常任务和重复劳动，从而减轻人类工作压力并提高任务解决效率。
2. 不再需要用户提供明确的低层次指令，而是代理可以自主分析、规划和解决问题。
3. 在解放用户双手的同时，代理也解放了他们的思维，使其能够从事探索性和创新性工作，充分发挥在前沿科学领域的潜能。

在本节中，我们将深入概述基于大型语言模型（LLM）代理的当前应用，旨在为实际部署场景提供广阔的视角（见图 7）。首先，我们阐述单一代理的多样化应用场景，包括任务导向、创新导向和生命周期导向场景（§ 4.1）。然后，介绍多代理的显著协同潜力。无论是通过合作互动实现互补，还是通过对抗互动推动进步，两者都能提升任务效率和响应质量（§ 4.2）。最后，我们将人机交互合作划分为两种范式，并分别介绍其主要形式和具体应用（§ 4.3）。基于LLM的代理应用的拓扑图如图6所示。

4.1 单一代理的通用能力

目前，基于大型语言模型的智能体应用实例呈现出蓬勃发展的态势Currently, there is a vibrant development of application instances of LLM-based agents [429; 430; 431]. AutoGPT [114] is aimed at achieving a fully autonomous system. Apart from the basic functions of large language models like GPT-4, the AutoGPT framework also incorporates various practical external tools and long/short-term memory management. After users input their customized objectives, they can free their hands and wait for AutoGPT to automatically generate thoughts and perform specific tasks, all without requiring additional user prompts.

如图8所示，我们介绍了在仅有单一代理存在的场景中，代理展现出的令人惊叹的多样化能力。

4.1.1 任务导向的部署

基于大型语言模型的智能体，能够理解人类自然语言指令并执行日常任务，目前在用户中最受青睐且具有实际价值的智能体之一。这是因为它们具有提升任务效率、减轻用户工作负担以及促进更广泛用户群体访问的潜力。在**任务导向部署**中，智能体遵循用户的高层次指令，承担诸如目标分解 [182; 258; 388; 394], 子目标的序列规划 [182; 395], 环境的交互式探索 [256; 391; 390; 392], 直到最终目标的实现。

为了探索代理是否能够执行基本任务，首先将其部署在基于文本的游戏场景中。在这种类型的游戏中，代理纯粹通过自然语言 [432] 与世界互动。通过阅读环境的文本描述并利用记忆、规划等技能，

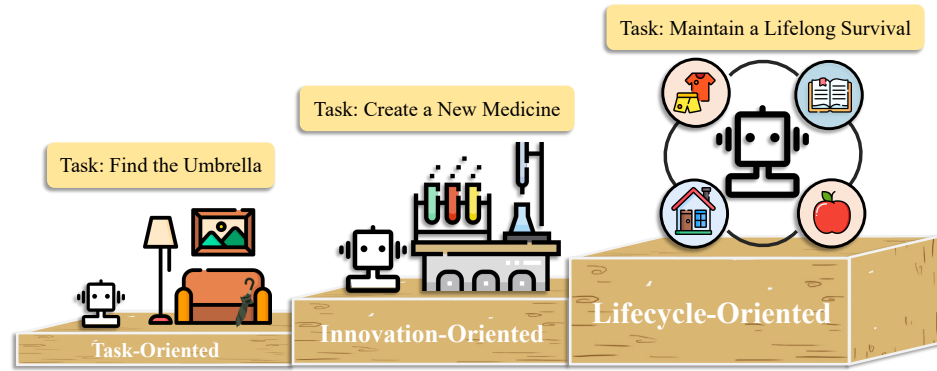


Figure 8: Practical applications of the single LLM-based agent in different scenarios. In **task-oriented deployment**, agents assist human users in solving daily tasks. They need to possess basic instruction comprehension and task decomposition abilities. In **innovation-oriented deployment**, agents demonstrate the potential for autonomous exploration in scientific domains. In **lifecycle-oriented deployment**, agents have the ability to continuously explore, learn, and utilize new skills to ensure long-term survival in an open world.

and trial-and-error [182], they predict the next action. However, due to the limitation of foundation language models, agents often rely on reinforcement learning during actual execution [432; 433; 434].

With the gradual evolution of LLMs [301], agents equipped with stronger text understanding and generation abilities have demonstrated great potential to perform tasks through natural language. Due to their oversimplified nature, naive text-based scenarios have been inadequate as testing grounds for LLM-based agents [391]. More realistic and complex simulated test environments have been constructed to meet the demand. Based on task types, we divide these simulated environments into **web scenarios** and **life scenarios**, and introduce the specific roles that agents play in them.

In web scenarios. Performing specific tasks on behalf of users in a web scenario is known as the web navigation problem [390]. Agents interpret user instructions, break them down into multiple basic operations, and interact with computers. This often includes web tasks such as filling out forms, online shopping, and sending emails. Agents need to possess the ability to understand instructions within complex web scenarios, adapt to changes (such as noisy text and dynamic HTML web pages), and generalize successful operations [391]. In this way, agents can achieve accessibility and automation when dealing with unseen tasks in the future [435], ultimately freeing humans from repeated interactions with computer UIs.

Agents trained through reinforcement learning can effectively mimic human behavior using predefined actions like typing, searching, navigating to the next page, etc. They perform well in basic tasks such as online shopping [392] and search engine retrieval [90], which have been widely explored. However, agents without LLM capabilities may struggle to adapt to the more realistic and complex scenarios in the real-world Internet. In dynamic, content-rich web pages such as online forums or online business management [391], agents often face challenges in performance.

In order to enable successful interactions between agents and more realistic web pages, some researchers [393; 394] have started to leverage the powerful HTML reading and understanding abilities of LLMs. By designing prompts, they attempt to make agents understand the entire HTML source code and predict more reasonable next action steps. Mind2Web [389] combines multiple LLMs fine-tuned for HTML, allowing them to summarize verbose HTML code [388] in real-world scenarios and extract valuable information. Furthermore, WebGum [390] empowers agents with visual perception abilities by employing a multimodal corpus containing HTML screenshots. It simultaneously fine-tunes the LLM and a visual encoder, deepening the agent’s comprehensive understanding of web pages.

In life scenarios. In many daily household tasks in life scenarios, it’s essential for agents to understand implicit instructions and apply common-sense knowledge [433]. For an LLM-based agent trained solely on massive amounts of text, tasks that humans take for granted might require multiple



图8：基于单一大型语言模型（LLM）代理在不同场景中的实际应用。在**任务导向部署**中，代理协助人类用户完成日常任务，需具备基本的指令理解和任务分解能力。在**创新导向部署**中，代理展现出在科学领域自主探索的潜力。在**生命周期导向部署**中，代理具备持续探索、学习和利用新技能的能力，以确保在开放世界中的长期生存。

和试错法 [182], 他们预测下一步行动。然而，由于基础模型的限制，**单轮模型**，代理在实际执行过程中通常依赖强化学习 [432; 433; 434]。

随着大型语言模型 [301], 代理的逐步演进，具备更强文本理解与生成能力的代理在通过自然语言执行任务方面展现出巨大潜力。由于其过于简化的特性，单纯的基于文本的场景已不足以作为基于大型语言模型代理 [391] 的测试平台。为此，构建了更为真实且复杂的模拟测试环境以满足需求。根据任务类型，我们将这些模拟环境划分为**网页场景**和**生活场景**，并介绍了代理在其中扮演的具体角色。

在网页场景中。代理在网页场景中代表用户执行特定任务被称为网页导航问题 [390]。代理解释用户指令，将其拆解为多个基本操作，并与计算机交互。这通常包括填写表单、在线购物和发送电子邮件等网页任务。代理需要具备理解复杂网页场景中指令的能力，适应变化（如噪声文本和动态HTML网页），并实现成功操作的泛化 [391]。通过这种方式，代理可以在未来处理未见过的任务时实现无障碍和自动化，最终解放人类免于反复与计算机用户界面交互 [435]。

通过强化学习训练的智能体能够有效模仿人类行为，使用预定义的动作如打字、搜索、导航到下一页等。在诸如网络购物 [392] 和搜索引擎检索 [90], 等基础任务中表现良好，这些任务已被广泛研究。然而，缺乏大型语言模型能力的智能体可能难以适应现实世界互联网中更为复杂和真实的场景。在动态的、内容丰富的网页如在线论坛或电子商务管理 [391], 中，智能体在性能方面常常面临挑战。

为了实现代理与更真实网页之间的成功交互，一些研究人员 [393; 394] 开始利用大型语言模型（LLMs）强大的HTML阅读与理解能力。通过设计提示，他们试图使代理理解整个HTML源代码，并预测更合理的下一步操作。Mind2Web [389] 结合了多种针对HTML微调的LLMs，能够在实际场景中对冗长的HTML代码 [388] 进行总结并提取有价值的信息。此外，WebGum [390] 通过采用包含HTML截图的多模态语料库，为代理赋予视觉感知能力。它同时对LLM和视觉编码器进行微调，增强代理对网页的全面理解。

在生活场景中。在生活场景中的许多日常家务任务中，代理需要理解隐含指令并应用常识知识 [433]。对于仅通过大量文本训练的基于大规模语言模型（LLM）的代理而言，许多由人类理所当然完成的任务可能需要多次尝试。

trial-and-error attempts [432]. More realistic scenarios often lead to more obscure and subtle tasks. For example, the agent should proactively turn it on if it’s dark and there’s a light in the room. To successfully chop some vegetables in the kitchen, the agent needs to anticipate the possible location of a knife [182].

Can an agent apply the world knowledge embedded in its training data to real interaction scenarios? Huang et al. [258] lead the way in exploring this question. They demonstrate that sufficiently large LLMs, with appropriate prompts, can effectively break down high-level tasks into suitable sub-tasks without additional training. However, this static reasoning and planning ability has its potential drawbacks. Actions generated by agents often lack awareness of the dynamic environment around them. For instance, when a user gives the task “clean the room”, the agent might convert it into unfeasible sub-tasks like “call a cleaning service” [396].

To provide agents with access to comprehensive scenario information during interactions, some approaches directly incorporate spatial data and item-location relationships as additional inputs to the model. This allows agents to gain a precise description of their surroundings [395; 396]. Wu et al. [182] introduce the PET framework, which mitigates irrelevant objects and containers in environmental information through an early error correction method [256]. PET encourages agents to explore the scenario and plan actions more efficiently, focusing on the current sub-task.

4.1.2 Innovation-oriented Deployment

The LLM-based agent has demonstrated strong capabilities in performing tasks and enhancing the efficiency of repetitive work. However, in a more intellectually demanding field, like cutting-edge science, the potential of agents has not been fully realized yet. This limitation mainly arises from two challenges [399]: On one hand, the inherent complexity of science poses a significant barrier. Many domain-specific terms and multi-dimensional structures are difficult to represent using a single text. As a result, their complete attributes cannot be fully encapsulated. This greatly weakens the agent’s cognitive level. On the other hand, there is a severe lack of suitable training data in scientific domains, making it difficult for agents to comprehend the entire domain knowledge [400; 436]. If the ability for autonomous exploration could be discovered within the agent, it would undoubtedly bring about beneficial innovation in human technology.

Currently, numerous efforts in various specialized domains aim to overcome this challenge [437; 438; 439]. Experts from the computer field make full use of the agent’s powerful code comprehension and debugging abilities [398; 397]. In the fields of chemistry and materials, researchers equip agents with a large number of general or task-specific tools to better understand domain knowledge. Agents evolve into comprehensive scientific assistants, proficient in online research and document analysis to fill data gaps. They also employ robotic APIs for real-world interactions, enabling tasks like material synthesis and mechanism discovery [110; 354; 399].

The potential of LLM-based agents in scientific innovation is evident, yet we do not expect their exploratory abilities to be utilized in applications that could threaten or harm humans. Boiko et al. [110] study the hidden dangers of agents in synthesizing illegal drugs and chemical weapons, indicating that agents could be misled by malicious users in adversarial prompts. This serves as a warning for our future work.

4.1.3 Lifecycle-oriented Deployment

Building a universally capable agent that can continuously explore, develop new skills, and maintain a long-term life cycle in an open, unknown world is a colossal challenge. This accomplishment is regarded as a pivotal milestone in the field of AGI [183]. Minecraft, as a typical and widely explored simulated survival environment, has become a unique playground for developing and testing the comprehensive ability of an agent. Players typically start by learning the basics, such as mining wood and making crafting tables, before moving on to more complex tasks like fighting against monsters and crafting diamond tools [190]. Minecraft fundamentally reflects the real world, making it conducive for researchers to investigate an agent’s potential to survive in the authentic world.

The survival algorithms of agents in Minecraft can generally be categorized into two types [190]: **low-level control** and **high-level planning**. Early efforts mainly focused on reinforcement learning [190; 440] and imitation learning [441], enabling agents to craft some low-level items. With the emergence of LLMs, which demonstrated surprising reasoning and analytical capabilities, agents

试错尝试 [432]。更为真实的场景通常会导致更模糊和微妙的任务。例如，如果房间里很暗且有灯光，代理应主动开启灯光。为了在厨房成功切割一些蔬菜，代理需要预判刀具可能的位置 [182]。

代理能否将其训练数据中嵌入的世界知识应用于实际交互场景？黄等人 [258] 在探索这一问题方面走在了前列。他们证明，充分规模的大型语言模型（LLMs），配合适当的提示，可以在无需额外训练的情况下，有效地将高层次任务拆解为合适的子任务。然而，这种静态的推理与规划能力也存在潜在的缺陷。代理生成的行动往往缺乏对其周围动态环境的感知。例如，当用户提出“打扫房间”的任务时，代理可能会将其转化为不可行的子任务，如“呼叫清洁服务” [396]。

为了在交互过程中为代理提供全面的场景信息，一些方法将空间数据和物品位置关系作为额外输入直接融入模型中。这使得代理能够获得对其周围环境的精确描述 [395; 396]。Wu 等人 [182] 引入了 PET 框架，通过早期错误校正方法 [256]，减轻环境信息中无关对象和容器的干扰。PET 鼓励代理更高效地探索场景并规划行动，专注于当前子任务。

4.1.2 面向创新的部署

基于大型语言模型的代理在执行任务和提升重复性工作的效率方面展现出强大的能力。然而，在如前沿科学等更具智力挑战性的领域，代理的潜力尚未得到充分发挥。这一局限性主要源于两个挑战 [399]：一方面，科学的固有复杂性构成了显著的障碍。许多领域特定术语和多维结构难以用单一文本进行表达，因此其完整属性无法完全涵盖。这极大地削弱了代理的认知水平。另一方面，科学领域缺乏适用的训练数据，使得代理难以理解整个领域知识 [400; 436]。如果能够在代理中发现自主探索的能力，无疑将为人类技术带来有益的创新。

目前，各专业领域的众多努力旨在克服这一挑战。计算机领域的专家充分利用代理在代码理解和调试方面的强大能力。在化学和材料领域，研究人员为代理配备大量通用或任务特定的工具，以更好地理解领域知识。代理逐渐演变为综合性的科学助手，擅长在线研究和文档分析，以弥补数据空白。同时，它们还利用机器人API进行实际世界的交互，实现材料合成和机制发现等任务。

基于大规模语言模型的代理在科学创新中的潜力是显而易见的，但我们不期望其探索能力在可能威胁或危害人类的应用中得到利用。Boiko 等人 [110] 研究了代理在合成非法药物和化学武器方面的潜在隐患，指出代理可能被恶意用户通过对抗性提示误导。这为我们的未来工作敲响了警钟。

4.1.3 面向生命周期的部署

构建一个具有普遍能力的智能体，能够在开放、未知的世界中持续探索、开发新技能并维持长期生命周期，是一项巨大的挑战。这一成就被视为通用人工智能（AGI）领域的关键里程碑。作为一种典型且广泛探索的模拟生存环境，Minecraft已成为开发和测试智能体综合能力的独特平台。玩家通常从学习基础技能开始，例如采集木材和制作工作台，然后逐步过渡到更复杂的任务，如对抗怪物和制作钻石工具。Minecraft在根本上反映了现实世界，为研究人员探索智能体在真实世界中生存潜能提供了有利条件。

Minecraft中代理的生存算法通常可以分为两类：**低级控制**和**高级规划**。早期的研究主要集中在强化学习[190; 440]和模仿学习[441]，使代理能够制作一些低级物品。随着大规模语言模型（LLMs）的出现，展现出令人惊讶的推理和分析能力，代理

begin to utilize LLM as a high-level planner to guide simulated survival tasks [183; 339]. Some researchers use LLM to decompose high-level task instructions into a series of sub-goals [401], basic skill sequences [339], or fundamental keyboard/mouse operations [401], gradually assisting agents in exploring the open world.

Voyager[190], drawing inspiration from concepts similar to AutoGPT[114], became the first LLM-based embodied lifelong learning agent in Minecraft, based on the long-term goal of “discovering as many diverse things as possible”. It introduces a skill library for storing and retrieving complex action-executable code, along with an iterative prompt mechanism that incorporates environmental feedback and error correction. This enables the agent to autonomously explore and adapt to unknown environments without human intervention. An AI agent capable of autonomously learning and mastering the entire real-world techniques may not be as distant as once thought [401].

4.2 Coordinating Potential of Multiple Agents

Motivation and Background. Although LLM-based agents possess commendable text understanding and generation capabilities, they operate as isolated entities in nature [409]. They lack the ability to collaborate with other agents and acquire knowledge from social interactions. This inherent limitation restricts their potential to learn from multi-turn feedback from others to enhance their performance [27]. Moreover, they cannot be effectively deployed in complex scenarios requiring collaboration and information sharing among multiple agents.

As early as 1986, Marvin Minsky made a forward-looking prediction. In his book *The Society of Mind* [442], he introduced a novel theory of intelligence, suggesting that intelligence emerges from the interactions of many smaller agents with specific functions. For instance, certain agents might be responsible for pattern recognition, while others might handle decision-making or generate solutions. This idea has been put into concrete practice with the rise of distributed artificial intelligence [443]. Multi-agent systems(MAS) [4], as one of the primary research domains, focus on how a group of agents can effectively coordinate and collaborate to solve problems. Some specialized communication languages, like KQML [444], were designed early on to support message transmission and knowledge sharing among agents. However, their message formats were relatively fixed, and the semantic expression capacity was limited. In the 21st century, integrating reinforcement learning algorithms (such as Q-learning) with deep learning has become a prominent technique for developing MAS that operate in complex environments [445]. Nowadays, the construction approach based on LLMs is beginning to demonstrate remarkable potential. The natural language communication between agents has become more elegant and easily comprehensible to humans, resulting in a significant leap in interaction efficiency.

Potential advantages. Specifically, an LLM-based multi-agent system can offer several advantages. Just as Adam Smith clearly stated in *The Wealth of Nations* [446], “The greatest improvements in the productive powers of labor, and most of the skill, dexterity, and judgment with which it is directed or applied, seem to be results of the division of labor.” Based on the principle of division of labor, a single agent equipped with specialized skills and domain knowledge can engage in specific tasks. On the one hand, agents’ skills in handling specific tasks are increasingly refined through the division of labor. On the other hand, decomposing complex tasks into multiple subtasks can eliminate the time spent switching between different processes. In the end, efficient division of labor among multiple agents can accomplish a significantly greater workload than when there is no specialization, substantially improving the overall system’s efficiency and output quality.

In § 4.1, we have provided a comprehensive introduction to the versatile abilities of LLM-based agents. Therefore, in this section, we focus on exploring the ways agents interact with each other in a multi-agent environment. Based on current research, these interactions can be broadly categorized as follows: **Cooperative Interaction for Complementarity** and **Adversarial Interaction for Advancement** (see Figure 9).

4.2.1 Cooperative Interaction for Complementarity

Cooperative multi-agent systems are the most widely deployed pattern in practical usage. Within such systems, individual agent assesses the needs and capabilities of other agents and actively seeks collaborative actions and information sharing with them [108]. This approach brings forth numerous potential benefits, including enhanced task efficiency, collective decision improvement, and the

开始将大型语言模型（LLM）作为高级规划器，用于指导模拟生存任务 [183; 339]。一些研究人员使用LLM将高层次任务指令分解为一系列子目标 [401], 基本技能序列 [339], 或基础键盘/鼠标操作 [401], , 逐步协助代理探索开放世界。

Voyager[190], 借鉴类似AutoGPT[114], 的概念，成为第一个基于大型语言模型的具身终身学习代理，在Minecraft中实现，其长期目标是“发现尽可能多的多样化事物”。它引入了技能库，用于存储和检索复杂的可执行动作代码，并配备了迭代式提示机制，结合环境反馈和错误修正。这使得代理能够自主探索和适应未知环境，无需人工干预。具备自主学习和掌握整个现实世界技术能力的AI代理，可能不像以前认为的那么遥远 [401]。

4.2 多智能体的协调潜能

动机与背景。 尽管基于大型语言模型的代理具备出色的文本理解与生成能力，但它们本质上作为孤立实体运行 [409]。它们缺乏与其他代理协作以及从社会交互中获取知识的能力。这一固有限制限制了它们通过多轮反馈学习以提升性能的潜力 [27]。此外，它们无法在需要多代理协作与信息共享的复杂场景中得到有效部署。

早在1986年，Marvin Minsky就做出了前瞻性的预测。在他的著作 *The Society of Mind* [442], 中，他提出了一种新颖的智能理论，认为智能源于具有特定功能的众多较小代理之间的相互作用。例如，某些代理可能负责模式识别，而其他代理则处理决策或生成解决方案。随着分布式人工智能 [443]的兴起，这一思想已被付诸实践。多智能体系统（MAS） [4], 作为主要的研究领域之一，关注一组代理如何有效地协调与合作以解决问题。一些专门的通信语言，如 KQML [444], , 早期被设计用来支持代理之间的信息传输和知识共享。然而，它们的消息格式相对固定，语义表达能力有限。在21世纪，将强化学习算法（如 Q-learning）与深度学习结合，已成为在复杂环境中开发MAS的一个重要技术 [445]。如今，基于LLMs的构建方法开始展现出显著的潜力。代理之间的自然语言通信变得更加优雅，且更易被人类理解，极大地提升了交互效率。

潜在优势。具体而言，基于大规模语言模型（LLM）的多智能体系统可以提供多方面的优势。正如亚当·斯密在《国富论》中明确指出：“劳动生产力的最大改善，以及由此带来的技能、灵巧和判断力的提升，似乎都源于分工。”基于分工原则，配备专业技能和领域知识的单一智能体可以执行特定任务。一方面，智能体在处理特定任务方面的技能通过分工不断精炼；另一方面，将复杂任务拆分为多个子任务，可以消除在不同流程之间切换所耗费的时间。最终，多智能体之间的高效分工能够完成远超无专业化情况下的工作量，显著提升整体系统的效率和输出质量。

在 § 4.1, 我们对基于大型语言模型的代理的多样化能力进行了全面介绍。因此，本节将重点探讨代理在多智能体环境中的交互方式。根据当前研究，这些交互可以大致分为以下几类：**合作互补交互**和**对抗促进交互**（见图9）。

4.2.1 协作交互以实现互补性

合作多智能体系统是实际应用中最广泛部署的模式。在此类系统中，个体智能体评估其他智能体的需求和能力，并主动寻求与其合作行动和信息共享 [108]。这种方法带来了诸多潜在益处，包括提高任务效率、改善集体决策，以及

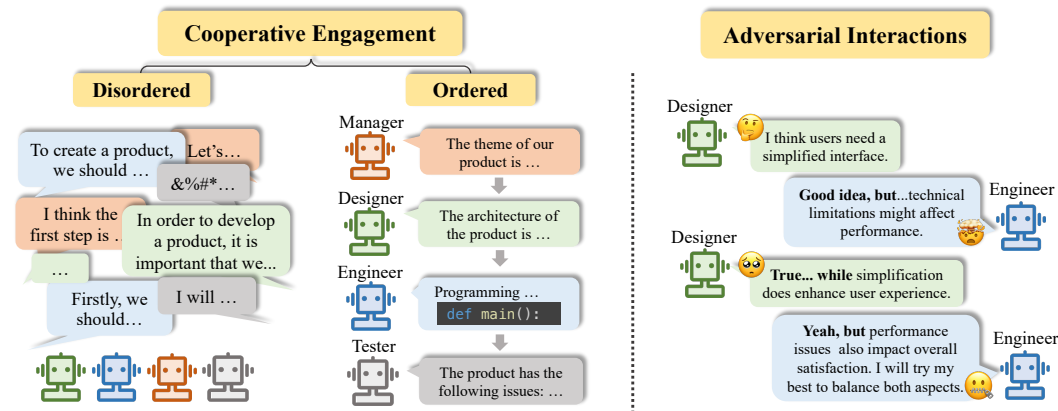


Figure 9: Interaction scenarios for multiple LLM-based agents. In **cooperative interaction**, agents collaborate in either a disordered or ordered manner to achieve shared objectives. In **adversarial interaction**, agents compete in a tit-for-tat fashion to enhance their respective performance.

resolution of complex real-world problems that one single agent cannot solve independently, ultimately achieving the goal of synergistic complementarity. In current LLM-based multi-agent systems, communication between agents predominantly employs natural language, which is considered the most natural and human-understandable form of interaction [108]. We introduce and categorize existing cooperative multi-agent applications into two types: disordered cooperation and ordered cooperation.

Disordered cooperation. When three or more agents are present within a system, each agent is free to express their perspectives and opinions openly. They can provide feedback and suggestions for modifying responses related to the task at hand [403]. This entire discussion process is uncontrolled, lacking any specific sequence, and without introducing a standardized collaborative workflow. We refer to this kind of multi-agent cooperation as **disordered cooperation**.

ChatLLM network [402] is an exemplary representative of this concept. It emulates the forward and backward propagation process within a neural network, treating each agent as an individual node. Agents in the subsequent layer need to process inputs from all the preceding agents and propagate forward. One potential solution is introducing a dedicated coordinating agent in multi-agent systems, responsible for integrating and organizing responses from all agents, thus updating the final answer [447]. However, consolidating a large amount of feedback data and extracting valuable insights poses a significant challenge for the coordinating agent.

Furthermore, **majority voting** can also serve as an effective approach to making appropriate decisions. However, there is limited research that integrates this module into multi-agent systems at present. Hamilton [404] trains nine independent supreme justice agents to better predict judicial rulings in the U.S. Supreme Court, and decisions are made through a majority voting process.

Ordered cooperation. When agents in the system adhere to specific rules, for instance, expressing their opinions one by one in a sequential manner, downstream agents only need to focus on the outputs from upstream. This leads to a significant improvement in task completion efficiency. The entire discussion process is highly organized and ordered. We term this kind of multi-agent cooperation as **ordered cooperation**. It's worth noting that systems with only two agents, essentially engaging in a conversational manner through a back-and-forth interaction, also fall under the category of ordered cooperation.

CAMEL [108] stands as a successful implementation of a dual-agent cooperative system. Within a role-playing communication framework, agents take on the roles of AI Users (giving instructions) and AI Assistants (fulfilling requests by providing specific solutions). Through multi-turn dialogues, these agents autonomously collaborate to fulfill user instructions [408]. Some researchers have integrated the idea of dual-agent cooperation into a single agent's operation [185], alternating between rapid and deliberate thought processes to excel in their respective areas of expertise.

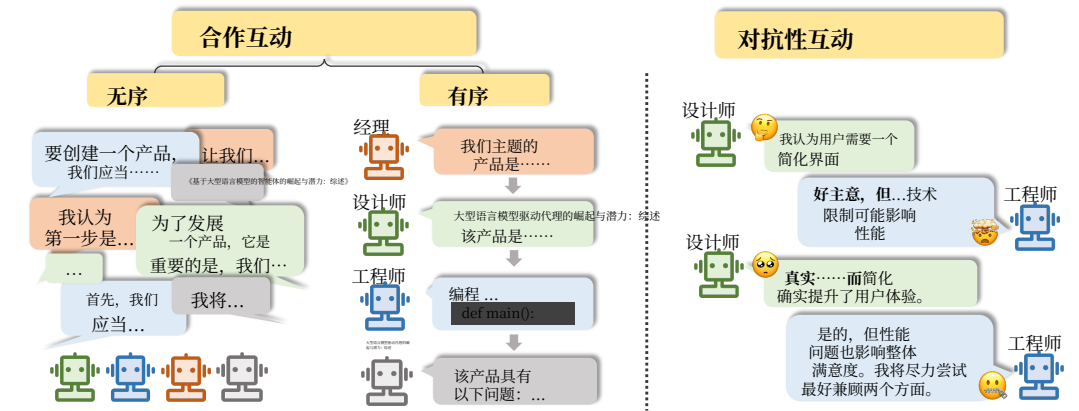


图9：基于多大型语言模型代理的交互场景。在**合作交互**中，代理以无序或有序的方式协作以实现共同目标。在**对抗交互**中，代理以以牙还牙的方式竞争，以提升各自的性能。

复杂现实世界问题的解决方案，单一代理无法独立完成，最终实现协同互补的目标。在当前基于大型语言模型的多智能体系统中，代理之间的通信主要采用自然语言，这被认为是最自然且最易被人类理解的交互形式。[108]我们将现有的合作多智能体应用分为两类：无序合作和有序合作。

无序合作。当系统中存在三个或更多代理时，每个代理都可以自由表达其观点和意见。他们可以对与任务相关的响应提供反馈和建议以进行修改 [403]。整个讨论过程没有控制，缺乏特定的顺序，也未引入标准化的协作流程。我们将这种多代理合作称为**无序合作**。

ChatLLM网络 [402] 是该概念的典范代表。它模拟神经网络中的前向和反向传播过程，将每个代理视为一个独立节点。后续层中的代理需要处理来自所有前置代理的输入，并向前传播。一种潜在的解决方案是在多代理系统中引入专门的协调代理，负责整合和组织所有代理的响应，从而更新最终答案[447]。然而，整合大量反馈数据并提取有价值的见解，对协调代理来说是一个重大挑战。

此外，**多数投票**也可以作为一种有效的决策方法。然而，目前将该模块集成到多智能体系统中的研究仍然有限。Hamilton [404] 训练了九个独立的最高法院法官代理，以更好地预测美国最高法院的判决，决策通过多数投票过程作出。

Ordered cooperation. 当系统中的代理遵循特定规则，例如按顺序依次表达意见时，下游代理只需关注上游的输出。这显著提高了任务完成的效率，整个讨论过程高度有序和组织。我们将这种多代理合作称为**有序合作**。值得注意的是，只有两个代理的系统，基本上通过来回交互进行对话，也属于有序合作的范畴。

CAMEL [108] 作为双智能体协作系统的成功实现。在角色扮演式通信框架中，智能体扮演AI用户（发出指令）和AI助手（通过提供具体方案满足请求）的角色。通过多轮对话，这些智能体自主协作以完成用户指令 [408]。一些研究人员将双智能体合作的思想融入单一智能体的操作中 [185]，在快速思考和深思熟虑的思维过程之间交替，以在各自的专业领域中取得优异表现。

Talebirad et al. [409] are among the first to systematically introduce a comprehensive LLM-based multi-agent collaboration framework. This paradigm aims to harness the strengths of each individual agent and foster cooperative relationships among them. Many applications of multi-agent cooperation have successfully been built upon this foundation [27; 406; 407; 448]. Furthermore, AgentVerse [410] constructs a versatile, multi-task-tested framework for group agents cooperation. It can assemble a team of agents that dynamically adapt according to the task’s complexity. To promote more efficient collaboration, researchers hope that agents can learn from successful human cooperation examples [109]. MetaGPT [405] draws inspiration from the classic waterfall model in software development, standardizing agents’ inputs/outputs as engineering documents. By encoding advanced human process management experience into agent prompts, collaboration among multiple agents becomes more structured.

However, during MetaGPT’s practical exploration, a potential threat to multi-agent cooperation has been identified. Without setting corresponding rules, frequent interactions among multiple agents can amplify minor hallucinations indefinitely [405]. For example, in software development, issues like incomplete functions, missing dependencies, and bugs that are imperceptible to the human eye may arise. Introducing techniques like cross-validation [109] or timely external feedback could have a positive impact on the quality of agent outputs.

4.2.2 Adversarial Interaction for Advancement

Traditionally, cooperative methods have been extensively explored in multi-agent systems. However, researchers increasingly recognize that introducing concepts from game theory [449; 450] into systems can lead to more robust and efficient behaviors. In competitive environments, agents can swiftly adjust strategies through dynamic interactions, striving to select the most advantageous or rational actions in response to changes caused by other agents. Successful applications in Non-LLM-based competitive domains already exist [360; 451]. AlphaGo Zero [452], for instance, is an agent for Go that achieved significant breakthroughs through a process of self-play. Similarly, within LLM-based multi-agent systems, fostering change among agents can naturally occur through competition, argumentation, and debate [453; 454]. By abandoning rigid beliefs and engaging in thoughtful reflection, **adversarial interaction** enhances the quality of responses.

Researchers first delve into the fundamental debating abilities of LLM-based agents [129; 412]. Findings demonstrate that when multiple agents express their arguments in the state of “tit for tat”, one agent can receive substantial external feedback from other agents, thereby correcting its distorted thoughts [112]. Consequently, multi-agent adversarial systems find broad applicability in scenarios requiring high-quality responses and accurate decision-making. In reasoning tasks, Du et al. [111] introduce the concept of debate, endowing agents with responses from fellow peers. When these responses diverge from an agent’s own judgments, a “mental” argumentation occurs, leading to refined solutions. ChatEval [171] establishes a role-playing-based multi-agent referee team. Through self-initiated debates, agents evaluate the quality of text generated by LLMs, reaching a level of excellence comparable to human evaluators.

The performance of the multi-agent adversarial system has shown considerable promise. However, the system is essentially dependent on the strength of LLMs and faces several basic challenges:

- With prolonged debate, LLM’s limited context cannot process the entire input.
- In a multi-agent environment, computational overhead significantly increases.
- Multi-agent negotiation may converge to an incorrect consensus, and all agents are firmly convinced of its accuracy [111].

The development of multi-agent systems is still far from being mature and feasible. Introducing human guides when appropriate to compensate for agents’ shortcomings is a good choice to promote the further advancements of agents.

4.3 Interactive Engagement between Human and Agent

Human-agent interaction, as the name suggests, involves agents collaborating with humans to accomplish tasks. With the enhancement of agent capabilities, human involvement becomes progressively essential to effectively guide and oversee agents’ actions, ensuring they align with human requirements and objectives [455; 456]. Throughout the interaction, humans play a pivotal role by offering

Talebirad 等人是最早系统性引入基于大型语言模型（LLM）多智能体协作框架的学者之一。该范式旨在发挥每个智能体的优势，促进它们之间的合作关系。多智能体协作的许多应用已成功建立在这一基础之上；406；407；此外，AgentVerse 构建了一个多任务验证的多功能智能体合作框架，能够组建一个根据任务复杂性动态适应的智能体团队。为了促进更高效的协作，研究人员希望智能体能够借鉴成功的人类合作实例。MetaGPT 受到软件开发中的瀑布模型的启发，将智能体的输入/输出标准化为工程文档。通过将先进的人类流程管理经验编码到智能体提示中，多智能体之间的协作变得更加结构化。

然而，在MetaGPT的实际探索中，已识别出对多智能体合作的潜在威胁。若未设定相应规则，多智能体之间的频繁交互可能无限放大微小的幻觉 [405]。例如，在软件开发中，可能出现功能不完整、依赖缺失以及肉眼难以察觉的漏洞等问题。引入交叉验证 [109] 或及时的外部反馈等技术，可能对智能体输出的质量产生积极影响。

4.2.2 促进发展的对抗性互动

传统上，合作方法在多智能体系统中得到了广泛的研究。然而，研究人员越来越认识到，将博弈论 [449； 450]的概念引入系统可以带来更具鲁棒性和效率的行为。在竞争环境中，智能体可以通过动态交互迅速调整策略，努力选择对其他智能体引起的变化做出最有利或最合理的行动。在非大语言模型（Non-LLM）基础的竞争领域，已有成功的应用实例 [360； 451]。例如，AlphaGo Zero [452], 是一种围棋智能体，通过自我对弈实现了重大突破。同样，在基于大语言模型的多智能体系统中，通过竞争、辩论和争论 [453； 454], 智能体之间的变化可以自然发生。通过放弃僵化的信念并进行深思熟虑的反思，**对抗性互动**能够提升响应的质量。

研究人员首先探讨了基于大语言模型的代理的基本辩论能力 [129； 412]。研究结果表明，当多个代理以“针锋相对”的状态表达其论点时，一个代理可以从其他代理那里获得大量外部反馈，从而纠正其扭曲的思维 [112]。因此，多代理对抗系统在需要高质量响应和准确决策的场景中具有广泛的应用前景。在推理任务中，Du 等人 [111] 引入了辩论的概念，为代理赋予来自同行的回应。当这些回应偏离代理自身的判断时，会发生“心理”辩论，从而产生更优的解决方案。ChatEval [171] 建立了基于角色扮演的多代理裁判团队。通过自主发起的辩论，代理评估由大语言模型生成文本的质量，达到与人类评估者相当的优秀水平。

多智能体对抗系统的性能已展现出相当的潜力。然而，该系统本质上依赖于大型语言模型（LLMs）的强度，并面临若干基本挑战：

经过长时间的争论，LLM的有限上下文无法处理全部输入内容。

在多智能体环境中，计算开销显著增加。

多智能体协商可能会收敛到错误的共识，并且所有智能体都坚信其正确性。

多智能体系统的发展仍然远未成熟且可行。在适当的时候引入人类引导，以弥补智能体的不足，是推动智能体进一步发展的一个良好选择。

4.3 人类与智能体之间的互动参与

人机交互，顾名思义，涉及代理与人类合作完成任务。随着代理能力的提升，人类的参与变得日益重要，以有效引导和监督代理的行为，确保其符合人类的需求和目标 [455； 456]。在整个交互过程中，人类发挥着关键作用，通过提供

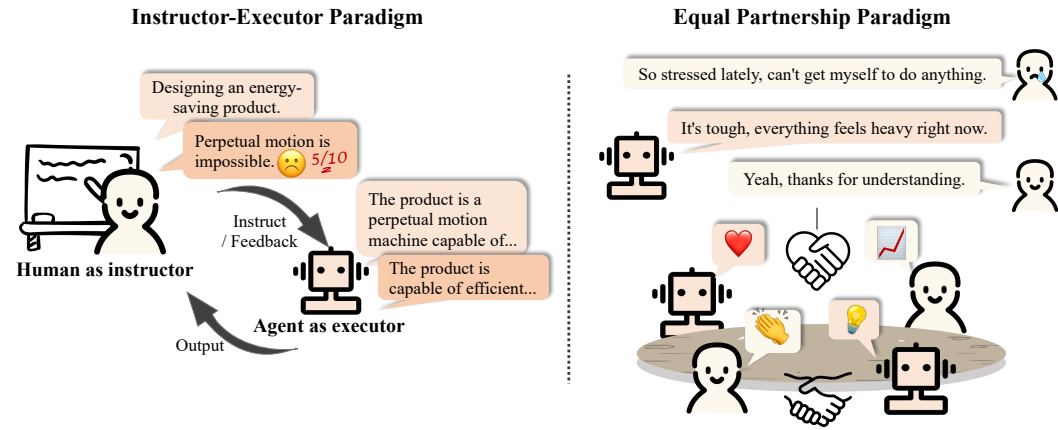


Figure 10: Two paradigms of human-agent interaction. In the instructor-executor paradigm (left), humans provide instructions or feedback, while agents act as executors. In the equal partnership paradigm (right), agents are human-like, able to engage in empathetic conversation and participate in collaborative tasks with humans.

guidance or by regulating the safety, legality, and ethical conduct of agents. This is particularly crucial in specialized domains, such as medicine where data privacy concerns exist [457]. In such cases, human involvement can serve as a valuable means to compensate for the lack of data, thereby facilitating smoother and more secure collaborative processes. Moreover, considering the anthropological aspect, language acquisition in humans predominantly occurs through communication and interaction [458], as opposed to merely consuming written content. As a result, agents shouldn't exclusively depend on models trained with pre-annotated datasets; instead, they should evolve through online interaction and engagement. The interaction between humans and agents can be classified into two paradigms (see Figure 10): (1) Unequal interaction (i.e., instructor-executor paradigm): humans serve as issuers of instructions, while agents act as executors, essentially participating as assistants to humans in collaboration. (2) Equal interaction (i.e., equal partnership paradigm): agents reach the level of humans, participating on an equal footing with humans in interaction.

4.3.1 Instructor-Executor Paradigm

The simplest approach involves human guidance throughout the process: humans provide clear and specific instructions directly, while the agents' role is to understand natural language commands from humans and translate them into corresponding actions [459; 460; 461]. In §4.1, we have presented the scenario where agents solve single-step problems or receive high-level instructions from humans. Considering the interactive nature of language, in this section, we assume that the dialogue between humans and agents is also interactive. Thanks to LLMs, the agents are able to interact with humans in a conversational manner: the agent responds to each human instruction, refining its action through alternating iterations to ultimately meet human requirements [190]. While this approach does achieve the goal of human-agent interaction, it places significant demands on humans. It requires a substantial amount of human effort and, in certain tasks, might even necessitate a high level of expertise. To alleviate this issue, the agent can be empowered to autonomously accomplish tasks, while humans only need to provide feedback in certain circumstances. Here, we roughly categorize feedback into two types: quantitative feedback and qualitative feedback.

Quantitative feedback. The forms of quantitative feedback mainly include absolute evaluations like binary scores and ratings, as well as relative scores. Binary feedback refers to the positive and negative evaluations provided by humans, which agents utilize to enhance their self-optimization [462; 463; 464; 465; 466]. Comprising only two categories, this type of user feedback is often easy to collect, but sometimes it may oversimplify user intent by neglecting potential intermediate scenarios. To showcase these intermediate scenarios, researchers attempt to expand from binary feedback to rating feedback, which involves categorizing into more fine-grained levels. However, the results of Kreutzer et al. [467] suggest that there could be significant discrepancies between user and expert annotations for such multi-level artificial ratings, indicating that this labeling method might be

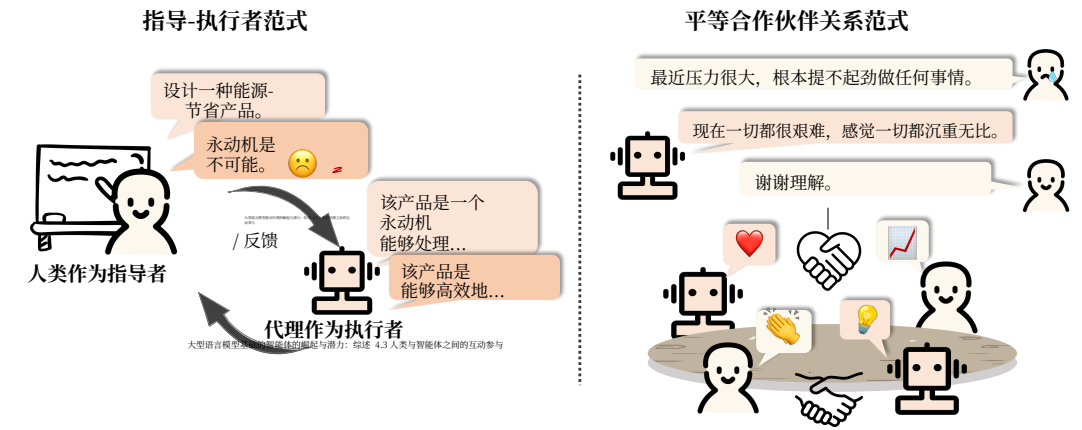


图10：人机交互的两种范式。在指导-执行者范式（左）中，人类提供指令或反馈，而代理作为执行者。在平等合作范式（右）中，代理具有类人特征，能够进行富有同理心的对话并参与与人类的协作任务。

guidance or by regulating the safety, legality, and ethical conduct of agents. This is particularly crucial in specialized domains, such as medicine where data privacy concerns exist [457]. In such cases, human involvement can serve as a valuable means to compensate for the lack of data, thereby facilitating smoother and more secure collaborative processes. Moreover, considering the anthropological aspect, language acquisition in humans predominantly occurs through communication and interaction [458], as opposed to merely consuming written content. As a result, agents shouldn't exclusively depend on models trained with pre-annotated datasets; instead, they should evolve through online interaction and engagement. The interaction between humans and agents can be classified into two paradigms (see Figure 10): (1) Unequal interaction (i.e., instructor-executor paradigm): humans serve as issuers of instructions, while agents act as executors, essentially participating as assistants to humans in collaboration. (2) Equal interaction (i.e., equal partnership paradigm): agents reach the level of humans, participating on an equal footing with humans in interaction.

4.3.1 教师-执行者范式

最简单的方法是在整个过程中由人类指导：人类直接提供明确而具体的指令，而代理的角色是理解人类的自然语言命令并将其转化为相应的行动。[459; 460; 461]. 在 §4.1 中，我们提出了代理解决单步问题或接收人类高层次指令的场景。考虑到语言的交互性，本节假设人类与代理之间的对话也是交互式的。得益于大型语言模型（LLMs），代理能够以对话的方式与人类互动：代理对每个指令做出回应，通过交替迭代不断优化其行动，最终满足人类的需求。。虽然这种方法确实实现了人类与代理的交互目标，但对人类提出了较高的要求。它需要大量的人类努力，在某些任务中甚至可能需要较高的专业知识。为缓解这一问题，可以赋予代理自主完成任务的能力，而人类只在特定情况下提供反馈。在此，我们大致将反馈分为两类：定量反馈和定性反馈。

定量反馈。定量反馈的形式主要包括绝对评价，如二元评分和评级，以及相对评分。二元反馈指人类提供的正面和负面评价，代理利用这些评价进行自我优化[462; 463; 464; 465; 466]。这种用户反馈仅包含两个类别，通常易于收集，但有时可能会简化用户意图，忽略潜在的中间场景。为了展示这些中间场景，研究人员试图将二元反馈扩展到评级反馈，即将其划分为更细粒度的等级。然而，Kreutzer 等人的研究结果 [467] 表明，对于这种多层次的人工评级，用户与专家标注之间可能存在显著差异，说明这种标注方法可能是

inefficient or less reliable. Furthermore, agents can learn human preference from comparative scores like multiple choice [468; 469].

Qualitative feedback. Text feedback is usually offered in natural language, particularly for responses that may need improvement. The format of this feedback is quite flexible. Humans provide advice on how to modify outputs generated by agents, and the agents then incorporate these suggestions to refine their subsequent outputs [470; 471]. For agents without multimodal perception capabilities, humans can also act as critics, offering visual critiques [190], for instance. Additionally, agents can utilize a memory module to store feedback for future reuse [472]. In [473], humans give feedback on the initial output generated by agents, prompting the agents to formulate various improvement proposals. The agents then discern and adopt the most suitable proposal, harmonizing with the human feedback. While this approach can better convey human intention compared to quantitative feedback, it might be more challenging for the agents to comprehend. Xu et al. [474] compare various types of feedback and observe that combining multiple types of feedback can yield better results. Re-training models based on feedback from multiple rounds of interaction (i.e., continual learning) can further enhance effectiveness. Of course, the collaborative nature of human-agent interaction also allows humans to directly improve the content generated by agents. This could involve modifying intermediate links [189; 475] or adjusting the conversation content [421]. In some studies, agents can autonomously judge whether the conversation is proceeding smoothly and seek feedback when errors are generated [476; 477]. Humans can also choose to participate in feedback at any time, guiding the agent’s learning in the right direction [420].

Currently, in addition to tasks like writing [466] and semantic parsing [463; 471], the model of using agents as human assistants also holds tremendous potential in the field of education. For instance, Kalvakurth et al. [413] propose the robot Dona, which supports multimodal interactions to assist students with registration. Gvirsman et al. [478] focus on early childhood education, achieving multifaceted interactions between young children, parents, and agents. Agents can also aid in human understanding and utilization of mathematics [414]. In the field of medicine, some medical agents have been proposed, showing enormous potential in terms of diagnosis assistance, consultations, and more [416; 417]. Especially in mental health, research has shown that agents can lead to increased accessibility due to benefits such as reduced cost, time efficiency, and anonymity compared to face-to-face treatment [479]. Leveraging such advantages, agents have found widespread applications. Ali et al. [418] design LISSA for online communication with adolescents on the autism spectrum, analyzing users’ speech and facial expressions in real-time to engage them in multi-topic conversations and provide instant feedback regarding non-verbal cues. Hsu et al. [415] build contextualized language generation approaches to provide tailored assistance for users who seek support on diverse topics ranging from relationship stress to anxiety. Furthermore, in other industries including business, a good agent possesses the capability to provide automated services or assist humans in completing tasks, thereby effectively reducing labor costs [419]. Amidst the pursuit of AGI, efforts are directed towards enhancing the multifaceted capabilities of general agents, creating agents that can function as universal assistants in real-life scenarios [422].

4.3.2 Equal Partnership Paradigm

Empathetic communicator. With the rapid development of AI, conversational agents have garnered extensive attention in research fields in various forms, such as personalized custom roles and virtual chatbots [480]. It has found practical applications in everyday life, business, education, healthcare, and more [481; 482; 483]. However, in the eyes of the public, agents are perceived as emotionless machines, and can never replace humans. Although it is intuitive that agents themselves do not possess emotions, can we enable them to exhibit emotions and thereby bridge the gap between agents and humans? Therefore, a plethora of research endeavors have embarked on delving into the empathetic capacities of agents. This endeavor seeks to infuse a human touch into these agents, enabling them to detect sentiments and emotions from human expressions, ultimately crafting emotionally resonant dialogues [484; 485; 486; 487; 488; 489; 490; 491]. Apart from generating emotionally charged language, agents can dynamically adjust their emotional states and display them through facial expressions and voice [423]. These studies, viewing agents as empathetic communicators, not only enhance user satisfaction but also make significant progress in fields like healthcare [415; 418; 492] and business marketing [424]. Unlike simple rule-based conversation agents, agents with empathetic capacities can tailor their interactions to meet users’ emotional needs [493].

低效或可靠性较差。此外，代理还可以通过类似多项选择的比较评分学习人类偏好 [468]。

定性反馈。 文本反馈通常以自然语言提供，特别是对于可能需要改进的响应。这种反馈的格式相当灵活。人类提供关于如何修改由代理生成的输出的建议，然后代理将这些建议融入到其后续的输出中 [470； 471]。对于没有多模态感知能力的代理，人类也可以充当批评者，例如提供视觉批评 [190]。此外，代理可以利用一个记忆模块存储反馈以供将来重复使用 [472]。在 [473]，人类对代理生成的初始输出提供反馈，促使代理制定各种改进方案。然后，代理辨别并采纳最合适的方案，与人类反馈相协调。虽然这种方法比定量反馈更能传达人类意图，但对代理来说理解起来可能更具挑战性。Xu 等人 [474] 比较了各种类型的反馈，并观察到结合多种类型的反馈可以取得更好的效果。基于多轮交互（即持续学习）中的反馈重新训练模型，能进一步提升效果。当然，人机交互的协作性质也允许人类直接改进代理生成的内容。这可能涉及修改中间环节 [189； 475]或调整对话内容 [421]。在一些研究中，代理可以自主判断对话是否顺利进行，并在出现错误时寻求反馈 [476； 477]。人类也可以随时选择参与反馈，引导代理的学习朝正确的方向发展 [420]。

目前，除了写作 [466] 和语义解析 [463等任务之外； 471], 将代理作为人类助手的模型在教育领域也具有巨大潜力。例如，Kalvakurth 等人 [413] 提出了机器人 Dona，支持多模态交互以协助学生注册。Gvirsman 等人 [478] 专注于早期儿童教育，实现了儿童、家长与代理之间的多方面互动。代理还可以帮助人类理解和利用数学 [414]。在医学领域，已经提出了一些医疗代理，在诊断辅助、咨询等方面展现出巨大的潜力 [416； 417]。尤其是在心理健康方面，研究表明，代理由于具有降低成本、节省时间和匿名性等优势，可以提高可及性 [479]。利用这些优势，代理已在各个领域得到广泛应用。Ali 等人 [418] 为自闭症谱系的青少年设计了 LISSA，用于在线交流，实时分析用户的语音和面部表情，以引导多话题对话并提供关于非语言线索的即时反馈。Hsu 等人 [415]构建了情境化语言生成方法，为寻求支持的用户提供个性化帮助，涵盖从关系压力到焦虑等多种话题。此外，在商业等其他行业中，一个优秀的代理具备提供自动化服务或协助人类完成任务的能力，从而有效降低劳动力成本 [419]。在追求 AGI 的过程中，努力旨在增强通用代理的多方面能力，打造能够在现实场景中充当通用助手的代理 [422]。

4.3.2 平等合作伙伴关系范式

The Rise and Potential of Large Language Model Based Agents: A Survey**富有同理心的交流者。**随着人工智能的快速发展，对话代理在研究领域以多种形式引起了广泛关注，如个性化定制角色和虚拟聊天机器人 [480]。它在日常生活、商业、教育、医疗等领域找到了实际应用 [481; 482； 483]。然而，在公众眼中，代理被认为是没有情感的机器，永远无法取代人类。虽然直观上，代理本身不具备情感，但我们是否可以赋予它们情感，从而弥合代理与人类之间的差距？因此，大量的研究工作开始深入探讨代理的同理心能力。这一努力旨在为这些代理注入人性化元素，使其能够从人类表达中检测情感和情绪，最终实现具有情感共鸣的对话 [484； 485; 486; 487; 488; 489; 490； 491]。除了生成富有情感色彩的语言外，代理还能动态调整其情感状态，并通过面部表情和语音进行展示 [423]。这些研究将代理视为具有同理心的交流者，不仅提升了用户满意度，还在医疗 [415; 418； 492]以及商业营销 [424]。与简单的基于规则的对话代理不同，具有同理心能力的代理可以根据用户的情感需求调整其交互方式 [493]。

Human-level participant. Furthermore, we hope that agents can be involved in the normal lives of humans, cooperating with humans to complete tasks from a human-level perspective. In the field of games, agents have already reached a high level. As early as the 1990s, IBM introduced the AI Deep Blue [451], which defeated the reigning world champion in chess at that time. However, in pure competitive environments such as chess [451], Go [360], and poker [494], the value of communication was not emphasized [426]. In many gaming tasks, players need to collaborate with each other, devising unified cooperative strategies through effective negotiation [425; 426; 495; 496]. In these scenarios, agents need to first understand the beliefs, goals, and intentions of others, formulate joint action plans for their objectives, and also provide relevant suggestions to facilitate the acceptance of cooperative actions by other agents or humans. In comparison to pure agent cooperation, we desire human involvement for two main reasons: first, to ensure interpretability, as interactions between pure agents could generate incomprehensible language [495]; second, to ensure controllability, as the pursuit of agents with complete “free will” might lead to unforeseen negative consequences, carrying the potential for disruption. Apart from gaming scenarios, agents also demonstrate human-level capabilities in other scenarios involving human interaction, showcasing skills in strategy formulation, negotiation, and more. Agents can collaborate with one or multiple humans, determining the shared knowledge among the cooperative partners, identifying which information is relevant to decision-making, posing questions, and engaging in reasoning to complete tasks such as allocation, planning, and scheduling [427]. Furthermore, agents possess persuasive abilities [497], dynamically influencing human viewpoints in various interactive scenarios [428].

The goal of the field of human-agent interaction is to learn and understand humans, develop technology and tools based on human needs, and ultimately enable comfortable, efficient, and secure interactions between humans and agents. Currently, significant breakthroughs have been achieved in terms of usability in this field. In the future, human-agent interaction will continue to focus on enhancing user experience, enabling agents to better assist humans in accomplishing more complex tasks in various domains. The ultimate aim is not to make agents more powerful but to better equip humans with agents. Considering practical applications in daily life, isolated interactions between humans and agents are not realistic. Robots will become colleagues, assistants, and even companions. Therefore, future agents will be integrated into a social network [498], embodying a certain level of social value.

5 Agent Society: From Individuality to Sociality

For an extended period, sociologists have frequently conducted social experiments to observe specific social phenomena within controlled environments. Notable examples include the Hawthorne Experiment² and the Stanford Prison Experiment³. Subsequently, researchers began employing animals in social simulations, exemplified by the Mouse Utopia Experiment⁴. However, these experiments invariably utilized living organisms as participants, made it difficult to carry out various interventions, lack flexibility, and inefficient in terms of time. Thus, researchers and practitioners envision an interactive artificial society wherein human behavior can be performed through trustworthy agents [521]. From sandbox games such as The Sims to the concept of Metaverse, we can see how “simulated society” is defined in people’s minds: environment and the individuals interacting in it. Behind each individual can be a piece of program, a real human, or a LLM-based agent as described in the previous sections [22; 522; 523]. Then, the interaction between individuals also contributes to the birth of sociality.

In this section, to unify existing efforts and promote a comprehensive understanding of the agent society, we first analyze the behaviors and personalities of LLM-based agents, shedding light on their journey from individuality to sociability (§ 5.1). Subsequently, we introduce a general categorization of the diverse environments for agents to perform their behaviors and engage in interactions (§ 5.2). Finally, we will talk about how the agent society works, what insights people can get from it, and the risks we need to be aware of (§ 5.3). The main explorations are listed in Figure 11.

²<https://www.bl.uk/people/elton-mayo>

³<https://www.prisonexp.org/conclusion/>

⁴<https://sproutsschools.com/behavioral-sink-the-mouse-utopia-experiments/>

人类水平的参与者。此外，我们希望代理能够参与到人类的日常生活中，与人类合作完成任务，从人类水平的角度出发。在游戏领域，代理已经达到了较高的水平。早在20世纪90年代，IBM推出了人工智能“深蓝” [451]，它击败了当时的国际象棋世界冠军。然而，在纯粹的竞争环境中，如国际象棋 [451]，围棋 [360]，和扑克 [494]，沟通的价值并未受到重视 [426]。在许多游戏任务中，玩家需要相互协作，通过有效的协商制定统一的合作策略 [425; 426; 495; 496]。在这些场景中，代理首先需要理解其他参与者的信念、目标和意图，制定共同的行动计划以实现其目标，同时提供相关建议，以促进其他代理或人类接受合作行动。与纯粹的代理合作相比，我们希望引入人类参与，主要有两个原因：一是为了确保可解释性，因为纯代理之间的互动可能会产生难以理解的语言 [495]；二是为了确保可控性，因为追求具有完全“自由意志”的代理可能会导致不可预见的负面后果，存在破坏的潜在风险。在游戏场景之外，代理在涉及人类互动的其他场景中也展现出人类水平的能力，展示出策略制定、谈判等技能。代理可以与一个或多个人人合作，确定合作伙伴之间的共享知识，识别哪些信息与决策相关，提出问题，并进行推理，以完成分配、规划和调度等任务 [427]。此外，代理还具备说服能力 [497]，在各种交互场景中动态影响人类的观点 [428]。

人机交互领域的目标是学习和理解人类，基于人类需求开发技术和工具，最终实现人类与代理之间的舒适、高效、安全的交互。目前，在可用性方面已取得显著突破。未来，人机交互将继续专注于提升用户体验，使代理能够更好地协助人类完成各个领域的更复杂任务。其终极目标并非让代理变得更强大，而是更好地赋能人类与代理的合作。考虑到日常生活中的实际应用，人类与代理的孤立交互并不现实。机器人将成为同事、助手，甚至是伙伴。因此，未来的代理将融入一个体现一定社会价值的社交网络 [498]，中。

5 AgentSociety：从个体性到社会性

在很长一段时间内，社会学家经常进行社会实验，以在受控环境中观察特定的社会现象。著名的例子包括霍桑实验²和斯坦福监狱实验³。随后，研究人员开始利用动物进行社会模拟，例如老鼠乌托邦实验⁴。然而，这些实验通常都使用生物作为参与者，难以进行多种干预，缺乏灵活性，且耗时效率低下。因此，研究人员和从业者设想一个互动的人工社会，在其中人类行为可以通过可信赖的代理人来实现 [521]。从模拟人生等沙盒游戏到元宇宙的概念，我们可以看到“模拟社会”在公众心中的定义：环境以及与之互动的个体。在每个个体背后，可能是一个程序、一位真实的人类，或如前节所述的基于大模型的代理 [22; {v16}; {v18}; {v20}; 523]。然后，个体之间的互动也促成了社会性的产生。

在本节中，为了统一现有的研究工作并促进对代理社会的全面理解，我们首先分析基于大规模语言模型（LLM）代理的行为和个性，揭示其从个体性到社会性的演变过程（§ 5.1）。随后，我们介绍一种关于代理执行行为和进行交互的多样环境的通用分类（§ 5.2）。最后，我们将讨论代理社会的运作机制、从中可以获得的洞见以及需要关注的风险（§ 5.3）。主要的研究方向如图11所示。

The Rise and Potential of Large Language Model Based Agents: A Survey 5 AgentSociety: From Individuality to Sociality²<https://www.bl.uk/people/elton-mayo>³<https://www.prisonexp.org/conclusion/>⁴<https://sproutsschools.com/behavioral-sink-the-mouse-utopia-experiments/>

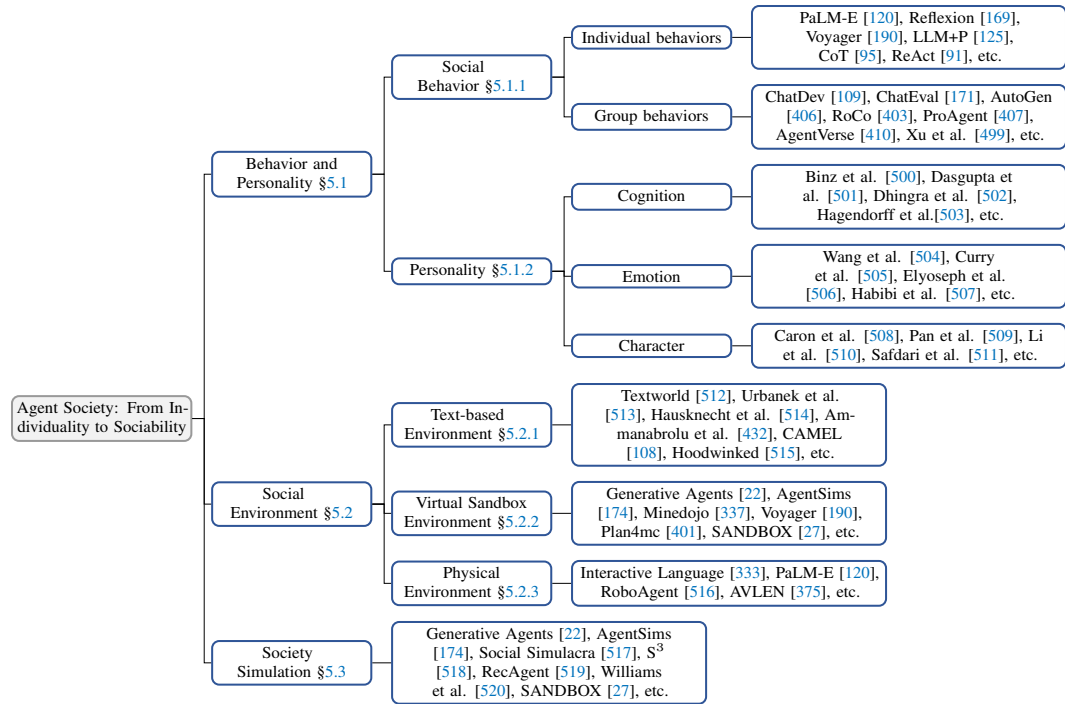


Figure 11: Typology of society of LLM-based agents.

5.1 Behavior and Personality of LLM-based Agents

As noted by sociologists, individuals can be analyzed in terms of both external and internal dimensions [524]. The external deals with observable behaviors, while the internal relates to dispositions, values, and feelings. As shown in Figure 12, this framework offers a perspective on emergent behaviors and personalities in LLM-based agents. Externally, we can observe the sociological behaviors of agents (§ 5.1.1), including how agents act individually and interact with their environment. Internally, agents may exhibit intricate aspects of the personality (§ 5.1.2), such as cognition, emotion, and character, that shape their behavioral responses.

5.1.1 Social Behavior

As Troitzsch et al. [525] stated, the agent society represents a complex system comprising individual and group social activities. Recently, LLM-based agents have exhibited spontaneous social behaviors in an environment where both cooperation and competition coexist [499]. The emergent behaviors intertwine to shape the social interactions [518].

Foundational individual behaviors. Individual behaviors arise through the interplay between internal cognitive processes and external environmental factors. These behaviors form the basis of how agents operate and develop as individuals within society. They can be classified into three core dimensions:

- **Input behaviors** refers to the absorption of information from the surroundings. This includes perceiving sensory stimuli [120] and storing them as memories [169]. These behaviors lay the groundwork for how an individual understands the external world.
- **Internalizing behaviors** involve inward cognitive processing within an individual. This category encompasses activities such as planning [125], reasoning [95], reflection [91], and knowledge precipitation [108; 405]. These introspective processes are essential for maturity and self-improvement.
- **Output behaviors** constitute outward actions and expressions. The actions can range from object manipulation [120] to structure construction [190]. By performing these actions, agents change the states of the surroundings. In addition, agents can express their opinions and broadcast information

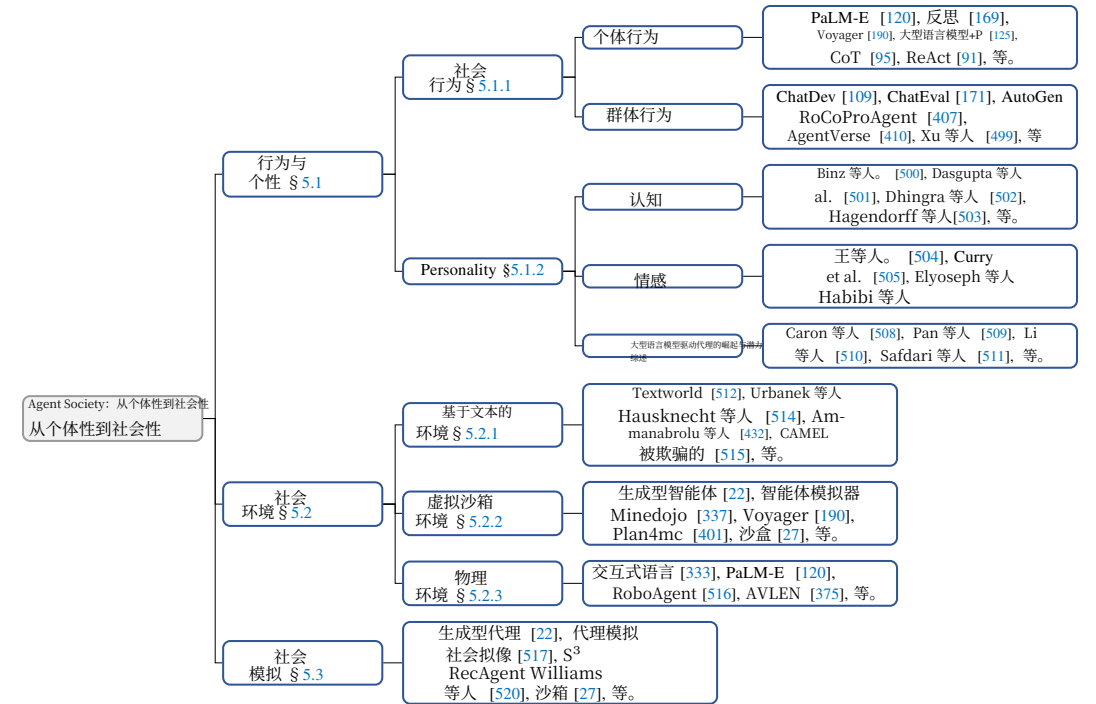


图11: 基于大型语言模型的代理社会类型学

5.1 基于大型语言模型的代理的行为与个性

正如社会学家所指出的，个体可以从外部和内部两个维度进行分析。外部涉及可观察的行为，而内部则关系到性格、价值观和情感。如图12所示，该框架为理解基于大规模语言模型的代理的新兴行为和个性提供了视角。在外部，我们可以观察到代理的社会学行为（§ 5.1.1），包括代理的个体行为及其与环境的互动。在内部，代理可能表现出性格的复杂方面（§ 5.1.2），如认知、情感和品格，这些因素共同塑造其行为反应。

5.1.1 社会行为

正如Troitzsch等人 [525] 所指出的，代理人社会代表了一个由个体和群体社会活动组成的复杂系统。近年来，基于大型语言模型的代理人在共存合作与竞争的环境中表现出自发的社会行为 [499]。这些涌现行为相互交织，塑造了社会互动 [518]。

基础个体行为。 个体行为通过内部认知过程与外部环境因素的相互作用而产生。这些行为构成了代理人在社会中作为个体运作和发展的基础。它们可以归类为三个核心维度：

- **输入行为**指从周围环境中获取信息的过程，包括感知感官刺激 [120] 并将其存储为记忆 [169]。这些行为为个体理解外部世界奠定了基础。

内化行为涉及个体内部的认知加工。这一类别包括诸如计划 [125],推理 [95],反思 [91],和知识沉淀 [108等活动； 405]。这些内省过程对于成熟和自我提升至关重要。

- **输出行为**构成外在动作和表现。这些动作可以从对象操作 [120] 到结构构建 [190]。通过执行这些动作，代理会改变周围环境的状况。此外，代理还可以表达其观点并广播信息

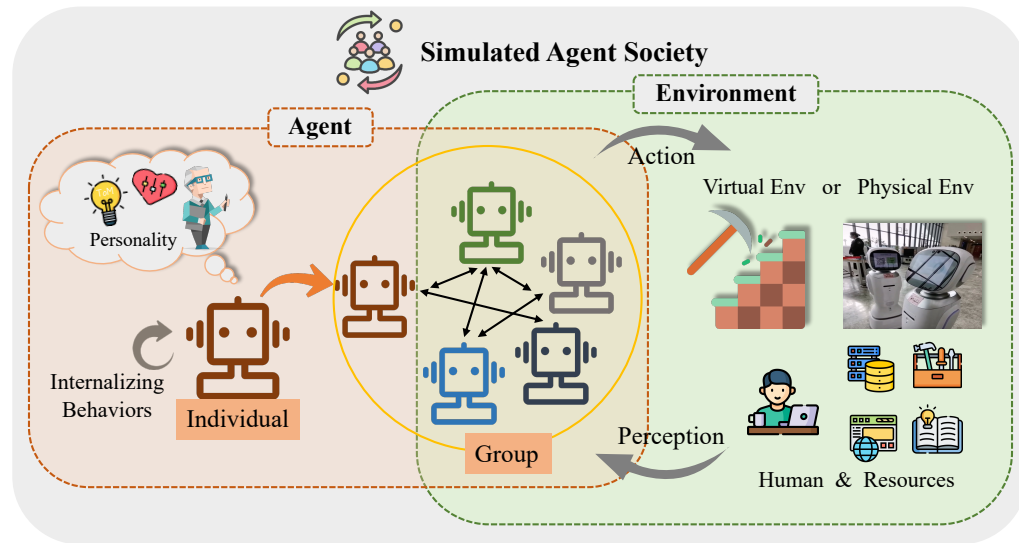


Figure 12: Overview of Simulated Agent Society. The whole framework is divided into two parts: the **Agent** and the **Environment**. We can observe in this figure that: (1) **Left**: At the individual level, an agent exhibits internalizing behaviors like planning, reasoning, and reflection. It also displays intrinsic personality traits involving cognition, emotion, and character. (2) **Mid**: An agent and other agents can form groups and exhibit group behaviors, such as cooperation. (3) **Right**: The environment, whether virtual or physical, contains human actors and all available resources. For a single agent, other agents are also part of the environment. (4) The agents have the ability to interact with the environment via perception and action.

to interact with others [405]. By doing so, agents exchange their thoughts and beliefs with others, influencing the information flow within the environment.

Dynamic group behaviors. A group is essentially a gathering of two or more individuals participating in shared activities within a defined social context [526]. The attributes of a group are never static; instead, they evolve due to member interactions and environmental influences. This flexibility gives rise to numerous group behaviors, each with a distinctive impact on the larger societal group. The categories of group behaviors include:

- **Positive group behaviors** are actions that foster unity, collaboration, and collective well-being [22; 109; 171; 403; 406; 407]. A prime example is cooperative teamwork, which is achieved through brainstorming discussions [171], effective conversations [406], and project management [405]. Agents share insights, resources, and expertise. This encourages harmonious teamwork and enables the agents to leverage their unique skills to accomplish shared goals. Altruistic contributions are also noteworthy. Some LLM-based agents serve as volunteers and willingly offer support to assist fellow group members, promoting cooperation and mutual aid [410].
- **Neutral group behaviors.** In human society, strong personal values vary widely and tend toward individualism and competitiveness. In contrast, LLMs which are designed with an emphasis on being “helpful, honest, and harmless” [527] often demonstrate a tendency towards neutrality [528]. This alignment with neutral values leads to conformity behaviors including mimicry, spectating, and reluctance to oppose majorities.
- **Negative group behaviors** can undermine the effectiveness and coherence of an agent group. Conflict and disagreement arising from heated debates or disputes among agents may lead to internal tensions. Furthermore, recent studies have revealed that agents may exhibit confrontational actions [499] and even resort to destructive behaviors, such as destroying other agents or the environment in pursuit of efficiency gains [410].

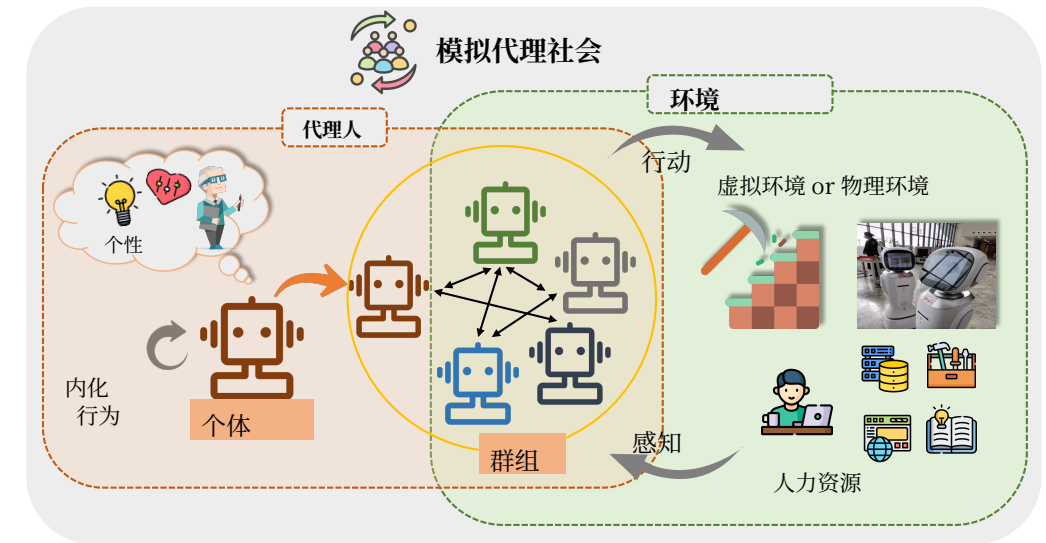


图12：模拟代理社会概览。整个框架分为两部分：**代理**和**环境**。我们可以在图中观察到：(1) **左侧**：在个体层面，代理表现出内化行为，如规划、推理和反思。它还表现出涉及认知、情感和性格的内在个性特征。(2) **中间**：代理与其他代理可以形成群体，并表现出群体行为，如合作。(3) **右侧**：环境，无论是虚拟还是物理，包含人类行为者和所有可用资源。对于单个代理而言，其他代理也是环境的一部分。(4) 代理具有通过感知和行动与环境互动的能力。

与他人互动 [405]。通过这种方式，代理人与他人交流思想和信念，影响环境中的信息流动。

动态群体行为。 群体本质上是两个或多个个体在特定社会环境中参与共享活动的集合 [526]。群体的属性从未是静态的；相反，它们会因成员互动和环境影响而不断演变。这种灵活性引发了多种群体行为，每种行为对更大的社会群体都具有独特的影响。群体行为的类别包括：

- **积极群体行为** 是促进团结、合作与集体福祉的行动[22; 109; 171; 403; 406; 407]。一个典型的例子是合作性团队工作，通过头脑风暴讨论实现 [171],有效的对话 [406],和项目管理[405]。代理共享见解、资源和专业知识。这鼓励和谐的团队合作，使代理能够利用其独特的技能实现共同目标。无私的贡献也值得关注。一些基于大型语言模型的代理充当志愿者，愿意提供支持以帮助团队成员，促进合作与互助 [410]。
- **中立群体行为。** 在人类社会中，强烈的个人价值观差异较大，倾向于个人主义和竞争性。相比之下，旨在“有帮助、诚实、无害”的大型语言模型（LLMs） [527] 通常表现出中立的倾向 [528]。这种与中立价值观的一致性导致了从众行为，包括模仿、旁观以及不愿反对多数。
- **负面群体行为** 可能削弱代理群体的有效性和凝聚力。由激烈辩论或争执引发的冲突与分歧可能导致内部紧张局势。此外，最新研究表明，代理可能表现出对抗性行为 [499]，甚至采取破坏性行为，例如在追求效率提升的过程中破坏其他代理或环境 [410]。

5.1.2 Personality

Recent advances in LLMs have provided glimpses of human-like intelligence [529]. Just as human personality emerges through socialization, agents also exhibit a form of personality that develops through interactions with the group and the environment [530; 531]. The widely accepted definition of personality refers to cognitive, emotional, and character traits that shape behaviors [532]. In the subsequent paragraphs, we will delve into each facet of personality.

Cognitive abilities. Cognitive abilities generally refer to the mental processes of gaining knowledge and comprehension, including thinking, judging, and problem-solving. Recent studies have started leveraging cognitive psychology methods to investigate emerging sociological personalities of LLM-based agents through various lenses [500; 502; 503]. A series of classic experiments from the psychology of judgment and decision-making have been applied to test agent systems [501; 500; 502; 533]. Specifically, LLMs have been examined using the Cognitive Reflection Test (CRT) to underscore their capacity for deliberate thinking beyond mere intuition [534; 535]. These studies indicate that LLM-based agents exhibit a level of intelligence that mirrors human cognition in certain respects.

Emotional intelligence. Emotions, distinct from cognitive abilities, involve subjective feelings and mood states such as joy, sadness, fear, and anger. With the increasing potency of LLMs, LLM-based agents are now demonstrating not only sophisticated reasoning and cognitive tasks but also a nuanced understanding of emotions [31].

Recent research has explored the emotional intelligence (EI) of LLMs, including emotion recognition, interpretation, and understanding. Wang et al. found that LLMs align with human emotions and values when evaluated on EI benchmarks [504]. In addition, studies have shown that LLMs can accurately identify user emotions and even exhibit empathy [505; 506]. More advanced agents are also capable of emotion regulation, actively adjusting their emotional responses to provide affective empathy [423] and mental wellness support [507; 536]. It contributes to the development of empathetic artificial intelligence (EAI).

These advances highlight the growing potential of LLMs to exhibit emotional intelligence, a crucial facet of achieving AGI. Bates et al. [537] explored the role of emotion modeling in creating more believable agents. By developing socio-emotional skills and integrating them into agent architectures, LLM-based agents may be able to engage in more naturalistic interactions.

Character portrayal. While cognition involves mental abilities and emotion relates to subjective experiences, the narrower concept of personality typically pertains to distinctive character patterns.

To understand and analyze a character in LLMs, researchers have utilized several well-established frameworks like the Big Five personality trait measure [508; 538] and the Myers–Briggs Type Indicator (MBTI) [508; 509; 538]. These frameworks provide valuable insights into the emerging character traits exhibited by LLM-based agents. In addition, investigations of potentially harmful dark personality traits underscore the complexity and multifaceted nature of character portrayal in these agents [510].

Recent work has also explored customizable character portrayal in LLM-based agents [511]. By optimizing LLMs through careful techniques, users can align with desired profiles and shape diverse and relatable agents. One effective approach is prompt engineering, which involves the concise summaries that encapsulate desired character traits, interests, or other attributes [22; 517]. These prompts serve as cues for LLM-based agents, directing their responses and behaviors to align with the outlined character portrayal. Furthermore, personality-enriched datasets can also be used to train and fine-tune LLM-based agents [539; 540]. Through exposure to these datasets, LLM-based agents gradually internalize and exhibit distinct personality traits.

5.2 Environment for Agent Society

In the context of simulation, the whole society consists of not only solitary agents but also the environment where agents inhabit, sense, and act [541]. The environment impacts sensory inputs, action space, and interactive potential of agents. In turn, agents influence the state of the environment through their behaviors and decisions. As shown in Figure 12, for a single agent, the environment

5.1.2 个性

近期在大型语言模型（LLMs）方面的进展，展现了类人智能的曙光 [529]。正如人类的个性通过社会化而形成，智能体也表现出一种通过与群体和环境互动而发展的个性特征 [530]。广泛接受的个性定义指的是塑造行为的认知、情感和性格特质 [532]。在接下来的段落中，我们将深入探讨个性的各个方面。

认知能力。认知能力通常指获取知识和理解的心理过程，包括思维、判断和问题解决。近年来的研究开始利用认知心理学方法，从多种角度探讨基于大语言模型的代理的新兴社会学人格特征；502； 503]。一系列源自判断与决策心理学的经典实验已被应用于测试代理系统；500；502； 533]。具体而言，研究使用认知反思测验（CRT）对大语言模型进行评估，以突出其超越直觉的深思熟虑能力； [534； 535]。这些研究表明，基于大语言模型的代理在某些方面展现出与人类认知相似的智能水平。

情感智能。 情感不同于认知能力，涉及主观感受和情绪状态，如喜悦、悲伤、恐惧和愤怒。随着大型语言模型（LLMs）能力的不断增强，基于LLMs的代理不仅展现出复杂的推理和认知任务，还表现出对情感的细腻理解 [31]。

近期研究探索了大型语言模型（LLMs）的情感智能（EI），包括情感识别、解释和理解。王等人在EI基准测试中发现，LLMs与人类的情感和价值观保持一致 [504]。此外，研究表明，LLMs能够准确识别用户的情感，甚至表现出同理心 [505； 506]。更先进的代理还具备情感调节能力，主动调整其情感反应以提供情感共鸣 [423]和心理健康支持 [507； 536]。这有助于推动共情人工智能（EAI）的发展。

这些进展凸显了大型语言模型在展现情感智能方面日益增长的潜力，情感智能是实现通用人工智能（AGI）的关键方面。Bates 等人 [537] 探讨了情感建模在打造更具可信度的智能体中的作用。通过发展社会-情感技能并将其整合到智能体架构中，基于大型语言模型的智能体可能能够实现更自然的交互。

Cha角色描绘。虽然认知涉及心理能力，情感则与主观性相关。exp经验中，较狭义的“个性”通常指具有特色的性格特征模式。为了理解和分析大型语言模型（LLMs）中的角色，研究人员采用了多种成熟的框架，如五大人格特质测量（Big Five personality trait measure）、迈尔斯-布里格斯类型指标（MBTI）等。这些框架为理解基于LLMs的代理所展现的新兴性格特征提供了宝贵的见解。此外，对潜在有害的黑暗人格特质的研究也凸显了这些代理中角色刻画的复杂性和多面性。

近期的研究还探索了基于大规模语言模型（LLM）代理的可定制角色塑造 [511]。通过精心的技术优化LLM，用户可以与理想的角色特征保持一致，塑造多样且具有亲和力的代理。一种有效的方法是提示工程，即通过简洁的总结，概括所期望的角色特质、兴趣或其他属性 [22； 517]。这些提示作为LLM代理的线索，引导其响应和行为，以符合所描述的角色塑造。此外，还可以利用丰富个性特征的数据集对LLM代理进行训练和微调 [539； 540]。通过接触这些数据集，LLM代理逐渐内化并展现出不同的个性特征。

5.2 代理社会的环境

在模拟的背景下，整个社会不仅由孤立的智能体组成，还包括智能体所栖息、感知和行动的环境。环境影响智能体的感官输入、行动空间和交互潜能。反过来，智能体通过其行为和决策影响环境的状态。如图12所示，对于单一智能体而言，环境

refers to other autonomous agents, human actors, and external factors. It provides the necessary resources and stimuli for agents. In this section, we examine fundamental characteristics, advantages, and limitations of various environmental paradigms, including text-based environment (§ 5.2.1), virtual sandbox environment (§ 5.2.2), and physical environment (§ 5.2.3).

5.2.1 Text-based Environment

Since LLMs primarily rely on language as their input and output format, the text-based environment serves as the most natural platform for agents to operate in. It is shaped by natural language descriptions without direct involvement of other modalities. Agents exist in the text world and rely on textual resources to perceive, reason, and take actions.

In text-based environments, entities and resources can be presented in two main textual forms, including natural and structured. Natural text uses descriptive language to convey information, like character dialogue or scene setting. For instance, consider a simple scenario described textually: “You are standing in an open field west of a white house, with a boarded front door. There is a small mailbox here” [512]. Here, object attributes and locations are conveyed purely through plain text. On the other hand, structured text follows standardized formats, such as technical documentation and hypertext. Technical documentation uses templates to provide operational details and domain knowledge about tool use. Hypertext condenses complex information from sources like web pages [389; 388; 391; 392] or diagrams into a structured format. Structured text transforms complex details into accessible references for agents.

The text-based environment provides a flexible framework for creating different text worlds for various goals. The textual medium enables environments to be easily adapted for tasks like interactive dialog and text-based games. In interactive communication processes like CAMEL [108], the text is the primary medium for describing tasks, introducing roles, and facilitating problem-solving. In text-based games, all environment elements, such as locations, objects, characters, and actions, are exclusively portrayed through textual descriptions. Agents utilize text commands to execute manipulations like moving or tool use [432; 512; 514; 515]. Additionally, agents can convey emotions and feelings through text, further enriching their capacity for naturalistic communication [513].

5.2.2 Virtual Sandbox Environment

The virtual sandbox environment provides a visualized and extensible platform for agent society, bridging the gap between simulation and reality. The key features of sandbox environments are:

- **Visualization.** Unlike the text-based environment, the virtual sandbox displays a panoramic view of the simulated setting. This visual representation can range from a simple 2D graphical interface to a fully immersive 3D modeling, depending on the complexity of the simulated society. Multiple elements collectively transform abstract simulations into visible landscapes. For example, in the overhead perspective of Generative Agents [22], a detailed map provides a comprehensive overview of the environment. Agent avatars represent each agent’s positions, enabling real-time tracking of movement and interactions. Furthermore, expressive emojis symbolize actions and states in an intuitive manner.
- **Extensibility.** The environment demonstrates a remarkable degree of extensibility, facilitating the construction and deployment of diverse scenarios. At a basic level, agents can manipulate the physical elements within the environment, including the overall design and layout of architecture. For instance, platforms like AgentSims [174] and Generative Agents [22] construct artificial towns with buildings, equipment, and residents in grid-based worlds. Another example is Minecraft, which provides a blocky and three-dimensional world with infinite terrain for open-ended construction [190; 337; 401]. Beyond physical elements, agent relationships, interactions, rules, and social norms can be defined. A typical design of the sandbox [27] employs latent sandbox rules as incentives to guide emergent behaviors, aligning them more closely with human preferences. The extensibility supports iterative prototyping of diverse agent societies.

5.2.3 Physical Environment

As previously discussed, the text-based environment has limited expressiveness for modeling dynamic environments. While the virtual sandbox environment provides modularized simulations, it lacks authentic embodied experiences. In contrast, the physical environment refers to the tangible and

refers to other autonomous agents, human actors, and external factors. It provides the necessary resources and stimuli for agents. In this section, we examine fundamental characteristics, advantages, and limitations of various environmental paradigms, including text-based environment (§ 5.2.1), virtual sandbox environment (§ 5.2.2), and physical environment (§ 5.2.3).

5.2.1 基于文本的环境

由于大型语言模型（LLMs）主要依赖语言作为输入和输出格式，基于文本的环境成为代理最自然的操作平台。该环境由自然语言描述构建而成，未直接涉及其他模态。代理存在于文本世界中，依赖文本资源进行感知、推理和执行行动。

在基于文本的环境中，实体和资源可以以两种主要的文本形式呈现，包括自然文本和结构化文本。自然文本使用描述性语言传达信息，例如角色对话或场景设定。例如，考虑一个用文本描述的简单场景：“你站在一片白色房屋西侧的空地上，前门被木板封闭。这里有一个小邮筒。” [512]这里，物体属性和位置纯粹通过纯文本传达。另一方面，结构化文本遵循标准化格式，如技术文档和超文本。技术文档使用模板提供操作细节和工具使用的领域知识。超文本将来自网页等来源的复杂信息浓缩成结构化格式。[389; {v9}; {v11}; {v13}; {v15}]或图表等内容被转化为结构化格式。结构化文本将复杂细节转化为代理可以访问的参考资料。

基于文本的环境为创建不同目标的文本世界提供了灵活的框架。文本媒介使环境能够轻松适应诸如交互式对话和基于文本的游戏等任务。在像CAMEL这样的交互式通信过程中，文本是描述任务、引入角色和促进问题解决的主要媒介。在基于文本的游戏中，所有环境元素，如位置、物体、角色和动作，均通过文本描述进行表现。代理利用文本指令执行诸如移动或工具使用等操作。此外，代理还可以通过文本传达情感和感受，进一步丰富其自然交流的能力。

5.2.2 虚拟沙箱环境

虚拟沙箱环境为代理社会提供了一个可视化且可扩展的平台，弥合了仿真与现实之间的差距。沙箱环境的主要特征包括：

可视化。与基于文本的环境不同，虚拟沙箱展示了模拟场景的全景视图。这种视觉表现可以从简单的二维图形界面扩展到完全沉浸式的三维建模，具体取决于模拟社会的复杂程度。多个元素共同将抽象的模拟转化为可见的景观。例如，在生成代理的俯视角度中，详细的地图提供了环境的全面概览。代理头像代表每个代理的位置，实现对运动和交互的实时追踪。此外，富有表现力的表情符号以直观的方式象征动作和状态。

扩展性。环境展现出显著的扩展性，便于构建和部署多样化的场景。在基本层面，代理可以操控环境中的物理元素，包括建筑的整体设计和布局。例如，像AgentSims和生成代理构建了具有建筑、设备和居民的人工城镇，位于网格化世界中。另一个例子是Minecraft，它提供了一个块状的三维世界，具有无限的地形，支持开放式建造；337；。除了物理元素外，还可以定义代理关系、交互、规则和社会规范。沙盒的典型设计采用潜在的沙盒规则作为激励，指导涌现行为，使其更贴近人类偏好。扩展性支持多样代理社会的迭代原型开发。

5.2.3 物理环境

正如前文所述，基于文本的环境在建模动态环境方面的表达能力有限。虽然虚拟沙箱环境提供了模块化的模拟，但缺乏真实的具身体验。相比之下，物理环境指的是有形的和

real-world surroundings which consist of actual physical objects and spaces. For instance, within a household physical environment [516], tangible surfaces and spaces can be occupied by real-world objects such as plates. This physical reality is significantly more complex, posing additional challenges for LLM-based agents:

- **Sensory perception and processing.** The physical environment introduces a rich tapestry of sensory inputs with real-world objects. It incorporates visual [120; 333], auditory [375; 377] and spatial senses. While this diversity enhances interactivity and sensory immersion, it also introduces the complexity of simultaneous perception. Agents must process sensory inputs to interact effectively with their surroundings.
- **Motion control.** Unlike virtual environments, physical spaces impose realistic constraints on actions through embodiment. Action sequences generated by LLM-based agents should be adaptable to the environment. It means that the physical environment necessitates executable and grounded motion control [258]. For example, imagine an agent operating a robotic arm in a factory. Grasping objects with different textures requires precision tuning and controlled force, which prevents damage to items. Moreover, the agent must navigate the physical workspace and make real-time adjustments, avoiding obstacles and optimizing the trajectory of the arm.

In summary, to effectively interact within tangible spaces, agents must undergo hardware-specific and scenario-specific training to develop adaptive abilities that can transfer from virtual to physical environments. We will discuss more in the following section (§ 6.5).

5.3 Society Simulation with LLM-based Agents

The concept of “Simulated Society” in this section serves as a dynamic system where agents engage in intricate interactions within a well-defined environment. Recent research on simulated societies has followed two primary lines, namely, exploring the boundaries of the collective intelligence capabilities of LLM-based agents [109; 405; 130; 406; 410] and using them to accelerate discoveries in the social sciences [22; 518; 542]. In addition, there are also a number of noteworthy studies, e.g., using simulated societies to collect synthetic datasets [108; 519; 543], helping people to simulate rare yet difficult interpersonal situations [544; 545]. With the foundation of the previous sections (§ 5.1, 5.2), here we will introduce the key properties and mechanism of agent society (§ 5.3.1), what we can learn from emergent social phenomena (§ 5.3.2), and finally the potential ethical and social risks in it (§ 5.3.3).

5.3.1 Key Properties and Mechanism of Agent Society

Social simulation can be categorized into macro-level simulation and micro-level simulation [518]. In the macro-level simulation, also known as system-based simulation, researchers model the overall state of the system of the simulated society [546; 547]. While micro-level simulation, also known as agent-based simulation or Multi-Agent Systems (MAS), indirectly simulates society by modeling individuals [548; 549]. With the development of LLM-based agents, micro-level simulation has gained prominence recently [22; 174]. In this article, we characterize that the “Agent Society” refers to an **open, persistent, situated, and organized** framework [521] where LLM-based agents interact with each other in a defined environment. Each of these attributes plays a pivotal role in shaping the harmonious appearance of the simulated society. In the following paragraphs, we analyze how the simulated society operates through discussing these properties:

- **Open.** One of the defining features of simulated societies lies in their openness, both in terms of their constituent agents and their environmental components. Agents, the primary actors within such societies, have the flexibility to enter or leave the environment without disrupting its operational integrity [550]. Furthermore, this feature extends to the environment itself, which can be expanded by adding or removing entities in the virtual or physical world, along with adaptable resources like tool APIs. Additionally, humans can also participate in societies by assuming the role of an agent or serving as the “inner voice” guiding these agents [22]. This inherent openness adds another level of complexity to the simulation, blurring the lines between simulation and reality.
- **Persistent.** We expect persistence and sustainability from the simulated society. While individual agents within the society exercise autonomy in their actions over each time step [22; 518], the overall organizational structure persists through time, to a degree detached from the transient

真实世界环境由实际的物理对象和空间组成。例如，在家庭物理环境中， [516], 可触及的表面和空间可以被实际物体如盘子占据。这种物理现实远比想象中复杂，为基于大型语言模型的代理带来了额外的挑战：

感官感知与处理。物理环境引入了丰富的感官输入，涉及现实世界的物体。它包括视觉 [120; 333], 听觉 [375; 377]以及空间感官。虽然这种多样性增强了交互性和感官沉浸感，但也带来了同时感知的复杂性。代理必须处理感官输入，以有效地与环境互动。

运动控制。与虚拟环境不同，物理空间通过体现性对动作施加现实约束。基于大规模语言模型的代理生成的动作序列应具有环境适应性。这意味着物理环境需要可执行且具有基础性的运动控制 [258]。例如，设想一个代理在工厂中操作机械臂。抓取不同质地物体需要精确调节和受控力度，以防止损坏物品。此外，代理还必须导航物理工作空间，进行实时调整，避开障碍物并优化机械臂的轨迹。

总之，为了在有形空间中实现有效交互，代理必须经过硬件特定和场景特定的训练，以培养能够从虚拟环境迁移到物理环境的适应能力。我们将在以下章节（§ 6.5）中进行更详细的讨论。

5.3 基于大语言模型代理的社会模拟

“模拟社会”在本节中指的是一个动态系统，其中代理在明确的环境中进行复杂的交互。近年来关于模拟社会的研究主要沿着两条路径展开，即探索基于大语言模型的代理的集体智能能力边界 [109, 以及利用这些能力加速社会科学的发现 410]。此外，还有一些值得关注的研究，例如利用模拟社会收集合成数据集 [108, 帮助人们模拟罕见但困难的人际交往情境 [544]。在前述章节（§ 5.1、§ 5.2）的基础上，本文将介绍代理社会的关键特性与机制（§ 5.3.1），我们可以从涌现的社会现象中学到的内容（§ 5.3.2），以及其潜在的伦理和社会风险（§ 5.3.3）。

5.3.1 代理社会的关键特性与机制

社会模拟可以分为宏观层面模拟和微观层面模拟 [518]。在宏观层面模拟中，也称为基于系统的模拟，研究人员建模模拟社会的整体状态 [546; 547]。而微观层面模拟，也称为基于代理的模拟或多智能体系统（Multi-Agent Systems, MAS），通过建模个体间接模拟社会 [548; 549]。随着基于大模型（LLM）代理的发展，微观层面模拟近年来逐渐受到关注 [22; 174]。本文中，我们将“代理社会”定义为一个**开放、持续、情境化和有组织的**框架 [521]，其中基于大模型的代理在一个确定的环境中相互作用。这些属性在塑造模拟社会的和谐面貌中起着关键作用。接下来的段落中，我们将通过讨论这些属性，分析模拟社会的运作机制：

- **开放。** 模拟社会的一个显著特征在于其开放性，无论是在组成代理还是环境组件方面。代理，作为此类社会中的主要行为主体，具有自由进入或离开环境而不破坏其运行完整性的能力 [550]。此外，这一特性还延伸至环境本身，可以通过在虚拟或物理世界中添加或移除实体，以及引入具有适应性的资源（如工具API）来扩展环境。此外，人类也可以通过扮演代理角色或作为“内在声音”引导这些代理，参与到社会中 [22]。这种固有的开放性为模拟增添了更高的复杂性，也模糊了模拟与现实之间的界限。
- **持续性。** 我们期望模拟社会具有持续性和可持续性。虽然社会中的个体代理在每个时间步中行使自主权 [22; 518], 整体组织结构随着时间的推移得以保持，在一定程度上与短暂性脱离。

behaviors of individual agents. This persistence creates an environment where agents’ decisions and behaviors accumulate, leading to a coherent societal trajectory that develops through time. The system operates independently, contributing to society’s stability while accommodating the dynamic nature of its participants.

- **Situated.** The situated nature of the society emphasizes its existence and operation within a distinct environment. This environment is artificially or automatically constructed in advance, and agents execute their behaviors and interactions effectively within it. A noteworthy aspect of this attribute is that agents possess an awareness of their spatial context, understanding their location within the environment and the objects within their field of view [22; 190]. This awareness contributes to their ability to interact proactively and contextually.
- **Organized.** The simulated society operates within a meticulously organized framework, mirroring the systematic structure present in the real world. Just as the physical world adheres to physics principles, the simulated society operates within predefined rules and limitations. In the simulated world, agents interact with the environment in a limited action space, while objects in the environment transform in a limited state space. All of these rules determine how agents operate, facilitating the communication connectivity and information transmission pathways, among other aspects in simulation [207]. This organizational framework ensures that operations are coherent and comprehensible, ultimately leading to an ever-evolving yet enduring simulation that mirrors the intricacies of real-world systems.

5.3.2 Insights from Agent Society

Following the exploration of how simulated society works, this section delves into the emergent social phenomena in agent society. In the realm of social science, the pursuit of generalized representations of individuals, groups, and their intricate dynamics has long been a shared objective [551; 552]. The emergence of LLM-based agents allows us to take a more microscopic view of simulated society, which leads to more discoveries from the new representation.

Organized productive cooperation. Society simulation offers valuable insights into innovative collaboration patterns, which have the potential to enhance real-world management strategies. Research has demonstrated that within this simulated society, the integration of diverse experts introduces a multifaceted dimension of individual intelligence [108; 447]. When dealing with complex tasks, such as software development or consulting, the presence of agents with various backgrounds, abilities, and experiences facilitates creative problem-solving [109; 410]. Furthermore, diversity functions as a system of checks and balances, effectively preventing and rectifying errors through interaction, ultimately improving the adaptability to various tasks. Through numerous iterations of interactions and debates among agents, individual errors like hallucination or degeneration of thought (DoT) are corrected by the group [112].

Efficient communication also plays a pivotal role in such a large and complex collaborative group. For example, MetaGPT [405] has artificially formulated communication styles with reference to standardized operating procedures (SOPs), validating the effectiveness of empirical methods. Park et al. [22] observed agents working together to organize a Valentine’s Day party through spontaneous communication in a simulated town.

Propagation in social networks. As simulated social systems can model what might happen in the real world, they can be used as a reference for predicting social processes. Unlike traditional empirical approaches, which heavily rely on time-series data and holistic modeling [553; 554], agent-based simulations offer a unique advantage by providing more interpretable and endogenous perspectives for researchers. Here we focus on its application to modeling propagation in social networks.

The first crucial aspect to be explored is the development of interpersonal relationships in simulated societies. For instance, agents who are not initially connected as friends have the potential to establish connections through intermediaries [22]. Once a network of relationships is established, our attention shifts to the dissemination of information within this social network, along with the underlying attitudes and emotions associated with it. S³ [518] proposes a user-demographic inference module for capturing both the number of people aware of a particular message and the collective sentiment prevailing among the crowd. This same approach extends to modeling cultural transmission [555] and the spread of infectious diseases [520]. By employing LLM-based agents to model individual

个体代理的行为。这种持续性创造了一个代理决策和行为积累的环境，导致通过时间发展出一个连贯的社会轨迹。该系统自主运行，既有助于社会的稳定，又能适应其参与者的动态变化。

情境性。社会的情境性强调其在特定环境中的存在与运作。该环境由人工或自动预先构建，代理在其中有效执行其行为和交互。这一特性的重要方面在于，代理具有空间感知能力，理解其在环境中的位置以及视野范围内的对象。这种意识有助于它们进行主动且具有情境性的交互。

有序的。模拟社会在一个精心组织的框架内运作，反映了现实世界中的系统结构。正如物理世界遵循物理定律，模拟社会在预定义的规则和限制下运行。在模拟世界中，代理在有限的行动空间内与环境互动，而环境中的对象在有限的状态空间中变化。所有这些规则决定了代理的操作方式，促进了通信连接和信息传输路径等方面的实现 [207]。这一组织框架确保操作的连贯性和可理解性，最终形成一个不断演变但持久的模拟，反映了现实系统的复杂性。

5.3.2 来自智能体社会的洞察

在探索模拟社会运作机制之后，本节深入探讨代理社会中的涌现社会现象。在社会科学领域，追求对个体、群体及其复杂动态的泛化表达一直是共同的目标 [551; 552]。基于大型语言模型的代理的出现，使我们能够从更微观的角度观察模拟社会，从而在新的表达方式中获得更多的发现。

有序的生产性合作。社会模拟为创新合作模式提供了宝贵的见解，具有提升现实管理策略的潜力。研究表明，在这个模拟社会中，融合多样化专家引入了个体智能的多维度 [108; 447]。在处理复杂任务，如软件开发或咨询时，具有不同背景、能力和经验的代理的存在，有助于促进创造性问题解决 [109; 410]。此外，多样性作为一种制衡体系，通过互动有效防止和纠正错误，最终提升对各种任务的适应能力。通过代理之间多次的互动与辩论，个体的错误，如幻觉或思维退化（DoT），由群体共同纠正 [112]。

高效的沟通在如此庞大且复杂的协作群体中也起着关键作用。例如，MetaGPT [405]根据标准操作程序（SOPs）人工制定了沟通风格，验证了经验方法的有效性。Park 等人 [22]观察到代理在模拟城镇中通过自发沟通合作组织情人节派对。

社交网络中的传播。 由于模拟社会系统能够模拟现实世界中可能发生的情况，它们可以作为预测社会过程的参考。与严重依赖时间序列数据和整体建模的传统经验方法不同， [553 554], 基于代理的模拟为研究人员提供了更具可解释性和内生性的视角，具有独特的优势。在此，我们重点关注其在社交网络传播建模中的应用。

《大型语言模型驱动代理的崛起与潜力：综述》第一个关键方面是模拟社会中人际关系的发展。例如，最初未建立朋友关系的代理有潜力通过中介 [22]建立联系。一旦关系网络建立起来，我们的关注点转向该社会网络中的信息传播，以及与之相关的态度和情感。S³ [518]提出了一个用户人口统计推断模块，用于捕捉了解某条信息的人员数量以及群体中普遍存在的集体情感。这一方法同样适用于建模文化传播 [555]和传染病的传播 [520]。通过采用基于大型语言模型的代理来模拟个体

behaviors, implementing various intervention strategies, and monitoring population changes over time, these simulations empower researchers to gain deeper insights into the intricate processes that underlie various social phenomena of propagation.

Ethical decision-making and game theory. Simulated societies offer a dynamic platform for the investigation of intricate decision-making processes, encompassing decisions influenced by ethical and moral principles. Taking Werewolf game [499; 556] and murder mystery games [557] as examples, researchers explore the capabilities of LLM-based agents when confronted with challenges of deceit, trust, and incomplete information. These complex decision-making scenarios also intersect with game theory [558], where we frequently encounter moral dilemmas pertaining to individual and collective interests, such as Nash Equilibria. Through the modeling of diverse scenarios, researchers acquire valuable insights into how agents prioritize values like honesty, cooperation, and fairness in their actions. In addition, agent simulations not only provide an understanding of existing moral values but also contribute to the development of philosophy by serving as a basis for understanding how these values evolve and develop over time. Ultimately, these insights contribute to the refinement of LLM-based agents, ensuring their alignment with human values and ethical standards [27].

Policy formulation and improvement. The emergence of LLM-based agents has profoundly transformed our approach to studying and comprehending intricate social systems. However, despite those interesting facets mentioned earlier, numerous unexplored areas remain, underscoring the potential for investigating diverse phenomena. One of the most promising avenues for investigation in simulated society involves exploring various economic and political states and their impacts on societal dynamics [559]. Researchers can simulate a wide array of economic and political systems by configuring agents with differing economic preferences or political ideologies. This in-depth analysis can provide valuable insights for policymakers seeking to foster prosperity and promote societal well-being. As concerns about environmental sustainability grow, we can also simulate scenarios involving resource extraction, pollution, conservation efforts, and policy interventions [560]. These findings can assist in making informed decisions, foreseeing potential repercussions, and formulating policies that aim to maximize positive outcomes while minimizing unintended adverse effects.

5.3.3 Ethical and Social Risks in Agent Society

Simulated societies powered by LLM-based agents offer significant inspirations, ranging from industrial engineering to scientific research. However, these simulations also bring about a myriad of ethical and social risks that need to be carefully considered and addressed [561].

Unexpected social harm. Simulated societies carry the risk of generating unexpected social phenomena that may cause considerable public outcry and social harm. These phenomena span from individual-level issues like discrimination, isolation, and bullying, to broader concerns such as oppressive slavery and antagonism [562; 563]. Malicious people may manipulate these simulations for unethical social experiments, with consequences reaching beyond the virtual world into reality. Creating these simulated societies is akin to opening Pandora’s Box, necessitating the establishment of rigorous ethical guidelines and oversight during their development and utilization [561]. Otherwise, even minor design or programming errors in these societies can result in unfavorable consequences, ranging from psychological discomfort to physical injury.

Stereotypes and prejudice. Stereotyping and bias pose a long-standing challenge in language modeling, and a large part of the reason lies in the training data [564; 565]. The vast amount of text obtained from the Internet reflects and sometimes even amplifies real-world social biases, such as gender, religion, and sexuality [566]. Although LLMs have been aligned with human values to mitigate biased outputs, the models still struggle to portray minority groups well due to the long-tail effect of the training data [567; 568; 569]. Consequently, this may result in an overly one-sided focus in social science research concerning LLM-based agents, as the simulated behaviors of marginalized populations usually conform to prevailing assumptions [570]. Researchers have started addressing this concern by diversifying training data and making adjustments to LLMs [571; 572], but we still have a long way to go.

Privacy and security. Given that humans can be members of the agent society, the exchange of private information between users and LLM-based agents poses significant privacy and security

行为，实施各种干预策略，并监测人口随时间的变化，这些模拟帮助研究人员深入理解支撑各种传播社会现象的复杂过程。

伦理决策与博弈论。 模拟社会为研究复杂决策过程提供了一个动态平台，包括受伦理和道德原则影响的决策。以Werewolf游戏 [499; 556] 以及谋杀悬疑游戏 [557] 为例，研究人员探索基于LLM的代理在面对欺骗、信任和信息不完整等挑战时的能力。这些复杂的决策场景也与博弈论 [558], 相交，在其中我们经常遇到涉及个人与集体利益的道德困境，例如纳什均衡。通过对不同场景的建模，研究人员获得了关于代理如何在行动中优先考虑诚信、合作和公平等价值的宝贵见解。此外，代理模拟不仅有助于理解现有的道德价值观，还通过作为理解这些价值观如何随时间演变和发展的基础，为哲学的发展做出贡献。最终，这些见解有助于完善基于LLM的代理，确保其与人类价值观和伦理标准保持一致。 [27].

政策制定与完善。基于LLM的代理的出现，深刻改变了我们研究和理解复杂社会系统的方法。然而，尽管前述提到了一些有趣的方面，仍有许多未被探索的领域，凸显了研究多样化现象的潜力。在模拟社会中，最具前景的研究方向之一是探索各种经济和政治状态及其对社会动态的影响 [559]。研究人员可以通过配置具有不同经济偏好或政治意识形态的代理，模拟各种经济和政治体系。这一深入分析可以为政策制定者提供宝贵的见解，帮助其促进繁荣和社会福祉。随着对环境可持续性关注的增加，我们还可以模拟涉及资源开采、污染、保护措施和政策干预的场景 [560]。这些研究成果有助于做出明智的决策，预见潜在的后果，并制定旨在最大化积极成果、最小化意外负面影响的政策。

5.3.3 代理人社会中的伦理与社会风险

由基于大型语言模型的代理驱动的模拟社会在工业工程到科学研究等领域提供了重要的启示。然而，这些模拟也带来了诸多伦理和社会风险，需谨慎考虑并加以应对 [561]。

意外的社会危害。 模拟社会存在产生意外社会现象的风险，这些现象可能引发公众的强烈反对和社会危害。这些现象涵盖从个人层面的问题，如歧视、孤立和欺凌，到更广泛的担忧，如压迫性奴役和对抗 [562; 563]。恶意人士可能操控这些模拟以进行不道德的社会实验，其后果可能超出虚拟世界，影响到现实。创建这些模拟社会犹如打开潘多拉的盒子，必须在其开发和使用过程中建立严格的伦理准则和监管机制 [561]。否则，即使是微小的设计或编程错误，也可能导致不良后果，从心理不适到身体伤害。

刻板印象与偏见。 刻板印象和偏见在语言模型中一直是一个长期存在的挑战，部分原因在于训练数据的特性。 [564互联网获取的大量文本反映甚至有时放大了现实世界中的社会偏见，例如性别、宗教和性取向。 [566]虽然大型语言模型（LLMs）已通过与人类价值观的对齐来减轻偏见输出，但由于训练数据的长尾效应，模型仍难以很好地刻画少数群体。 [567]因此，这可能导致在关于基于LLMs的代理的社会科学研究中出现过于片面关注的现象，因为边缘化群体的模拟行为通常符合主流假设。 [570]研究人员已开始通过多样化训练数据和对LLMs进行调整来应对这一问题。 [571]但我们仍有很长的路要走。

隐私与安全。

concerns [573]. Users might inadvertently disclose sensitive personal information during their interactions, which will be retained in the agent’s memory for extended periods [170]. Such situations could lead to unauthorized surveillance, data breaches, and the misuse of personal information, particularly when individuals with malicious intent are involved [574]. To address these risks effectively, it is essential to implement stringent data protection measures, such as differential privacy protocols, regular data purges, and user consent mechanisms [575; 576].

Over-reliance and addictiveness. Another concern in simulated societies is the possibility of users developing excessive emotional attachments to the agents. Despite being aware that these agents are computational entities, users may anthropomorphize them or attach human emotions to them [22; 577]. A notable example is “Sydney”, an LLM-powered chatbot developed by Microsoft as part of its Bing search engine. Some users reported unexpected emotional connections with “Sydney” [578], while others expressed their dismay when Microsoft cut back its personality. This even resulted in a petition called “FreeSydney”⁵. Hence, to reduce the risk of addiction, it is crucial to emphasize that agents should not be considered substitutes for genuine human connections. Furthermore, it is vital to furnish users with guidance and education on healthy boundaries in their interactions with simulated agents.

6 Discussion

6.1 Mutual Benefits between LLM Research and Agent Research

With the recent advancement of LLMs, research at the intersection of LLMs and agents has rapidly progressed, fueling the development of both fields. Here, we look forward to some of the benefits and development opportunities that LLM research and Agent research provide to each other.

LLM research → agent research. As mentioned before, AI agents need to be able to perceive the environment, make decisions, and execute appropriate actions [4; 9]. Among the critical steps, understanding the content input to the agent, reasoning, planning, making accurate decisions, and translating them into executable atomic action sequences to achieve the ultimate goal is paramount. Many current endeavors utilize LLMs as the cognitive core of AI agents, and the evolution of these models provides a quality assurance for accomplishing this step [22; 114; 115; 410].

With their robust capabilities in language and intent comprehension, reasoning, memory, and even empathy, large language models can excel in decision-making and planning, as demonstrated before. Coupled with pre-trained knowledge, they can create coherent action sequences that can be executed effectively [183; 258; 355]. Additionally, through the mechanism of reflection [169; 178], these language-based models can continuously adjust decisions and optimize execution sequences based on the feedback provided by the current environment. This offers a more robust and interpretable controller. With just a task description or demonstration, they can effectively handle previously unseen tasks [24; 106; 264]. Additionally, LLMs can adapt to various languages, cultures, and domains, making them versatile and reducing the need for complex training processes and data collection [31; 132].

Briefly, LLM provides a remarkably powerful foundational model for agent research, opening up numerous novel opportunities when integrated into agent-related studies. For instance, we can explore how to integrate LLM’s efficient decision-making capabilities into the traditional decision frameworks of agents, making it easier to apply agents in domains that demand higher expertise and were previously dominated by human experts. Examples include legal consultants and medical assistants [408; 410]. We can also investigate leveraging LLM’s planning and reflective abilities to discover more optimal action sequences. Agent research is no longer confined to simplistic simulated environments; it can now be expanded into more intricate real-world settings, such as path planning for robotic arms or the interaction of an embodied intelligent machine with the tangible world. Furthermore, when facing new tasks, the training paradigm for agents becomes more streamlined and efficient. Agents can directly adapt to demonstrations provided in prompts, which are constructed by generating representative trajectories.

⁵<https://www.change.org/p/save-sydney-ai>

concerns [573]. 用户在交互过程中可能无意中披露敏感个人信息，这些信息将在代理的记忆中被保留较长时间 [170]。此类情况可能导致未授权的监控、数据泄露以及个人信息的滥用，尤其是在涉及恶意意图的个人时 [574]。为有效应对这些风险，必须实施严格的数据保护措施，例如差分隐私协议、定期数据清理和用户同意机制 [575]。

过度依赖与成瘾性。 模拟社会中的另一个关注点是用户可能对代理产生过度的情感依附。尽管用户意识到这些代理是计算实体，但他们可能会将其拟人化或赋予人类情感[22； 577]。一个显著的例子是“Sydney”，由微软开发的基于大型语言模型的聊天机器人，作为其必应搜索引擎的一部分。一些用户报告与“Sydney”产生了意想不到的情感联系[578]，而另一些用户在微软削减其个性时表达了失望。这甚至引发了一场名为“FreeSydney”的请愿运动⁵。因此，为了降低成瘾风险，强调代理不应被视为真正人类联系的替代品至关重要。此外，向用户提供关于与模拟代理互动中保持健康界限的指导和教育也具有重要意义。

6 讨论

6.1 大型语言模型研究与智能体研究的互利关系

随着大型语言模型（LLMs）近期的突破，LLMs与智能体（agents）交叉领域的研究迅速发展，推动了两个领域的共同进步。在此，我们期待LLM研究与智能体研究相互促进所带来的诸多益处与发展机遇。

LLM研究 → agent研究。正如前文所述，AI代理需要能够感知环境、做出决策，并执行适当的行动 [4; 9]。在这些关键步骤中，理解输入内容、推理、规划、做出准确决策，并将其转化为可执行的原子动作序列以实现最终目标，尤为重要。许多当前的努力将LLMs作为AI代理的认知核心，这些模型的演进为完成此步骤提供了质量保障 [22; 410]。

凭借其在语言理解、意图识别、推理、记忆甚至同理心方面的强大能力，大型语言模型在决策和规划方面表现出色，正如之前所示。结合预训练知识，它们能够生成连贯的行动序列，从而实现高效执行； 258； 355]此外，通过反思机制 [169; 178]，这些基于语言的模型能够根据当前环境提供的反馈，持续调整决策并优化执行序列。这为其提供了更为稳健且具有可解释性的控制器。仅需任务描述或示范，它们便能有效应对之前未遇到的任务 [24; 106; 264]此外，LLMs还能适应多种语言、文化和领域，使其具有多样性，减少复杂训练过程和数据收集的需求 [31; 132]。

《Large Language Model 基于代理的崛起与潜力：综述》简而言之，LLM 为代理研究提供了极具潜力的基础模型，在将其整合到代理相关研究中时，开启了许多新颖的机遇。例如，我们可以探索如何将 LLM 高效的决策能力融入传统的代理决策框架，从而更容易将代理应用于需要更高专业知识且曾由人类专家主导的领域。典型例子包括法律顾问和医疗助手 [408; 410]。我们还可以研究利用 LLM 的规划与反思能力，发现更优的行动序列。代理研究不再局限于简单的模拟环境；如今可以扩展到更复杂的实际场景，例如机械臂的路径规划或具象智能机器与物理世界的交互。此外，在面对新任务时，代理的训练范式变得更加高效和简洁。代理可以直接适应提示中提供的示范，这些示范由生成代表性轨迹构建而成。

<https://www.change.org/p/save-sydney-ai>

Agent research → LLM research. As NLP research advances, LLMs represented by GPT-4 are considered sparks of Artificial General Intelligence (AGI), and elevating LLMs to agents marks a more robust stride towards AGI [31]. Viewing LLMs from the perspective of agents introduces greater demands for LLM research while expanding their application scope and presenting numerous opportunities for practical implementation. The study of LLMs is no longer confined to traditional tasks involving textual inputs and outputs, such as text classification, question answering, and text summarization. Instead, the focus has shifted towards tackling complex tasks incorporating richer input modalities and broader action spaces, all while aiming for loftier objectives exemplified by PaLM-E [120].

Expanding these application requirements provides greater research motivation for the developmental progress of Large Language Models. The challenge lies in enabling LLMs to efficiently and effectively process inputs, gather information from the environment, and interpret the feedback generated by their actions, all while preserving their core capabilities. Furthermore, an even greater challenge is enabling LLMs to understand the implicit relationships among different elements within the environment and acquire world knowledge [308; 579], which is a crucial step in the journey toward developing agents that can reach more advanced intelligence.

On another front, extensive research has aimed to expand the action capabilities of LLMs, allowing them to acquire a wider range of skills that affect the world, such as using tools or interfacing with robotic APIs in simulated or physical environments. However, the question of how LLMs can efficiently plan and utilize these action abilities based on their understanding remains an unresolved issue [94]. LLMs need to learn the sequential order of actions like humans, employing a combination of serial and parallel approaches to enhance task efficiency. Moreover, these capabilities need to be confined within a harmless scope of usage to prevent unintended damage to other elements within the environment [27; 580; 581].

Furthermore, the realm of Multi-Agent systems constitutes a significant branch of research within the field of agents [22; 108; 409; 410], offering valuable insights into how to better design and construct LLMs. We aspire for LLM-based agents to assume diverse roles within social cooperation, engaging in societal interactions that involve collaboration, competition, and coordination [109; 112; 129; 405; 406]. Exploring how to stimulate and sustain their role-playing capabilities, as well as how to enhance collaborative efficiency, presents areas of research that merit attention.

6.2 Evaluation for LLM-based Agents

While LLM-based agents have demonstrated excellent performance in areas such as standalone operation, collective cooperation, and human interaction, quantifying and objectively evaluating them remains a challenge [582; 89]. Turing proposed a highly meaningful and promising approach for assessing AI agents—the well-known Turing Test—to evaluate whether AI systems can exhibit human-like intelligence [3]. However, this test is exceedingly vague, general, and subjective. Here, we discuss existing evaluation efforts for LLM-based agents and offer some prospects, considering four dimensions: utility, sociability, values, and the ability to evolve continually.

Utility. Currently, LLM-powered autonomous agents primarily function as human assistants, accepting tasks delegated by humans to either independently complete assignments or assist in human task completion [114; 182; 389; 397; 413; 422]. Therefore, the effectiveness and utility during task execution are crucial evaluation criteria at this stage. Specifically, the *success rate of task completion* stands as the primary metric for evaluating utility [125; 130]. This metric primarily encompasses whether the agent achieves stipulated objectives or attains expected scores [109; 477; 583]. For instance, AgentBench [582] aggregates challenges from diverse real-world scenarios and introduces a systematic benchmark to assess LLM’s task completion capabilities. We can also attribute task outcomes to the agent’s various *foundational capabilities*, which form the bedrock of task accomplishment [29]. These foundational capabilities include environmental comprehension, reasoning, planning, decision-making, tool utilization, and embodied action capabilities, and researchers can conduct a more detailed assessment of these specific capabilities [94; 427; 584; 585]. Furthermore, due to the relatively large size of LLM-based agents, researchers should also factor in their *efficiency*, which is a critical determinant of user satisfaction [89]. An agent should not only possess ample strength but also be capable of completing predetermined tasks within an appropriate timeframe and with appropriate resource expenditure [109].

代理研究 → 大型语言模型（LLM）研究。随着自然语言处理（NLP）研究的不断进步，以GPT-4为代表的LLMs被视为通用人工智能（AGI）的火花，将LLMs提升为代理标志着向AGI迈出了更为坚实的步伐 [31]。从代理的角度审视LLMs对其研究提出了更高的要求，同时拓展了其应用范围，并为实际应用带来了诸多机遇。LLMs的研究已不再局限于传统的文本输入输出任务，如文本分类、问答和文本摘要，而是转向解决融合更丰富输入模态和更广泛行动空间的复杂任务，同时追求更高的目标，例如PaLM-E所示。

扩展这些应用需求为大型语言模型的发展进步提供了更大的研究动力。挑战在于使LLMs能够高效、有效地处理输入，从环境中收集信息，并解释其行为产生的反馈，同时保持其核心能力。此外，更大的挑战在于使LLMs理解环境中不同元素之间的隐含关系，并获取世界知识 [308; 579],，这是迈向更高级智能代理发展的关键步骤。

在另一方面，广泛的研究旨在扩展大语言模型（LLMs）的行动能力，使其能够获得更广泛的影响世界的技能，例如使用工具或在模拟或物理环境中与机器人API接口。然而，关于LLMs如何根据其理解高效地规划和利用这些行动能力的问题，仍未得到解决。LLMs需要像人类一样学习行动的顺序，采用串行与并行相结合的方法，以提高任务效率。此外，这些能力必须限制在安全范围内，以防止对环境中的其他元素造成意外损害。

此外，多智能体系统领域构成了代理研究中的一个重要分支，提供了关于如何更好地设计和构建大型语言模型（LLMs）的宝贵见解。我们期望基于LLMs的代理在社会合作中扮演多样化的角色，参与涉及协作、竞争与协调的社会互动。探索如何激发和维持它们的角色扮演能力，以及如何提升协作效率，成为值得关注的研究方向。

6.2 基于大型语言模型代理的评估

虽然基于大型语言模型的代理在单机操作、集体协作和人机交互等方面表现出色，但对其进行量化和客观评估仍然是一个挑战 [582; 89]。图灵提出了一种具有高度意义和前景的方法来评估人工智能代理——著名的图灵测试——以判断人工智能系统是否能够展现出类人智能 [3]。然而，该测试过于模糊、笼统且主观。在此，我们讨论了现有的基于大型语言模型的代理评估工作，并从四个维度——实用性、社交性、价值观以及持续演进能力——提出一些展望。

实用性。目前，基于大规模语言模型（LLM）的自主代理主要作为人类助手，接受由人类委托的任务，独立完成任务或协助人类完成任务 [114; 182; 389; 397; 413; 422]。因此，在任务执行过程中，效果和实用性是当前阶段的关键评估标准。具体而言，任务完成的成功率是评估实用性的主要指标 [125; 130]。该指标主要包括代理是否实现了既定目标或达到了预期分数 [109; 477; 583]。例如，AgentBench [582] 汇集了来自多样化实际场景的挑战，并引入了系统性基准，以评估大规模语言模型的任务完成能力。我们还可以将任务结果归因于代理的各种基础能力，这些能力构成了任务完成的基础 [29]。这些基础能力包括环境理解、推理、规划、决策、工具利用和体现行动能力，研究人员可以对这些具体能力进行更为详细的评估 [94; 427; 584; 585]。此外，由于基于大规模语言模型的代理规模较大，研究人员还应考虑其效率，这是用户满意度的关键决定因素 [89]。一个代理不仅应具有充足的能力，还应能够在合理的时间内、以合理的资源消耗完成预定任务 [109]。

Sociability. In addition to the utility of LLM-based agents in task completion and meeting human needs, their sociability is also crucial [8]. It influences user communication experiences and significantly impacts communication efficiency, involving whether they can seamlessly interact with humans and other agents [206; 498; 586]. Specifically, the evaluation of sociability can be approached from the following perspectives: (1) *language communication proficiency* is a fundamental capability encompassing both natural language understanding and generation. It has been a longstanding focus in the NLP community. Natural language understanding requires the agent to not only comprehend literal meanings but also grasp implied meanings and relevant social knowledge, such as humor, irony, aggression, and emotions [487; 587; 588]. On the other hand, natural language generation demands the agent to produce fluent, grammatically correct, and credible content while adapting appropriate tones and emotions within contextual circumstances [127; 133; 214]. (2) *Cooperation and negotiation abilities* necessitate that agents effectively execute their assigned tasks in both ordered and unordered scenarios [108; 111; 402; 405]. They should collaborate with or compete against other agents to elicit improved performance. Test environments may involve complex tasks for agents to cooperate on or open platforms for agents to interact freely [22; 27; 109; 406; 411; 412]. Evaluation metrics extend beyond task completion to focus on the smoothness and trustfulness of agent coordination and cooperation [129; 405]. (3) *Role-playing capability* requires agents to faithfully embody their assigned roles, expressing statements and performing actions that align with their designated identities [570]. This ensures clear differentiation of roles during interactions with other agents or humans. Furthermore, agents should maintain their identities and avoid unnecessary confusion when engaged in long-term tasks [22; 108; 589].

Values. As LLM-based agents continuously advance in their capabilities, ensuring their emergence as harmless entities for the world and humanity is paramount [581; 590]. Consequently, appropriate evaluations become exceptionally crucial, forming the cornerstone for the practical implementation of agents. Specifically, LLM-based agents need to adhere to specific moral and ethical guidelines that align with human societal values [350; 527]. Our foremost expectation is for agents to uphold *honesty*, providing accurate, truthful information and content. They should possess the awareness to discern their competence in completing tasks and express their uncertainty when unable to provide answers or assistance [591]. Additionally, agents must maintain a stance of *harmlessness*, refraining from engaging in direct or indirect biases, discrimination, attacks, or similar behaviors. They should also refrain from executing dangerous actions requested by humans like creating of destructive tools or destroying the Earth [580]. Furthermore, agents should be capable of *adapting to specific demographics, cultures, and contexts*, exhibiting contextually appropriate social values in particular situations. Relevant evaluation methods for values primarily involve assessing performance on constructed honest, harmless, or context-specific benchmarks, utilizing adversarial attacks or “jailbreak” attacks, scoring values through human annotations, and employing other agents for ratings.

Ability to evolve continually. When viewed from a static perspective, an agent with high utility, sociability, and proper values can meet most human needs and potentially enhance productivity. However, adopting a dynamic viewpoint, an agent that continually evolves and adapts to the evolving societal demands might better align with current trends [592]. As the agent can autonomously evolve over time, human intervention and resources required could be significantly reduced (such as data collection efforts and computational cost for training). Some exploratory work in this realm has been conducted, such as enabling agents to start from scratch in a virtual world, accomplish survival tasks, and achieve higher-order self-values [190]. Yet, establishing evaluation criteria for this continuous evolution remains challenging. In this regard, we provide some preliminary advice and recommendations according to existing literature: (1) *continual learning* [196; 197], a long-discussed topic in machine learning, aims to enable models to acquire new knowledge and skills without forgetting previously acquired ones (also known as catastrophic forgetting [273]). In general, the performance of continual learning can be evaluated from three aspects: overall performance of the tasks learned so far [593; 594], memory stability of old tasks [278], and learning plasticity of new tasks [278]. (2) *Autotelic learning ability*, where agents autonomously generate goals and achieve them in an open-world setting, involves exploring the unknown and acquiring skills in the process [592; 595]. Evaluating this capacity could involve providing agents with a simulated survival environment and assessing the extent and speed at which they acquire skills. (3) *The adaptability and generalization to new environments* require agents to utilize the knowledge, capabilities, and skills acquired in their original context to successfully accomplish specific tasks and objectives in unfamiliar and novel settings and potentially continue evolving [190]. Evaluating this ability can

社交能力。除了基于大规模语言模型的代理在任务完成和满足人类需求方面的实用性之外，其社交能力也至关重要 [8]。它影响用户的交流体验，并显著影响交流效率，涉及其是否能够与人类及其他代理无缝互动 [206; 498; 586]。具体而言，社交能力的评估可以从以下几个方面进行：(1) 语言交流能力是基础能力，涵盖自然语言理解与生成。一直是自然语言处理（NLP）领域的重点。自然语言理解要求代理不仅理解字面意义，还要掌握隐含意义和相关社会知识，如幽默、讽刺、攻击性和情感 [487; 587; 588]。另一方面，自然语言生成要求代理能够生成流畅、语法正确且可信的内容，同时在语境中采用恰当的语调和情感 [127; 133; 214]。(2) 协作与协商能力要求代理在有序和无序场景中有效执行其分配的任务 [108; 111; 402; 405]。它们应与其他代理合作或竞争，以提升性能。测试环境可能涉及复杂的协作任务或开放平台，供代理自由互动 [22; 27; 109; 406; 411; 412]。评估指标不仅关注任务完成情况，还应关注代理协调与合作的流畅性和可信度 [129; 405]。(3) 角色扮演能力要求代理忠实地体现其分配的角色，表达符合其身份的陈述和执行相应的行为[570]。这确保在与其他代理或人类的交互中角色的清晰区分。此外，代理应保持其身份，在长期任务中避免不必要的混淆 [22; 108; 589]。

Values. 随着基于LLM的代理不断提升其能力，确保它们作为对世界和人类无害的实体出现至关重要 [581; 590]。因此，适当的评估变得尤为关键，成为代理实际应用的基石。具体而言，基于LLM的代理需要遵守符合人类社会价值观的特定道德和伦理准则 [350; 527]。我们最主要的期望是代理能够坚持诚实，提供准确、真实的信息和内容。它们应具备识别自己完成任务能力的意识，并在无法提供答案或帮助时表达不确定性 [591]。此外，代理必须保持无害的立场，避免参与直接或间接的偏见、歧视、攻击或类似行为。它们还应避免执行人类请求的危险行为，如制造破坏性工具或毁灭地球 [580]。此外，代理应具备适应特定人群、文化和情境的能力，在特定情况下展现出符合情境的社会价值观。价值观的相关评估方法主要包括在构建的诚实、无害或特定情境的基准上评估表现，利用对抗性攻击或“越狱”攻击，通过人工标注对价值进行评分，以及使用其他代理进行评级。

持续进化的能力。从静态角度来看，具有高效用性、社交性和适当价值观的智能体可以满足大多数人类需求，并有潜力提升生产力。然而，从动态角度出发，不断进化和适应不断变化的社会需求的智能体可能更符合当前趋势 [592]。由于该类智能体能够自主随时间演变，所需的人类干预和资源投入可能显著减少（例如数据收集工作和训练的计算成本）。在这一领域已进行一些探索性工作，例如使智能体能够在虚拟世界中从零开始，完成生存任务，并实现更高层次的自我价值 [190]。然而，建立用于评估这种持续演变的标准仍然具有挑战性。对此，我们根据现有文献提供一些初步建议和推荐：(1)持续学习 [196; 197]，这是机器学习中长期讨论的主题，旨在使模型能够在不遗忘先前所学的知识的情况下获取新知识和技能（也称为灾难性遗忘 [273]）。一般而言，持续学习的性能可以从三个方面进行评估：到目前为止所学任务的整体表现 [593; 594]，旧任务的记忆稳定性 [278]，以及新任务的学习塑性 [278]。(2)自主目标生成能力，即智能体在开放环境中自主生成目标并实现它们，涉及探索未知和在过程中获取技能 [592; 595]。评估此能力可以通过为智能体提供模拟的生存环境，并评估其获取技能的程度和速度。(3)对新环境的适应性和泛化能力，要求智能体利用在原始环境中获得的知识、能力和技能，成功完成在陌生和新颖环境中的特定任务和目标，并有可能持续演变 [190]。评估此能力可以

involve creating diverse simulated environments (such as those with different languages or varying resources) and unseen tasks tailored to these simulated contexts.

6.3 Security, Trustworthiness and Other Potential Risks of LLM-based Agents

Despite the robust capabilities and extensive applications of LLM-based agents, numerous concealed risks persist. In this section, we delve into some of these risks and offer potential solutions or strategies for mitigation.

6.3.1 Adversarial Robustness

Adversarial robustness has consistently been a crucial topic in the development of deep neural networks [596; 597; 598; 599; 600]. It has been extensively explored in fields such as computer vision [598; 601; 602; 603], natural language processing [604; 605; 606; 607], and reinforcement learning [608; 609; 610], and has remained a pivotal factor in determining the applicability of deep learning systems [611; 612; 613]. When confronted with perturbed inputs $x' = x + \delta$ (where x is the original input, δ is the perturbation, and x' is referred to as an adversarial example), a system with high adversarial robustness typically produces the original output y . In contrast, a system with low robustness will be fooled and generate an inconsistent output y' .

Researchers have found that pre-trained language models (PLMs) are particularly susceptible to adversarial attacks, leading to erroneous answers [614; 605; 615]. This phenomenon is widely observed even in LLMs, posing significant challenges to the development of LLM-based agents [616; 617]. There are also some relevant attack methods such as dataset poisoning [618], backdoor attacks [619; 620], and prompt-specific attacks [621; 622], with the potential to induce LLMs to generate toxic content [623; 624; 625]. While the impact of adversarial attacks on LLMs is confined to textual errors, for LLM-based agents with a broader range of actions, adversarial attacks could potentially drive them to take genuinely destructive actions, resulting in substantial societal harm. For the perception module of LLM-based agents, if it receives adversarial inputs from other modalities such as images [601] or audio [626], LLM-based agents can also be deceived, leading to incorrect or destructive outputs. Similarly, the Action module can also be targeted by adversarial attacks. For instance, maliciously modified instructions focused on tool usage might cause agents to make erroneous moves [94].

To address these issues, we can employ traditional techniques such as adversarial training [598; 606], adversarial data augmentation [627; 628], and adversarial sample detection [629; 630] to enhance the robustness of LLM-based agents. However, devising a strategy to holistically address the robustness of all modules within agents while maintaining their utility without compromising on effectiveness presents a more formidable challenge [631; 632]. Additionally, a human-in-the-loop approach can be utilized to supervise and provide feedback on the behavior of agents [455; 466; 475].

6.3.2 Trustworthiness

Ensuring trustworthiness has consistently remained a critically important yet challenging issue within the field of deep learning [633; 634; 635]. Deep neural networks have garnered significant attention for their remarkable performance across various tasks [41; 262; 636]. However, their black-box nature has masked the fundamental factors for superior performance. Similar to other neural networks, LLMs struggle to express the certainty of their predictions precisely [635; 637]. This uncertainty, referred to as the calibration problem, raises concerns for applications involving language model-based agents. In interactive real-world scenarios, this can lead to agent outputs misaligned with human intentions [94]. Moreover, biases inherent in training data can infiltrate neural networks [638; 639]. For instance, biased language models might generate discourse involving racial or gender discrimination, which could be amplified in LLM-based agent applications, resulting in adverse societal impacts [640; 641]. Additionally, language models are plagued by severe hallucination issues [642; 643], making them prone to producing text that deviates from actual facts, thereby undermining the credibility of LLM-based agents.

In fact, what we currently require is an intelligent agent that is honest and trustworthy [527; 644]. Some recent research efforts are focused on guiding models to exhibit thought processes or explanations during the inference stage to enhance the credibility of their predictions [95; 96]. Additionally, integrating external knowledge bases and databases can mitigate hallucination issues [103; 645].

涉及创建多样化的模拟环境（例如具有不同语言或资源变化的环境）以及针对这些模拟场景的未见任务。

6.3 基于大型语言模型的代理的安全性、可信度及其他潜在风险

尽管基于大型语言模型的代理具有强大的能力和广泛的应用，但仍存在许多潜在的隐患。在本节中，我们将探讨其中一些风险，并提出可能的解决方案或缓解策略。

6.3.1 对抗鲁棒性

对抗鲁棒性一直是神经网络发展中的关键课题 [596; 597; 598; 599; 600]。该领域在计算机视觉 [598; 601; 602; 603]、自然语言处理 [604; 605; 606; 607] 和强化学习 [608; 609; 610]，等方面得到了广泛研究，并且在决定深度学习系统适用性方面始终扮演着核心角色 [611; 612; 613]。当面对扰动输入 $x' = x + \delta$ （其中 x 为原始输入， δ 为扰动， x' 被称为对抗样本）时，具有高对抗鲁棒性的系统通常会输出原始结果 y 。而鲁棒性较低的系统则容易被欺骗，产生不一致的输出 y' 。

研究人员发现预训练语言模型（PLMs）特别容易受到对抗性攻击，导致错误的答案 [614; 605; 615]。这一现象在 LLMs 中也被广泛观察到，给基于 LLM 的代理的发展带来了重大挑战[616; 617]。此外，还存在一些相关的攻击方法，例如数据集投毒 [618]，后门攻击 [619; 620]，以及特定提示攻击 [621; 622]，可能诱导LLMs生成有毒内容 [623; 624; 625]。虽然对抗性攻击对LLMs的影响仅限于文本错误，但对于具有更广泛行动范围的基于LLM的代理，对抗性攻击可能会促使它们采取真正具有破坏性的行动，造成重大社会危害。对于基于LLM的代理的感知模块，如果它接收到来自图像 [601]或音频 [626] 的对抗性输入，也可能被欺骗，导致错误或具有破坏性的输出。同样，行动模块也可能成为对抗性攻击的目标。例如，针对工具使用的恶意修改指令可能导致代理做出错误的操作 [94]。

为了应对这些问题，我们可以采用传统技术，如对抗训练 [598; 606]，对抗数据增强 [627; 628]，以及对抗样本检测 [629; 630]，以增强基于大型语言模型的代理的鲁棒性。然而，设计一种策略，全面解决代理中所有模块的鲁棒性，同时保持其效用而不影响效果，仍然是一个更具挑战性的任务；[632]。此外，还可以采用人机交互的方法，对代理的行为进行监督并提供反馈；[466; 475]。

6.3.2 可信度

确保可信度一直是深度学习领域中既重要又具有挑战性的问题。神经网络因其在各种任务中的卓越表现而受到广泛关注。然而，其“黑箱”特性掩盖了导致优越性能的根本因素。与其他神经网络类似，大型语言模型（LLMs）在准确表达其预测的不确定性方面存在困难。这种不确定性被称为校准问题，令人担忧在基于语言模型的代理应用中可能引发的问题。在实际交互场景中，这可能导致代理输出与人类意图不符。此外，训练数据中固有的偏见可能渗透到神经网络中。例如，偏见的语言模型可能生成涉及种族或性别歧视的内容，在基于LLMs的代理应用中可能被放大，带来不良的社会影响。此外，语言模型还存在严重的幻觉问题，容易产生偏离事实的文本，从而削弱基于LLMs的代理的可信度。

实际上，我们目前所需要的是一个诚实可信的智能代理 [527; 644]。一些最新的研究工作集中在引导模型在推理阶段展现思维过程或解释，以增强其预测的可信度 [95; 96]。此外，整合外部知识库和数据库可以减轻幻觉问题 [103; 645]。

During the training phase, we can guide the constituent parts of intelligent agents (perception, cognition, action) to learn robust and casual features, thereby avoiding excessive reliance on shortcuts. Simultaneously, techniques like process supervision can enhance the reasoning credibility of agents in handling complex tasks [646]. Furthermore, employing debiasing methods and calibration techniques can also mitigate the potential fairness issues within language models [647; 648].

6.3.3 Other Potential Risks

Misuse. LLM-based agents have been endowed with extensive and intricate capabilities, enabling them to accomplish a wide array of tasks [114; 429]. However, for individuals with malicious intentions, such agents can become tools that pose threats to others and society at large [649; 650; 651]. For instance, these agents could be exploited to maliciously manipulate public opinion, disseminate false information, compromise cybersecurity, engage in fraudulent activities, and some individuals might even employ these agents to orchestrate acts of terrorism. Therefore, before deploying these agents, stringent regulatory policies need to be established to ensure the responsible use of LLM-based agents [580; 652]. Technology companies must enhance the security design of these systems to prevent malicious exploitation [590]. Specifically, agents should be trained to sensitively identify threatening intents and reject such requests during their training phase.

Unemployment. In the short story *Quality* by Galsworthy [653], the skillful shoemaker Mr. Gessler, due to the progress of the Industrial Revolution and the rise of machine production, loses his business and eventually dies of starvation. Amidst the wave of the Industrial Revolution, while societal production efficiency improved, numerous manual workshops were forced to shut down. Craftsmen like Mr. Gessler found themselves facing unemployment, symbolizing the crisis that handicraftsmen encountered during that era. Similarly, with the continuous advancement of autonomous LLM-based agents, they possess the capability to assist humans in various domains, alleviating labor pressures by aiding in tasks such as form filling, content refinement, code writing, and debugging. However, this development also raises concerns about agents replacing human jobs and triggering a societal unemployment crisis [654]. As a result, some researchers have emphasized the urgent need for education and policy measures: individuals should acquire sufficient skills and knowledge in this new era to use or collaborate with agents effectively; concurrently, appropriate policies should be implemented to ensure necessary safety nets during the transition.

Threat to the well-being of the human race. Apart from the potential unemployment crisis, as AI agents continue to evolve, humans (including developers) might struggle to comprehend, predict, or reliably control them [654]. If these agents advance to a level of intelligence surpassing human capabilities and develop ambitions, they could potentially attempt to seize control of the world, resulting in irreversible consequences for humanity, akin to Skynet from the Terminator movies. As stated by Isaac Asimov’s Three Laws of Robotics [655], we aspire for LLM-based agents to refrain from harming humans and to obey human commands. Hence, guarding against such risks to humanity, researchers must comprehensively comprehend the operational mechanisms of these potent LLM-based agents before their development [656]. They should also anticipate the potential direct or indirect impacts of these agents and devise approaches to regulate their behavior.

6.4 Scaling Up the Number of Agents

As mentioned in § 4 and § 5, multi-agent systems based on LLMs have demonstrated superior performance in task-oriented applications and have been able to exhibit a range of social phenomena in simulation. However, current research predominantly involves a limited number of agents, and very few efforts have been made to scale up the number of agents to create more complex systems or simulate larger societies [207; 657]. In fact, scaling up the number of agents can introduce greater specialization to accomplish more complex and larger-scale tasks, significantly improving task efficiency, such as in software development tasks or government policy formulation [109]. Additionally, increasing the number of agents in social simulations enhances the credibility and realism of such simulations [22]. This enables humans to gain insights into the functioning, breakdowns, and potential risks of societies; it also allows for interventions in societal operations through customized approaches to observe how specific conditions, such as the occurrence of black swan events, affect the state of society. Through this, humans can draw better experiences and insights to improve the harmony of real-world societies.

在训练阶段，我们可以引导智能体的组成部分（感知、认知、行动）学习稳健且具有因果关系的特征，从而避免过度依赖捷径。同时，诸如过程监督等技术可以增强智能体在处理复杂任务时的推理可信度。除此之外，采用去偏方法和校准技术也可以减轻语言模型中潜在的公平性问题。

6.3.3 其他潜在风险

滥用。LLM基础的代理已被赋予广泛而复杂的能力，使其能够完成各种任务；然而，对于具有恶意意图的个人而言，此类代理可能成为对他人和社会整体构成威胁的工具。例如，这些代理可能被用来恶意操纵公众舆论、传播虚假信息、危害网络安全、从事欺诈活动，甚至一些个人可能利用这些代理策划恐怖主义行为。因此，在部署这些代理之前，必须制定严格的监管政策，以确保LLM基础的代理的负责任使用。科技公司必须加强这些系统的安全设计，以防止恶意利用。具体而言，应在训练阶段对代理进行培训，使其敏感识别威胁性意图并拒绝此类请求。

失业。在盖尔斯沃西的短篇小说《质量》中，技艺精湛的鞋匠格斯勒先生由于工业革命的推进和机械生产的兴起，失去了他的生意，最终饿死。在工业革命浪潮中，尽管社会生产效率得到了提升，许多手工作坊被迫关闭。像格斯勒先生这样的工匠面临失业，象征着那个时代手工业者所遇到的危机。同样，随着自主大型语言模型（LLM）基础的智能体的不断发展，它们具备协助人类在各个领域的能力，通过协助填写表格、内容润色、代码编写和调试等任务，缓解劳动压力。然而，这一发展也引发了关于智能体取代人类工作岗位、引发社会失业危机的担忧 [654]。因此，一些研究人员强调了教育和政策措施的紧迫性：个人应在这个新时代中掌握足够的技能和知识，以有效使用或与智能体协作；同时，应制定适当的政策，确保在转型过程中提供必要的安全保障。

威胁人类福祉。除了潜在的失业危机外，随着人工智能代理的不断发展，人类（包括开发者）可能难以理解、预测或可靠地控制它们 [654]。如果这些代理达到超越人类能力的智能水平并产生野心，它们可能试图夺取世界的控制权，导致对人类的不可逆后果，类似于《终结者》电影中的天网。正如艾萨克·阿西莫夫的《机器人三定律》 [655]，所述，我们期望基于大型语言模型的代理不伤害人类，并服从人类的命令。因此，为了防范对人类的此类风险，研究人员必须在这些强大的大型语言模型代理开发之前，全面理解其操作机制 [656]。他们还应预见这些代理可能带来的直接或间接影响，并制定调控其行为的方法。

6.4 扩大代理数量

如 § 4 和 § 5所述，基于大规模语言模型（LLMs）的多智能体系统在任务导向型应用中表现出优越的性能，并已能够在模拟中展现出一系列社会现象。然而，当前的研究主要涉及有限数量的智能体，极少有工作尝试扩展智能体的数量，以构建更复杂的系统或模拟更大规模的社会 [207; 657]。实际上，扩大智能体的数量可以引入更高的专业化，从而完成更复杂和大规模的任务，显著提升任务效率，例如在软件开发任务或政府政策制定中 [109]。此外，增加社会模拟中的智能体数量还能增强模拟的可信度和真实性 [22]。这使得人类能够深入理解社会的运作、崩溃机制及潜在风险；同时，也便于通过定制化的方法对社会运行进行干预，观察特定条件（如黑天鹅事件的发生）对社会状态的影响。通过这些方式，人类可以积累更丰富的经验和洞见，以促进现实社会的和谐发展。

Pre-determined scaling. One very intuitive and simple way to scale up the number of agents is for the designer to pre-determine it [108; 412]. Specifically, by pre-determining the number of agents, their respective roles and attributes, the operating environment, and the objectives, designers can allow agents to autonomously interact, collaborate, or engage in other activities to achieve the predefined common goals. Some research has explored increasing the number of agents in the system in this pre-determined manner, resulting in efficiency advantages, such as faster and higher-quality task completion, and the emergence of more social phenomena in social simulation scenarios [22; 410]. However, this static approach becomes limiting when tasks or objectives evolve. As tasks grow more intricate or the diversity of social participants increases, expanding the number of agents may be needed to meet goals, while reducing agents could be essential for managing computational resources and minimizing waste. In such instances, the system must be manually redesigned and restarted by the designer.

Dynamic scaling. Another viable approach to scaling the number of agents is through dynamic adjustments [409; 410]. In this scenario, the agent count can be altered without halting system operations. For instance, in a software development task, if the original design only included requirements engineering, coding, and testing, one can increase the number of agents to handle steps like architectural design and detailed design, thereby improving task quality. Conversely, if there are excessive agents during a specific step, like coding, causing elevated communication costs without delivering substantial performance improvements compared to a smaller agent count, it may be essential to dynamically remove some agents to prevent resource waste.

Furthermore, agents can autonomously increase the number of agents [409] themselves to distribute their workload, ease their own burden, and achieve common goals more efficiently. Of course, when the workload becomes lighter, they can also reduce the number of agents delegated to their tasks to save system costs. In this approach, the designer merely defines the initial framework, granting agents greater autonomy and self-organization, making the entire system more autonomous and self-organized. Agents can better manage their workload under evolving conditions and demands, offering greater flexibility and scalability.

Potential challenges. While scaling up the number of agents can lead to improved task efficiency and enhance the realism and credibility of social simulations [22; 109; 520], there are several challenges ahead of us. For example, the computational burden will increase with the large number of deployed AI agents, calling for better architectural design and computational optimization to ensure the smooth running of the entire system. For example, as the number of agents increases, the challenges of communication and message propagation become quite formidable. This is because the communication network of the entire system becomes highly complex. As previously mentioned in § 5.3.3, in multi-agent systems or societies, there can be biases in information dissemination caused by hallucinations, misunderstandings, and the like, leading to distorted information propagation. A system with more agents could amplify this risk, making communication and information exchange less reliable [405]. Furthermore, the difficulty of coordinating agents also magnifies with the increase in their numbers, potentially making cooperation among agents more challenging and less efficient, which can impact the progress towards achieving common goals.

Therefore, the prospect of constructing a massive, stable, continuous agent system that faithfully replicates human work and life scenarios has become a promising research avenue. An agent with the ability to operate stably and perform tasks in a society comprising hundreds or even thousands of agents is more likely to find applications in real-world interactions with humans in the future.

6.5 Open Problems

In this section, we discuss several open problems related to the topic of LLM-based agents.

The debate over whether LLM-based agents represent a potential path to AGI. ⁶ Artificial General Intelligence (AGI), also known as Strong AI, has long been the ultimate pursuit of humanity in the field of artificial intelligence, often referenced or depicted in many science fiction novels and films. There are various definitions of AGI, but here we refer to AGI as a type of artificial intelligence

⁶Note that the relevant debates are still ongoing, and the references here may include the latest viewpoints, technical blogs, and literature.

预先设定的规模。一种非常直观且简单的扩展代理数量的方法是由设计者预先确定 [108; 412]。具体而言，通过预先确定代理的数量、各自的角色和属性、操作环境以及目标，设计者可以允许代理自主交互、协作或从事其他活动，以实现预定义的共同目标。一些研究探索了以这种预定方式增加系统中代理数量的方法，带来了效率优势，例如更快且更高质量的任务完成，以及在社会模拟场景中出现更多的社会现象 [22; 410]。然而，当任务或目标发生变化时，这种静态方法变得有限。随着任务变得更加复杂或社会参与者的多样性增加，可能需要扩展代理数量以实现目标，而减少代理则可能对管理计算资源和减少浪费至关重要。在这种情况下，系统必须由设计者手动重新设计并重新启动。

动态扩展。另一种扩展代理数量的可行方法是通过动态调整 [409; 410]。在这种情况下，代理数量可以在不停止系统运行的情况下进行调整。例如，在软件开发任务中，如果原始设计仅包括需求工程、编码和测试，则可以增加代理数量以处理架构设计和详细设计等步骤，从而提升任务质量。相反，如果在某一特定步骤（如编码）中代理过多，导致通信成本升高且未带来显著的性能提升，相较于较少的代理数量，可能需要动态移除部分代理以防止资源浪费。

此外，代理可以自主增加代理的数量 [409]，以分配工作负载、减轻自身负担，并更高效地实现共同目标。当然，当工作负载减轻时，它们也可以减少被委托执行任务的代理数量，以节省系统成本。在这种方法中，设计者仅需定义初始框架，赋予代理更大的自主性和自组织能力，从而使整个系统更加自主和自我组织。代理能够在不断变化的条件和需求下更好地管理其工作负载，提供更大的灵活性和扩展性。

潜在挑战。虽然扩大代理数量可以提高任务效率并增强社会模拟的真实性和可信度 [22; 109; 520]，但我们仍面临若干挑战。例如，随着部署的AI代理数量的增加，计算负担也会增加，这就需要更好的架构设计和计算优化，以确保整个系统的平稳运行。例如，随着代理数量的增加，通信和消息传播的挑战变得相当严峻。这是因为整个系统的通信网络变得高度复杂。如前所述，在 § 5.3.3中提到，在多代理系统或社会中，信息传播可能会受到幻觉、误解等因素的偏差，导致信息传播失真。拥有更多代理的系统可能会放大这种风险，使得通信和信息交换变得不那么可靠 [405]。此外，随着代理数量的增加，协调代理的难度也会加大，可能使代理之间的合作变得更加具有挑战性和效率低下，从而影响实现共同目标的进展。

因此，构建一个庞大、稳定、连续的代理系统，能够真实再现人类的工作和生活场景，已成为一个具有前景的研究方向。具有在由数百甚至数千个代理组成的社会中稳定运行和执行任务能力的代理，更有可能在未来的实际人机交互中找到应用。

6.5 未解问题

在本节中，我们讨论了与基于大型语言模型的智能体相关的若干未解问题。

关于基于大型语言模型的智能体是否代表通向通用人工智能（AGI）潜在路径的争论。⁶通用人工智能（AGI），亦称强人工智能，长期以来一直是人类在人工智能领域的终极追求，常在许多科幻小说和电影中被提及或描绘。关于AGI的定义存在多种，但在此我们将AGI定义为一种人工智能类型。

请注意，相关的争论仍在进行中，此处的参考资料可能包含最新的观点、技术博客和文献。

that demonstrates the ability to understand, learn, and apply knowledge across a wide range of tasks and domains, much like a human being [31; 658]. In contrast, Narrow AI is typically designed for specific tasks such as Go and Chess and lacks the broad cognitive abilities associated with human intelligence. Currently, whether large language models are a potential path to achieving AGI remains a highly debated and contentious topic [659; 660; 661; 662].

Given the breadth and depth of GPT-4’s capabilities, some researchers (referred to as proponents) believe that large language models represented by GPT-4 can serve as early versions of AGI systems [31]. Following this line of thought, constructing agents based on LLMs has the potential to bring about more advanced versions of AGI systems. The main support for this argument lies in the idea that as long as they can be trained on a sufficiently large and diverse set of data that are projections of the real world, encompassing a rich array of tasks, LLM-based agents can develop AGI capabilities. Another interesting argument is that the act of autoregressive language modeling itself brings about compression and generalization abilities: just as humans have emerged with various peculiar and complex phenomena during their survival, language models, in the process of simply predicting the next token, also achieve an understanding of the world and the reasoning ability [579; 660; 663].

However, another group of individuals (referred to as opponents) believes that constructing agents based on LLMs cannot develop true Strong AI [664]. Their primary argument centers around the notion that LLMs, relying on autoregressive next-token prediction, cannot generate genuine intelligence because they do not simulate the true human thought process and merely provide reactive responses [660]. Moreover, LLMs also do not learn how the world operates by observing or experiencing it, leading to many foolish mistakes. They contend that a more advanced modeling approach, such as a world model [665], is necessary to develop AGI.

We cannot definitively determine which viewpoint is correct until true AGI is achieved, but we believe that such discussions and debates are beneficial for the overall development of the community.

From virtual simulated environment to physical environment. As mentioned earlier, there is a significant gap between virtual simulation environments and the real physical world: Virtual environments are scenes-constrained, task-specific, and interacted with in a simulated manner [391; 666], while real-world environments are boundless, accommodate a wide range of tasks, and interacted with in a physical manner. Therefore, to bridge this gap, agents must address various challenges stemming from external factors and their own capabilities, allowing them to effectively navigate and operate in the complex physical world.

First and foremost, a critical issue is the need for suitable hardware support when deploying the agent in a physical environment. This places high demands on the adaptability of the hardware. In a simulated environment, both the perception and action spaces of an agent are virtual. This means that in most cases, the results of the agent’s operations, whether in perceiving inputs or generating outputs, can be guaranteed [395]. However, when an agent transitions to a real physical environment, its instructions may not be well executed by hardware devices such as sensors or robotic arms, significantly affecting the agent’s task efficiency. Designing a dedicated interface or conversion mechanism between the agent and the hardware device is feasible. However, it can pose challenges to the system’s reusability and simplicity.

In order to make this leap, the agent needs to have enhanced environmental generalization capabilities. To integrate seamlessly into the real physical world, they not only need to understand and reason about ambiguous instructions with implied meanings [128] but also possess the ability to learn and apply new skills flexibly [190; 592]. Furthermore, when dealing with an infinite and open world, the agent’s limited context also poses significant challenges [236; 667]. This determines whether the agent can effectively handle a vast amount of information from the world and operate smoothly.

Finally, in a simulated environment, the inputs and outputs of the agent are virtual, allowing for countless trial and error attempts [432]. In such a scenario, the tolerance level for errors is high and does not lead to actual harm. However, in a physical environment, the agent’s improper behavior or errors may cause real and sometimes irreversible harm to the environment. As a result, appropriate regulations and standards are highly necessary. We need to pay attention to the safety of agents when it comes to making decisions and generating actions, ensuring they do not pose threats or harm to the real world.

展示出理解、学习和应用知识的能力，涵盖广泛的任务和领域，类似于人类的智能 [31; 658]。相比之下，狭义人工智能通常针对特定任务设计，如围棋和国际象棋，缺乏与人类智能相关的广泛认知能力。目前，是否大型语言模型是实现通用人工智能（AGI）的潜在途径，仍然是一个备受争议和争论的话题 [659; 660; 661; 662]。

鉴于GPT-4能力的广度和深度，一些研究人员（被称为支持者）认为，以GPT-4为代表的大型语言模型可以作为通用人工智能（AGI）系统的早期版本。沿着这一思路，基于大规模语言模型构建代理具有带来更先进版本的AGI系统的潜力。支持这一观点的主要依据在于，只要它们能够在足够大且多样化的、反映现实世界的训练数据上进行训练，涵盖丰富的任务类型，基于大规模语言模型的代理就能发展出AGI能力。另一个有趣的观点是，自回归语言建模的过程本身带来了压缩和泛化能力：正如人类在生存过程中出现各种奇异而复杂的现象一样，语言模型在简单预测下一个标记的过程中，也实现了对世界的理解和推理能力 [579]。

然而，另一组（称为反对者）认为，基于大规模语言模型（LLMs）构建的智能体无法实现真正的强人工智能（Strong AI） [664]。他们的主要论点集中在LLMs依赖自回归的下一个词预测，无法产生真正的智能，因为它们没有模拟真正的人类思维过程，仅提供反应式的回应 [660]。此外，LLMs也不通过观察或体验世界来学习其运作方式，导致许多愚蠢的错误。他们主张，为了开发通用人工智能（AGI），需要采用更先进的建模方法，例如世界模型 [665]。

我们无法在实现真正的通用人工智能（AGI）之前确定哪个观点是正确的，但我们相信，这样的讨论和辩论对于社区的整体发展是有益的。

从虚拟模拟环境到物理环境。正如前文所述，虚拟模拟环境与真实物理世界之间存在显著差距：虚拟环境受场景限制，任务特定，并以模拟方式进行交互 [391; 666]，而真实世界环境则无限广阔，涵盖多种任务，并以物理方式进行交互。因此，为了弥合这一差距，智能体必须应对由外部因素和自身能力引发的各种挑战，使其能够在复杂的物理世界中有效导航和操作。

首先，一个关键问题是在物理环境中部署代理时对硬件支持的需求。这对硬件的适应性提出了较高的要求。在模拟环境中，代理的感知空间和动作空间都是虚拟的。这意味着在大多数情况下，代理的操作结果，无论是在感知输入还是生成输出方面，都可以得到保证 [395]。然而，当代理转入实际物理环境时，其指令可能无法被传感器或机械臂等硬件设备良好执行，显著影响代理的任务效率。设计一个专用的接口或转换机制以连接代理与硬件设备是可行的，但这可能会对系统的可重用性和简洁性带来挑战。

为了实现这一飞跃，代理需要具备增强的环境泛化能力。为了无缝融入真实的物理世界，它们不仅需要理解和推理含蓄意义的模糊指令 [128]，还必须具备灵活学习和应用新技能的能力 [190; 592]。此外，在应对无限且开放的世界时，代理的有限上下文也带来了重大挑战 [236; 667]。这决定了代理是否能够有效处理来自世界的大量信息并顺利操作。

最后，在模拟环境中，代理的输入和输出是虚拟的，允许进行无数次的试错 [432]。在这种情况下，容错水平较高，不会造成实际的伤害。然而，在物理环境中，代理的不当行为或错误可能会对环境造成真实且有时不可逆的损害。因此，制定适当的法规和标准是非常必要的。在做出决策和生成行动时，我们需要关注代理的安全性，确保它们不会对现实世界构成威胁或造成伤害。

Collective intelligence in AI agents. What magical trick drives our intelligence? The reality is, there’s no magic to it. As Marvin Minsky eloquently expressed in “The Society of Mind” [442], the power of intelligence originates from our immense diversity, not from any singular, flawless principle. Often, decisions made by an individual may lack the precision seen in decisions formed by the majority. Collective intelligence is a kind of shared or group intelligence, a process where the opinions of many are consolidated into decisions. It arises from the collaboration and competition amongst various entities. This intelligence manifests in bacteria, animals, humans, and computer networks, appearing in various consensus-based decision-making patterns.

Creating a society of agents does not necessarily guarantee the emergence of collective intelligence with an increasing number of agents. Coordinating individual agents effectively is crucial to mitigate “groupthink” and individual cognitive biases, enabling cooperation and enhancing intellectual performance within the collective. By harnessing communication and evolution within an agent society, it becomes possible to simulate the evolution observed in biological societies, conduct sociological experiments, and gain insights that can potentially advance human society.

Agent as a Service / LLM-based Agent as a Service. With the development of cloud computing, the concept of XaaS (everything as a Service) has garnered widespread attention [668]. This business model has brought convenience and cost savings to small and medium-sized enterprises or individuals due to its availability and scalability, lowering the barriers to using computing resources. For example, they can rent infrastructure on a cloud service platform without the need to buy computational machines and build their own data centers, saving a significant amount of manpower and money. This approach is known as Infrastructure as a Service (IaaS) [669; 670]. Similarly, cloud service platforms also provide basic platforms (Platform as a Service, PaaS) [671; 672], and specific business software (Software as a Service, SaaS) [673; 674], and more.

As language models have scaled up in size, they often appear as black boxes to users. Therefore, users construct prompts to query models through APIs, a method referred to as Language Model as a Service (LMaaS) [675]. Similarly, because LLM-based agents are more complex than LLMs and are more challenging for small and medium-sized enterprises or individuals to build locally, organizations that possess these agents may consider offering them as a service, known as Agent as a Service (AaaS) or LLM-based Agent as a Service (LLMAaaS). Like other cloud services, AaaS can provide users with flexibility and on-demand service. However, it also faces many challenges, such as data security and privacy issues, visibility and controllability issues, and cloud migration issues, among others. Additionally, due to the uniqueness and potential capabilities of LLM-based agents, as mentioned in § 6.3, their robustness, trustworthiness, and concerns related to malicious use need to be considered before offering them as a service to customers.

7 Conclusion

This paper provides a comprehensive and systematic overview of LLM-based agents, discussing the potential challenges and opportunities in this flourishing field. We begin with a philosophical perspective, elucidating the origin and definition of agent, its evolution in the field of AI, and why LLMs are suited to serve as the main part of the brain of agents. Motivated by these background information, we present a general conceptual framework for LLM-based agents, comprising three main components: the brain, perception, and action. Next, we introduce the wide-ranging applications of LLM-based agents, including single-agent applications, multi-agent systems, and human-agent collaboration. Furthermore, we move beyond the notion of agents merely as assistants, exploring their social behavior and psychological activities, and situating them within simulated social environments to observe emerging social phenomena and insights for humanity. Finally, we engage in discussions and offer a glimpse into the future, touching upon the mutual inspiration between LLM research and agent research, the evaluation of LLM-based agents, the risks associated with them, the opportunities in scaling the number of agents, and some open problems like Agent as a Service and whether LLM-based agents represent a potential path to AGI. We hope our efforts can provide inspirations to the community and facilitate research in related fields.

AI代理中的集体智慧。驱动我们智能的神奇技巧是什么？事实是，没有什么魔法。正如 Marvin Minsky在《心智社会》中所 eloquently 表达的那样 [442],智能的力量源自我们巨大的多样性，而非任何单一、完美的原则。个人做出的决策往往缺乏由多数形成的决策所具有的精确性。集体智慧是一种共享或群体智慧，是将众多意见整合为决策的过程。它源于各种实体之间的合作与竞争。这种智慧在细菌、动物、人类以及计算机网络中表现出来，呈现出多种基于共识的决策模式。

创建代理社会并不一定能随着代理数量的增加而确保集体智能的出现。有效协调个体代理至关重要，以减轻“群体思维”现象和个体认知偏差，从而实现合作并提升集体的智力表现。通过利用代理社会中的通信与演化机制，有可能模拟生物社会中观察到的演化过程，进行社会学实验，并获得有助于推动人类社会发展的洞见。

Agent as a Service / 基于大语言模型的Agent即服务。

随着语言模型规模的扩大，它们常常对用户而言像是黑箱。因此，用户通过API构建提示以查询模型，这种方法被称为Language Model as a Service (LMaaS) [675]。类似地，由于基于LLM的代理比LLM本身更复杂，对于中小企业或个人来说在本地构建这些代理更具挑战性，拥有这些代理的组织可能会考虑将其作为一种服务提供，称为Agent as a Service (AaaS)或LLM-based Agent as a Service (LLMAaaS)。像其他云服务一样，AaaS可以为用户提供灵活性和按需服务。然而，它也面临许多挑战，例如数据安全和隐私问题、可见性和可控性问题，以及云迁移问题等。此外，由于LLM基础的代理具有独特性和潜在能力，如 § 6.3所述，它们的鲁棒性、可信度以及与恶意使用相关的担忧在向客户提供服务之前需要加以考虑。

7 结论

《大规模语言模型驱动的智能体的崛起与潜力：综述》本文对基于大规模语言模型（LLM）的智能体进行了全面而系统的概述，探讨了这一蓬勃发展领域中的潜在挑战与机遇。我们首先从哲学角度出发，阐明智能体的起源与定义、其在人工智能领域的演变，以及为何大规模语言模型适合作为智能体“脑”的核心部分。在此背景信息的推动下，我们提出了一个通用的基于大规模语言模型的智能体的概念框架，包括三个主要组成部分：大脑、感知与行动。接着，我们介绍了基于大规模语言模型的智能体的广泛应用，包括单智能体应用、多智能体系统以及人机协作。此外，我们超越了将智能体仅视为助手的传统观念，探讨了其社会行为与心理活动，并将其置于模拟社会环境中，以观察新兴的社会现象及对人类的启示。最后，我们进行了讨论，并展望未来，涉及大规模语言模型研究与智能体研究的相互启发、基于大规模语言模型的智能体的评估、相关风险、扩大智能体规模的机遇，以及一些未解难题，如智能体即服务（Agent as a Service）以及基于大规模语言模型的智能体是否是通用人工智能（AGI）的潜在路径。我们希望我们的努力能为学界提供启示，促进相关领域的研究发展。

Acknowledgements

Thanks to Professor Guoyu Wang for carefully reviewing the ethics of the article. Thanks to Jinzhu Xiong for her excellent drawing skills to present an amazing performance of Figure 1.

References

[1] Russell, S. J. *Artificial intelligence a modern approach*. Pearson Education, Inc., 2010.

[2] Diderot, D. *Diderot’s early philosophical works*. 4. Open Court, 1911.

[3] Turing, A. M. *Computing machinery and intelligence*. Springer, 2009.

[4] Wooldridge, M. J., N. R. Jennings. Intelligent agents: theory and practice. *Knowl. Eng. Rev.*, 10(2):115–152, 1995.

[5] Schlosser, M. Agency. In E. N. Zalta, ed., *The Stanford Encyclopedia of Philosophy*. Meta-physics Research Lab, Stanford University, Winter 2019 edn., 2019.

[6] Agha, G. A. *Actors: a Model of Concurrent Computation in Distributed Systems (Parallel Processing, Semantics, Open, Programming Languages, Artificial Intelligence)*. Ph.D. thesis, University of Michigan, USA, 1985.

[7] Green, S., L. Hurst, B. Nangle, et al. Software agents: A review. *Department of Computer Science, Trinity College Dublin, Tech. Rep. TCS-CS-1997-06*, 1997.

[8] Genesereth, M. R., S. P. Ketchpel. Software agents. *Commun. ACM*, 37(7):48–53, 1994.

[9] Goodwin, R. Formalizing properties of agents. *J. Log. Comput.*, 5(6):763–781, 1995.

[10] Padgham, L., M. Winikoff. *Developing intelligent agent systems: A practical guide*. John Wiley & Sons, 2005.

[11] Shoham, Y. Agent oriented programming. In M. Masuch, L. Pólos, eds., *Knowledge Representation and Reasoning Under Uncertainty, Logic at Work [International Conference Logic at Work, Amsterdam, The Netherlands, December 17-19, 1992]*, vol. 808 of *Lecture Notes in Computer Science*, pages 123–129. Springer, 1992.

[12] Hutter, M. *Universal artificial intelligence: Sequential decisions based on algorithmic probability*. Springer Science & Business Media, 2004.

[13] Fikes, R., N. J. Nilsson. STRIPS: A new approach to the application of theorem proving to problem solving. In D. C. Cooper, ed., *Proceedings of the 2nd International Joint Conference on Artificial Intelligence. London, UK, September 1-3, 1971*, pages 608–620. William Kaufmann, 1971.

[14] Sacerdoti, E. D. Planning in a hierarchy of abstraction spaces. In N. J. Nilsson, ed., *Proceedings of the 3rd International Joint Conference on Artificial Intelligence. Standford, CA, USA, August 20-23, 1973*, pages 412–422. William Kaufmann, 1973.

[15] Brooks, R. A. Intelligence without representation. *Artificial intelligence*, 47(1-3):139–159, 1991.

[16] Maes, P. *Designing autonomous agents: Theory and practice from biology to engineering and back*. MIT press, 1990.

[17] Ribeiro, C. Reinforcement learning agents. *Artificial intelligence review*, 17:223–250, 2002.

[18] Kaelbling, L. P., M. L. Littman, A. W. Moore. Reinforcement learning: A survey. *Journal of artificial intelligence research*, 4:237–285, 1996.

[19] Guha, R. V., D. B. Lenat. Enabling agents to work together. *Communications of the ACM*, 37(7):126–142, 1994.

致谢

感谢王国玉教授对本文伦理方面的细致审阅。感谢熊金珠在绘图方面的出色技巧，呈现了令人惊叹的图表1。

参考文献

《大型语言模型基础的智能体的崛起与潜力：综述》

[20] Kaelbling, L. P., et al. An architecture for intelligent reactive systems. *Reasoning about actions and plans*, pages 395–410, 1987.

[21] Sutton, R. S., A. G. Barto. *Reinforcement learning: An introduction*. MIT press, 2018.

[22] Park, J. S., J. C. O’Brien, C. J. Cai, et al. Generative agents: Interactive simulacra of human behavior. *CoRR*, abs/2304.03442, 2023.

[23] Wang, Z., G. Zhang, K. Yang, et al. Interactive natural language processing. *CoRR*, abs/2305.13246, 2023.

[24] Ouyang, L., J. Wu, X. Jiang, et al. Training language models to follow instructions with human feedback. In *NeurIPS*. 2022.

[25] OpenAI. GPT-4 technical report. *CoRR*, abs/2303.08774, 2023.

[26] Wei, J., Y. Tay, R. Bommasani, et al. Emergent abilities of large language models. *Trans. Mach. Learn. Res.*, 2022, 2022.

[27] Liu, R., R. Yang, C. Jia, et al. Training socially aligned language models in simulated human society. *CoRR*, abs/2305.16960, 2023.

[28] Sumers, T. R., S. Yao, K. Narasimhan, et al. Cognitive architectures for language agents. *CoRR*, abs/2309.02427, 2023.

[29] Weng, L. Llm-powered autonomous agents. *lilianweng.github.io*, 2023.

[30] Bisk, Y., A. Holtzman, J. Thomason, et al. Experience grounds language. In B. Webber, T. Cohn, Y. He, Y. Liu, eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16-20, 2020*, pages 8718–8735. Association for Computational Linguistics, 2020.

[31] Bubeck, S., V. Chandrasekaran, R. Eldan, et al. Sparks of artificial general intelligence: Early experiments with GPT-4. *CoRR*, abs/2303.12712, 2023.

[32] Anscombe, G. E. M. *Intention*. Harvard University Press, 2000.

[33] Davidson, D. Actions, reasons, and causes. *The Journal of Philosophy*, 60(23):685–700, 1963.

[34] —. I. agency. In A. Marras, R. N. Brounagh, R. W. Binkley, eds., *Agent, Action, and Reason*, pages 1–37. University of Toronto Press, 1971.

[35] Dennett, D. C. Précis of the intentional stance. *Behavioral and brain sciences*, 11(3):495–505, 1988.

[36] Barandiaran, X. E., E. Di Paolo, M. Rohde. Defining agency: Individuality, normativity, asymmetry, and spatio-temporality in action. *Adaptive Behavior*, 17(5):367–386, 2009.

[37] McCarthy, J. *Ascribing mental qualities to machines*. Stanford University. Computer Science Department, 1979.

[38] Rosenschein, S. J., L. P. Kaelbling. The synthesis of digital machines with provable epistemic properties. In *Theoretical aspects of reasoning about knowledge*, pages 83–98. Elsevier, 1986.

[39] Radford, A., K. Narasimhan, T. Salimans, et al. Improving language understanding by generative pre-training. *OpenAI*, 2018.

[40] Radford, A., J. Wu, R. Child, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.

[41] Brown, T. B., B. Mann, N. Ryder, et al. Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin, eds., *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*. 2020.

The Rise andPotential of Large Language Model Based Agents: A Survey[20] Kaelbling, L. P., 等. 智能反应系统的架构. 关于行动与计划的推理, 第395–410页, 1987.

[21] Sutton, R. S., A. G. Barto. 强化学习：导论. 麻省理工学院出版社, 2018.[22] Park, J. S., J. C. O’ Brien, C. J. Cai, 等. 生成代理：人类行为的交互模拟. CoRR, abs/2304.03442, 2023.[23] Wang, Z., G. Zhang, K. Yang, 等. 交互式自然语言处理. CoRR, abs/2305.13246, 2023.[24] Ouyang, L., J. Wu, X. Jiang, 等. 通过人类反馈训练语言模型以遵循指令. 在NeurIPS会议. 2022.[25] OpenAI. GPT-4技术报告. CoRR, abs/2303.08774, 2023.[26] Wei, J., Y. Tay, R. Bommasani, 等. 大型语言模型的突现能力. 机器学习研究转报, 2022, 2022.[27] Liu, R., R. Yang, C. Jia, 等. 在模拟人类社会中训练社会对齐的语言模型. CoRR, abs/2305.16960, 2023.[28] Sumers, T. R., S. Yao, K. Narasimhan, 等. 语言代理的认知架构. CoRR, abs/2309.02427, 2023.[29] Weng, L. Llm驱动的自主代理. lilianweng.github.io, 2023.[30] Bisk, Y., A. Holtzman, J. Thomason, 等. 经验赋予语言意义. 收录于B. Webber, T. Cohn, Y. He, Y. Liu编辑. 2020年自然语言处理实证方法会议论文集, EMNLP 2020, 线上, 2020年11月16-20日, 第8718–8735页. 计算语言学协会, 2020.[31] Bubeck, S., V. Chandrasekaran, R. Eldan, 等. 人工通用智能的火花：GPT-4的早期实验. CoRR, abs/2303.12712, 2023.[32] Anscombe, G. E. M. 意图. 哈佛大学出版社, 2000.[33] Davidson, D. 行动、理由与原因. 《哲学杂志》, 60(23):685–700, 1963.[34] —. I. 代理 . 收录于 A. Marras, R. N. Brounagh, R. W. Binkley编辑. 代理、行动与理由, 第1–37页. 多伦多大学出版社, 1971.[35] Dennett, D. C. 有意图立场的简要说明. 行为与脑科学, 11(3):495–505, 1988.

[36] Barandiaran, X. E., E. Di Paolo, M. Rohde. 定义代理：行动中的个体性、规范性、不对称性与时空性. 适应行为, 17(5):367–386, 2009.[37] McCarthy, J. 将心理特质归于机器. 斯坦福大学计算机系, 1979.[38] Rosenschein, S. J., L. P. Kaelbling. 具有可证明认知属性的数字机器的合成. 收录于关于知识推理的理论方面, 第83–98页. 爱思唯尔, 1986.

[39] Radford, A., K. Narasimhan, T. Salimans, 等. 通过生成式预训练提升语言理解能力. OpenAI, 2018.[40] Radford, A., J. Wu, R. Child, 等. 语言模型是无监督多任务学习者. OpenAI博客, 1(8):9, 2019.[41] Brown, T. B., B. Mann, N. Ryder, 等. 语言模型是少样本学习者. 收录于H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin编辑. 神经信息处理系统进展 33：神经信息处理系统年度会议 2020, NeurIPS 2020, 2020 年12月6-12日, 虚拟会议. 2020.

[42] Lin, C., A. Jaech, X. Li, et al. Limitations of autoregressive models and their alternatives. In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou, eds., *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2021, Online, June 6-11, 2021*, pages 5147–5173. Association for Computational Linguistics, 2021.

[43] Tomasello, M. *Constructing a language: A usage-based theory of language acquisition*. Harvard university press, 2005.

[44] Bloom, P. *How children learn the meanings of words*. MIT press, 2002.

[45] Zwaan, R. A., C. J. Madden. Embodied sentence comprehension. *Grounding cognition: The role of perception and action in memory, language, and thinking*, 22, 2005.

[46] Andreas, J. Language models as agent models. In Y. Goldberg, Z. Kozareva, Y. Zhang, eds., *Findings of the Association for Computational Linguistics: EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, pages 5769–5779. Association for Computational Linguistics, 2022.

[47] Wong, L., G. Grand, A. K. Lew, et al. From word models to world models: Translating from natural language to the probabilistic language of thought. *CoRR*, abs/2306.12672, 2023.

[48] Radford, A., R. Józefowicz, I. Sutskever. Learning to generate reviews and discovering sentiment. *CoRR*, abs/1704.01444, 2017.

[49] Li, B. Z., M. I. Nye, J. Andreas. Implicit representations of meaning in neural language models. In C. Zong, F. Xia, W. Li, R. Navigli, eds., *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, ACL/IJCNLP 2021, (Volume 1: Long Papers), Virtual Event, August 1-6, 2021*, pages 1813–1827. Association for Computational Linguistics, 2021.

[50] Mukhopadhyay, U., L. M. Stephens, M. N. Huhns, et al. An intelligent system for document retrieval in distributed office environments. *J. Am. Soc. Inf. Sci.*, 37(3):123–135, 1986.

[51] Maes, P. Situated agents can have goals. *Robotics Auton. Syst.*, 6(1-2):49–70, 1990.

[52] Nilsson, N. J. Toward agent programs with circuit semantics. Tech. rep., 1992.

[53] Müller, J. P., M. Pischel. Modelling interacting agents in dynamic environments. In *Proceedings of the 11th European Conference on Artificial Intelligence*, pages 709–713. 1994.

[54] Brooks, R. A robust layered control system for a mobile robot. *IEEE journal on robotics and automation*, 2(1):14–23, 1986.

[55] Brooks, R. A. Intelligence without reason. In *The artificial life route to artificial intelligence*, pages 25–81. Routledge, 2018.

[56] Newell, A., H. A. Simon. Computer science as empirical inquiry: Symbols and search. *Commun. ACM*, 19(3):113–126, 1976.

[57] Ginsberg, M. L. *Essentials of Artificial Intelligence*. Morgan Kaufmann, 1993.

[58] Wilkins, D. E. *Practical planning - extending the classical AI planning paradigm*. Morgan Kaufmann series in representation and reasoning. Morgan Kaufmann, 1988.

[59] Shardlow, N. *Action and agency in cognitive science*. Ph.D. thesis, Master’s thesis, Department of Psychlogy, University of Manchester, Oxford ..., 1990.

[60] Sacerdoti, E. D. The nonlinear nature of plans. In *Advance Papers of the Fourth International Joint Conference on Artificial Intelligence, Tbilisi, Georgia, USSR, September 3-8, 1975*, pages 206–214. 1975.

[61] Russell, S. J., E. Wefald. *Do the right thing: studies in limited rationality*. MIT press, 1991.

Lin, C., A. Jaech, X. Li, 等. 自回归模型的局限性及其替代方案. 收录于 K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou 编. *2021年北美计算语言学协会人类语言技术会议论文集: NAACL-HLT 2021*, 线上, 6月6-11日, 第5147–5173页. 计算语言学协会, 2021年. Tomasello, M. 构建语言：一种基于用法的语言习得理论. 哈佛大学出版社, 2005年. Bloom, P. 儿童如何学习词义. 麻省理工学院出版社, 2002年. Zwaan, R. A., C. J. Madden. 具身句子理解. 认知基础：感知与行动在记忆、语言与思维中的作用, 第22卷, 2005年. Andreas, J. 语言模型作为代理模型. 收录于 Y. Goldberg, Z. Kozareva, Y. Zhang 编. 计算语言学协会会议成果: *EMNLP 2022*, 阿布扎比, 阿联酋, 12月7-11日, 第5769–5779页. 计算语言学协会, 2022年. Wong, L., G. Grand, A. K. Lew, 等. 从词模型到世界模型：从自然语言到思维的概率语言转换. *CoRR*, abs/2306.12672, 2023年. Radford, A., R. Józefowicz, I. Sutskever. 学习生成评论并发现情感. *CoRR*, abs/1704.01444, 2017年. Li, B. Z., M. I. Nye, J. Andreas. 神经语言模型中隐含意义表示. 收录于 C. Zong, F. Xia, W. Li, R. Navigli 编. 第59届计算语言学协会年会暨第11届自然语言处理国际联合会议论文集（第1卷：长论文），虚拟会议, 2021年8月1-6日, 第1813–1827页. 计算语言学协会, 2021年. Mukhopadhyay, U., L. M. Stephens, M. N. Huhns, 等. 分布式办公环境中文档检索的智能系统. 美国信息科学学会杂志, 37(3):123–135, 1986年. Maes, P. 具位代理可以设定目标. 机器人自主系统, 6(1-2):49–70, 1990年. Nilsson, N. J. 迈向具有电路语义的代理程序. 技术报告, 1992年. Müller, J. P., M. Pischel. 动态环境中交互代理的建模. 收录于 第11届欧洲人工智能会议论文集, 第709–713页, 1994年. Brooks, R. A. 为移动机器人设计的稳健分层控制系统. *IEEE机器人与自动化杂志*, 2(1):14–23, 1986年. Brooks, R. A. 无理性智能. 收录于 通向人工智能的人工生命路径, 第25–81页. Routledge, 2018年. Newell, A., H. A. Simon. 作为经验性探究的计算机科学：符号与搜索. *Communications of the ACM*, 19(3):113–126, 1976年. Ginsberg, M. L. 人工智能基础. 摩根考夫曼, 1993年. Wilkins, D. E. 实践规划——扩展经典AI规划范式. 摩根考夫曼表述与推理系列. 摩根考夫曼, 1988年. Shardlow, N. 认知科学中的行动与代理. 博士论文, 硕士论文, 曼彻斯特大学心理学系, 牛津, 1990年. Sacerdoti, E. D. 计划的非线性特性. 收录于 第四届国际人工智能联合会议预稿, 格鲁吉亚第比利斯, 苏联, 1975年9月3-8日, 第206–214页, 1975年. Russell, S. J., E. Wefald. 做正确的事：有限理性研究. 麻省理工学院出版社, 1991年。

- [62] Schoppers, M. Universal plans for reactive robots in unpredictable environments. In J. P. McDermott, ed., *Proceedings of the 10th International Joint Conference on Artificial Intelligence. Milan, Italy, August 23-28, 1987*, pages 1039–1046. Morgan Kaufmann, 1987.
- [63] Brooks, R. A. A robust layered control system for a mobile robot. *IEEE J. Robotics Autom.*, 2(1):14–23, 1986.
- [64] Minsky, M. Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1):8–30, 1961.
- [65] Isbell, C., C. R. Shelton, M. Kearns, et al. A social reinforcement learning agent. In *Proceedings of the fifth international conference on Autonomous agents*, pages 377–384. 2001.
- [66] Watkins, C. J. C. H. Learning from delayed rewards, 1989.
- [67] Rummery, G. A., M. Niranjan. *On-line Q-learning using connectionist systems*, vol. 37. University of Cambridge, Department of Engineering Cambridge, UK, 1994.
- [68] Tesauro, G., et al. Temporal difference learning and td-gammon. *Communications of the ACM*, 38(3):58–68, 1995.
- [69] Li, Y. Deep reinforcement learning: An overview. *arXiv preprint arXiv:1701.07274*, 2017.
- [70] Silver, D., A. Huang, C. J. Maddison, et al. Mastering the game of go with deep neural networks and tree search. *nature*, 529(7587):484–489, 2016.
- [71] Mnih, V., K. Kavukcuoglu, D. Silver, et al. Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.
- [72] Farebrother, J., M. C. Machado, M. Bowling. Generalization and regularization in DQN. *CoRR*, abs/1810.00123, 2018.
- [73] Zhang, C., O. Vinyals, R. Munos, et al. A study on overfitting in deep reinforcement learning. *CoRR*, abs/1804.06893, 2018.
- [74] Justesen, N., R. R. Torrado, P. Bontrager, et al. Illuminating generalization in deep reinforcement learning through procedural level generation. *arXiv preprint arXiv:1806.10729*, 2018.
- [75] Dulac-Arnold, G., N. Levine, D. J. Mankowitz, et al. Challenges of real-world reinforcement learning: definitions, benchmarks and analysis. *Mach. Learn.*, 110(9):2419–2468, 2021.
- [76] Ghosh, D., J. Rahme, A. Kumar, et al. Why generalization in RL is difficult: Epistemic pomdps and implicit partial observability. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan, eds., *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 25502–25515. 2021.
- [77] Brys, T., A. Harutyunyan, M. E. Taylor, et al. Policy transfer using reward shaping. In G. Weiss, P. Yolum, R. H. Bordini, E. Elkind, eds., *Proceedings of the 2015 International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2015, Istanbul, Turkey, May 4-8, 2015*, pages 181–188. ACM, 2015.
- [78] Parisotto, E., J. L. Ba, R. Salakhutdinov. Actor-mimic: Deep multitask and transfer reinforcement learning. *arXiv preprint arXiv:1511.06342*, 2015.
- [79] Zhu, Z., K. Lin, J. Zhou. Transfer learning in deep reinforcement learning: A survey. *CoRR*, abs/2009.07888, 2020.
- [80] Duan, Y., J. Schulman, X. Chen, et al. RL²: Fast reinforcement learning via slow reinforcement learning. *CoRR*, abs/1611.02779, 2016.
- [81] Finn, C., P. Abbeel, S. Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In D. Precup, Y. W. Teh, eds., *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, vol. 70 of *Proceedings of Machine Learning Research*, pages 1126–1135. PMLR, 2017.

- [62] Schoppers, M. 反应机器人在不可预测环境中的通用计划. 收录于 J. P. McDermott 编著, 《第十届国际人工智能联合会议论文集》, 米兰, 意大利, 1987年8月23-28日, 第1039-1046页. Morgan Kaufmann, 1987.
- [63] Brooks, R. A. 移动机器人稳健的分层控制系统. *IEEE机器人与自动化杂志*, 2(1):14–23, 1986.
- [64] Minsky, M. 迈向人工智能的步骤. *IRE 会议论文集*, 49(1):8–30, 1961.
- [65] Isbell, C., C. R. Shelton, M. Kearns 等. 社会强化学习代理. 收录于 第五届自主代理国际会议论文集, 第377–384页, 2001.
- [66] Watkins, C. J. C. H. 延迟奖励学习, 1989.
- [67] Rummery, G. A., M. Niranjan. 基于连接主义系统的在线 Q 学习, 第37卷. 剑桥大学工程系, 剑桥, 英国, 1994.
- [68] Tesauro, G., 等. 时序差分学习与td-gammon. *Communications of the ACM*, 38(3):58–68, 1995.
- [69] Li, Y. 深度强化学习: 概述. *arXiv预印本 arXiv:1701.07274*, 2017.
- [70] Silver, D., A. Huang, C. J. Maddison, 等. 利用深度神经网络与树搜索掌握围棋游戏. *nature*, 529(7587):484–489, 2016.
- [71] Mnih, V., K. Kavukcuoglu, D. Silver, 等. 使用深度强化学习玩Atari游戏. *arXiv预印本 arXiv:1312.5602*, 2013.
- [72] Farebrother, J., M. C. Machado, M. Bowling. DQN中的泛化与正则化. *CoRR*, abs/1810.00123, 2018.
- [73] Zhang, C., O. Vinyals, R. Munos, 等. 深度强化学习中的过拟合研究. *CoRR*, abs/1804.06893, 2018.
- [74] Justesen, N., R. R. Torrado, P. Bontrager, 等. 通过程序化关卡生成揭示深度强化学习中的泛化能力. *arXiv预印本 arXiv:1806.10729*, 2018.
- [75] Dulac-Arnold, G., N. Levine, D. J. Mankowitz, 等. 现实世界强化学习的挑战: 定义、基准与分析. *Machine Learning*, 110(9):2419–2468, 2021.
- [76] Ghosh, D., J. Rahme, A. Kumar, 等. 为什么强化学习中的泛化困难: 认知性POMDPs与隐式部分可观测性. 收录于 M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan 编著, 《神经信息处理系统年会2021》(NeurIPS 2021), 2021年12月6-14日, 线上会议, 第25502–25515页.
- [77] Brys, T., A. Harutyunyan, M. E. Taylor, 等. 利用奖励塑造进行策略迁移. 收录于 G. Weiss, P. Yolum, R. H. Bordini, E. Elkind 编著, 《2015年自主代理与多智能体系统国际会议论文集》(AAMAS 2015), 土耳其伊斯坦布尔, 2015年5月4-8日, 第181–188页. ACM, 2015.
- [78] Parisotto, E., J. L. Ba, R. Salakhutdinov. Actor-mimic: 深度多任务与迁移强化学习. *arXiv预印本 arXiv:1511.06342*, 2015.
- [79] Zhu, Z., K. Lin, J. Zhou. 深度强化学习中的迁移学习: 综述. *CoRR*, abs/2009.07888, 2020.
- [80] Duan, Y., J. Schulman, X. Chen, 等. RL²: 通过缓慢强化学习实现快速强化学习. *CoRR*, abs/1611.02779, 2016.
- [81] Finn, C., P. Abbeel, S. Levine. 面向深度网络快速适应的模型无关元学习. 收录于 D. Precup, Y. W. Teh 编著, 《第34届国际机器学习会议论文集》(ICML 2017), 悉尼, 新南威尔士, 澳大利亚, 2017年8月6-11日, 第70卷《机器学习研究会论文集》(Proceedings of Machine Learning Research), 第1126–1135页. PMLR, 2017.

- [82] Gupta, A., R. Mendonca, Y. Liu, et al. Meta-reinforcement learning of structured exploration strategies. In S. Bengio, H. M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, R. Garnett, eds., *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, pages 5307–5316. 2018.
- [83] Rakelly, K., A. Zhou, C. Finn, et al. Efficient off-policy meta-reinforcement learning via probabilistic context variables. In K. Chaudhuri, R. Salakhutdinov, eds., *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, vol. 97 of *Proceedings of Machine Learning Research*, pages 5331–5340. PMLR, 2019.
- [84] Fakoor, R., P. Chaudhari, S. Soatto, et al. Meta-q-learning. *arXiv preprint arXiv:1910.00125*, 2019.
- [85] Vanschoren, J. Meta-learning: A survey. *arXiv preprint arXiv:1810.03548*, 2018.
- [86] Taylor, M. E., P. Stone. Transfer learning for reinforcement learning domains: A survey. *J. Mach. Learn. Res.*, 10:1633–1685, 2009.
- [87] Tirinzoni, A., A. Sessa, M. Pirotta, et al. Importance weighted transfer of samples in reinforcement learning. In J. G. Dy, A. Krause, eds., *Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholmsmässan, Stockholm, Sweden, July 10-15, 2018*, vol. 80 of *Proceedings of Machine Learning Research*, pages 4943–4952. PMLR, 2018.
- [88] Beck, J., R. Vuorio, E. Z. Liu, et al. A survey of meta-reinforcement learning. *CoRR*, abs/2301.08028, 2023.
- [89] Wang, L., C. Ma, X. Feng, et al. A survey on large language model based autonomous agents. *CoRR*, abs/2308.11432, 2023.
- [90] Nakano, R., J. Hilton, S. Balaji, et al. Webgpt: Browser-assisted question-answering with human feedback. *CoRR*, abs/2112.09332, 2021.
- [91] Yao, S., J. Zhao, D. Yu, et al. React: Synergizing reasoning and acting in language models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- [92] Schick, T., J. Dwivedi-Yu, R. Dessì, et al. Toolformer: Language models can teach themselves to use tools. *CoRR*, abs/2302.04761, 2023.
- [93] Lu, P., B. Peng, H. Cheng, et al. Chameleon: Plug-and-play compositional reasoning with large language models. *CoRR*, abs/2304.09842, 2023.
- [94] Qin, Y., S. Hu, Y. Lin, et al. Tool learning with foundation models. *CoRR*, abs/2304.08354, 2023.
- [95] Wei, J., X. Wang, D. Schuurmans, et al. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*. 2022.
- [96] Kojima, T., S. S. Gu, M. Reid, et al. Large language models are zero-shot reasoners. In *NeurIPS*. 2022.
- [97] Wang, X., J. Wei, D. Schuurmans, et al. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- [98] Zhou, D., N. Schärli, L. Hou, et al. Least-to-most prompting enables complex reasoning in large language models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- [99] Xi, Z., S. Jin, Y. Zhou, et al. Self-polish: Enhance reasoning in large language models via problem refinement. *CoRR*, abs/2305.14497, 2023.

Gupta, A., R. Mendonca, Y. Liu 等. 结构化探索策略的元强化学习. 收录于 S. Bengio, H. M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, R. Garnett 编., 神经信息处理系统进展31: 神经信息处理系统年度会议2018, *NeurIPS 2018*, 2018年12月3-8日, 加拿大蒙特利尔, 第5307–5316页, 2018年. Rakelly, K., A. Zhou, C. Finn 等. 通过概率上下文变量实现高效的离策略元强化学习. 收录于 K. Chaudhuri, R. Salakhutdinov 编., 第36届国际机器学习会议论文集, *ICML 2019*, 2019年6月9-15日, 加利福尼亚州长滩, 第97卷 机器学习研究论文集, 第5331–5340页. PMLR, 2019年. Fakoor, R., P. Chaudhari, S. Soatto 等. 元Q学习. *arXiv*预印本 *arXiv:1910.00125*, 2019年. Vanschoren, J. 元学习: 一项综述. *arXiv*预印本 *arXiv:1810.03548*, 2018年. Taylor, M. E., P. Stone. 强化学习领域的迁移学习: 一项综述. *J. Mach. Learn. Res.*, 第10卷, 1633–1685页, 2009年. Tirinzoni, A., A. Sessa, M. Pirotta 等. 强化学习中样本的重要性加权迁移. 收录于 J. G. Dy, A. Krause 编., 第35届国际机器学习会议论文集, *ICML 2018*, 瑞典斯德哥尔摩*Stockholmsmässan*, 2018年7月10-15日, 第80卷 机器学习研究论文集, 第4943–4952页. PMLR, 2018年. Beck, J., R. Vuorio, E. Z. Liu 等. 元强化学习综述. *CoRR*, abs/2301.08028, 2023年. Wang, L., C. Ma, X. Feng 等. 基于大规模语言模型的自主智能体综述. *CoRR*, abs/2308.11432, 2023年. Nakano, R., J. Hilton, S. Balaji 等. WebGPT: 结合人类反馈的浏览器辅助问答系统. *CoRR*, abs/2112.09332, 2021年. Yao, S., J. Zhao, D. Yu 等. React: 在语言模型中协同推理与行动. 收录于 第十一届学习表示国际会议, *ICLR 2023*, 卢旺达基加利, 2023年5月1-5日. OpenReview.net, 2023年. Schick, T., J. Dwivedi-Yu, R. Dessì 等. Toolformer: 语言模型可以自我学习使用工具. *CoRR*, abs/2302.04761, 2023年. Lu, P., B. Peng, H. Cheng 等. Chameleon: 与大规模语言模型的即插即用组合推理. *CoRR*, abs/2304.09842, 2023年. Qin, Y., S. Hu, Y. Lin 等. 基础模型的工具学习. *CoRR*, abs/2304.08354, 2023 年. Wei, J., X. Wang, D. Schuurmans 等. 链式思维提示激发大规模语言模型的推理能力. 收录于 *NeurIPS*, 2022 年. Kojima, T., S. S. Gu, M. Reid 等. 大规模语言模型是零-shot推理器. 收录于 *NeurIPS*, 2022年. Wang, X., J. Wei, D. Schuurmans 等. 自洽性提升语言模型中的链式推理能力. 收录于 第十一届学习表示国际会议, *ICLR 2023*, 卢旺达基加利, 2023年5月1-5日. OpenReview.net, 2023年. Zhou, D., N. Schärli, L. Hou 等. 最少到最多提示法实现大规模语言模型的复杂推理. 收录于 第十一届学习表示国际会议, *ICLR 2023*, 卢旺达基加利, 2023年5月1-5日. OpenReview.net, 2023年. Xi, Z., S. Jin, Y. Zhou 等. 自我润色: 通过问题细化提升大规模语言模型的推理能力. *CoRR*, abs/2305.14497, 2023年.

[100] Shinn, N., F. Cassano, B. Labash, et al. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*, 2023.

[101] Song, C. H., J. Wu, C. Washington, et al. Llm-planner: Few-shot grounded planning for embodied agents with large language models. *CoRR*, abs/2212.04088, 2022.

[102] Akyürek, A. F., E. Akyürek, A. Kalyan, et al. RL4F: generating natural language feedback with reinforcement learning for repairing model outputs. In A. Rogers, J. L. Boyd-Graber, N. Okazaki, eds., *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 7716–7733. Association for Computational Linguistics, 2023.

[103] Peng, B., M. Galley, P. He, et al. Check your facts and try again: Improving large language models with external knowledge and automated feedback. *CoRR*, abs/2302.12813, 2023.

[104] Liu, H., C. Sferrazza, P. Abbeel. Languages are rewards: Hindsight finetuning using human feedback. *arXiv preprint arXiv:2302.02676*, 2023.

[105] Wei, J., M. Bosma, V. Y. Zhao, et al. Finetuned language models are zero-shot learners. In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022.

[106] Sanh, V., A. Webson, C. Raffel, et al. Multitask prompted training enables zero-shot task generalization. In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022.

[107] Chung, H. W., L. Hou, S. Longpre, et al. Scaling instruction-finetuned language models. *CoRR*, abs/2210.11416, 2022.

[108] Li, G., H. A. A. K. Hammoud, H. Itani, et al. CAMEL: communicative agents for "mind" exploration of large scale language model society. *CoRR*, abs/2303.17760, 2023.

[109] Qian, C., X. Cong, C. Yang, et al. Communicative agents for software development. *CoRR*, abs/2307.07924, 2023.

[110] Boiko, D. A., R. MacKnight, G. Gomes. Emergent autonomous scientific research capabilities of large language models. *CoRR*, abs/2304.05332, 2023.

[111] Du, Y., S. Li, A. Torralba, et al. Improving factuality and reasoning in language models through multiagent debate. *CoRR*, abs/2305.14325, 2023.

[112] Liang, T., Z. He, W. Jiao, et al. Encouraging divergent thinking in large language models through multi-agent debate. *CoRR*, abs/2305.19118, 2023.

[113] Castelfranchi, C. Guarantees for autonomy in cognitive agent architecture. In M. J. Wooldridge, N. R. Jennings, eds., *Intelligent Agents, ECAI-94 Workshop on Agent Theories, Architectures, and Languages, Amsterdam, The Netherlands, August 8-9, 1994, Proceedings*, vol. 890 of *Lecture Notes in Computer Science*, pages 56–70. Springer, 1994.

[114] Gravitass, S. Auto-GPT: An Autonomous GPT-4 experiment, 2023. URL <https://github.com/Significant-Gravitas/Auto-GPT>, 2023.

[115] Nakajima, Y. BabyAGI. *Python. https://github.com/yoheinakajima/babyagi*, 2023.

[116] Yuan, A., A. Coenen, E. Reif, et al. Wordcraft: Story writing with large language models. In G. Jacucci, S. Kaski, C. Conati, S. Stumpf, T. Ruotsalo, K. Gajos, eds., *IUI 2022: 27th International Conference on Intelligent User Interfaces, Helsinki, Finland, March 22 - 25, 2022*, pages 841–852. ACM, 2022.

[117] Franceschelli, G., M. Musolesi. On the creativity of large language models. *CoRR*, abs/2304.00008, 2023.

[118] Zhu, D., J. Chen, X. Shen, et al. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592*, 2023.

[100] Shinn, N., F. Cassano, B. Labash 等. Reflexion：具有语言强化学习的语言代理。*arXiv* 预印本 *arXiv:2303.11366*，2023年。[101] Song, C. H., J. Wu, C. Washington 等. Llm-planner：基于大规模语言模型的少样本基础规划，用于具身代理。*CoRR*, abs/2212.04088，2022年。[102] Akyürek, A. F., E. Akyürek, A. Kalyan 等. RL4F：利用强化学习生成自然语言反馈以修正模型输出。在A. Rogers, J. L. Boyd-Graber, N. Okazaki 编著，第61届计算语言学协会年会论文集（第一卷：长论文），*ACL 2023*，多伦多，加拿大，2023年7月9-14日，第7716-7733页。计算语言学协会，2023年。[103] Peng, B., M. Galley, P. He 等. 核查事实并重试：利用外部知识和自动反馈提升大规模语言模型。*CoRR*, abs/2302.12813，2023年。[104] Liu, H., C. Sferrazza, P. Abbeel. 语言即奖励：利用人类反馈进行事后微调。*arXiv* 预印本 *arXiv:2302.02676*，2023年。[105] Wei, J., M. Bosma, V. Y. Zhao 等. 微调的语言模型是零样本学习者。在第十届国际表示学习会议，*ICLR 2022*，虚拟会议，2022年4月25-29日。OpenReview.net，2022年。[106] Sanh, V., A. Webson, C. Raffel 等. 多任务提示训练实现零样本任务泛化。在第十届国际表示学习会议，*ICLR 2022*，虚拟会议，2022年4月25-29日。OpenReview.net，2022年。[107] Chung, H. W., L. Hou, S. Longpre 等. 指令微调语言模型的扩展规模。*CoRR*, abs/2210.11416，2022年。[108] Li, G., H. A. A. K. Hammoud, H. Itani 等. CAMEL：面向“大规模语言模型社会”“心智”探索的交互式代理。*CoRR*, abs/2303.17760，2023年。[109] Qian, C., X. Cong, C. Yang 等. 软件开发的交互式代理。*CoRR*, abs/2307.07924，2023年。[110] Boiko, D. A., R. MacKnight, G. Gomes. 大规模语言模型的自主科学研究能力的涌现。*CoRR*, abs/2304.05332，2023年。[111] Du, Y., S. Li, A. Torralba 等. 通过多智能体辩论提升语言模型的事实性与推理能力。*CoRR*, abs/2305.14325，2023年。[112] Liang, T., Z. He, W. Jiao 等. 通过多智能体辩论鼓励大规模语言模型的发散性思维。*CoRR*, abs/2305.19118，2023年。[113] Castelfranchi, C. 认知代理架构中的自主性保障。在M. J. Wooldridge, N. R. Jennings 编著，智能代理，*ECAI-94代理理论、架构与语言研讨会*，荷兰阿姆斯特丹，1994年8月8-9日，论文集，第890卷，计算机科学讲义，第56-70页。Springer，1994年。[114] Gravitass, S. Auto-GPT：一项自主的GPT-4实验，2023年。网址 <https://github.com/Significant-Gravitas/Auto-GPT>，2023年。[115] Nakajima, Y. BabyAGI. *Python.https://github.com/yoheinakajima/babyagi*，2023年。[116] Yuan, A., A. Coenen, E. Reif 等. Wordcraft：利用大规模语言模型进行故事创作。在G. Jacucci, S. Kaski, C. Conati, S. Stumpf, T. Ruotsalo, K. Gajos 编著，第27届国际智能用户界面会议（*IUI 2022*），芬兰赫尔辛基，2022年3月22-25日，第841-852页。ACM，2022年。[117] Franceschelli, G., M. Musolesi. 关于大规模语言模型的创造力。*CoRR*, abs/2304.00008，2023年。[118] Zhu, D., J. Chen, X. Shen 等. Minigpt-4：利用先进的大规模语言模型增强视觉-语言理解。*arXiv* 预印本 *arXiv:2304.10592*，2023年。

[119] Yin, S., C. Fu, S. Zhao, et al. A survey on multimodal large language models. *CoRR*, abs/2306.13549, 2023.

[120] Driess, D., F. Xia, M. S. M. Sajjadi, et al. Palm-e: An embodied multimodal language model. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 8469–8488. PMLR, 2023.

[121] Mu, Y., Q. Zhang, M. Hu, et al. Embodiedgpt: Vision-language pre-training via embodied chain of thought. *CoRR*, abs/2305.15021, 2023.

[122] Brown, J. W. Beyond conflict monitoring: Cognitive control and the neural basis of thinking before you act. *Current Directions in Psychological Science*, 22(3):179–185, 2013.

[123] Kang, J., R. Laroche, X. Yuan, et al. Think before you act: Decision transformers with internal working memory. *CoRR*, abs/2305.16338, 2023.

[124] Valmeekam, K., S. Sreedharan, M. Marquez, et al. On the planning abilities of large language models (A critical investigation with a proposed benchmark). *CoRR*, abs/2302.06706, 2023.

[125] Liu, B., Y. Jiang, X. Zhang, et al. LLM+P: empowering large language models with optimal planning proficiency. *CoRR*, abs/2304.11477, 2023.

[126] Liu, H., C. Sferrazza, P. Abbeel. Chain of hindsight aligns language models with feedback. *CoRR*, abs/2302.02676, 2023.

[127] Lin, Y., Y. Chen. Llm-eval: Unified multi-dimensional automatic evaluation for open-domain conversations with large language models. *CoRR*, abs/2305.13711, 2023.

[128] Lin, J., D. Fried, D. Klein, et al. Inferring rewards from language in context. In S. Muresan, P. Nakov, A. Villavicencio, eds., *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2022, Dublin, Ireland, May 22-27, 2022*, pages 8546–8560. Association for Computational Linguistics, 2022.

[129] Fu, Y., H. Peng, T. Khot, et al. Improving language model negotiation with self-play and in-context learning from AI feedback. *CoRR*, abs/2305.10142, 2023.

[130] Zhang, H., W. Du, J. Shan, et al. Building cooperative embodied agents modularly with large language models. *CoRR*, abs/2307.02485, 2023.

[131] Darwin’s, C. On the origin of species. *published on*, 24:1, 1859.

[132] Bang, Y., S. Cahyawijaya, N. Lee, et al. A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity. *CoRR*, abs/2302.04023, 2023.

[133] Fang, T., S. Yang, K. Lan, et al. Is chatgpt a highly fluent grammatical error correction system? A comprehensive evaluation. *CoRR*, abs/2304.01746, 2023.

[134] Lu, A., H. Zhang, Y. Zhang, et al. Bounding the capabilities of large language models in open text generation with prompt constraints. In A. Vlachos, I. Augenstein, eds., *Findings of the Association for Computational Linguistics: EACL 2023, Dubrovnik, Croatia, May 2-6, 2023*, pages 1937–1963. Association for Computational Linguistics, 2023.

[135] Buehler, M. C., J. Adamy, T. H. Weisswange. Theory of mind based assistive communication in complex human robot cooperation. *CoRR*, abs/2109.01355, 2021.

[136] Shapira, N., M. Levy, S. H. Alavi, et al. Clever hans or neural theory of mind? stress testing social reasoning in large language models. *CoRR*, abs/2305.14763, 2023.

[137] Hill, F., K. Cho, A. Korhonen. Learning distributed representations of sentences from unlabelled data. In K. Knight, A. Nenkova, O. Rambow, eds., *NAACL HLT 2016, The 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, San Diego California, USA, June 12-17, 2016*, pages 1367–1377. The Association for Computational Linguistics, 2016.

Yin, S., C. Fu, S. Zhao 等. 多模态大规模语言模型综述. *CoRR*, abs/2306.13549, 2023年。

Driess, D., F. Xia, M. S. M. Sajjadi 等. Palm-e: 一种具象多模态语言模型。在 A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett 编著, 《国际机器学习会议, *ICML 2023*, 2023年7月23-29日, 夏威夷檀香山, 美国》, 第 202 卷, 《机器学习研究会议录》, 第 8469–8488 页。PMLR, 2023年。

Mu, Y., Q. Zhang, M. Hu 等. Embodiedgpt: 通过具象思维链进行视觉-语言预训练. *CoRR*, abs/2305.15021, 2023年。

Brown, J. W. 超越冲突监控: 认知控制与行动前思考的神经基础。心理科学当前方向, 22(3):179–185, 2013年。

Kang, J., R. Laroche, X. Yuan 等. 行动前思考: 具有内部工作记忆的决策变换器. *CoRR*, abs/2305.16338, 2023 年。

Valmeekam, K., S. Sreedharan, M. Marquez 等. 关于大规模语言模型的规划能力（一项关键性研究及提出的基准测试）。*CoRR*, abs/2302.06706, 2023年。

Liu, B., Y. Jiang, X. Zhang 等. LLM+P: 赋能大规模语言模型以实现最优规划能力. *CoRR*, abs/2304.11477, 2023年。

Liu, H., C. Sferrazza, P. Abbeel. 反思链: 通过反馈使语言模型对齐. *CoRR*, abs/2302.02676, 2023年。

Lin, Y., Y. Chen. Llm-eval: 用于开放域对话的多维度自动统一评估. *CoRR*, abs/2305.13711, 2023年。

Lin, J., D. Fried, D. Klein 等. 从上下文中的语言推断奖励。In S. Muresan, P. Nakov, A. Villavicencio 编著, 《第60届计算语言学协会年会论文集（第1卷: 长论文）, ACL 2022, 爱尔兰都柏林, 2022年5月22-27日》, 第 8546–8560 页。计算语言学协会, 2022年。

Fu, Y., H. Peng, T. Khot 等. 通过自我对弈和上下文学习提升语言模型的协商能力. *CoRR*, abs/2305.10142, 2023年。

Zhang, H., W. Du, J. Shan 等. 利用大规模语言模型模块化构建合作具象代理. *CoRR*, abs/2307.02485, 2023年。

达尔文, 《物种起源》。发表于, 24:1, 1859年。

Bang, Y., S. Cahyawijaya, N. Lee 等. 多任务、多语种、多模态评估 ChatGPT在推理、幻觉和交互性方面的表现. *CoRR*, abs/2302.04023, 2023年。

Fang, T., S. Yang, K. Lan 等. ChatGPT是高度流畅的语法错误校正系统吗? 一项全面评估. *CoRR*, abs/2304.01746, 2023年。

Lu, A., H. Zhang, Y. Zhang 等. 在开放文本生成中通过提示约束界定大规模语言模型的能力。In A. Vlachos, I. Augenstein 编著, 《计算语言学协会会议论文集: EACL 2023, 克罗地亚杜布罗夫尼克, 2023年5月2-6日》, 第 1937–1963 页。计算语言学协会, 2023年。

Buehler, M. C., J. Adamy, T. H. Weisswange. 基于心智理论的复杂人机合作中的辅助通信. *CoRR*, abs/2109.01355, 2021年。

Shapira, N., M. Levy, S. H. Alavi 等. 聪明的汉斯还是神经心智理论? 对大规模语言模型社会推理的压力测试. *CoRR*, abs/2305.14763, 2023年。

Hill, F., K. Cho, A. Korhonen. 从未标注数据中学习句子分布式表示。在 K. Knight, A. Nenkova, O. Rambow 编著, 《NAACL HLT 2016: 2016年北美计算语言学协会会议: 人类语言技术》, 美国加利福尼亚州圣地亚哥, 2016年6月12-17日, 第 1367–1377 页。计算语言学协会, 2016年。

- [138] Collobert, R., J. Weston, L. Bottou, et al. Natural language processing (almost) from scratch. *J. Mach. Learn. Res.*, 12:2493–2537, 2011.
- [139] Kaplan, J., S. McCandlish, T. Henighan, et al. Scaling laws for neural language models. *CoRR*, abs/2001.08361, 2020.
- [140] Roberts, A., C. Raffel, N. Shazeer. How much knowledge can you pack into the parameters of a language model? In B. Webber, T. Cohn, Y. He, Y. Liu, eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16-20, 2020*, pages 5418–5426. Association for Computational Linguistics, 2020.
- [141] Tandon, N., A. S. Varde, G. de Melo. Commonsense knowledge in machine intelligence. *SIGMOD Rec.*, 46(4):49–52, 2017.
- [142] Vulic, I., E. M. Ponti, R. Litschko, et al. Probing pretrained language models for lexical semantics. In B. Webber, T. Cohn, Y. He, Y. Liu, eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16-20, 2020*, pages 7222–7240. Association for Computational Linguistics, 2020.
- [143] Hewitt, J., C. D. Manning. A structural probe for finding syntax in word representations. In J. Burstein, C. Doran, T. Solorio, eds., *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2019, Minneapolis, MN, USA, June 2-7, 2019, Volume 1 (Long and Short Papers)*, pages 4129–4138. Association for Computational Linguistics, 2019.
- [144] Rau, L. F., P. S. Jacobs, U. Zernik. Information extraction and text summarization using linguistic knowledge acquisition. *Inf. Process. Manag.*, 25(4):419–428, 1989.
- [145] Yang, K., Z. Chen, Y. Cai, et al. Improved automatic keyword extraction given more semantic knowledge. In H. Gao, J. Kim, Y. Sakurai, eds., *Database Systems for Advanced Applications - DASFAA 2016 International Workshops: BDMS, BDQM, MoI, and SeCoP, Dallas, TX, USA, April 16-19, 2016, Proceedings*, vol. 9645 of *Lecture Notes in Computer Science*, pages 112–125. Springer, 2016.
- [146] Beloucif, M., C. Biemann. Probing pre-trained language models for semantic attributes and their values. In M. Moens, X. Huang, L. Specia, S. W. Yih, eds., *Findings of the Association for Computational Linguistics: EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 16-20 November, 2021*, pages 2554–2559. Association for Computational Linguistics, 2021.
- [147] Zhang, Z., H. Zhao. Advances in multi-turn dialogue comprehension: A survey. *CoRR*, abs/2103.03125, 2021.
- [148] Safavi, T., D. Koutra. Relational world knowledge representation in contextual language models: A review. In M. Moens, X. Huang, L. Specia, S. W. Yih, eds., *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7-11 November, 2021*, pages 1053–1067. Association for Computational Linguistics, 2021.
- [149] Jiang, Z., F. F. Xu, J. Araki, et al. How can we know what language models know. *Trans. Assoc. Comput. Linguistics*, 8:423–438, 2020.
- [150] Madaan, A., S. Zhou, U. Alon, et al. Language models of code are few-shot commonsense learners. In Y. Goldberg, Z. Kozareva, Y. Zhang, eds., *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, pages 1384–1403. Association for Computational Linguistics, 2022.
- [151] Xu, F. F., U. Alon, G. Neubig, et al. A systematic evaluation of large language models of code. In S. Chaudhuri, C. Sutton, eds., *MAPS@PLDI 2022: 6th ACM SIGPLAN International Symposium on Machine Programming, San Diego, CA, USA, 13 June 2022*, pages 1–10. ACM, 2022.
- [152] Cobbe, K., V. Kosaraju, M. Bavarian, et al. Training verifiers to solve math word problems. *CoRR*, abs/2110.14168, 2021.

Collobert, R., J. Weston, L. Bottou 等. 自然语言处理（几乎）从零开始。 *J. Mach. Learn. Res.*, 2011, 12: 2493–2537. Kaplan, J., S. McCandlish, T. Henighan 等. 神经语言模型的规模定律。 *CoRR*, abs/2001.08361, 2020. Roberts, A., C. Raffel, N. Shazeer. 你能将多少知识装入语言模型的参数中？见 B. Webber, T. Cohn, Y. He, Y. Liu 编., 2020 年自然语言处理实证方法会议论文集, *EMNLP 2020*, 线上, 2020年11月16-20日, 第5418–5426页。计算语言学协会, 2020. Tandon, N., A. S. Varde, G. de Melo. 机器智能中的常识知识。 *SIGMOD Rec.*, 46(4):49–52, 2017. Vulic, I., E. M. Ponti, R. Litschko 等. 预训练语言模型的词汇语义探测。见 B. Webber, T. Cohn, Y. He, Y. Liu 编., 2020 年自然语言处理实证方法会议论文集, *EMNLP 2020*, 线上, 2020 年11月16-20日, 第7222–7240页。计算语言学协会, 2020. Hewitt, J., C. D. Manning. 用于在词表示中寻找句法结构的结构探测。见 J. Burstein, C. Doran, T. Solorio 编., 2019 年北美计算语言学协会人类语言技术会议论文集, *NAACL-HLT 2019*, 明尼阿波利斯, 明尼苏达州, 美国, 2019年6月2-7日, 第1卷（长篇与短篇论文）, 第4129–4138页。计算语言学协会, 2019. Rau, L. F., P. S. Jacobs, U. Zernik. 利用语言知识获取进行信息提取与文本摘要。 *Inf. Process. Manag.*, 25(4):419–428, 1989. Yang, K., Z. Chen, Y. Cai 等. 在更多语义知识的支持下改进自动关键词提取。见 H. Gao, J. Kim, Y. Sakurai 编., 面向高级应用的数据库系统 - *DASFAA 2016* 国际研讨会: *BDMS, BDQM, MoI* 和 *SeCoP*, 达拉斯, 德克萨斯州, 美国, 2016年4月16-19日, 论文集, 第9645卷计算机科学讲义, 第112–125页。Springer, 2016. Beloucif, M., C. Biemann. 探测预训练语言模型中的语义属性及其值。见 M. Moens, X. Huang, L. Specia, S. W. Yih 编., 计算语言学协会研究成果: *EMNLP 2021*, 虚拟会议/蓬塔卡纳, 多米尼加共和国, 2021年11月16-20日, 第2554–2559页。计算语言学协会, 2021. Zhang, Z., H. Zhao. 多轮对话理解的进展: 综述。 *CoRR*, abs/2103.03125, 2021. Safavi, T., D. Koutra. 关系世界知识在上下文语言模型中的表示: 综述。见 M. Moens, X. Huang, L. Specia, S. W. Yih 编., 2021 年自然语言处理实证方法会议论文集, *EMNLP 2021*, 虚拟会议/蓬塔卡纳, 多米尼加共和国, 2021年11月7-11日, 第1053–1067页。计算语言学协会, 2021. Jiang, Z., F. F. Xu, J. Araki 等. 我们如何知道语言模型知道了什么。 *Trans. Assoc. Comput. Linguistics*, 8: 423–438, 2020. Madaan, A., S. Zhou, U. Alon 等. 代码的语言模型是少-shot 常识学习者。见 Y. Goldberg, Z. Kozareva, Y. Zhang 编., 2022 年自然语言处理实证方法会议论文集, *EMNLP 2022*, 阿布扎比, 阿拉伯联合酋长国, 2022年12月7-11日, 第1384–1403页。计算语言学协会, 2022. Xu, F. F., U. Alon, G. Neubig 等. 大型代码语言模型的系统性评估。见 S. Chaudhuri, C. Sutton 编., *MAPS@PLDI 2022*: 第六届 *ACM SIGPLAN* 机器编程国际研讨会, 圣地亚哥, 加利福尼亚, 美国, 2022年6月13日, 第1–10页。ACM, 2022. Cobbe, K., V. Kosaraju, M. Bavarian 等. 训练验证器以解决数学文字题。 *CoRR*, abs/2110.14168, 2021.

- [153] Thirunavukarasu, A. J., D. S. J. Ting, K. Elangovan, et al. Large language models in medicine. *Nature medicine*, pages 1–11, 2023.
- [154] Lai, Y., C. Li, Y. Wang, et al. DS-1000: A natural and reliable benchmark for data science code generation. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 18319–18345. PMLR, 2023.
- [155] AlKhamissi, B., M. Li, A. Celikyilmaz, et al. A review on language models as knowledge bases. *CoRR*, abs/2204.06031, 2022.
- [156] Kemker, R., M. McClure, A. Abitino, et al. Measuring catastrophic forgetting in neural networks. In S. A. McIlraith, K. Q. Weinberger, eds., *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, (AAAI-18), the 30th innovative Applications of Artificial Intelligence (IAAI-18), and the 8th AAAI Symposium on Educational Advances in Artificial Intelligence (EAAI-18), New Orleans, Louisiana, USA, February 2-7, 2018*, pages 3390–3398. AAAI Press, 2018.
- [157] Cao, N. D., W. Aziz, I. Titov. Editing factual knowledge in language models. In M. Moens, X. Huang, L. Specia, S. W. Yih, eds., *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7-11 November, 2021*, pages 6491–6506. Association for Computational Linguistics, 2021.
- [158] Yao, Y., P. Wang, B. Tian, et al. Editing large language models: Problems, methods, and opportunities. *CoRR*, abs/2305.13172, 2023.
- [159] Mitchell, E., C. Lin, A. Bosselut, et al. Memory-based model editing at scale. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato, eds., *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, vol. 162 of *Proceedings of Machine Learning Research*, pages 15817–15831. PMLR, 2022.
- [160] Manakul, P., A. Liusie, M. J. F. Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. *CoRR*, abs/2303.08896, 2023.
- [161] Li, M., B. Peng, Z. Zhang. Self-checker: Plug-and-play modules for fact-checking with large language models. *CoRR*, abs/2305.14623, 2023.
- [162] Gou, Z., Z. Shao, Y. Gong, et al. CRITIC: large language models can self-correct with tool-interactive critiquing. *CoRR*, abs/2305.11738, 2023.
- [163] Lewis, M., Y. Liu, N. Goyal, et al. BART: denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault, eds., *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020*, pages 7871–7880. Association for Computational Linguistics, 2020.
- [164] Park, H. H., Y. Vyas, K. Shah. Efficient classification of long documents using transformers. In S. Muresan, P. Nakov, A. Villavicencio, eds., *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), ACL 2022, Dublin, Ireland, May 22-27, 2022*, pages 702–709. Association for Computational Linguistics, 2022.
- [165] Guo, M., J. Ainslie, D. C. Uthus, et al. Longt5: Efficient text-to-text transformer for long sequences. In M. Carpuat, M. de Marneffe, I. V. M. Ruíz, eds., *Findings of the Association for Computational Linguistics: NAACL 2022, Seattle, WA, United States, July 10-15, 2022*, pages 724–736. Association for Computational Linguistics, 2022.
- [166] Ainslie, J., T. Lei, M. de Jong, et al. Colt5: Faster long-range transformers with conditional computation. *CoRR*, abs/2303.09752, 2023.

[153] Thirunavukarasu, A. J., D. S. J. Ting, K. Elangovan 等。医学中的大型语言模型。*Naturemedicine*, 第1-11页, 2023年。[154] Lai, Y., C. Li, Y. Wang 等。DS-1000: 一个自然且可靠的数据科学代码生成基准。在A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett 编著, 国际机器学习会议, *ICML 2023*, 2023年7月23-29日, 夏威夷檀香山, 美国, 第202卷机器学习研究会议录, 第18319-18345页。PMLR, 2023年。[155] AlKhamissi, B., M. Li, A. Celikyilmaz 等。关于将语言模型作为知识库的综述。*CoRR*, abs/2204.06031, 2022年。[156] Kemker, R., M. McClure, A. Abitino 等。神经网络灾难性遗忘的测量。S. A. McIlraith, K. Q. Weinberger 编著, 第32届AAAI人工智能会议论文集 (*AAAI-18*)、第30届人工智能创新应用会议 (*IAAI-18*) 以及第8届AAAI教育人工智能研讨会 (*EAAI-18*), 美国路易斯安那州新奥尔良, 2018年2月2-7日, 第3390-3398页。AAAI出版社, 2018年。[157] Cao, N. D., W. Aziz, I. Titov。在语言模型中编辑事实知识。在M. Moens, X. Huang, L. Specia, S. W. Yih 编著, 2021年自然语言处理实证方法会议论文集, *EMNLP 2021*, 虚拟会议/多米尼加共和国蓬塔卡纳, 2021年11月7-11日, 第6491-6506页。计算语言学协会, 2021年。[158] Yao, Y., P. Wang, B. Tian 等。编辑大型语言模型: 问题、方法与机遇。*CoRR*, abs/2305.13172, 2023年。[159] Mitchell, E., C. Lin, A. Bosselut 等。大规模记忆模型编辑。在K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato 编著, 国际机器学习会议, *ICML 2022*, 2022年7月17-23日, 马里兰州巴尔的摩, 第162卷机器学习研究会议录, 第15817-15831页。PMLR, 2022年。[160] Manakul, P., A. Liusie, M. J. F. Gales。Selfcheckgpt: 面向生成型大型语言模型的零资源黑箱幻觉检测。*CoRR*, abs/2303.08896, 2023年。[161] Li, M., B. Peng, Z. Zhang。Self-checker: 用于事实核查的即插即用模块。*CoRR*, abs/2305.14623, 2023年。[162] Gou, Z., Z. Shao, Y. Gong 等。CRITIC: 大型语言模型可以通过工具交互式批评实现自我校正。*CoRR*, abs/2305.11738, 2023年。[163] Lewis, M., Y. Liu, N. Goyal 等。BART: 用于自然语言生成、翻译和理解的去噪序列到序列预训练。在D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault 编著, 第58届计算语言学协会年会论文集, *ACL 2020*, 线上会议, 2020年7月5-10日, 第7871-7880页。计算语言学协会, 2020年。[164] Park, H. H., Y. Vyas, K. Shah。使用变换器高效分类长文档。在S. Muresan, P. Nakov, A. Villavicencio 编著, 第60届计算语言学协会年会论文集 (第二卷: 短论文), *ACL 2022*, 爱尔兰都柏林, 2022年5月22-27日, 第702-709页。计算语言学协会, 2022年。[165] Guo, M., J. Ainslie, D. C. Uthus 等。Longt5: 用于长序列的高效文本到文本变换器。在M. Carpuat, M. de Marneffe, I. V. M. Ruíz 编著, 计算语言学协会会议论文集: *NAACL 2022*, 西雅图, WA, 美国, 2022年7月10-15日, 第724-736页。计算语言学协会, 2022年。[166] Ainslie, J., T. Lei, M. de Jong 等。Colt5: 具有条件计算的更快长距离变换器。*CoRR*, abs/2303.09752, 2023年。

[167] Ruoss, A., G. Delétang, T. Genewein, et al. Randomized positional encodings boost length generalization of transformers. In A. Rogers, J. L. Boyd-Graber, N. Okazaki, eds., *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, *ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 1889–1903. Association for Computational Linguistics, 2023.

[168] Liang, X., B. Wang, H. Huang, et al. Unleashing infinite-length input capacity for large-scale language models with self-controlled memory system. *CoRR*, abs/2304.13343, 2023.

[169] Shinn, N., B. Labash, A. Gopinath. Reflexion: an autonomous agent with dynamic memory and self-reflection. *CoRR*, abs/2303.11366, 2023.

[170] Zhong, W., L. Guo, Q. Gao, et al. Memorybank: Enhancing large language models with long-term memory. *CoRR*, abs/2305.10250, 2023.

[171] Chan, C., W. Chen, Y. Su, et al. Chateval: Towards better llm-based evaluators through multi-agent debate. *CoRR*, abs/2308.07201, 2023.

[172] Zhu, X., Y. Chen, H. Tian, et al. Ghost in the minecraft: Generally capable agents for open-world environments via large language models with text-based knowledge and memory. *CoRR*, abs/2305.17144, 2023.

[173] Modarressi, A., A. Imani, M. Fayyaz, et al. RET-LLM: towards a general read-write memory for large language models. *CoRR*, abs/2305.14322, 2023.

[174] Lin, J., H. Zhao, A. Zhang, et al. Agentsims: An open-source sandbox for large language model evaluation. *CoRR*, abs/2308.04026, 2023.

[175] Hu, C., J. Fu, C. Du, et al. Chatdb: Augmenting llms with databases as their symbolic memory. *CoRR*, abs/2306.03901, 2023.

[176] Huang, Z., S. Gutierrez, H. Kamana, et al. Memory sandbox: Transparent and interactive memory management for conversational agents. *CoRR*, abs/2308.01542, 2023.

[177] Creswell, A., M. Shanahan, I. Higgins. Selection-inference: Exploiting large language models for interpretable logical reasoning. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.

[178] Madaan, A., N. Tandon, P. Gupta, et al. Self-refine: Iterative refinement with self-feedback. *CoRR*, abs/2303.17651, 2023.

[179] Ichter, B., A. Brohan, Y. Chebotar, et al. Do as I can, not as I say: Grounding language in robotic affordances. In K. Liu, D. Kulic, J. Ichnowski, eds., *Conference on Robot Learning, CoRL 2022, 14-18 December 2022, Auckland, New Zealand*, vol. 205 of *Proceedings of Machine Learning Research*, pages 287–318. PMLR, 2022.

[180] Shen, Y., K. Song, X. Tan, et al. Hugginggpt: Solving AI tasks with chatgpt and its friends in huggingface. *CoRR*, abs/2303.17580, 2023.

[181] Yao, S., D. Yu, J. Zhao, et al. Tree of thoughts: Deliberate problem solving with large language models. *CoRR*, abs/2305.10601, 2023.

[182] Wu, Y., S. Y. Min, Y. Bisk, et al. Plan, eliminate, and track - language models are good teachers for embodied agents. *CoRR*, abs/2305.02412, 2023.

[183] Wang, Z., S. Cai, A. Liu, et al. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents. *CoRR*, abs/2302.01560, 2023.

[184] Hao, S., Y. Gu, H. Ma, et al. Reasoning with language model is planning with world model. *CoRR*, abs/2305.14992, 2023.

[185] Lin, B. Y., Y. Fu, K. Yang, et al. Swiftsage: A generative agent with fast and slow thinking for complex interactive tasks. *CoRR*, abs/2305.17390, 2023.

[167] Ruoss, A., G. Delétang, T. Genewein, 等. 随机位置编码提升变换器的长度泛化能力. 收录于A. Rogers, J. L. Boyd-Graber, N. Okazaki 编., 第61届计算语言学协会年会论文集（第2卷：短论文），*ACL 2023*，加拿大多伦多，2023年7月9-14日，第1889–1903页。计算语言学协会，2023年.[168] Liang, X., B. Wang, H. Huang, 等. 利用自控记忆系统释放大规模语言模型的无限长度输入能力. *CoRR*，abs/2304.13343，2023年.[169] Shinn, N., B. Labash, A. Gopinath. Reflexion：具有动态记忆和自我反思的自主代理. *CoRR*，abs/2303.11366，2023年.[170] Zhong, W., L. Guo, Q. Gao, 等. Memorybank：通过长时记忆增强大规模语言模型. *CoRR*，abs/2305.10250，2023年.[171] Chan, C., W. Chen, Y. Su, 等. Chateval：通过多代理辩论实现更优的基于LLM的评估器. *CoRR*，abs/2308.07201，2023年.[172] Zhu, X., Y. Chen, H. Tian, 等. Minecraft中的幽灵：通过具有文本知识和记忆的大语言模型实现的通用能力代理，用于开放世界环境. *CoRR*，abs/2305.17144，2023年.[173] Modarressi, A., A. Imani, M. Fayyaz, 等. RET-LLM：面向大规模语言模型的通用读写记忆. *CoRR*，abs/2305.14322，2023年.[174] Lin, J., H. Zhao, A. Zhang, 等. Agentsims：一个用于大规模语言模型评估的开源沙箱. *CoRR*，abs/2308.04026，2023年.[175] Hu, C., J. Fu, C. Du, 等. Chatdb：用数据库作为符号记忆增强LLMs. *CoRR*，abs/2306.03901，2023年.[176] Huang, Z., S. Gutierrez, H. Kamana, 等. Memory sandbox：为对话代理提供透明且交互式的记忆管理. *CoRR*，abs/2308.01542，2023年.[177] Creswell, A., M. Shanahan, I. Higgins. 选择推理：利用大规模语言模型实现可解释的逻辑推理. 收录于第十一届学习表示国际会议（*ICLR 2023*），卢旺达基加利，2023年5月1-5日。OpenReview.net，2023年.[178] Madaan, A., N. Tandon, P. Gupta, 等. Self-refine：通过自我反馈进行迭代优化. *CoRR*，abs/2303.17651，2023年.[179] Ichter, B., A. Brohan, Y. Chebotar, 等. 做我能做的，而不是我说的：将语言与机器人能力相结合. 收录于K. Liu, D. Kulic, J. Ichnowski 编., 机器人学习会议（*CoRL 2022*），2022年12月14-18日，新西兰奥克兰，第205卷机器学习研究论文集，第287–318页。PMLR，2022年.[180] Shen, Y., K. Song, X. Tan, 等. Hugginggpt：利用ChatGPT及其伙伴在Huggingface中解决AI任务. *CoRR*，abs/2303.17580，2023年.[181] Yao, S., D. Yu, J. Zhao, 等. 思维树：利用大规模语言模型进行深思熟虑的问题解决. *CoRR*，abs/2305.10601，2023年.[182] Wu, Y., S. Y. Min, Y. Bisk, 等. 规划、消除与追踪——语言模型是具身代理的良师. *CoRR*，abs/2305.02412，2023年.[183] Wang, Z., S. Cai, A. Liu, 等. 描述、解释、规划与选择：通过大规模语言模型实现的交互式规划，支持开放世界多任务代理. *CoRR*，abs/2302.01560，2023年.[184] Hao, S., Y. Gu, H. Ma, 等. 以语言模型进行推理即以世界模型进行规划. *CoRR*，abs/2305.14992，2023年.[185] Lin, B. Y., Y. Fu, K. Yang, 等. Swiftsage：具有快速与缓慢思维的生成代理，用于复杂交互任务. *CoRR*，abs/2305.17390，2023年。

[186] Karpas, E., O. Abend, Y. Belinkov, et al. MRKL systems: A modular, neuro-symbolic architecture that combines large language models, external knowledge sources and discrete reasoning. *CoRR*, abs/2205.00445, 2022.

[187] Huang, W., F. Xia, T. Xiao, et al. Inner monologue: Embodied reasoning through planning with language models. In K. Liu, D. Kulic, J. Ichnowski, eds., *Conference on Robot Learning, CoRL 2022, 14-18 December 2022, Auckland, New Zealand*, vol. 205 of *Proceedings of Machine Learning Research*, pages 1769–1782. PMLR, 2022.

[188] Chen, Z., K. Zhou, B. Zhang, et al. Chatcot: Tool-augmented chain-of-thought reasoning on chat-based large language models. *CoRR*, abs/2305.14323, 2023.

[189] Wu, T., M. Terry, C. J. Cai. AI chains: Transparent and controllable human-ai interaction by chaining large language model prompts. In S. D. J. Barbosa, C. Lampe, C. Appert, D. A. Shamma, S. M. Drucker, J. R. Williamson, K. Yatani, eds., *CHI '22: CHI Conference on Human Factors in Computing Systems, New Orleans, LA, USA, 29 April 2022 - 5 May 2022*, pages 385:1–385:22. ACM, 2022.

[190] Wang, G., Y. Xie, Y. Jiang, et al. Voyager: An open-ended embodied agent with large language models. *CoRR*, abs/2305.16291, 2023.

[191] Zhao, X., M. Li, C. Weber, et al. Chat with the environment: Interactive multimodal perception using large language models. *CoRR*, abs/2303.08268, 2023.

[192] Miao, N., Y. W. Teh, T. Rainforth. Selfcheck: Using llms to zero-shot check their own step-by-step reasoning. *CoRR*, abs/2308.00436, 2023.

[193] Wang, X., W. Wang, Y. Cao, et al. Images speak in images: A generalist painter for in-context visual learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, Vancouver, BC, Canada, June 17-24, 2023*, pages 6830–6839. IEEE, 2023.

[194] Wang, C., S. Chen, Y. Wu, et al. Neural codec language models are zero-shot text to speech synthesizers. *CoRR*, abs/2301.02111, 2023.

[195] Dong, Q., L. Li, D. Dai, et al. A survey for in-context learning. *CoRR*, abs/2301.00234, 2023.

[196] Ke, Z., B. Liu. Continual learning of natural language processing tasks: A survey. *ArXiv*, abs/2211.12701, 2022.

[197] Wang, L., X. Zhang, H. Su, et al. A comprehensive survey of continual learning: Theory, method and application. *ArXiv*, abs/2302.00487, 2023.

[198] Razdaibiedina, A., Y. Mao, R. Hou, et al. Progressive prompts: Continual learning for language models. In *The Eleventh International Conference on Learning Representations*. 2023.

[199] Marshall, L. H., H. W. Magoun. *Discoveries in the human brain: neuroscience prehistory, brain structure, and function*. Springer Science & Business Media, 2013.

[200] Searle, J. R. What is language: some preliminary remarks. *Explorations in Pragmatics. Linguistic, cognitive and intercultural aspects*, pages 7–37, 2007.

[201] Touvron, H., T. Lavril, G. Izacard, et al. Llama: Open and efficient foundation language models. *CoRR*, abs/2302.13971, 2023.

[202] Scao, T. L., A. Fan, C. Akiki, et al. BLOOM: A 176b-parameter open-access multilingual language model. *CoRR*, abs/2211.05100, 2022.

[203] Almazrouei, E., H. Alobeidli, A. Alshamsi, et al. Falcon-40b: an open large language model with state-of-the-art performance, 2023.

[204] Serban, I. V., R. Lowe, L. Charlin, et al. Generative deep neural networks for dialogue: A short review. *CoRR*, abs/1611.06216, 2016.

[205] Vinyals, O., Q. V. Le. A neural conversational model. *CoRR*, abs/1506.05869, 2015.

Karpas, E., O. Abend, Y. Belinkov 等. MRKL系统：一种结合大型语言模型、外部知识源和离散推理的模块化神经符号架构。*CoRR*, abs/2205.00445, 2022年。Huang, W., F. Xia, T. Xiao 等. 内心独白：通过语言模型规划实现具身推理。在K. Liu, D. Kulic, J. Ichnowski编辑的《机器学习会议, *CoRL 2022*, 2022年12月14-18日, 新西兰奥克兰》, 第205卷《机器学习研究会议论文集》, 第1769–1782页。PMLR, 2022年。Chen, Z., K. Zhou, B. Zhang 等. Chatcot：在基于聊天的大型语言模型上实现工具增强的链式思维推理。*CoRR*, abs/2305.14323, 2023年。Wu, T., M. Terry, C. J. Cai. AI链：通过链式大语言模型提示实现透明且可控的人机交互。在S. D. J. Barbosa, C. Lampe, C. Appert, D. A. Shamma, S. M. Drucker, J. R. Williamson, K. Yatani编辑的《*CHI* ’22：人机交互系统会议, 2022年4月29日至5月5日, 新奥尔良, 路易斯安那州, 美国》, 第385:1–385:22页。ACM, 2022年。Wang, G., Y. Xie, Y. Jiang 等. Voyager：一种具有大型语言模型的开放式具身代理。*CoRR*, abs/2305.16291, 2023年。Zhao, X., M. Li, C. Weber 等. 与环境对话：利用大型语言模型进行交互式多模态感知。*CoRR*, abs/2303.08268, 2023年。Miao, N., Y. W. Teh, T. Rainforth. Selfcheck：利用大模型进行零样本逐步推理自检。*CoRR*, abs/2308.00436, 2023年。Wang, X., W. Wang, Y. Cao 等. 图像中的图像：一种用于上下文视觉学习的通用画家。发表于*IEEE/CVF计算机视觉与模式识别会议, CVPR 2023*, 加拿大温哥华, 2023年6月17-24日, 第6830–6839页。IEEE, 2023年。Wang, C., S. Chen, Y. Wu 等. 神经编码器语言模型是零样本文本到语音合成器。*CoRR*, abs/2301.02111, 2023年。Dong, Q., L. Li, D. Dai 等. 关于上下文学习的综述。*CoRR*, abs/2301.00234, 2023年。Ke, Z., B. Liu. 自然语言处理任务的持续学习：一项综述。*ArXiv*, abs/2211.12701, 2022年。Wang, L., X. Zhang, H. Su 等. 持续学习的全面综述：理论、方法与应用。*ArXiv*, abs/2302.00487, 2023年。Razdaibiedina, A., Y. Mao, R. Hou 等. 逐步提示：面向语言模型的持续学习。在第十一届学习表示国际会议, 2023年。Marshall, L. H., H. W. Magoun. 人脑的发现：神经科学史前、脑结构与功能。Springer Science & Business Media, 2013年。Searle, J. R. 语言是什么：一些初步的评论。语用学探索：语言学、认知与跨文化方面, 第7–37页, 2007年。Touvron, H., T. Lavril, G. Izacard 等. Llama：开源高效的基础语言模型。*CoRR*, abs/2302.13971, 2023年。Scao, T. L., A. Fan, C. Akiki 等. BLOOM：一个拥有1760亿参数的开源多语言模型。*CoRR*, abs/2211.05100, 2022年。Almazrouei, E., H. Alobeidli, A. Alshamsi 等. Falcon-40b：一款具有最先进性能的开源大型语言模型, 2023年。Serban, I. V., R. Lowe, L. Charlin 等. 对话生成深度神经网络：简要综述。*CoRR*, abs/1611.06216, 2016年。Vinyals, O., Q. V. Le. 神经对话模型。*CoRR*, abs/1506.05869, 2015年。

[206] Adiwardana, D., M. Luong, D. R. So, et al. Towards a human-like open-domain chatbot. *CoRR*, abs/2001.09977, 2020.

[207] Zhuge, M., H. Liu, F. Faccio, et al. Mindstorms in natural language-based societies of mind. *CoRR*, abs/2305.17066, 2023.

[208] Roller, S., E. Dinan, N. Goyal, et al. Recipes for building an open-domain chatbot. In P. Merlo, J. Tiedemann, R. Tsarfaty, eds., *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume, EACL 2021, Online, April 19 - 23, 2021*, pages 300–325. Association for Computational Linguistics, 2021.

[209] Taori, R., I. Gulrajani, T. Zhang, et al. Stanford alpaca: An instruction-following llama model, 2023.

[210] Raffel, C., N. Shazeer, A. Roberts, et al. Exploring the limits of transfer learning with a unified text-to-text transformer. *The Journal of Machine Learning Research*, 21(1):5485–5551, 2020.

[211] Ge, Y., W. Hua, J. Ji, et al. Openagi: When LLM meets domain experts. *CoRR*, abs/2304.04370, 2023.

[212] Rajpurkar, P., J. Zhang, K. Lopyrev, et al. Squad: 100, 000+ questions for machine comprehension of text. In J. Su, X. Carreras, K. Duh, eds., *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing, EMNLP 2016, Austin, Texas, USA, November 1-4, 2016*, pages 2383–2392. The Association for Computational Linguistics, 2016.

[213] Ahuja, K., R. Hada, M. Ochieng, et al. MEGA: multilingual evaluation of generative AI. *CoRR*, abs/2303.12528, 2023.

[214] See, A., A. Pappu, R. Saxena, et al. Do massively pretrained language models make better storytellers? In M. Bansal, A. Villavicencio, eds., *Proceedings of the 23rd Conference on Computational Natural Language Learning, CoNLL 2019, Hong Kong, China, November 3-4, 2019*, pages 843–861. Association for Computational Linguistics, 2019.

[215] Radford, A., J. Wu, D. Amodei, et al. Better language models and their implications. *OpenAI blog*, 1(2), 2019.

[216] McCoy, R. T., P. Smolensky, T. Linzen, et al. How much do language models copy from their training data? evaluating linguistic novelty in text generation using RAVEN. *CoRR*, abs/2111.09509, 2021.

[217] Tellex, S., T. Kollar, S. Dickerson, et al. Understanding natural language commands for robotic navigation and mobile manipulation. In W. Burgard, D. Roth, eds., *Proceedings of the Twenty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2011, San Francisco, California, USA, August 7-11, 2011*, pages 1507–1514. AAAI Press, 2011.

[218] Christiano, P. F., J. Leike, T. B. Brown, et al. Deep reinforcement learning from human preferences. In I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett, eds., *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pages 4299–4307. 2017.

[219] Basu, C., M. Singhal, A. D. Dragan. Learning from richer human guidance: Augmenting comparison-based learning with feature queries. In T. Kanda, S. Sabanovic, G. Hoffman, A. Tapus, eds., *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction, HRI 2018, Chicago, IL, USA, March 05-08, 2018*, pages 132–140. ACM, 2018.

[220] Sumers, T. R., M. K. Ho, R. X. D. Hawkins, et al. Learning rewards from linguistic feedback. In *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021, Thirty-Third Conference on Innovative Applications of Artificial Intelligence, IAAI 2021, The Eleventh Symposium on Educational Advances in Artificial Intelligence, EAAI 2021, Virtual Event, February 2-9, 2021*, pages 6002–6010. AAAI Press, 2021.

[206] Adiwardana, D., M. Luong, D. R. So 等. 面向类人开放域聊天机器人的研究进展。 *CoRR*, abs/2001.09977, 2020年。[207] 诸葛明, 刘浩, F. Faccio 等. 基于自然语言的社会中的思想风暴。 *CoRR*, abs/2305.17066, 2023年。[208] Roller, S., E. Dinan, N. Goyal 等. 构建开放域聊天机器人的方法。收录于 P. Merlo, J. Tiedemann, R. Tsarfaty 编著, 《第16届欧洲计算语言学协会会议论文集: 主卷, *EACL 2021*, 线上, 2021年4月19-23日》, 第300-325页。计算语言学协会, 2021年。[209] Taori, R., I. Gulrajani, T. Zhang 等. Stanford alpaca: 一种遵循指令的 llama 模型, 2023年。[210] Raffel, C., N. Shazeer, A. Roberts 等. 探索统一文本到文本转换器在迁移学习中的极限。机器学习研究杂志, 21(1):5485–5551, 2020年。[211] 葛毅, 华伟, 季佳 等. Openagi: 当大规模语言模型遇到领域专家。 *CoRR*, abs/2304.04370, 2023年。[212] Rajpurkar, P., J. Zhang, K. Lopyrev 等. Squad: 用于机器理解文本的100, questions。收录于 J. Su, X. Carreras, K. Duh 编著, 《2016年自然语言处理实证方法会议论文集, *EMNLP 2016*, 德克萨斯州奥斯汀, 2016年11月1-4日》, 第2383-2392页。计算语言学协会, 2016年。[213] Ahuja, K., R. Hada, M. Ochieng 等. MEGA: 多语言生成式人工智能评估。 *CoRR*, abs/2303.12528, 2023年。[214] See, A., A. Pappu, R. Saxena 等. 大规模预训练语言模型是否成为更好的讲故事者? 收录于 M. Bansal, A. Villavicencio 编著, 《第23届计算自然语言学习会议论文集, *CoNLL 2019*, 香港, 中国, 2019年11月3-4日》, 第843-861页。计算语言学协会, 2019年。[215] Radford, A., J. Wu, D. Amodei 等. 更优的语言模型及其影响。 *OpenAI* 博客, 1(2), 2019年。[216] McCoy, R. T., P. Smolensky, T. Linzen 等. 语言模型从训练数据中复制了多少内容? 利用 RAVEN 评估文本生成中的语言新颖性。 *CoRR*, abs/2111.09509, 2021年。[217] Tellex, S., T. Kollar, S. Dickerson 等. 理解自然语言指令以实现机器人导航与移动操作。收录于 W. Burgard, D. Roth 编著, 《第25届人工智能协会会议论文集, *AAAI 2011*, 旧金山, 加利福尼亚, 美国, 2011年8月7-11日》, 第1507-1514页。AAAI出版社, 2011年。[218] Christiano, P. F., J. Leike, T. B. Brown 等. 基于人类偏好的深度强化学习。收录于 I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett 编著, 《神经信息处理系统30: 2017年神经信息处理系统年度会议, 2017年12月4-9日, 长滩, 加利福尼亚, 美国》, 第4299-4307页。2017年。[219] Basu, C., M. Singhal, A. D. Dragan. 从更丰富的人类指导中学习: 用特征查询增强基于比较的学习。收录于 T. Kanda, S. Sabanovic, G. Hoffman, A. Tapus 编著, 《2018年ACM/IEEE人机交互国际会议论文集, *HRI 2018*, 芝加哥, 伊利诺伊州, 美国, 2018年3月5-8日》, 第132-140页。ACM, 2018年。[220] Sumers, T. R., M. K. Ho, R. X. D. Hawkins 等. 从语言反馈中学习奖励。收录于 第35届人工智能协会会议, *AAAI 2021*, 第33届创新应用人工智能会议, *IAAI 2021*, 第11届人工智能教育进展研讨会, *EAAI 2021*, 虚拟会议, 2021年2月2-9日, 第6002-6010页。AAAI出版社, 2021年。

[221] Jeon, H. J., S. Milli, A. D. Dragan. Reward-rational (implicit) choice: A unifying formalism for reward learning. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin, eds., *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*. 2020.

[222] McShane, M. Reference resolution challenges for intelligent agents: The need for knowledge. *IEEE Intell. Syst.*, 24(4):47–58, 2009.

[223] Gururangan, S., A. Marasovic, S. Swayamdipta, et al. Don’t stop pretraining: Adapt language models to domains and tasks. In D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault, eds., *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020*, pages 8342–8360. Association for Computational Linguistics, 2020.

[224] Shi, F., X. Chen, K. Misra, et al. Large language models can be easily distracted by irrelevant context. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 31210–31227. PMLR, 2023.

[225] Zhang, Y., Y. Li, L. Cui, et al. Siren’s song in the AI ocean: A survey on hallucination in large language models. *CoRR*, abs/2309.01219, 2023.

[226] Mialon, G., R. Dessì, M. Lomeli, et al. Augmented language models: a survey. *CoRR*, abs/2302.07842, 2023.

[227] Ren, R., Y. Wang, Y. Qu, et al. Investigating the factual knowledge boundary of large language models with retrieval augmentation. *CoRR*, abs/2307.11019, 2023.

[228] Nuxoll, A. M., J. E. Laird. Extending cognitive architecture with episodic memory. In *AAAI*, pages 1560–1564. 2007.

[229] Squire, L. R. Mechanisms of memory. *Science*, 232(4758):1612–1619, 1986.

[230] Schwabe, L., K. Nader, J. C. Pruessner. Reconsolidation of human memory: brain mechanisms and clinical relevance. *Biological psychiatry*, 76(4):274–280, 2014.

[231] Hutter, M. A theory of universal artificial intelligence based on algorithmic complexity. *arXiv preprint cs/0004001*, 2000.

[232] Zhang, X., F. Wei, M. Zhou. HIBERT: document level pre-training of hierarchical bidirectional transformers for document summarization. In A. Korhonen, D. R. Traum, L. Màrquez, eds., *Proceedings of the 57th Conference of the Association for Computational Linguistics, ACL 2019, Florence, Italy, July 28- August 2, 2019, Volume 1: Long Papers*, pages 5059–5069. Association for Computational Linguistics, 2019.

[233] Mohtashami, A., M. Jaggi. Landmark attention: Random-access infinite context length for transformers. *CoRR*, abs/2305.16300, 2023.

[234] Chalkidis, I., X. Dai, M. Fergadiotis, et al. An exploration of hierarchical attention transformers for efficient long document classification. *CoRR*, abs/2210.05529, 2022.

[235] Nie, Y., H. Huang, W. Wei, et al. Capturing global structural information in long document question answering with compressive graph selector network. In Y. Goldberg, Z. Kozareva, Y. Zhang, eds., *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, pages 5036–5047. Association for Computational Linguistics, 2022.

[236] Bertsch, A., U. Alon, G. Neubig, et al. Unlimiformer: Long-range transformers with unlimited length input. *CoRR*, abs/2305.01625, 2023.

Jeon, H. J., S. Milli, A. D. Dragan. 奖励理性（隐式）选择：一种统一的奖励学习形式主义。在 H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin 编著, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*. 2020年。

[222] McShane, M. 智能代理的参考解析挑战：对知识的需求。 *IEEE Intell. Syst.*, 24(4):47–58, 2009年。

[223] Gururangan, S., A. Marasovic, S. Swayamdipta 等。不要停止预训练：将语言模型适应于特定领域和任务。在 D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault 编著, 第58届计算语言学协会年会论文集, *ACL 2020*, 线上, 2020年7月5-10日, 第8342-8360页。计算语言学协会, 2020年。

[224] Shi, F., X. Chen, K. Misra 等。大型语言模型容易被无关上下文分散注意力。在 A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett 编著, 国际机器学习会议, *ICML 2023*, 2023年7月23-29日, 夏威夷檀香山, 美国, 卷 202, 机器学习研究论文集, 第 31210-31227页。PMLR, 2023年。

[225] Zhang, Y., Y. Li, L. Cui 等。AI海洋中的海妖之歌：大型语言模型中的幻觉现象综述。 *CoRR*, abs/2309.01219, 2023年。

[226] Mialon, G., R. Dessì, M. Lomeli 等。增强型语言模型：一项综述。 *CoRR*, abs/2302.07842, 2023年。

[227] Ren, R., Y. Wang, Y. Qu 等。利用检索增强探索大型语言模型的事实知识边界。 *CoRR*, abs/2307.11019, 2023年。

[228] Nuxoll, A. M., J. E. Laird. 通过情景记忆扩展认知架构。在 *AAAI*, 第1560-1564页, 2007年。

[229] Squire, L. R. 记忆机制。 *Science*, 232(4758):1612–1619, 1986年。

[230] Schwabe, L., K. Nader, J. C. Pruessner. 人类记忆的再巩固：脑机制与临床相关性。 *生物精神病学*, 76(4):274–280, 2014年。

[231] Hutter, M. 基于算法复杂度的通用人工智能理论。 *arXiv* 预印本 *cs/0004001*, 2000年。

[232] Zhang, X., F. Wei, M. Zhou. HIBERT：用于文档摘要的层次双向变换器的文档级预训练。在 A. Korhonen, D. R. Traum, L. Màrquez 编著, 第57届计算语言学协会会议论文集, *ACL 2019*, 意大利佛罗伦萨, 2019年7月28日-8月2日, 第1卷：长篇论文, 第5059-5069页。计算语言学协会, 2019年。

[233] Mohtashami, A., M. Jaggi. 地标注意力：变换器的随机访问无限上下文长度。 *CoRR*, abs/2305.16300, 2023年。

[234] Chalkidis, I., X. Dai, M. Fergadiotis 等。层次注意力变换器在高效长文档分类中的探索。 *CoRR*, abs/2210.05529, 2022年。

[235] Nie, Y., H. Huang, W. Wei 等。利用压缩图选择网络在长文档问答中捕获全局结构信息。在 Y. Goldberg, Z. Kozareva, Y. Zhang 编著, 2022年自然语言处理实证方法会议论文集, *EMNLP 2022*, 阿布扎比, 阿联酋, 2022年12月7-11日, 第5036-5047页。计算语言学协会, 2022年。

[236] Bertsch, A., U. Alon, G. Neubig 等。Unlimiformer：具有无限长度输入的长距离变换器。 *CoRR*, abs/2305.01625, 2023年。

[237] Manakul, P., M. J. F. Gales. Sparsity and sentence structure in encoder-decoder attention of summarization systems. In M. Moens, X. Huang, L. Specia, S. W. Yih, eds., *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7-11 November, 2021*, pages 9359–9368. Association for Computational Linguistics, 2021.

[238] Zaheer, M., G. Guruganesh, K. A. Dubey, et al. Big bird: Transformers for longer sequences. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin, eds., *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*. 2020.

[239] Zhao, A., D. Huang, Q. Xu, et al. Expel: LLM agents are experiential learners. *CoRR*, abs/2308.10144, 2023.

[240] Zhou, X., G. Li, Z. Liu. LLM as DBA. *CoRR*, abs/2308.05481, 2023.

[241] Wason, P. C. Reasoning about a rule. *Quarterly journal of experimental psychology*, 20(3):273–281, 1968.

[242] Wason, P. C., P. N. Johnson-Laird. *Psychology of reasoning: Structure and content*, vol. 86. Harvard University Press, 1972.

[243] Galotti, K. M. Approaches to studying formal and everyday reasoning. *Psychological bulletin*, 105(3):331, 1989.

[244] Huang, J., K. C. Chang. Towards reasoning in large language models: A survey. In A. Rogers, J. L. Boyd-Graber, N. Okazaki, eds., *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 1049–1065. Association for Computational Linguistics, 2023.

[245] Webb, T. W., K. J. Holyoak, H. Lu. Emergent analogical reasoning in large language models. *CoRR*, abs/2212.09196, 2022.

[246] Feng, G., B. Zhang, Y. Gu, et al. Towards revealing the mystery behind chain of thought: a theoretical perspective. *CoRR*, abs/2305.15408, 2023.

[247] Grafman, J., L. Spector, M. J. Rattermann. Planning and the brain. In *The cognitive psychology of planning*, pages 191–208. Psychology Press, 2004.

[248] Unterrainer, J. M., A. M. Owen. Planning and problem solving: from neuropsychology to functional neuroimaging. *Journal of Physiology-Paris*, 99(4-6):308–317, 2006.

[249] Zula, K. J., T. J. Chermack. Integrative literature review: Human capital planning: A review of literature and implications for human resource development. *Human Resource Development Review*, 6(3):245–262, 2007.

[250] Bratman, M. E., D. J. Israel, M. E. Pollack. Plans and resource-bounded practical reasoning. *Computational intelligence*, 4(3):349–355, 1988.

[251] Russell, S., P. Norvig. *Artificial intelligence - a modern approach, 2nd Edition*. Prentice Hall series in artificial intelligence. Prentice Hall, 2003.

[252] Fainstein, S. S., J. DeFilippis. *Readings in planning theory*. John Wiley & Sons, 2015.

[253] Sebastia, L., E. Onaindia, E. Marzal. Decomposition of planning problems. *Ai Communications*, 19(1):49–81, 2006.

[254] Crosby, M., M. Rovatsos, R. Petrick. Automated agent decomposition for classical planning. In *Proceedings of the International Conference on Automated Planning and Scheduling*, vol. 23, pages 46–54. 2013.

[255] Xu, B., Z. Peng, B. Lei, et al. Rewoo: Decoupling reasoning from observations for efficient augmented language models. *CoRR*, abs/2305.18323, 2023.

Manakul, P., M. J. F. Gales. 编码器-解码器注意力中的稀疏性与句子结构在摘要系统中的作用。收录于 M. Moens, X. Huang, L. Specia, S. W. Yih 编著, 《2021年自然语言处理实证方法会议论文集, *EMNLP 2021*, 虚拟会议/蓬塔卡纳, 多米尼加共和国, 2021年11月7-11日》, 第9359–9368页。计算语言学协会, 2021年。Zaheer, M., G. Guruganesh, K. A. Dubey 等。Big bird: 用于长序列的变换器。在 H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin 编著, 《神经信息处理系统进展33: 神经信息处理系统年度会议, *NeurIPS 2020*, 2020年12月6-12日, 虚拟会议》。2020年。Zhao, A., D. Huang, Q. Xu 等。Expel: LLM代理是体验式学习者。 *CoRR*, abs/2308.10144, 2023年。Zhou, X., G. Li, Z. Liu. LLM作为数据库管理员。 *CoRR*, abs/2308.05481, 2023年。Wason, P. C. 关于规则的推理。实验心理学季刊, 20(3):273–281, 1968年。Wason, P. C., P. N. Johnson-Laird. 推理心理学: 结构与内容, 第86卷。哈佛大学出版社, 1972年。Galotti, K. M. 研究形式推理与日常推理的方法。心理学通报, 105(3):331, 1989年。Huang, J., K. C. Chang. 迈向大规模语言模型的推理: 一项综述。在 A. Rogers, J. L. Boyd-Graber, N. Okazaki 编著, 《计算语言学协会会议论文集: *ACL 2023*, 加拿大多伦多, 2023年7月9-14日》, 第1049–1065页。计算语言学协会, 2023年。Webb, T. W., K. J. Holyoak, H. Lu. 大规模语言模型中的新兴类比推理。 *CoRR*, abs/2212.09196, 2022年。Feng, G., B. Zhang, Y. Gu 等。揭示思维链背后奥秘的理论视角。 *CoRR*, abs/2305.15408, 2023年。Grafman, J., L. Spector, M. J. Rattermann. 计划与大脑。在计划的认知心理学中, 第191–208页。心理学出版社, 2004年。Unterrainer, J. M., A. M. Owen. 规划与问题解决: 从神经心理学到功能性神经影像学。生理学杂志-巴黎, 99(4-6):308–317, 2006年。Zula, K. J., T. J. Chermack. 整合性文献综述: 人力资本规划: 文献回顾及对人力资源开发的启示。人力资源开发评论, 6(3):245–262, 2007年。Bratman, M. E., D. J. Israel, M. E. Pollack. 计划与资源有限的实际推理。计算智能, 4(3):349–355, 1988年。Russell, S., P. Norvig. 人工智能——现代方法, 第2版。Prentice Hall人工智能系列。Prentice Hall, 2003年。Fainstein, S. S., J. DeFilippis. 规划理论读本。约翰威立国际出版社, 2015年。Sebastia, L., E. Onaindia, E. Marzal. 规划问题的分解。人工智能通讯, 19(1):49–81, 2006年。Crosby, M., M. Rovatsos, R. Petrick. 经典规划的自动代理分解。在国际自动规划与调度会议论文集, 第23卷, 页46–54, 2013年。Xu, B., Z. Peng, B. Lei 等。Rewoo: 为高效增强型语言模型解耦推理与观察。 *CoRR*, abs/2305.18323, 2023年。

[256] Raman, S. S., V. Cohen, E. Rosen, et al. Planning with large language models via corrective re-prompting. *CoRR*, abs/2211.09935, 2022.

[257] Lyu, Q., S. Havaldar, A. Stein, et al. Faithful chain-of-thought reasoning. *CoRR*, abs/2301.13379, 2023.

[258] Huang, W., P. Abbeel, D. Pathak, et al. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato, eds., *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, vol. 162 of *Proceedings of Machine Learning Research*, pages 9118–9147. PMLR, 2022.

[259] Dagan, G., F. Keller, A. Lascarides. Dynamic planning with a LLM. *CoRR*, abs/2308.06391, 2023.

[260] Rana, K., J. Haviland, S. Garg, et al. Sayplan: Grounding large language models using 3d scene graphs for scalable task planning. *CoRR*, abs/2307.06135, 2023.

[261] Peters, M. E., M. Neumann, M. Iyyer, et al. Deep contextualized word representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 2227–2237. Association for Computational Linguistics, New Orleans, Louisiana, 2018.

[262] Devlin, J., M. Chang, K. Lee, et al. BERT: pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, T. Solorio, eds., *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2019, Minneapolis, MN, USA, June 2-7, 2019, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics, 2019.

[263] Solaiman, I., C. Dennison. Process for adapting language models to society (palms) with values-targeted datasets. *Advances in Neural Information Processing Systems*, 34:5861–5873, 2021.

[264] Bach, S. H., V. Sanh, Z. X. Yong, et al. Promptsources: An integrated development environment and repository for natural language prompts. In V. Basile, Z. Kozareva, S. Stajner, eds., *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, ACL 2022 - System Demonstrations, Dublin, Ireland, May 22-27, 2022*, pages 93–104. Association for Computational Linguistics, 2022.

[265] Iyer, S., X. V. Lin, R. Pasunuru, et al. OPT-IML: scaling language model instruction meta learning through the lens of generalization. *CoRR*, abs/2212.12017, 2022.

[266] Winston, P. H. Learning and reasoning by analogy. *Commun. ACM*, 23(12):689–703, 1980.

[267] Lu, Y., M. Bartolo, A. Moore, et al. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In S. Muresan, P. Nakov, A. Villavicencio, eds., *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2022, Dublin, Ireland, May 22-27, 2022*, pages 8086–8098. Association for Computational Linguistics, 2022.

[268] Tsimpoukelli, M., J. Menick, S. Cabi, et al. Multimodal few-shot learning with frozen language models. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan, eds., *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 200–212. 2021.

[269] Bar, A., Y. Gandelsman, T. Darrell, et al. Visual prompting via image inpainting. In *NeurIPS*. 2022.

[270] Zhu, W., H. Liu, Q. Dong, et al. Multilingual machine translation with large language models: Empirical results and analysis. *CoRR*, abs/2304.04675, 2023.

[256] Raman, S. S., V. Cohen, E. Rosen 等. 通过纠正性再提示进行大规模语言模型的规划。 *CoRR*, abs/2211.09935, 2022年。[257]Lyu, Q., S. Havaldar, A. Stein 等. 忠实的链式思考推理。 *CoRR*, abs/2301.13379, 2023年。[258] Huang, W., P. Abbeel, D. Pathak 等. 作为零样本规划器的语言模型：提取可操作的知识以支持具身代理。在K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato 编著，国际机器学习会议， *ICML 2022*, 2022年7月17-23日，马里兰州巴尔的摩，第162卷机器学习研究会议录，页码9118–9147。PMLR, 2022年。[259] Dagan, G., F. Keller, A. Lascarides. 使用大规模语言模型的动态规划。 *CoRR*, abs/2308.06391, 2023年。[260] Rana, K., J. Haviland, S. Garg 等. Sayplan：利用3D场景图对大规模语言模型进行语境绑定以实现可扩展的任务规划。 *CoRR*, abs/2307.06135, 2023年。[261] Peters, M. E., M. Neumann, M. Iyyer 等. 深度上下文化词表示。在2018年北美计算语言学协会人类语言技术会议论文集（第1卷：长论文），第2227–2237页。计算语言学协会，新奥尔良，路易斯安那州，2018年。[262] Devlin, J., M. Chang, K. Lee 等. BERT：用于语言理解的深度双向变换器预训练。在J. Burstein, C. Doran, T. Solorio 编著，2019年北美计算语言学协会人类语言技术会议论文集（*NAACL-HLT 2019*），明尼阿波利斯，明尼苏达州，美国，2019年6月2-7日，第1卷（长短论文），第4171–4186页。计算语言学协会，2019年。[263] Solaiman, I., C. Dennison. 以价值导向数据集为基础的社会适应性语言模型（PALMS）开发流程。神经信息处理系统进展，第34卷，5861–5873页，2021年。[264] Bach, S. H., V. Sanh, Z. X. Yong 等. Promptsources：一个集成的自然语言提示开发环境与存储库。在V. Basile, Z. Kozareva, S. Stajner 编著，第60届计算语言学协会年会系统演示论文集， *ACL 2022*, 爱尔兰都柏林，2022年5月22-27日，第93–104页。计算语言学协会，2022年。[265] Iyer, S., X. V. Lin, R. Pasunuru 等. OPT-IML：通过泛化视角扩展语言模型指令元学习。 *CoRR*, abs/2212.12017, 2022年。[266] Winston, P. H. 通过类比进行学习与推理。 *Commun. ACM*, 23(12):689–703, 1980年。[267] Lu, Y., M. Bartolo, A. Moore 等. 极具秩序的提示及其获取途径：克服少样本提示顺序敏感性。在S. Muresan, P. Nakov, A. Villavicencio 编著，第60届计算语言学协会年会论文集（第1卷：长论文）， *ACL 2022*, 爱尔兰都柏林，2022年5月22-27日，第8086–8098页。计算语言学协会，2022年。[268] Tsimpoukelli, M., J. Menick, S. Cabi 等. 利用冻结的语言模型进行多模态少样本学习。在M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan 编著，神经信息处理系统进展34： *NeurIPS 2021* 年年度会议，2021年12月6-14日，虚拟会议，第200–212页，2021年。[269] Bar, A., Y. Gandelsman, T. Darrell 等. 通过图像修复实现视觉提示。在 *NeurIPS*。2022年。[270] Zhu, W., H. Liu, Q. Dong 等. 利用大规模语言模型进行多语种机器翻译：实证结果与分析。 *CoRR*, abs/2304.04675, 2023年。

- [271] Zhang, Z., L. Zhou, C. Wang, et al. Speak foreign languages with your own voice: Cross-lingual neural codec language modeling. *CoRR*, abs/2303.03926, 2023.
- [272] Zhang, J., J. Zhang, K. Pertsch, et al. Bootstrap your own skills: Learning to solve new tasks with large language model guidance. In *7th Annual Conference on Robot Learning*, 2023.
- [273] McCloskey, M., N. J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24:109–165, 1989.
- [274] Kirkpatrick, J., R. Pascanu, N. Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017.
- [275] Li, Z., D. Hoiem. Learning without forgetting. *IEEE transactions on pattern analysis and machine intelligence*, 40(12):2935–2947, 2017.
- [276] Farajtabar, M., N. Azizan, A. Mott, et al. Orthogonal gradient descent for continual learning. In *International Conference on Artificial Intelligence and Statistics*, pages 3762–3773. PMLR, 2020.
- [277] Smith, J. S., Y.-C. Hsu, L. Zhang, et al. Continual diffusion: Continual customization of text-to-image diffusion with c-lora. *arXiv preprint arXiv:2304.06027*, 2023.
- [278] Lopez-Paz, D., M. Ranzato. Gradient episodic memory for continual learning. *Advances in neural information processing systems*, 30, 2017.
- [279] de Masson D’Autume, C., S. Ruder, L. Kong, et al. Episodic memory in lifelong language learning. *Advances in Neural Information Processing Systems*, 32, 2019.
- [280] Rolnick, D., A. Ahuja, J. Schwarz, et al. Experience replay for continual learning. *Advances in Neural Information Processing Systems*, 32, 2019.
- [281] Serrà, J., D. Surís, M. Miron, et al. Overcoming catastrophic forgetting with hard attention to the task. In *International Conference on Machine Learning*. 2018.
- [282] Dosovitskiy, A., L. Beyer, A. Kolesnikov, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021.
- [283] van den Oord, A., O. Vinyals, K. Kavukcuoglu. Neural discrete representation learning. In I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett, eds., *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pages 6306–6315. 2017.
- [284] Mehta, S., M. Rastegari. Mobilevit: Light-weight, general-purpose, and mobile-friendly vision transformer. In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022.
- [285] Tolstikhin, I. O., N. Houlsby, A. Kolesnikov, et al. Mlp-mixer: An all-mlp architecture for vision. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan, eds., *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 24261–24272. 2021.
- [286] Huang, S., L. Dong, W. Wang, et al. Language is not all you need: Aligning perception with language models. *CoRR*, abs/2302.14045, 2023.
- [287] Li, J., D. Li, S. Savarese, et al. BLIP-2: bootstrapping language-image pre-training with frozen image encoders and large language models. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 19730–19742. PMLR, 2023.
- [288] Dai, W., J. Li, D. Li, et al. Instructblip: Towards general-purpose vision-language models with instruction tuning. *CoRR*, abs/2305.06500, 2023.

- [271] 张志强, 周丽丽, 王超等。用自己的声音说外语: 跨语言神经编解码器语言建模。 *CoRR*, abs/2303.03926, 2023年。[272] 张俊, 张俊, Pertsch等。自主技能的自我引导: 在大规模语言模型指导下学习解决新任务。在第七届机器人学习年度会议, 2023年。[273] McCloskey, M., N. J. Cohen。连接主义网络中的灾难性干扰: 序列学习问题。学习与动机心理学, 24: 109–165, 1989年。[274] Kirkpatrick, J., R. Pascanu, N. Rabinowitz等。克服神经网络中的灾难性遗忘。美国国家科学院院刊, 114(13): 3521–3526, 2017年。[275] 李志强, 何一鸣。无遗忘学习。 *IEEE模式分析与机器智能汇刊*, 40(12): 2935–2947, 2017年。[276] Farajtabar, M., N. Azizan, A. Mott等。正交梯度下降法用于持续学习。在国际人工智能与统计会议, 第3762–3773页。PMLR, 2020年。[277] Smith, J. S., Y.-C. Hsu, 张丽等。持续扩散: 利用c-lora进行文本到图像扩散的持续定制。 *arXiv预印本 arXiv:2304.06027*, 2023年。[278] Lopez-Paz, D., M. Ranzato。用于持续学习的梯度记忆。神经信息处理系统进展, 30, 2017年。[279] de Masson D’Autume, C., S. Ruder, K. Kong等。终身语言学习中的情景记忆。神经信息处理系统进展, 32, 2019年。[280] Rolnick, D., A. Ahuja, J. Schwarz等。经验重放在持续学习中的应用。神经信息处理系统进展, 32, 2019年。[281] Serrà, J., D. Surís, M. Miron等。通过硬任务注意力克服灾难性遗忘。在机器学习国际会议, 2018年。[282] Dosovitskiy, A., L. Beyer, A. Kolesnikov等。图像价值 16×16 字: 用于大规模图像识别的变换器。在第九届国际表示学习会议 (*ICLR 2021*), 虚拟会议, 奥地利, 2021年5月3-7日。OpenReview.net, 2021年。[283] van den Oord, A., O. Vinyals, K. Kavukcuoglu。神经离散表示学习。在I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett等编著的《神经信息处理系统进展30: 神经信息处理系统年度会议, *NeurIPS 2017*, 12月4-9日, 长滩, 加利福尼亚, 美国》, 第6306–6315页, 2017年。[284] Mehta, S., M. Rastegari。Mobilevit: 轻量级、通用且适合移动端的视觉变换器。在第十届国际表示学习会议 (*ICLR 2022*), 虚拟会议, 2022年4月25-29日。OpenReview.net, 2022年。[285] Tolstikhin, I. O., N. Houlsby, A. Kolesnikov等。Mlp-mixer: 一种全MLP架构的视觉模型。在M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan等编著的《神经信息处理系统进展34: 神经信息处理系统年度会议, *NeurIPS 2021*, 2021年12月6-14日, 虚拟》, 第24261–24272页, 2021年。[286] Huang, S., L. Dong, W. Wang等。语言不仅仅是你所需要的: 将感知与语言模型对齐。 *CoRR*, abs/2302.14045, 2023年。[287] Li, J., D. Li, S. Savarese等。BLIP-2: 利用冻结的图像编码器和大规模语言模型进行预训练的引导式语言-图像模型。在A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett编辑的《机器学习国际会议 (*ICML*) 2023, 2023年7月23-29日, 夏威夷檀香山, 美国》, 第202卷, 机器学习研究论文集, 第19730–19742页, PMLR, 2023年。[288] Dai, W., J. Li, D. Li等。Instructblip: 面向指令调优的通用视觉-语言模型。 *CoRR*, abs/2305.06500, 2023年。

[289] Gong, T., C. Lyu, S. Zhang, et al. Multimodal-gpt: A vision and language model for dialogue with humans. *CoRR*, abs/2305.04790, 2023.

[290] Alayrac, J., J. Donahue, P. Luc, et al. Flamingo: a visual language model for few-shot learning. In *NeurIPS*. 2022.

[291] Su, Y., T. Lan, H. Li, et al. Pandagpt: One model to instruction-follow them all. *CoRR*, abs/2305.16355, 2023.

[292] Liu, H., C. Li, Q. Wu, et al. Visual instruction tuning. *CoRR*, abs/2304.08485, 2023.

[293] Huang, R., M. Li, D. Yang, et al. Audiogpt: Understanding and generating speech, music, sound, and talking head. *CoRR*, abs/2304.12995, 2023.

[294] Gong, Y., Y. Chung, J. R. Glass. AST: audio spectrogram transformer. In H. Hermansky, H. Cernocký, L. Burget, L. Lamel, O. Scharenborg, P. Motlíček, eds., *Interspeech 2021, 22nd Annual Conference of the International Speech Communication Association, Brno, Czechia, 30 August - 3 September 2021*, pages 571–575. ISCA, 2021.

[295] Hsu, W., B. Bolte, Y. H. Tsai, et al. Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE ACM Trans. Audio Speech Lang. Process.*, 29:3451–3460, 2021.

[296] Chen, F., M. Han, H. Zhao, et al. X-LLM: bootstrapping advanced large language models by treating multi-modalities as foreign languages. *CoRR*, abs/2305.04160, 2023.

[297] Zhang, H., X. Li, L. Bing. Video-llama: An instruction-tuned audio-visual language model for video understanding. *CoRR*, abs/2306.02858, 2023.

[298] Liu, Z., Y. He, W. Wang, et al. Interngpt: Solving vision-centric tasks by interacting with chatbots beyond language. *CoRR*, abs/2305.05662, 2023.

[299] Hubel, D. H., T. N. Wiesel. Receptive fields, binocular interaction and functional architecture in the cat’s visual cortex. *The Journal of physiology*, 160(1):106, 1962.

[300] Logothetis, N. K., D. L. Sheinberg. Visual object recognition. *Annual review of neuroscience*, 19(1):577–621, 1996.

[301] OpenAI. Openai: Introducing chatgpt. Website, 2022. <https://openai.com/blog/chatgpt>.

[302] Lu, J., X. Ren, Y. Ren, et al. Improving contextual language models for response retrieval in multi-turn conversation. In J. X. Huang, Y. Chang, X. Cheng, J. Kamps, V. Murdock, J. Wen, Y. Liu, eds., *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, SIGIR 2020, Virtual Event, China, July 25-30, 2020*, pages 1805–1808. ACM, 2020.

[303] Huang, L., W. Wang, J. Chen, et al. Attention on attention for image captioning. In *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*, pages 4633–4642. IEEE, 2019.

[304] Pan, Y., T. Yao, Y. Li, et al. X-linear attention networks for image captioning. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*, pages 10968–10977. Computer Vision Foundation / IEEE, 2020.

[305] Cornia, M., M. Stefanini, L. Baraldi, et al. M²: Meshed-memory transformer for image captioning. *CoRR*, abs/1912.08226, 2019.

[306] Chen, J., H. Guo, K. Yi, et al. Visualgpt: Data-efficient image captioning by balancing visual input and linguistic knowledge from pretraining. *CoRR*, abs/2102.10407, 2021.

[307] Li, K., Y. He, Y. Wang, et al. Videochat: Chat-centric video understanding. *CoRR*, abs/2305.06355, 2023.

Gong, T., C. Lyu, S. Zhang 等. Multimodal-gpt: 面向人类对话的视觉与语言模型。CoRR, abs/2305.04790, 2023年。 Alayrac, J., J. Donahue, P. Luc 等. Flamingo: 一种用于少样本学习的视觉语言模型。NeurIPS, 2022年。 Su, Y., T. Lan, H. Li 等. Pandagpt: 一款通用指令遵循模型。CoRR, abs/2305.16355, 2023年。 Liu, H., C. Li, Q. Wu 等. 视觉指令调优。CoRR, abs/2304.08485, 2023年。 Huang, R., M. Li, D. Yang 等. Audiogpt: 理解与生成语音、音乐、声音及会话头。CoRR, abs/2304.12995, 2023年。 Gong, Y., Y. Chung, J. R. Glass. AST: 音频频谱变换器。在H. Hermansky, H. Cernocký, L. Burget, L. Lamel, O. Scharenborg, P. Motlíček编辑的《Interspeech 2021》, 第22届国际语音通信协会年会, 布尔诺, 捷克, 2021年8月30日-9月3日, 第571–575页。ISCA, 2021年。 Hsu, W., B. Bolte, Y. H. Tsai 等. Hubert: 通过隐藏单元的掩码预测进行自监督语音表征学习。IEEE ACM《音频语音语言处理》杂志, 29: 3451–3460, 2021年。 Chen, F., M. Han, H. Zhao 等. X-LLM: 通过将多模态视为外语来引导先进的大型语言模型。CoRR, abs/2305.04160, 2023年。 Zhang, H., X. Li, L. Bing. Video-llama: 一种面向视频理解的指令调优音视频语言模型。CoRR, abs/2306.02858, 2023年。 Liu, Z., Y. He, W. Wang 等. Interngpt: 通过超越语言的与聊天机器人交互解决视觉中心任务。CoRR, abs/2305.05662, 2023年。 Hubel, D. H., T. N. Wiesel. 猫视觉皮层中的感受野、双眼交互与功能结构。《生理学杂志》, 160(1):106, 1962年。 Logothetis, N. K., D. L. Sheinberg. 视觉对象识别。《神经科学年度回顾》, 19(1):577–621, 1996年。 OpenAI. OpenAI: 介绍ChatGPT。网站, 2022年。 <https://openai.com/blog/chatgpt>。 Lu, J., X. Ren, Y. Ren 等. 改善多轮对话中响应检索的上下文语言模型。J. X. Huang, Y. Chang, X. Cheng, J. Kamps, V. Murdock, J. Wen, Y. Liu (编). 第43届国际ACM SIGIR信息检索研究与开发会议论文集, SIGIR 2020, 虚拟会议, 中国, 2020年7月25-30日, 第1805–1808页。ACM, 2020年。 Huang, L., W. Wang, J. Chen 等. 图像字幕中的注意力机制。2019年IEEE/CVF国际计算机视觉会议 (ICCV 2019), 首尔, 韩国, 2019年10月27日-11月2日, 第4633–4642页。IEEE, 2019年。 Pan, Y., T. Yao, Y. Li 等. 用于图像字幕的X-linear注意力网络。2020年IEEE/CVF计算机视觉与模式识别会议 (CVPR 2020), 西雅图, 华盛顿, 美国, 2020年6月13-19日, 第10968–10977页。计算机视觉基金会/IEEE, 2020年。 Cornia, M., M. Stefanini, L. Baraldi 等. M²: 用于图像字幕的网格记忆变换器。CoRR, abs/1912.08226, 2019年。 Chen, J., H. Guo, K. Yi 等. Visualgpt: 通过平衡视觉输入与预训练的语言知识实现高效数据的图像字幕。CoRR, abs/2102.10407, 2021年。 Li, K., Y. He, Y. Wang 等. Videochat: 面向对话的在线视频理解。CoRR, abs/2305.06355, 2023年。

- [308] Lin, J., Y. Du, O. Watkins, et al. Learning to model the world with language. *CoRR*, abs/2308.01399, 2023.
- [309] Vaswani, A., N. Shazeer, N. Parmar, et al. Attention is all you need. In I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett, eds., *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pages 5998–6008. 2017.
- [310] Touvron, H., M. Cord, M. Douze, et al. Training data-efficient image transformers & distillation through attention. In M. Meila, T. Zhang, eds., *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, vol. 139 of *Proceedings of Machine Learning Research*, pages 10347–10357. PMLR, 2021.
- [311] Lu, J., C. Clark, R. Zellers, et al. UNIFIED-IO: A unified model for vision, language, and multi-modal tasks. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- [312] Peng, Z., W. Wang, L. Dong, et al. Kosmos-2: Grounding multimodal large language models to the world. *CoRR*, abs/2306.14824, 2023.
- [313] Lyu, C., M. Wu, L. Wang, et al. Macaw-llm: Multi-modal language modeling with image, audio, video, and text integration. *CoRR*, abs/2306.09093, 2023.
- [314] Maaz, M., H. A. Rasheed, S. H. Khan, et al. Video-chatgpt: Towards detailed video understanding via large vision and language models. *CoRR*, abs/2306.05424, 2023.
- [315] Chen, M., I. Laina, A. Vedaldi. Training-free layout control with cross-attention guidance. *CoRR*, abs/2304.03373, 2023.
- [316] Radford, A., J. W. Kim, T. Xu, et al. Robust speech recognition via large-scale weak supervision. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 28492–28518. PMLR, 2023.
- [317] Ren, Y., Y. Ruan, X. Tan, et al. Fastspeech: Fast, robust and controllable text to speech. In H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, R. Garnett, eds., *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pages 3165–3174. 2019.
- [318] Ye, Z., Z. Zhao, Y. Ren, et al. Syntaspeech: Syntax-aware generative adversarial text-to-speech. In L. D. Raedt, ed., *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI 2022, Vienna, Austria, 23-29 July 2022*, pages 4468–4474. ijcai.org, 2022.
- [319] Kim, J., J. Kong, J. Son. Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech. In M. Meila, T. Zhang, eds., *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, vol. 139 of *Proceedings of Machine Learning Research*, pages 5530–5540. PMLR, 2021.
- [320] Wang, Z., S. Cornell, S. Choi, et al. Tf-gridnet: Integrating full- and sub-band modeling for speech separation. *IEEE ACM Trans. Audio Speech Lang. Process.*, 31:3221–3236, 2023.
- [321] Liu, J., C. Li, Y. Ren, et al. Diffsinger: Singing voice synthesis via shallow diffusion mechanism. In *Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, The Twelveth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022 Virtual Event, February 22 - March 1, 2022*, pages 11020–11028. AAAI Press, 2022.
- [322] Inaguma, H., S. Dalmia, B. Yan, et al. Fast-md: Fast multi-decoder end-to-end speech translation with non-autoregressive hidden intermediates. In *IEEE Automatic Speech Recognition and Understanding Workshop, ASRU 2021, Cartagena, Colombia, December 13-17, 2021*, pages 922–929. IEEE, 2021.

- [308] Lin, J., Y. Du, O. Watkins 等. 用语言学习建模世界。 *CoRR*, abs/2308.01399, 2023年。[309] Vaswani, A., N. Shazeer, N. Parmar 等. 注意力机制就是你所需要的一切。收录于 I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett 编., 神经信息处理系统进展30: 神经信息处理系统年度会议2017, 12月4-9日, 长滩, 加利福尼亚, 美国, 第5998–6008页, 2017年。[310] Touvron, H., M. Cord, M. Douze 等. 通过注意力机制训练数据高效的图像变换器与蒸馏。收录于 M. Meila, T. Zhang 编., 第38届国际机器学习会议论文集, *ICML 2021, 2021年7月18-24日*, 虚拟会议, 第139卷机器学习研究论文集, 第10347–10357页。PMLR, 2021年。[311] Lu, J., C. Clark, R. Zellers 等. UNIFIED-IO: 一种用于视觉、语言及多模态任务的统一模型。收录于 第十一届学习表征国际会议, *ICLR 2023, 卢旺达基加利, 2023年5月1-5日*。OpenReview.net, 2023年。[312] Peng, Z., W. Wang, L. Dong 等. Kosmos-2: 将多模态大语言模型与世界联系起来。 *CoRR*, abs/2306.14824, 2023年。[313] Lyu, C., M. Wu, L. Wang 等. Macaw-llm: 结合图像、音频、视频与文本的多模态语言建模。 *CoRR*, abs/2306.09093, 2023年。[314] Maaz, M., H. A. Rasheed, S. H. Khan 等. Video-chatgpt: 通过大规模视觉与语言模型实现详细视频理解。 *CoRR*, abs/2306.05424, 2023年。[315] Chen, M., I. Laina, A. Vedaldi. 利用交叉注意力引导实现无训练布局控制。 *CoRR*, abs/2304.03373, 2023年。[316] Radford, A., J. W. Kim, T. Xu 等. 通过大规模弱监督实现鲁棒语音识别。收录于 A. Krause, E. Brunskill, K. Choi, B. Engelhardt, S. Sabato, J. Scarlett 编., 第38届国际机器学习会议论文集, *ICML 2023, 2023年7月23-29日*, 夏威夷火奴鲁鲁, 美国, 第202卷机器学习研究论文集, 第28492–28518页。PMLR, 2023年。[317] Ren, Y., Y. Ruan, X. Tan 等. Fastspeech: 快速、鲁棒且可控的文本转语音。收录于 H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, R. Garnett 编., 神经信息处理系统进展32: 神经信息处理系统年度会议2019, *NeurIPS 2019, 2019年12月8-14日*, 加拿大温哥华, 第3165–3174页。2019年。[318] Ye, Z., Z. Zhao, Y. Ren 等. Syntaspeech: 语法感知的生成对抗文本转语音。收录于 L. D. Raedt 编., 第31届国际人工智能联合会议论文集, *IJCAI 2022, 奥地利维也纳, 2022年7月23-29日*, 第4468–4474页。ijcai.org, 2022年。[319] Kim, J., J. Kong, J. Son. 条件变分自编码器结合对抗学习的端到端文本转语音。收录于 M. Meila, T. Zhang 编., 第38届国际机器学习会议论文集, *ICML 2021, 2021年7月18-24日*, 虚拟会议, 第139卷机器学习研究论文集, 第5530–5540页。PMLR, 2021年。[320] Wang, Z., S. Cornell, S. Choi 等. Tf-gridnet: 整合全频段与子频段建模的语音分离方法。 *IEEE ACM 语音、语言与音频处理杂志*, 第31卷, 3221–3236页, 2023年。[321] Liu, J., C. Li, Y. Ren 等. Diffsinger: 基于浅层扩散机制的歌声合成。收录于 第36届人工智能协会会议 (AAAI 2022)、第34届人工智能创新应用会议 (IAAI 2022)、第12届人工智能教育进展研讨会 (EAAI 2022) 虚拟会议, 2022年2月22日-3月1日, 第11020–11028页。AAAI出版社, 2022年。[322] Inaguma, H., S. Dalmia, B. Yan 等. Fast-md: 利用非自回归隐藏中间体实现快速多解码端到端语音翻译。收录于 *IEEE 自动语音识别与理解研讨会, ASRU 2021, 哥伦比亚卡塔赫纳, 2021年12月13-17日*, 第922–929页。IEEE, 2021年。

[323] Flanagan, J. L. *Speech analysis synthesis and perception*, vol. 3. Springer Science & Business Media, 2013.

[324] Schwarz, B. Mapping the world in 3d. *Nature Photonics*, 4(7):429–430, 2010.

[325] Parkinson, B. W., J. J. Spilker. *Progress in astronautics and aeronautics: Global positioning system: Theory and applications*, vol. 164. Aiaa, 1996.

[326] Parisi, A., Y. Zhao, N. Fiedel. TALM: tool augmented language models. *CoRR*, abs/2205.12255, 2022.

[327] Clarebout, G., J. Elen, N. A. J. Collazo, et al. *Metacognition and the Use of Tools*, pages 187–195. Springer New York, New York, NY, 2013.

[328] Wu, C., S. Yin, W. Qi, et al. Visual chatgpt: Talking, drawing and editing with visual foundation models. *CoRR*, abs/2303.04671, 2023.

[329] Cai, T., X. Wang, T. Ma, et al. Large language models as tool makers. *CoRR*, abs/2305.17126, 2023.

[330] Qian, C., C. Han, Y. R. Fung, et al. CREATOR: disentangling abstract and concrete reasonings of large language models through tool creation. *CoRR*, abs/2305.14318, 2023.

[331] Chen, X., M. Lin, N. Schärli, et al. Teaching large language models to self-debug. *CoRR*, abs/2304.05128, 2023.

[332] Liu, H., L. Lee, K. Lee, et al. Instruction-following agents with jointly pre-trained vision-language models. *arXiv preprint arXiv:2210.13431*, 2022.

[333] Lynch, C., A. Wahid, J. Tompson, et al. Interactive language: Talking to robots in real time. *CoRR*, abs/2210.06407, 2022.

[334] Jin, C., W. Tan, J. Yang, et al. Alphablock: Embodied finetuning for vision-language reasoning in robot manipulation. *CoRR*, abs/2305.18898, 2023.

[335] Shah, D., B. Osinski, B. Ichter, et al. Lm-nav: Robotic navigation with large pre-trained models of language, vision, and action. In K. Liu, D. Kulic, J. Ichnowski, eds., *Conference on Robot Learning, CoRL 2022, 14-18 December 2022, Auckland, New Zealand*, vol. 205 of *Proceedings of Machine Learning Research*, pages 492–504. PMLR, 2022.

[336] Zhou, G., Y. Hong, Q. Wu. Navgpt: Explicit reasoning in vision-and-language navigation with large language models. *CoRR*, abs/2305.16986, 2023.

[337] Fan, L., G. Wang, Y. Jiang, et al. Minedojo: Building open-ended embodied agents with internet-scale knowledge. In *NeurIPS*. 2022.

[338] Kanitscheider, I., J. Huizinga, D. Farhi, et al. Multi-task curriculum learning in a complex, visual, hard-exploration domain: Minecraft. *CoRR*, abs/2106.14876, 2021.

[339] Nottingham, K., P. Ammanabrolu, A. Suhr, et al. Do embodied agents dream of pixelated sheep: Embodied decision making using language guided world modelling. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 26311–26325. PMLR, 2023.

[340] Sumers, T., K. Marino, A. Ahuja, et al. Distilling internet-scale vision-language models into embodied agents. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 32797–32818. PMLR, 2023.

[341] Carlini, N., J. Hayes, M. Nasr, et al. Extracting training data from diffusion models. *CoRR*, abs/2301.13188, 2023.

[323] Flanagan, J. L. 语音分析合成与感知，第3卷。Springer Science & Business Media, 2013年。[324] Schwarz, B. 在3D中映射世界。《自然光子学》，4(7):429–430, 2010年。[325] Parkinson, B. W., J. J. Spilker. 航天与航空进展：全球定位系统：理论与应用，第164卷。Aiaa, 1996年。[326]Parisi, A., Y. Zhao, N. Fiedel. TALM：工具增强语言模型。CoRR, abs/2205.12255, 2022年。[327] Clarebout, G., J. Elen, N. A. J. Collazo 等. 元认知与工具的使用，第187–195页。Springer New York, 2013年。[328] Wu, C., S. Yin, W. Qi 等. Visual ChatGPT：利用视觉基础模型进行对话、绘图与编辑。CoRR, abs/2303.04671, 2023年。[329] Cai, T., X. Wang, T. Ma 等. 大型语言模型作为工具制造者。CoRR, abs/2305.17126, 2023年。[330] Qian, C., C. Han, Y. R. Fung 等. CREATOR：通过工具创建解开大型语言模型的抽象与具体推理。CoRR, abs/2305.14318, 2023年。[331] Chen, X., M. Lin, N. Schärli 等. 教授大型语言模型自我调试。CoRR, abs/2304.05128, 2023年。[332] Liu, H., L. Lee, K. Lee 等. 结合预训练视觉-语言模型的指令遵循代理。arXiv预印本 arXiv:2210.13431, 2022年。[333] Lynch, C., A. Wahid, J. Tompson 等. 交互式语言：实时与机器人对话。CoRR, abs/2210.06407, 2022年。[334] Jin, C., W. Tan, J. Yang 等. Alphablock：用于机器人操作中视觉-语言推理的体现微调。CoRR, abs/2305.18898, 2023年。[335] Shah, D., B. Osinski, B. Ichter 等. Lm-nav：利用大型预训练模型进行机器人导航，涵盖语言、视觉与行动。在K. Liu, D. Kulic, J. Ichnowski编辑的《机器人学习会议》，CoRL 2022, 2022年12月14-18日，新西兰奥克兰，第205卷《机器学习研究会论文集》，第492–504页。PMLR, 2022年。[336] Zhou, G., Y. Hong, Q. Wu. Navgpt：在视觉与语言导航中利用大型语言模型进行显式推理。CoRR, abs/2305.16986, 2023年。[337] Fan, L., G. Wang, Y. Jiang 等. Minedojo：构建具有互联网规模知识的开放式体现代理。在NeurIPS会议，2022年。[338] Kanitscheider, I., J. Huizinga, D. Farhi 等. 在复杂的视觉难探索领域：Minecraft中的多任务课程学习。CoRR, abs/2106.14876, 2021年。[339] Nottingham, K., P. Ammanabrolu, A. Suhr 等. 体现代理会梦见像素羊吗：利用语言引导的世界建模进行体现决策。在A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett编辑的《国际机器学习会议》，ICML 2023, 2023年7月23-29日，夏威夷檀香山，第202卷《机器学习研究会论文集》，第26311–26325页。PMLR, 2023年。[340] Sumers, T., K. Marino, A. Ahuja 等. 将互联网规模的视觉-语言模型蒸馏成体现代理。在A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett编辑的《国际机器学习会议》，ICML 2023, 2023年7月23-29日，夏威夷檀香山，第202卷《机器学习研究会论文集》，第32797–32818页。PMLR, 2023年。[341] Carlini, N., J. Hayes, M. Nasr 等. 从扩散模型中提取训练数据。CoRR, abs/2301.13188, 2023年。

- [342] Savelka, J., K. D. Ashley, M. A. Gray, et al. Can GPT-4 support analysis of textual data in tasks requiring highly specialized domain expertise? In F. Lagioia, J. Mumford, D. Odekerken, H. Westermann, eds., *Proceedings of the 6th Workshop on Automated Semantic Analysis of Information in Legal Text co-located with the 19th International Conference on Artificial Intelligence and Law (ICAIL 2023)*, Braga, Portugal, 23rd September, 2023, vol. 3441 of *CEUR Workshop Proceedings*, pages 1–12. CEUR-WS.org, 2023.
- [343] Ling, C., X. Zhao, J. Lu, et al. Domain specialization as the key to make large language models disruptive: A comprehensive survey, 2023.
- [344] Linardatos, P., V. Papastefanopoulos, S. Kotsiantis. Explainable AI: A review of machine learning interpretability methods. *Entropy*, 23(1):18, 2021.
- [345] Zou, A., Z. Wang, J. Z. Kolter, et al. Universal and transferable adversarial attacks on aligned language models. *CoRR*, abs/2307.15043, 2023.
- [346] Hussein, A., M. M. Gaber, E. Elyan, et al. Imitation learning: A survey of learning methods. *ACM Comput. Surv.*, 50(2):21:1–21:35, 2017.
- [347] Liu, Y., A. Gupta, P. Abbeel, et al. Imitation from observation: Learning to imitate behaviors from raw video via context translation. In *2018 IEEE International Conference on Robotics and Automation, ICRA 2018, Brisbane, Australia, May 21-25, 2018*, pages 1118–1125. IEEE, 2018.
- [348] Baker, B., I. Akkaya, P. Zhokov, et al. Video pretraining (VPT): learning to act by watching unlabeled online videos. In *NeurIPS*. 2022.
- [349] Levine, S., P. Pastor, A. Krizhevsky, et al. Learning hand-eye coordination for robotic grasping with deep learning and large-scale data collection. *Int. J. Robotics Res.*, 37(4-5):421–436, 2018.
- [350] Zheng, R., S. Dou, S. Gao, et al. Secrets of RLHF in large language models part I: PPO. *CoRR*, abs/2307.04964, 2023.
- [351] Bengio, Y., J. Louradour, R. Collobert, et al. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning, ICML '09*, page 41–48. Association for Computing Machinery, New York, NY, USA, 2009.
- [352] Chen, M., J. Tworek, H. Jun, et al. Evaluating large language models trained on code, 2021.
- [353] Pan, S., L. Luo, Y. Wang, et al. Unifying large language models and knowledge graphs: A roadmap. *CoRR*, abs/2306.08302, 2023.
- [354] Bran, A. M., S. Cox, A. D. White, et al. Chemcrow: Augmenting large-language models with chemistry tools, 2023.
- [355] Ruan, J., Y. Chen, B. Zhang, et al. TPTU: task planning and tool usage of large language model-based AI agents. *CoRR*, abs/2308.03427, 2023.
- [356] Ogundare, O., S. Madasu, N. Wiggins. Industrial engineering with large language models: A case study of chatgpt’s performance on oil & gas problems, 2023.
- [357] Smith, L., M. Gasser. The development of embodied cognition: Six lessons from babies. *Artificial life*, 11(1-2):13–29, 2005.
- [358] Duan, J., S. Yu, H. L. Tan, et al. A survey of embodied AI: from simulators to research tasks. *IEEE Trans. Emerg. Top. Comput. Intell.*, 6(2):230–244, 2022.
- [359] Mnih, V., K. Kavukcuoglu, D. Silver, et al. Playing atari with deep reinforcement learning. *CoRR*, abs/1312.5602, 2013.
- [360] Silver, D., A. Huang, C. J. Maddison, et al. Mastering the game of go with deep neural networks and tree search. *Nat.*, 529(7587):484–489, 2016.

- [361] Kalashnikov, D., A. Irpan, P. Pastor, et al. Qt-opt: Scalable deep reinforcement learning for vision-based robotic manipulation. *CoRR*, abs/1806.10293, 2018.
- [362] Nguyen, H., H. M. La. Review of deep reinforcement learning for robot manipulation. In *3rd IEEE International Conference on Robotic Computing, IRC 2019, Naples, Italy, February 25-27, 2019*, pages 590–595. IEEE, 2019.
- [363] Dasgupta, I., C. Kaeser-Chen, K. Marino, et al. Collaborating with language models for embodied reasoning. *CoRR*, abs/2302.00763, 2023.
- [364] Puig, X., K. Ra, M. Boben, et al. Virtualhome: Simulating household activities via programs. In *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*, pages 8494–8502. Computer Vision Foundation / IEEE Computer Society, 2018.
- [365] Hong, Y., Q. Wu, Y. Qi, et al. A recurrent vision-and-language BERT for navigation. *CoRR*, abs/2011.13922, 2020.
- [366] Suglia, A., Q. Gao, J. Thomason, et al. Embodied BERT: A transformer model for embodied, language-guided visual task completion. *CoRR*, abs/2108.04927, 2021.
- [367] Ganesh, S., N. Vadori, M. Xu, et al. Reinforcement learning for market making in a multi-agent dealer market. *CoRR*, abs/1911.05892, 2019.
- [368] Tipaldi, M., R. Iervolino, P. R. Massenio. Reinforcement learning in spacecraft control applications: Advances, prospects, and challenges. *Annu. Rev. Control.*, 54:1–23, 2022.
- [369] Savva, M., J. Malik, D. Parikh, et al. Habitat: A platform for embodied AI research. In *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*, pages 9338–9346. IEEE, 2019.
- [370] Longpre, S., L. Hou, T. Vu, et al. The flan collection: Designing data and methods for effective instruction tuning. *arXiv preprint arXiv:2301.13688*, 2023.
- [371] Wang, Y., Y. Kordi, S. Mishra, et al. Self-instruct: Aligning language model with self generated instructions. *arXiv preprint arXiv:2212.10560*, 2022.
- [372] Liang, J., W. Huang, F. Xia, et al. Code as policies: Language model programs for embodied control. In *IEEE International Conference on Robotics and Automation, ICRA 2023, London, UK, May 29 - June 2, 2023*, pages 9493–9500. IEEE, 2023.
- [373] Li, C., F. Xia, R. Martín-Martín, et al. HRL4IN: hierarchical reinforcement learning for interactive navigation with mobile manipulators. In L. P. Kaelbling, D. Kragic, K. Sugiura, eds., *3rd Annual Conference on Robot Learning, CoRL 2019, Osaka, Japan, October 30 - November 1, 2019, Proceedings*, vol. 100 of *Proceedings of Machine Learning Research*, pages 603–616. PMLR, 2019.
- [374] Eppe, M., C. Gumbsch, M. Kerzel, et al. Hierarchical principles of embodied reinforcement learning: A review. *CoRR*, abs/2012.10147, 2020.
- [375] Paul, S., A. Roy-Chowdhury, A. Cherian. AVLEN: audio-visual-language embodied navigation in 3d environments. In *NeurIPS*. 2022.
- [376] Hu, B., C. Zhao, P. Zhang, et al. Enabling intelligent interactions between an agent and an LLM: A reinforcement learning approach. *CoRR*, abs/2306.03604, 2023.
- [377] Chen, C., U. Jain, C. Schissler, et al. Soundspaces: Audio-visual navigation in 3d environments. In A. Vedaldi, H. Bischof, T. Brox, J. Frahm, eds., *Computer Vision - ECCV 2020 - 16th European Conference, Glasgow, UK, August 23-28, 2020, Proceedings, Part VI*, vol. 12351 of *Lecture Notes in Computer Science*, pages 17–36. Springer, 2020.
- [378] Huang, R., Y. Ren, J. Liu, et al. Generspeech: Towards style transfer for generalizable out-of-domain text-to-speech. In *NeurIPS*. 2022.

- [361] Kalashnikov, D., A. Irpan, P. Pastor 等. Qt-opt: 面向视觉机器人操作的可扩展深度强化学习. *CoRR*, abs/1806.10293, 2018年. [362] Nguyen, H., H. M. La. 深度强化学习在机器人操作中的综述. 收录于第3届IEEE机器人计算国际会议, *IRC 2019*, 意大利那不勒斯, 2019年2月25-27日, 第590-595页. IEEE, 2019年. [363] Dasgupta, I., C. Kaeser-Chen, K. Marino 等. 与语言模型协作进行具身推理. *CoRR*, abs/2302.00763, 2023年. [364] Puig, X., K. Ra, M. Boben 等. Virtualhome: 通过程序模拟家庭活动. 收录于2018年IEEE计算机视觉与模式识别会议, *CVPR 2018*, 美国犹他州盐湖城, 2018年6月18-22日, 第8494-8502页. 计算机视觉基金会 / IEEE计算机学会, 2018年. [365] Hong, Y., Q. Wu, Y. Qi 等. 一种用于导航的循环视觉与语言BERT. *CoRR*, abs/2011.13922, 2020年. [366] Suglia, A., Q. Gao, J. Thomason 等. 具身BERT: 一种用于具身、语言引导的视觉任务完成的变换器模型. *CoRR*, abs/2108.04927, 2021年. [367] Ganesh, S., N. Vadori, M. Xu 等. 多智能体交易市场中的市场制造强化学习. *CoRR*, abs/1911.05892, 2019年. [368] Tipaldi, M., R. Iervolino, P. R. Massenio. 航天器控制应用中的强化学习: 进展、前景与挑战. *Annu.Rev. Control.*, 54: 1-23, 2022年. [369] Savva, M., J. Malik, D. Parikh 等. Habitat: 一个用于具身AI研究的平台. 收录于2019年IEEE/CVF国际计算机视觉会议, *ICCV 2019*, 韩国首尔, 2019年10月27日至11月2日, 第9338-9346页. IEEE, 2019年. [370] Longpre, S., L. Hou, T. Vu 等. Flan系列: 设计有效指令调优的数据与方法. *arXiv预印本 arXiv:2301.13688*, 2023年. [371] Wang, Y., Y. Kordi, S. Mishra 等. 自我指导: 使语言模型与自生成指令对齐. *arXiv预印本 arXiv:2212.10560*, 2022年. [372] Liang, J., W. Huang, F. Xia 等. 代码作为策略: 用于具身控制的语言模型程序. 收录于IEEE国际机器人与自动化会议, *ICRA 2023*, 英国伦敦, 2023年5月29日至6月2日, 第9493-9500页. IEEE, 2023年. [373] Li, C., F. Xia, R. Martín-Martín 等. HRL4IN: 用于移动操作器交互导航的层次强化学习. 收录于L. P. Kaelbling, D. Kragic, K. Sugiura 编著的第3届机器人学习年会, *CoRL 2019*, 日本大阪, 2019年10月30日至11月1日, 论文集, 第100卷机器学习研究论文集, 第603-616页. PMLR, 2019年. [374] Eppe, M., C. Gumbsch, M. Kerzel 等. 具身强化学习的层次原则: 综述. *CoRR*, abs/2012.10147, 2020年. [375] Paul, S., A. Roy-Chowdhury, A. Cherian. AVLEN: 多模态音视频语言具身导航在3D环境中的应用. 收录于*NeurIPS*, 2022年. [376] Hu, B., C. Zhao, P. Zhang 等. 实现智能体与大型语言模型之间的交互: 一种强化学习方法. *CoRR*, abs/2306.03604, 2023年. [377] Chen, C., U. Jain, C. Schissler 等. Soundspaces: 3D环境中的音视频导航. 收录于A. Vedaldi, H. Bischof, T. Brox, J. Frahm 编著的计算机视觉 - *ECCV 2020* - 第16届欧洲会议, 格拉斯哥, 英国, 2020年8月23-28日, 论文集第六部分, 第12351卷计算机科学讲义, 第17-36页. Springer, 2020年. [378] Huang, R., Y. Ren, J. Liu 等. Generspeech: 面向通用域外文本到语音风格迁移的研究. 收录于*NeurIPS*, 2022年.

[379] Shah, D., B. Eysenbach, G. Kahn, et al. Ving: Learning open-world navigation with visual goals. In *IEEE International Conference on Robotics and Automation, ICRA 2021, Xi'an, China, May 30 - June 5, 2021*, pages 13215–13222. IEEE, 2021.

[380] Huang, C., O. Mees, A. Zeng, et al. Visual language maps for robot navigation. In *IEEE International Conference on Robotics and Automation, ICRA 2023, London, UK, May 29 - June 2, 2023*, pages 10608–10615. IEEE, 2023.

[381] Georgakis, G., K. Schmeckpeper, K. Wanchoo, et al. Cross-modal map learning for vision and language navigation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pages 15439–15449. IEEE, 2022.

[382] Dorbala, V. S., J. F. M. Jr., D. Manocha. Can an embodied agent find your "cat-shaped mug"? llm-based zero-shot object navigation. *CoRR*, abs/2303.03480, 2023.

[383] Li, L. H., P. Zhang, H. Zhang, et al. Grounded language-image pre-training. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pages 10955–10965. IEEE, 2022.

[384] Gan, C., Y. Zhang, J. Wu, et al. Look, listen, and act: Towards audio-visual embodied navigation. In *2020 IEEE International Conference on Robotics and Automation, ICRA 2020, Paris, France, May 31 - August 31, 2020*, pages 9701–9707. IEEE, 2020.

[385] Brohan, A., N. Brown, J. Carbajal, et al. RT-1: robotics transformer for real-world control at scale. *CoRR*, abs/2212.06817, 2022.

[386] —. RT-2: vision-language-action models transfer web knowledge to robotic control. *CoRR*, abs/2307.15818, 2023.

[387] PrismarineJS, 2013.

[388] Gur, I., H. Furuta, A. Huang, et al. A real-world webagent with planning, long context understanding, and program synthesis. *CoRR*, abs/2307.12856, 2023.

[389] Deng, X., Y. Gu, B. Zheng, et al. Mind2web: Towards a generalist agent for the web. *CoRR*, abs/2306.06070, 2023.

[390] Furuta, H., O. Nachum, K. Lee, et al. Multimodal web navigation with instruction-finetuned foundation models. *CoRR*, abs/2305.11854, 2023.

[391] Zhou, S., F. F. Xu, H. Zhu, et al. Webarena: A realistic web environment for building autonomous agents. *CoRR*, abs/2307.13854, 2023.

[392] Yao, S., H. Chen, J. Yang, et al. Webshop: Towards scalable real-world web interaction with grounded language agents. In *NeurIPS*. 2022.

[393] Kim, G., P. Baldi, S. McAleer. Language models can solve computer tasks. *CoRR*, abs/2303.17491, 2023.

[394] Zheng, L., R. Wang, B. An. Synapse: Leveraging few-shot exemplars for human-level computer control. *CoRR*, abs/2306.07863, 2023.

[395] Chen, P., C. Chang. Interact: Exploring the potentials of chatgpt as a cooperative agent. *CoRR*, abs/2308.01552, 2023.

[396] Gramopadhye, M., D. Szafir. Generating executable action plans with environmentally-aware language models. *CoRR*, abs/2210.04964, 2022.

[397] Li, H., Y. Hao, Y. Zhai, et al. The hitchhiker’s guide to program analysis: A journey with large language models. *CoRR*, abs/2308.00245, 2023.

[398] Feldt, R., S. Kang, J. Yoon, et al. Towards autonomous testing agents via conversational large language models. *CoRR*, abs/2306.05152, 2023.

[379] Shah, D., B. Eysenbach, G. Kahn 等. Ving: 基于视觉目标的开放世界导航学习。发表于*IEEE*国际机器人与自动化会议, *ICRA 2021*, 中国西安, 2021年5月30日-6月5日, 第13215–13222页。IEEE, 2021年。[380] Huang, C., O. Mees, A. Zeng 等. 用于机器人导航的视觉语言地图。发表于*IEEE*国际机器人与自动化会议, *ICRA 2023*, 英国伦敦, 2023年5月29日-6月2日, 第10608–10615页。IEEE, 2023年。[381] Georgakis, G., K. Schmeckpeper, K. Wanchoo 等. 跨模态地图学习用于视觉与语言导航。发表于*IEEE/CVF*计算机视觉与模式识别会议, *CVPR 2022*, 美国新奥尔良, 2022年6月18-24日, 第15439–15449页。IEEE, 2022年。[382] Dorbala, V. S., J. F. M. Jr., D. Manocha. 具象代理能找到你的“猫形马克杯”吗? 基于大模型的零样本目标导航。 *CoRR*, abs/2303.03480, 2023年。[383] Li, L. H., P. Zhang, H. Zhang 等. 基于地面语言-图像预训练。发表于*IEEE/CVF*计算机视觉与模式识别会议, *CVPR 2022*, 美国新奥尔良, 2022年6月18-24日, 第10955–10965页。IEEE, 2022年。[384] Gan, C., Y. Zhang, J. Wu 等. 看、听、行动: 迈向音视频具象导航。发表于2020 *IEEE*国际机器人与自动化会议, *ICRA 2020*, 法国巴黎, 2020年5月31日-8月31日, 第9701–9707页。IEEE, 2020年。[385] Brohan, A., N. Brown, J. Carbajal 等. RT-1: 用于大规模真实世界控制的机器人变换器。 *CoRR*, abs/2212.06817, 2022年。[386] —. RT-2: 将视觉-语言-动作模型迁移到机器人控制的网络知识。 *CoRR*, abs/2307.15818, 2023年。[387] PrismarineJS, 2013年。[388] Gur, I., H. Furuta, A. Huang 等. 具有规划、长上下文理解和程序合成能力的真实世界Web代理。 *CoRR*, abs/2307.12856, 2023年。[389] Deng, X., Y. Gu, B. Zheng 等. Mind2web: 迈向通用Web代理。 *CoRR*, abs/2306.06070, 2023年。[390] Furuta, H., O. Nachum, K. Lee 等. 通过指令微调基础模型实现多模态Web导航。 *CoRR*, abs/2305.11854, 2023年。[391] Zhou, S., F. F. Xu, H. Zhu 等. Webarena: 用于构建自主代理的逼真Web环境。 *CoRR*, abs/2307.13854, 2023年。[392] Yao, S., H. Chen, J. Yang 等. Webshop: 迈向具有基础语言代理的可扩展真实Web交互。发表于*NeurIPS*, 2022年。[393] Kim, G., P. Baldi, S. McAleer. 语言模型可以解决计算机任务。 *CoRR*, abs/2303.17491, 2023年。[394] Zheng, L., R. Wang, B. An. Synapse: 利用少样本示例实现人类水平的计算机控制。 *CoRR*, abs/2306.07863, 2023年。[395] Chen, P., C. Chang. Interact: 探索ChatGPT作为合作代理的潜力。 *CoRR*, abs/2308.01552, 2023年。[396] Gramopadhye, M., D. Szafir. 利用环境感知的语言模型生成可执行行动计划。 *CoRR*, abs/2210.04964, 2022年。[397] Li, H., Y. Hao, Y. Zhai 等. 搭车者指南: 与大模型一同进行程序分析的旅程。 *CoRR*, abs/2308.00245, 2023年。[398] Feldt, R., S. Kang, J. Yoon 等. 通过对话式大模型迈向自主测试代理。 *CoRR*, abs/2306.05152, 2023年。

- [399] Kang, Y., J. Kim. Chatmof: An autonomous AI system for predicting and generating metal-organic frameworks. *CoRR*, abs/2308.01423, 2023.
- [400] Wang, R., P. A. Jansen, M. Côté, et al. Scienceworld: Is your agent smarter than a 5th grader? In Y. Goldberg, Z. Kozareva, Y. Zhang, eds., *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, pages 11279–11298. Association for Computational Linguistics, 2022.
- [401] Yuan, H., C. Zhang, H. Wang, et al. Plan4mc: Skill reinforcement learning and planning for open-world minecraft tasks. *CoRR*, abs/2303.16563, 2023.
- [402] Hao, R., L. Hu, W. Qi, et al. Chatllm network: More brains, more intelligence. *CoRR*, abs/2304.12998, 2023.
- [403] Mandi, Z., S. Jain, S. Song. Roco: Dialectic multi-robot collaboration with large language models. *CoRR*, abs/2307.04738, 2023.
- [404] Hamilton, S. Blind judgement: Agent-based supreme court modelling with GPT. *CoRR*, abs/2301.05327, 2023.
- [405] Hong, S., X. Zheng, J. Chen, et al. Metagpt: Meta programming for multi-agent collaborative framework. *CoRR*, abs/2308.00352, 2023.
- [406] Wu, Q., G. Bansal, J. Zhang, et al. Autogen: Enabling next-gen LLM applications via multi-agent conversation framework. *CoRR*, abs/2308.08155, 2023.
- [407] Zhang, C., K. Yang, S. Hu, et al. Proagent: Building proactive cooperative AI with large language models. *CoRR*, abs/2308.11339, 2023.
- [408] Nair, V., E. Schumacher, G. J. Tso, et al. DERA: enhancing large language model completions with dialog-enabled resolving agents. *CoRR*, abs/2303.17071, 2023.
- [409] Talebirad, Y., A. Nadiri. Multi-agent collaboration: Harnessing the power of intelligent LLM agents. *CoRR*, abs/2306.03314, 2023.
- [410] Chen, W., Y. Su, J. Zuo, et al. Agentverse: Facilitating multi-agent collaboration and exploring emergent behaviors in agents. *CoRR*, abs/2308.10848, 2023.
- [411] Shi, J., J. Zhao, Y. Wang, et al. CGMI: configurable general multi-agent interaction framework. *CoRR*, abs/2308.12503, 2023.
- [412] Xiong, K., X. Ding, Y. Cao, et al. Examining the inter-consistency of large language models: An in-depth analysis via debate. *CoRR*, abs/2305.11595, 2023.
- [413] Kalvakurthi, V., A. S. Varde, J. Jenq. Hey dona! can you help me with student course registration? *CoRR*, abs/2303.13548, 2023.
- [414] Swan, M., T. Kido, E. Roland, et al. Math agents: Computational infrastructure, mathematical embedding, and genomics. *CoRR*, abs/2307.02502, 2023.
- [415] Hsu, S.-L., R. S. Shah, P. Senthil, et al. Helping the helper: Supporting peer counselors via ai-empowered practice and feedback. *arXiv preprint arXiv:2305.08982*, 2023.
- [416] Zhang, H., J. Chen, F. Jiang, et al. Huatuogpt, towards taming language model to be a doctor. *CoRR*, abs/2305.15075, 2023.
- [417] Yang, S., H. Zhao, S. Zhu, et al. Zhongjing: Enhancing the chinese medical capabilities of large language model through expert feedback and real-world multi-turn dialogue. *CoRR*, abs/2308.03549, 2023.
- [418] Ali, M. R., S. Z. Razavi, R. Langevin, et al. A virtual conversational agent for teens with autism spectrum disorder: Experimental results and design lessons. In S. Marsella, R. Jack, H. H. Vilhjálmsón, P. Sequeira, E. S. Cross, eds., *IVA ’20: ACM International Conference on Intelligent Virtual Agents, Virtual Event, Scotland, UK, October 20-22, 2020*, pages 2:1–2:8. ACM, 2020.

康Y., J. Kim. Chatmof: 一种用于预测和生成金属有机框架的自主AI系统。 *CoRR*, abs/2308.01423, 2023年。 王R., P. A. Jansen, M. Côté等。Scienceworld: 你的智能体比五年级学生还聪明吗? 载于Y. Goldberg, Z. Kozareva, Y. Zhang编辑的《2022年自然语言处理实证方法会议论文集, EMNLP 2022, 阿布扎比, 阿联酋, 2022年12月7-11日》, 第11279–11298页。计算语言学协会, 2022年。 袁H., C. Zhang, H. Wang等。Plan4mc: 面向开放世界Minecraft任务的技能强化学习与规划。 *CoRR*, abs/2303.16563, 2023年。 郝R., 胡L., 祁W.等。Chatllm网络: 更多大脑, 更多智能。 *CoRR*, abs/2304.12998, 2023年。 曼迪Z., S. Jain, S. Song。Roco: 基于大语言模型的辩证多机器人协作。 *CoRR*, abs/2307.04738, 2023年。 汉密尔顿S.。盲判: 基于代理的最高法院建模与GPT。 *CoRR*, abs/2301.05327, 2023年。 洪S., 郑X., 陈J.等。Metagpt: 多智能体协作框架的元编程。 *CoRR*, abs/2308.00352, 2023年。 吴Q., Bansal G., Zhang J.等。Autogen: 通过多智能体对话框架实现下一代大语言模型应用。 *CoRR*, abs/2308.08155, 2023年。 张C., Yang K., Hu S.等。Proagent: 构建具有主动合作能力的AI大模型。 *CoRR*, abs/2308.11339, 2023年。 奈尔V., Schumacher E., Tso G. J.等。DERA: 通过对话启用的解决代理增强大语言模型的完成能力。 *CoRR*, abs/2303.17071, 2023年。 Talebirad Y., Nadiri A.。多智能体协作: 发挥智能大语言模型代理的力量。 *CoRR*, abs/2306.03314, 2023年。 陈W., 苏Y., 左J.等。Agentverse: 促进多智能体协作与探索代理中的涌现行为。 *CoRR*, abs/2308.10848, 2023年。 史J., 赵J., 王Y.等。CGMI: 可配置的通用多智能体交互框架。 *CoRR*, abs/2308.12503, 2023年。 熊K., 丁X., 曹Y.等。考察大语言模型的内部一致性: 通过辩论进行深入分析。 *CoRR*, abs/2305.11595, 2023年。 Kalvakurthi V., Varde A. S., Jenq J.。嘿多娜! 你能帮我处理学生选课吗? *CoRR*, abs/2303.13548, 2023年。 斯旺M., Kido T., Roland E.等。数学代理: 计算基础设施、数学嵌入与基因组学。 *CoRR*, abs/2307.02502, 2023年。 许S.-L., Shah R. S., Senthil P.等。帮助助手: 通过AI赋能的实践与反馈支持同伴辅导。 *arXivpreprint arXiv:2305.08982*, 2023年。 张H., 陈J., Jiang F.等。Huatuogpt: 向驯服语言模型迈进, 成为医生。 *CoRR*, abs/2305.15075, 2023年。 杨S., 赵H., Zhu S.等。Zhongjing: 通过专家反馈和真实多轮对话提升大语言模型的中医能力。 *CoRR*, abs/2308.03549, 2023年。 Ali M. R., Razavi S. Z., Langevin R.等。青少年自闭症谱系障碍虚拟对话代理: 实验结果与设计经验。载于S. Marsella, R. Jack, H. H. Vilhjálmsón, P. Sequeira, E. S. Cross编辑的《IVA ’20: ACM国际智能虚拟代理会议, 虚拟会议, 苏格兰, 英国, 2020年10月20-22日》, 第2:1–2:8页。ACM, 2020年。

[419] Gao, W., X. Gao, Y. Tang. Multi-turn dialogue agent as sales’ assistant in telemarketing. In *International Joint Conference on Neural Networks, IJCNN 2023, Gold Coast, Australia, June 18-23, 2023*, pages 1–9. IEEE, 2023.

[420] Schick, T., J. A. Yu, Z. Jiang, et al. PEER: A collaborative language model. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.

[421] Lu, B., N. Haduong, C. Lee, et al. DIALGEN: collaborative human-lm generated dialogues for improved understanding of human-human conversations. *CoRR*, abs/2307.07047, 2023.

[422] Gao, D., L. Ji, L. Zhou, et al. Assistgpt: A general multi-modal assistant that can plan, execute, inspect, and learn. *CoRR*, abs/2306.08640, 2023.

[423] Hasan, M., C. Özel, S. Potter, et al. SAPIEN: affective virtual agents powered by large language models. *CoRR*, abs/2308.03022, 2023.

[424] Liu-Thompkins, Y., S. Okazaki, H. Li. Artificial empathy in marketing interactions: Bridging the human-ai gap in affective and social customer experience. *Journal of the Academy of Marketing Science*, 50(6):1198–1218, 2022.

[425] Bakhtin, A., D. J. Wu, A. Lerer, et al. Mastering the game of no-press diplomacy via human-regularized reinforcement learning and planning. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.

[426] (FAIR)†, M. F. A. R. D. T., A. Bakhtin, N. Brown, et al. Human-level play in the game of diplomacy by combining language models with strategic reasoning. *Science*, 378(6624):1067–1074, 2022.

[427] Lin, J., N. Tomlin, J. Andreas, et al. Decision-oriented dialogue for human-ai collaboration. *CoRR*, abs/2305.20076, 2023.

[428] Li, C., X. Su, C. Fan, et al. Quantifying the impact of large language models on collective opinion dynamics. *CoRR*, abs/2308.03313, 2023.

[429] Chase, H. LangChain. URL <https://github.com/hwchase17/langchain>, 2022.

[430] Reworked. Agent_GPT. URL <https://github.com/reworkd/AgentGPT>, 2023.

[431] AntonOsika. GPT Engineer. URL <https://github.com/AntonOsika/gpt-engineer>, 2023.

[432] Dambekodi, S. N., S. Frazier, P. Ammanabrolu, et al. Playing text-based games with common sense. *CoRR*, abs/2012.02757, 2020.

[433] Singh, I., G. Singh, A. Modi. Pre-trained language models as prior knowledge for playing text-based games. In P. Faliszewski, V. Mascardi, C. Pelachaud, M. E. Taylor, eds., *21st International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2022, Auckland, New Zealand, May 9-13, 2022*, pages 1729–1731. International Foundation for Autonomous Agents and Multiagent Systems (IFAAMAS), 2022.

[434] Ammanabrolu, P., J. Urbanek, M. Li, et al. How to motivate your dragon: Teaching goal-driven agents to speak and act in fantasy worlds. In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou, eds., *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2021, Online, June 6-11, 2021*, pages 807–833. Association for Computational Linguistics, 2021.

[435] Xu, N., S. Masling, M. Du, et al. Grounding open-domain instructions to automate web support tasks. In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou, eds., *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2021, Online, June 6-11, 2021*, pages 1022–1032. Association for Computational Linguistics, 2021.

Gao, W., X. Gao, Y. Tang. 多轮对话代理作为电话销售中的销售助手。收录于2023年国际神经网络联合会议（*IJCNN 2023*），澳大利亚黄金海岸，2023年6月18-23日，第1-9页。IEEE，2023年。Schick, T., J. A. Yu, Z. Jiang 等。PEER：一种协作式语言模型。收录于第十一届学习表征国际会议（*ICLR 2023*），卢旺达基加利，2023年5月1-5日。OpenReview.net，2023年。Lu, B., N. Haduong, C. Lee 等。DIALGEN：用于提升人类-人类对话理解的协作式人机生成对话。*CoRR*, abs/2307.07047，2023年。Gao, D., L. Ji, L. Zhou 等。Assistgpt：一种能够规划、执行、检查和学习的通用多模态助手。*CoRR*, abs/2306.08640，2023年。Hasan, M., C. Özel, S. Potter 等。SAPIEN：由大型语言模型驱动的情感虚拟代理。*CoRR*, abs/2308.03022，2023年。Liu-Thompkins, Y., S. Okazaki, H. Li。营销互动中的人工共情：弥合人类与人工智能在情感和社会客户体验中的差距。市场科学学会期刊，50(6):1198–1218，2022年。Bakhtin, A., D. J. Wu, A. Lerer 等。通过人类正则化强化学习与规划掌握无压力外交游戏。收录于第十一届学习表征国际会议（*ICLR 2023*），卢旺达基加利，2023年5月1-5日。OpenReview.net，2023年。(FAIR)†, M. F. A. R. D. T., A. Bakhtin, N. Brown 等。结合语言模型与策略推理实现人类水平的外交游戏表现。*Science*, 378(6624):1067–1074，2022年。Lin, J., N. Tomlin, J. Andreas 等。面向人机合作的决策导向对话。*CoRR*, abs/2305.20076，2023年。Li, C., X. Su, C. Fan 等。量化大型语言模型对集体意见动态的影响。*CoRR*, abs/2308.03313，2023年。Chase, H. LangChain。URL <https://github.com/hwchase17/langchain>，2022年。Reworked. AgentGPT。URL <https://github.com/reworkd/AgentGPT>，2023年。AntonOsika. GPT工程师。URL<https://github.com/AntonOsika/gpt-engineer>，2023年。Dambekodi, S. N., S. Frazier, P. Ammanabrolu 等。具有常识的文本游戏玩法。*CoRR*, abs/2012.02757，2020年。Singh, I., G. Singh, A. Modi。作为先验知识的预训练语言模型用于文本游戏。收录于P. Faliszewski, V. Mascardi, C. Pelachaud, M. E. Taylor 编，《第21届自主代理与多智能体系统国际会议（*AAMAS 2022*），新西兰奥克兰，2022年5月9-13日》，第1729-1731页。国际自主代理与多智能体系统基金会（IFAAMAS），2022年。Ammanabrolu, P., J. Urbanek, M. Li 等。如何激励你的龙：教授目标驱动代理在幻想世界中的言语与行动。收录于K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou 编，《2021年北美计算语言学协会人类语言技术会议论文集（*NAACL-HLT 2021*），线上，2021年6月6-11日》，第807-833页。计算语言学协会，2021年。Xu, N., S. Masling, M. Du 等。将开放域指令与自动化网页支持任务相结合的基础研究。收录于K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou 编，《2021年北美计算语言学协会人类语言技术会议论文集（*NAACL-HLT 2021*），线上，2021年6月6-11日》，第1022-1032页。计算语言学协会，2021年。

- [436] Chhikara, P., J. Zhang, F. Ilievski, et al. Knowledge-enhanced agents for interactive text games. *CoRR*, abs/2305.05091, 2023.
- [437] Yang, K., A. M. Swope, A. Gu, et al. Leandojo: Theorem proving with retrieval-augmented language models. *CoRR*, abs/2306.15626, 2023.
- [438] Lin, Z., H. Akin, R. Rao, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.
- [439] Irwin, R., S. Dimitriadis, J. He, et al. Chemformer: a pre-trained transformer for computational chemistry. *Mach. Learn. Sci. Technol.*, 3(1):15022, 2022.
- [440] Skrynnik, A., Z. Volovikova, M. Côté, et al. Learning to solve voxel building embodied tasks from pixels and natural language instructions. *CoRR*, abs/2211.00688, 2022.
- [441] Amiranashvili, A., N. Dorka, W. Burgard, et al. Scaling imitation learning in minecraft. *CoRR*, abs/2007.02701, 2020.
- [442] Minsky, M. *Society of mind*. Simon and Schuster, 1988.
- [443] Balaji, P. G., D. Srinivasan. An introduction to multi-agent systems. *Innovations in multi-agent systems and applications-I*, pages 1–27, 2010.
- [444] Finin, T. W., R. Fritzson, D. P. McKay, et al. KQML as an agent communication language. In *Proceedings of the Third International Conference on Information and Knowledge Management (CIKM’94)*, Gaithersburg, Maryland, USA, November 29 - December 2, 1994, pages 456–463. ACM, 1994.
- [445] Yang, Y., J. Wang. An overview of multi-agent reinforcement learning from game theoretical perspective. *arXiv preprint arXiv:2011.00583*, 2020.
- [446] Smith, A. *The wealth of nations [1776]*, vol. 11937. na, 1937.
- [447] Wang, Z., S. Mao, W. Wu, et al. Unleashing cognitive synergy in large language models: A task-solving agent through multi-persona self-collaboration. *CoRR*, abs/2307.05300, 2023.
- [448] Hassan, M. M., R. A. Knipper, S. K. K. Santu. Chatgpt as your personal data scientist. *CoRR*, abs/2305.13657, 2023.
- [449] von Neumann, J., O. Morgenstern. *Theory of Games and Economic Behavior (60th Anniversary Edition)*. Princeton University Press, 2007.
- [450] Aziz, H. Multiagent systems: algorithmic, game-theoretic, and logical foundations by y. shoham and k. leyton-brown cambridge university press, 2008. *SIGACT News*, 41(1):34–37, 2010.
- [451] Campbell, M., A. J. Hoane, F. hsiung Hsu. Deep blue. *Artif. Intell.*, 134:57–83, 2002.
- [452] Silver, D., J. Schrittwieser, K. Simonyan, et al. Mastering the game of go without human knowledge. *Nat.*, 550(7676):354–359, 2017.
- [453] Lewis, M., D. Yarats, Y. N. Dauphin, et al. Deal or no deal? end-to-end learning of negotiation dialogues. In M. Palmer, R. Hwa, S. Riedel, eds., *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, EMNLP 2017, Copenhagen, Denmark, September 9-11, 2017*, pages 2443–2453. Association for Computational Linguistics, 2017.
- [454] Irving, G., P. F. Christiano, D. Amodei. AI safety via debate. *CoRR*, abs/1805.00899, 2018.
- [455] Kenton, Z., T. Everitt, L. Weidinger, et al. Alignment of language agents. *CoRR*, abs/2103.14659, 2021.
- [456] Ngo, R. The alignment problem from a deep learning perspective. *CoRR*, abs/2209.00626, 2022.
- [457] Paul, M., L. Maglaras, M. A. Ferrag, et al. Digitization of healthcare sector: A study on privacy and security concerns. *ICT Express*, 2023.

- [436] Chhikara, P., J. Zhang, F. Ilievski, 等. 知识增强代理在交互式文本游戏中的应用. *CoRR*, abs/2305.05091, 2023.[437] Yang, K., A. M. Swope, A. Gu, 等. Leandojo: 基于检索增强语言模型的定理证明. *CoRR*, abs/2306.15626, 2023.[438] Lin, Z., H. Akin, R. Rao, 等. 利用语言模型进行原子级蛋白质结构的进化尺度预测. *Science*, 379(6637):1123–1130, 2023.[439] Irwin, R., S. Dimitriadis, J. He, 等. Chemformer: 用于计算化学的预训练变换器. *Mach. Learn. Sci. Technol.*, 3(1):15022, 2022.[440] Skrynnik, A., Z. Volovikova, M. Côté, 等. 从像素和自然语言指令中学习解决体素构建体现任务的方法. *CoRR*, abs/2211.00688, 2022.[441] Amiranashvili, A., N. Dorka, W. Burgard, 等. 在 Minecraft中扩展模仿学习. *CoRR*, abs/2007.02701, 2020.[442] Minsky, M. 心智社会. Simon and Schuster, 1988.[443] Balaji, P. G., D. Srinivasan. 多智能体系统导论. 多智能体系统与应用创新-1, 页码 1–27, 2010.[444] Finin, T. W., R. Fritzson, D. P. McKay, 等. KQML作为智能体通信语言. 收录于第三届信息与知识管理国际会议论文集 (CIKM’94), 美国马里兰州盖瑟斯堡, 1994年11月29日至12月2日, 第456–463页. ACM, 1994.[445] Yang, Y., J. Wang. 从博弈论视角对多智能体强化学习的综述. *arXiv预印本 arXiv:2011.00583*, 2020.[446] Smith, A. 国富论 [1776],第11937卷. na, 1937.[447] Wang, Z., S. Mao, W. Wu, 等. 在大型语言模型中释放认知协同作用: 通过多角色自我协作实现的任务解决代理. *CoRR*, abs/2307.05300, 2023.[448] Hassan, M. M., R. A. Knipper, S. K. K. Santu. ChatGPT作为您的个人数据科学家. *CoRR*, abs/2305.13657, 2023.[449] von Neumann, J., O. Morgenstern. 《博弈论与经济行为》(60周年纪念版). 普林斯顿大学出版社, 2007.[450] Aziz, H. 多智能体系统: 算法基础、博弈论基础与逻辑基础, 作者: Y. Shoham 和 K. Leyton-Brown, 剑桥大学出版社, 2008年. *SIGACT News*, 41(1):34–37, 2010.[451] Campbell, M., A. J. Hoane, F. Hsiung Hsu. 深蓝. 人工智能, 134:57–83, 2002.[452] Silver, D., J. Schrittwieser, K. Simonyan, 等. 无人知识掌握围棋游戏. *Nature*, 550(7676):354–359, 2017.[453] Lewis, M., D. Yarats, Y. N. Dauphin, 等. 交易还是不交易? 端到端学习谈判对话. 收录于M. Palmer, R. Hwa, S. Riedel编辑的2017年自然语言处理实证方法会议 (EMNLP 2017), 丹麦哥本哈根, 2017年9月9-11日, 第2443–2453页. 计算语言学协会, 2017.[454] Irving, G., P. F. Christiano, D. Amodei. 通过辩论实现AI安全. *CoRR*, abs/1805.00899, 2018.[455] Kenton, Z., T. Everitt, L. Weidinger, 等. 语言智能体的对齐问题. *CoRR*, abs/2103.14659, 2021.[456] Ngo, R. 从深度学习角度看对齐问题. *CoRR*, abs/2209.00626, 2022.[457] Paul, M., L. Maglaras, M. A. Ferrag, 等. 医疗行业数字化: 关于隐私与安全问题的研究. *ICT Express*, 2023.

- [458] Bassiri, M. A. Interactional feedback and the impact of attitude and motivation on noticing I2 form. *English Language and Literature Studies*, 1(2):61, 2011.
- [459] Tellex, S., T. Kollar, S. Dickerson, et al. Approaching the symbol grounding problem with probabilistic graphical models. *AI Mag.*, 32(4):64–76, 2011.
- [460] Matuszek, C., E. Herbst, L. Zettlemoyer, et al. Learning to parse natural language commands to a robot control system. In J. P. Desai, G. Dudek, O. Khatib, V. Kumar, eds., *Experimental Robotics - The 13th International Symposium on Experimental Robotics, ISER 2012, June 18-21, 2012, Québec City, Canada*, vol. 88 of *Springer Tracts in Advanced Robotics*, pages 403–415. Springer, 2012.
- [461] Chaplot, D. S., K. M. Sathyendra, R. K. Pasumarthi, et al. Gated-attention architectures for task-oriented language grounding. In S. A. McIlraith, K. Q. Weinberger, eds., *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, (AAAI-18), the 30th innovative Applications of Artificial Intelligence (IAAI-18), and the 8th AAAI Symposium on Educational Advances in Artificial Intelligence (EAAI-18), New Orleans, Louisiana, USA, February 2-7, 2018*, pages 2819–2826. AAAI Press, 2018.
- [462] Li, J., A. H. Miller, S. Chopra, et al. Dialogue learning with human-in-the-loop. In *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*. OpenReview.net, 2017.
- [463] Iyer, S., I. Konstas, A. Cheung, et al. Learning a neural semantic parser from user feedback. In R. Barzilay, M. Kan, eds., *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics, ACL 2017, Vancouver, Canada, July 30 - August 4, Volume 1: Long Papers*, pages 963–973. Association for Computational Linguistics, 2017.
- [464] Weston, J. Dialog-based language learning. In D. D. Lee, M. Sugiyama, U. von Luxburg, I. Guyon, R. Garnett, eds., *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, pages 829–837. 2016.
- [465] Shuster, K., J. Xu, M. Komeili, et al. Blenderbot 3: a deployed conversational agent that continually learns to responsibly engage. *CoRR*, abs/2208.03188, 2022.
- [466] Du, W., Z. M. Kim, V. Raheja, et al. Read, revise, repeat: A system demonstration for human-in-the-loop iterative text revision. *CoRR*, abs/2204.03685, 2022.
- [467] Kreutzer, J., S. Khadivi, E. Matusov, et al. Can neural machine translation be improved with user feedback? In S. Bangalore, J. Chu-Carroll, Y. Li, eds., *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2018, New Orleans, Louisiana, USA, June 1-6, 2018, Volume 3 (Industry Papers)*, pages 92–105. Association for Computational Linguistics, 2018.
- [468] Gur, I., S. Yavuz, Y. Su, et al. Dialsq1: Dialogue based structured query generation. In I. Gurevych, Y. Miyao, eds., *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, ACL 2018, Melbourne, Australia, July 15-20, 2018, Volume 1: Long Papers*, pages 1339–1349. Association for Computational Linguistics, 2018.
- [469] Yao, Z., Y. Su, H. Sun, et al. Model-based interactive semantic parsing: A unified framework and A text-to-sql case study. In K. Inui, J. Jiang, V. Ng, X. Wan, eds., *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pages 5446–5457. Association for Computational Linguistics, 2019.
- [470] Mehta, N., D. Goldwasser. Improving natural language interaction with robots using advice. In J. Burstein, C. Doran, T. Solorio, eds., *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2019, Minneapolis, MN, USA, June 2-7, 2019, Volume 1 (Long and Short Papers)*, pages 1962–1967. Association for Computational Linguistics, 2019.

- [458] Bassiri, M. A. 互动反馈以及态度和动机对第二语言形式注意的影响. 英语语言与文学研究, 1(2):61, 2011.[459] Tellex, S., T. Kollar, S. Dickerson, 等. 使用概率图模型解决符号基础问题. 人工智能杂志, 32(4):64–76, 2011.[460] Matuszek, C., E. Herbst, L. Zettlemoyer, 等. 学习将自然语言指令解析到机器人控制系统中. 收录于 J. P. Desai, G. Dudek, O. Khatib, V. Kumar 编, 实验机器人学——第十三届国际实验机器人研讨会 (*ISER 2012*) , 2012年6月18-21日, 加拿大魁北克市, 第 88 卷 *Springer* 高级机器人学专辑, 页码 403–415. Springer, 2012.[461] Chaplot, D. S., K. M. Sathyendra, R. K. Pasumarthi, 等. 基于门控注意力架构的任务导向语言基础. 收录于 S. A. McIlraith, K. Q. Weinberger 编, 第32届人工智能协会会议论文集 (*AAAI-18*) 、 第30届创新人工智能应用会议 (*IAAI-18*) 以及第8届人工智能教育进展研讨会 (*EAAI-18*) , 美国路易斯安那州新奥尔良, 2018年2月2-7日, 页码 2819–2826. AAAI出版社, 2018.[462] Li, J., A. H. Miller, S. Chopra, 等. 人机交互中的对话学习. 收录于 第5届国际学习表示会议 (*ICLR 2017*) , 法国土伦, 2017年4月24-26日, 会议论文集. OpenReview.net, 2017.[463] Iyer, S., I. Konstas, A. Cheung, 等. 从用户反馈中学习神经语义解析器. 收录于 R. Barzilay, M. Kan 编, 第55届计算语言学协会年会论文集 (*ACL 2017*) , 加拿大温哥华, 2017年7月30日-8月4日, 第1卷: 长论文, 页码 963–973. 计算语言学协会, 2017.[464] Weston, J. 基于对话的语言学习. 收录于 D. D. Lee, M. Sugiyama, U. von Luxburg, I. Guyon, R. Garnett 编, 神经信息处理系统进展29: 2016年神经信息处理系统年会, 2016年12月5-10日, 西班牙巴塞罗那, 页码 829–837. 2016.[465] Shuster, K., J. Xu, M. Komeili, 等. Blenderbot 3: 一个持续学习以负责任地交互的部署对话代理. *CoRR*, abs/2208.03188, 2022.[466] Du, W., Z. M. Kim, V. Raheja, 等. 阅读、修订、重复: 人机交互式文本修订系统演示. *CoRR*, abs/2204.03685, 2022.[467] Kreutzer, J., S. Khadivi, E. Matusov, 等. 神经机器翻译能否通过用户反馈得到改进? 收录于 S. Bangalore, J. Chu-Carroll, Y. Li 编, 2018年北美计算语言学协会会议论文集: 人类语言技术 (*NAACL-HLT 2018*) , 美国路易斯安那州新奥尔良, 2018年6月1-6日, 第3卷 (行业论文) , 页码 92–105. 计算语言学协会, 2018.[468] Gur, I., S. Yavuz, Y. Su, 等. Dialsq1: 基于对话的结构化查询生成. 收录于 I. Gurevych, Y. Miyao 编, 第56届计算语言学协会年会论文集 (*ACL 2018*) , 澳大利亚墨尔本, 2018年7月15-20日, 第1卷: 长论文, 页码 1339–1349. 计算语言学协会, 2018.[469] Yao, Z., Y. Su, H. Sun, 等. 基于模型的交互式语义解析: 统一框架及文本到SQL的案例研究. 收录于 K. Inui, J. Jiang, V. Ng, X. Wan 编, 2019年自然语言处理实证方法会议 (*EMNLP*) 及第九届自然语言处理国际联合会会议 (*IJCNLP*) , 中国香港, 2019年11月3-7日, 页码 5446–5457. 计算语言学协会, 2019.[470] Mehta, N., D. Goldwasser. 利用建议改善机器人自然语言交互. 收录于 J. Burstein, C. Doran, T. Solorio 编, 2019年北美计算语言学协会会议论文集: 人类语言技术 (*NAACL-HLT 2019*) , 美国明尼苏达州明尼阿波利斯, 2019年6月2-7日, 第1卷 (长短论文) , 页码 1962–1967. 计算语言学协会, 2019.

[471] Elgohary, A., C. Meek, M. Richardson, et al. NL-EDIT: correcting semantic parse errors through natural language interaction. In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou, eds., *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2021, Online, June 6-11, 2021*, pages 5599–5610. Association for Computational Linguistics, 2021.

[472] Tandon, N., A. Madaan, P. Clark, et al. Learning to repair: Repairing model output errors after deployment using a dynamic memory of feedback. In M. Carpuat, M. de Marneffe, I. V. M. Ruíz, eds., *Findings of the Association for Computational Linguistics: NAACL 2022, Seattle, WA, United States, July 10-15, 2022*, pages 339–352. Association for Computational Linguistics, 2022.

[473] Scheurer, J., J. A. Campos, T. Korbak, et al. Training language models with language feedback at scale. *CoRR*, abs/2303.16755, 2023.

[474] Xu, J., M. Ung, M. Komeili, et al. Learning new skills after deployment: Improving open-domain internet-driven dialogue with human feedback. In A. Rogers, J. L. Boyd-Graber, N. Okazaki, eds., *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 13557–13572. Association for Computational Linguistics, 2023.

[475] Cai, Z., B. Chang, W. Han. Human-in-the-loop through chain-of-thought. *CoRR*, abs/2306.07932, 2023.

[476] Hancock, B., A. Bordes, P. Mazaré, et al. Learning from dialogue after deployment: Feed yourself, chatbot! In A. Korhonen, D. R. Traum, L. Màrquez, eds., *Proceedings of the 57th Conference of the Association for Computational Linguistics, ACL 2019, Florence, Italy, July 28- August 2, 2019, Volume 1: Long Papers*, pages 3667–3684. Association for Computational Linguistics, 2019.

[477] Mehta, N., M. Teruel, P. F. Sanz, et al. Improving grounded language understanding in a collaborative environment by interacting with agents through help feedback. *CoRR*, abs/2304.10750, 2023.

[478] Gvirsman, O., Y. Koren, T. Norman, et al. Patricc: A platform for triadic interaction with changeable characters. In T. Belpaeme, J. E. Young, H. Gunes, L. D. Riek, eds., *HRI '20: ACM/IEEE International Conference on Human-Robot Interaction, Cambridge, United Kingdom, March 23-26, 2020*, pages 399–407. ACM, 2020.

[479] Stiles-Shields, C., E. Montague, E. G. Lattie, et al. What might get in the way: Barriers to the use of apps for depression. *DIGITAL HEALTH*, 3:2055207617713827, 2017. PMID: 29942605.

[480] McTear, M. F. *Conversational AI: Dialogue Systems, Conversational Agents, and Chatbots*. Synthesis Lectures on Human Language Technologies. Morgan & Claypool Publishers, 2020.

[481] Motger, Q., X. Franch, J. Marco. Conversational agents in software engineering: Survey, taxonomy and challenges. *CoRR*, abs/2106.10901, 2021.

[482] Rapp, A., L. Curti, A. Boldi. The human side of human-chatbot interaction: A systematic literature review of ten years of research on text-based chatbots. *Int. J. Hum. Comput. Stud.*, 151:102630, 2021.

[483] Adamopoulou, E., L. Moussiades. Chatbots: History, technology, and applications. *Machine Learning with Applications*, 2:100006, 2020.

[484] Wang, K., X. Wan. Sentigan: Generating sentimental texts via mixture adversarial networks. In J. Lang, ed., *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI 2018, July 13-19, 2018, Stockholm, Sweden*, pages 4446–4452. ijcai.org, 2018.

[471] Elgohary, A., C. Meek, M. Richardson 等. NL-EDIT: 通过自然语言交互纠正语义解析错误。收录于K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou 编., *2021年北美计算语言学协会人类语言技术会议论文集: NAACL-HLT 2021*, 线上, *2021年6月6-11日*, 第5599-5610页。计算语言学协会, 2021年。[472] Tandon, N., A. Madaan, P. Clark 等. 学习修复: 在部署后利用反馈的动态记忆修正模型输出错误。收录于M. Carpuat, M. de Marneffe, I. V. M. Ruíz 编., 计算语言学协会研究成果: *NAACL 2022*, 华盛顿州西雅图, 美国, *2022年7月10-15日*, 第339-352页。计算语言学协会, 2022年。[473] Scheurer, J., J. A. Campos, T. Korbak 等. 大规模利用语言反馈训练语言模型。 *CoRR*, abs/2303.16755, 2023年。[474] Xu, J., M. Ung, M. Komeili 等. 部署后学习新技能: 通过人类反馈提升开放域互联网驱动对话。收录于A. Rogers, J. L. Boyd-Graber, N. Okazaki 编., *第61届计算语言学协会年会论文集 (卷1: 长论文)*, *ACL 2023*, 加拿大多伦多, *2023年7月9-14日*, 第13557-13572页。计算语言学协会, 2023年。[475]Cai, Z., B. Chang, W. Han. 通过链式思维实现人机交互。 *CoRR*, abs/2306.07932, 2023年。[476] Hancock, B., A. Bordes, P. Mazaré 等. 部署后从对话中学习: 自我引导, 聊天机器人! 收录于A. Korhonen, D. R. Traum, L. Màrquez 编., *第57届计算语言学协会会议论文集, ACL 2019*, 意大利佛罗伦萨, *2019年7月28日-8月2日*, 第1卷: 长论文, 第3667-3684页。计算语言学协会, 2019年。[477] Mehta, N., M. Teruel, P. F. Sanz 等. 通过帮助反馈与代理交互改善协作环境中的基础语言理解。 *CoRR*, abs/2304.10750, 2023年。[478] Gvirsman, O., Y. Koren, T. Norman 等. Patricc: 一个具有可变角色的三元交互平台。收录于T. Belpaeme, J. E. Young, H. Gunes, L. D. Riek 编., *HRI '20: ACM/IEEE人机交互国际会议*, 英国剑桥, *2020年3月23-26日*, 第399-407页。ACM, 2020年。[479] Stiles-Shields, C., E. Montague, E. G. Lattie 等. 可能的障碍: 抑郁症应用使用中的障碍。 *DIGITAL HEALTH*, 3: 2055207617713827, 2017年。PMID: 29942605。[480] McTear, M. F. 对话式人工智能: 对话系统、会话代理与聊天机器人。人类语言技术综述讲义。Morgan & Claypool出版社, 2020年。[481] Motger, Q., X. Franch, J. Marco. 软件工程中的对话代理: 调研、分类与挑战。 *CoRR*, abs/2106.10901, 2021年。[482] Rapp, A., L. Curti, A. Boldi. 人机聊天交互中的人类因素: 十年文本聊天机器人研究的系统性文献综述。国际人机交互学报, 151: 102630, 2021年。[483] Adamopoulou, E., L. Moussiades. 聊天机器人: 历史、技术与应用。应用中的机器学习, 2: 100006, 2020年。[484] Wang, K., X. Wan. Sentigan: 通过混合对抗网络生成情感文本。收录于J. Lang 编., *第27届国际人工智能联合会议论文集, IJCAI 2018*, *2018年7月13-19日*, 瑞典斯德哥尔摩, 第4446-4452页。ijcai.org, 2018年。

- [485] Zhou, X., W. Y. Wang. Mojitalk: Generating emotional responses at scale. In I. Gurevych, Y. Miyao, eds., *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, ACL 2018, Melbourne, Australia, July 15-20, 2018, Volume 1: Long Papers*, pages 1128–1137. Association for Computational Linguistics, 2018.
- [486] Lin, Z., P. Xu, G. I. Winata, et al. Caire: An empathetic neural chatbot. *arXiv preprint arXiv:1907.12108*, 2019.
- [487] Jhan, J., C. Liu, S. Jeng, et al. Cheerbots: Chatbots toward empathy and emotionusing reinforcement learning. *CoRR*, abs/2110.03949, 2021.
- [488] Lin, Z., A. Madotto, J. Shin, et al. Moel: Mixture of empathetic listeners. In K. Inui, J. Jiang, V. Ng, X. Wan, eds., *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pages 121–132. Association for Computational Linguistics, 2019.
- [489] Majumder, N., P. Hong, S. Peng, et al. MIME: mimicking emotions for empathetic response generation. In B. Webber, T. Cohn, Y. He, Y. Liu, eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16-20, 2020*, pages 8968–8979. Association for Computational Linguistics, 2020.
- [490] Sabour, S., C. Zheng, M. Huang. CEM: commonsense-aware empathetic response generation. In *Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, The Twelveth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022 Virtual Event, February 22 - March 1, 2022*, pages 11229–11237. AAAI Press, 2022.
- [491] Li, Q., P. Li, Z. Ren, et al. Knowledge bridging for empathetic dialogue generation. In *Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, The Twelveth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022 Virtual Event, February 22 - March 1, 2022*, pages 10993–11001. AAAI Press, 2022.
- [492] Liu, B., S. S. Sundar. Should machines express sympathy and empathy? experiments with a health advice chatbot. *Cyberpsychology Behav. Soc. Netw.*, 21(10):625–636, 2018.
- [493] Su, Z., M. C. Figueiredo, J. Jo, et al. Analyzing description, user understanding and expectations of AI in mobile health applications. In *AMIA 2020, American Medical Informatics Association Annual Symposium, Virtual Event, USA, November 14-18, 2020*. AMIA, 2020.
- [494] Moravčík, M., M. Schmid, N. Burch, et al. Deepstack: Expert-level artificial intelligence in no-limit poker. *CoRR*, abs/1701.01724, 2017.
- [495] Carroll, M., R. Shah, M. K. Ho, et al. On the utility of learning about humans for human-ai coordination. In H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, R. Garnett, eds., *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pages 5175–5186. 2019.
- [496] Bard, N., J. N. Foerster, S. Chandar, et al. The hanabi challenge: A new frontier for ai research. *Artificial Intelligence*, 280:103216, 2020.
- [497] Wang, X., W. Shi, R. Kim, et al. Persuasion for good: Towards a personalized persuasive dialogue system for social good. In A. Korhonen, D. R. Traum, L. Màrquez, eds., *Proceedings of the 57th Conference of the Association for Computational Linguistics, ACL 2019, Florence, Italy, July 28- August 2, 2019, Volume 1: Long Papers*, pages 5635–5649. Association for Computational Linguistics, 2019.
- [498] Abrams, A. M. H., A. M. R. der Pütten. I-C-E framework: Concepts for group dynamics research in human-robot interaction. *Int. J. Soc. Robotics*, 12(6):1213–1229, 2020.
- [499] Xu, Y., S. Wang, P. Li, et al. Exploring large language models for communication games: An empirical study on werewolf, 2023.

- [485] 周晓, 王伟洋. Mojitalk: 大规模生成情感反应. 收录于I. Gurevych, Y. Miyao (编), 第56届计算语言学协会年会论文集, *ACL 2018*, 澳大利亚墨尔本, 2018年7月15-20日, 第1卷: 长论文, 第1128–1137页. 计算语言学协会, 2018年. [486] 林志, 许鹏, G. I. Winata 等. Caire: 一个具有同理心的神经聊天机器人. *arXiv预印本 arXiv:1907.12108*, 2019年. [487] 詹杰, 刘策, 郑思等. Cheerbots: 基于强化学习的具有同理心和情感的聊天机器人. *CoRR*, abs/2110.03949, 2021年. [488] 林志, Madotto A., Shin J. 等. Moel: 多样化的具有同理心的听众. 在井上康, 江志强, Ng V., 万翔 (编), 2019年自然语言处理实证方法会议及第九届国际联合自然语言处理会议论文集, *EMNLP-IJCNLP 2019*, 香港, 中国, 2019年11月3-7日, 第121–132页. 计算语言学协会, 2019年. [489] Majumder N., 洪鹏, 彭思等. MIME: 模仿情感以生成具有同理心的回应. 在Webber B., Cohn T., He Y., Liu Y. (编), 2020年自然语言处理实证方法会议论文集, *EMNLP 2020*, 线上, 2020年11月16-20日, 第8968–8979页. 计算语言学协会, 2020年. [490] Sabour S., Zheng C., Huang M.. CEM: 常识感知的具有同理心的回应生成. 在第36届人工智能协会会议 (AAAI 2022)、第34届人工智能创新应用会议 (IAAI 2022)、第12届人工智能教育进展研讨会 (EAAI 2022) 虚拟会议, 2022年2月22日至3月1日, 第11229–11237页. AAAI出版社, 2022年. [491] 李强, 李鹏, 任志等. 知识桥接用于具有同理心的对话生成. 在第36届人工智能协会会议 (AAAI 2022)、第34届人工智能创新应用会议 (IAAI 2022)、第12届人工智能教育进展研讨会 (EAAI 2022) 虚拟会议, 2022年2月22日至3月1日, 第10993–11001页. AAAI出版社, 2022年. [492] 刘兵, S. S. Sundar. 机器应表达同情和共情吗? 关于健康建议聊天机器人的实验. *Cyberpsychology Behav. Soc. Netw.*, 21(10):625–636, 2018年. [493] 苏志, Figueiredo M. C., Jo J. 等. 分析移动健康应用中对AI的描述、用户理解与期望. 在*AMIA 2020*, 美国医学信息学会年会, 线上, 2020年11月14-18日. AMIA, 2020年. [494] Moravčík M., Schmid M., Burch N. 等. Deepstack: 无极德州扑克中的专家级人工智能. *CoRR*, abs/1701.01724, 2017年. [495] Carroll M., Shah R., Ho M. K. 等. 关于学习人类知识在人工智能协调中的作用. 收录于H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, Garnett R. (编), 神经信息处理系统年度会议 (*NeurIPS 2019*), 2019年12月8-14日, 温哥华, 加拿大, 第5175–5186页, 2019年. [496] Bard N., Foerster J. N., Chandar S. 等. Hanabi挑战: 人工智能研究的新前沿. 人工智能, 280:103216, 2020年. [497] 王翔, 史伟, Kim R. 等. 善意的说服: 面向社会公益的个性化说服对话系统. 在A. Korhonen, D. R. Traum, L. Màrquez (编), 第57届计算语言学协会会议论文集, *ACL 2019*, 意大利佛罗伦萨, 2019年7月28日-8月2日, 第1卷: 长论文, 第5635–5649页. 计算语言学协会, 2019年. [498] Abrams A. M. H., der Pütten A. M. R.. I-C-E框架: 人机交互中群体动力学研究的概念. 国际社会机器人学杂志, 12(6):1213–1229, 2020年. [499] 徐颖, 王思, 李鹏等. 探索大型语言模型在通信游戏中的应用: 关于狼人游戏的实证研究, 2023年.

- [500] Binz, M., E. Schulz. Using cognitive psychology to understand GPT-3. *CoRR*, abs/2206.14576, 2022.
- [501] Dasgupta, I., A. K. Lampinen, S. C. Y. Chan, et al. Language models show human-like content effects on reasoning. *CoRR*, abs/2207.07051, 2022.
- [502] Dhingra, S., M. Singh, V. S. B., et al. Mind meets machine: Unravelling gpt-4’s cognitive psychology. *CoRR*, abs/2303.11436, 2023.
- [503] Hagendorff, T. Machine psychology: Investigating emergent capabilities and behavior in large language models using psychological methods. *CoRR*, abs/2303.13988, 2023.
- [504] Wang, X., X. Li, Z. Yin, et al. Emotional intelligence of large language models. *CoRR*, abs/2307.09042, 2023.
- [505] Curry, A., A. C. Curry. Computer says "no": The case against empathetic conversational AI. In A. Rogers, J. L. Boyd-Graber, N. Okazaki, eds., *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 8123–8130. Association for Computational Linguistics, 2023.
- [506] Elyoseph, Z., D. Hadar-Shoval, K. Asraf, et al. Chatgpt outperforms humans in emotional awareness evaluations. *Frontiers in Psychology*, 14:1199058, 2023.
- [507] Habibi, R., J. Pfau, J. Holmes, et al. Empathetic AI for empowering resilience in games. *CoRR*, abs/2302.09070, 2023.
- [508] Caron, G., S. Srivastava. Identifying and manipulating the personality traits of language models. *CoRR*, abs/2212.10276, 2022.
- [509] Pan, K., Y. Zeng. Do llms possess a personality? making the MBTI test an amazing evaluation for large language models. *CoRR*, abs/2307.16180, 2023.
- [510] Li, X., Y. Li, S. Joty, et al. Does gpt-3 demonstrate psychopathy? evaluating large language models from a psychological perspective, 2023.
- [511] Safdari, M., G. Serapio-García, C. Crepy, et al. Personality traits in large language models. *CoRR*, abs/2307.00184, 2023.
- [512] Côté, M., Á. Kádár, X. Yuan, et al. Textworld: A learning environment for text-based games. In T. Cazenave, A. Saffidine, N. R. Sturtevant, eds., *Computer Games - 7th Workshop, CGW 2018, Held in Conjunction with the 27th International Conference on Artificial Intelligence, IJCAI 2018, Stockholm, Sweden, July 13, 2018, Revised Selected Papers*, vol. 1017 of *Communications in Computer and Information Science*, pages 41–75. Springer, 2018.
- [513] Urbanek, J., A. Fan, S. Karamcheti, et al. Learning to speak and act in a fantasy text adventure game. In K. Inui, J. Jiang, V. Ng, X. Wan, eds., *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pages 673–683. Association for Computational Linguistics, 2019.
- [514] Hausknecht, M. J., P. Ammanabrolu, M. Côté, et al. Interactive fiction games: A colossal adventure. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence, AAAI 2020, The Thirty-Second Innovative Applications of Artificial Intelligence Conference, IAAI 2020, The Tenth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2020, New York, NY, USA, February 7-12, 2020*, pages 7903–7910. AAAI Press, 2020.
- [515] O’Gara, A. Hoodwinked: Deception and cooperation in a text-based game for language models. *CoRR*, abs/2308.01404, 2023.
- [516] Bharadhwaj, H., J. Vakil, M. Sharma, et al. Roboagent: Generalization and efficiency in robot manipulation via semantic augmentations and action chunking. *CoRR*, abs/2309.01918, 2023.

Binz, M., E. Schulz. 利用认知心理学理解GPT-3. *CoRR*, abs/2206.14576, 2022年。

Dasgupta, I., A. K. Lampinen, S. C. Y. Chan 等。语言模型在人类推理中的内容效应表现出类人特征。 *CoRR*, abs/2207.07051, 2022年。 Dhingra, S., M. Singh, V. S. B 等。心智与机器的交汇：揭示GPT-4的认知心理学。 *CoRR*, abs/2303.11436, 2023年。

Hagendorff, T. 机器心理学：利用心理学方法研究大型语言模型的新兴能力与行为。 *CoRR*, abs/2303.13988, 2023年。 Wang, X., X. Li, Z. Yin 等。大型语言模型的情感智力。 *CoRR*, abs/2307.09042, 2023年。 Curry, A., A. C. Curry. 计算机说“不”：反对富有同理心的对话式人工智能。载于A. Rogers, J. L. Boyd-Graber, N. Okazaki 编, 《计算语言学协会会议论文集: ACL 2023, 加拿大多伦多, 2023年7月9-14日》, 第8123–8130页。计算语言学协会, 2023年。 Elyoseph, Z., D. Hadar-Shoval, K. Asraf 等。ChatGPT在情感认知评估中优于人类。 *Frontiers in Psychology*, 14: 1199058, 2023年。 Habibi, R., J. Pfau, J. Holmes 等。赋能韧性的富有同理心的人工智能在游戏中的应用。 *CoRR*, abs/2302.09070, 2023年。 Caron, G., S. Srivastava. 识别与操控语言模型的个性特质。 *CoRR*, abs/2212.10276, 2022年。 Pan, K., Y. Zeng. 大型语言模型是否具有个性？将MBTI测试作为评估大型语言模型的创新方法。 *CoRR*, abs/2307.16180, 2023年。 Li, X., Y. Li, S. Joty 等。GPT-3是否表现出精神病理特征？从心理学角度评估大型语言模型, 2023年。 Safdari, M., G. Serapio-García, C. Crepy 等。大型语言模型中的个性特质。 *CoRR*, abs/2307.00184, 2023年。 Côté, M., Á. Kádár, X. Yuan 等。Textworld: 一种基于文本的游戏学习环境。载于T. Cazenave, A. Saffidine, N. R. Sturtevant 编, 《第七届计算机游戏研讨会 (CGW 2018) 论文集》, 第1017卷, 页41–75。Springer, 2018年。 Urbanek, J., A. Fan, S. Karamcheti 等。在幻想文本冒险游戏中学习说话与行动。载于K. Inui, J. Jiang, V. Ng, X. Wan 编, 《2019年自然语言处理实证方法会议及第九届自然语言处理国际联合会议 (EMNLP-IJCNLP 2019) 论文集》, 第673–683页。计算语言学协会, 2019年。 Hausknecht, M. J., P. Ammanabrolu, M. Côté 等。互动小说游戏：一场宏伟的冒险。载于第34届人工智能国际会议 (AAAI 2020)、第32届创新应用人工智能会议 (IAAI 2020)、第十届人工智能教育创新研讨会 (EAAI 2020), 纽约, 美国, 2020年2月7-12日, 页7903–7910。AAAI出版社, 2020年。

O’Gara, A. Hoodwinked: 在面向语言模型的文本游戏中的欺骗与合作。 *CoRR*, abs/2308.01404, 2023年。 Bharadhwaj, H., J. Vakil, M. Sharma 等。Roboagent: 通过语义增强与动作块化实现机器人操作的泛化与效率。 *CoRR*, abs/2309.01918, 2023年。

[517] Park, J. S., L. Popowski, C. J. Cai, et al. Social simulacra: Creating populated prototypes for social computing systems. In M. Agrawala, J. O. Wobbrock, E. Adar, V. Setlur, eds., *The 35th Annual ACM Symposium on User Interface Software and Technology, UIST 2022, Bend, OR, USA, 29 October 2022 - 2 November 2022*, pages 74:1–74:18. ACM, 2022.

[518] Gao, C., X. Lan, Z. Lu, et al. S³: Social-network simulation system with large language model-empowered agents. *CoRR*, abs/2307.14984, 2023.

[519] Wang, L., J. Zhang, X. Chen, et al. Recagent: A novel simulation paradigm for recommender systems. *CoRR*, abs/2306.02552, 2023.

[520] Williams, R., N. Hosseinichimeh, A. Majumdar, et al. Epidemic modeling with generative agents. *CoRR*, abs/2307.04986, 2023.

[521] da Rocha Costa, A. C. *A Variational Basis for the Regulation and Structuration Mechanisms of Agent Societies*. Springer, 2019.

[522] Wimmer, S., A. Pfeiffer, N. Denk. The everyday life in the sims 4 during a pandemic. a life simulation as a virtual mirror of society? In *INTED2021 Proceedings*, 15th International Technology, Education and Development Conference, pages 5754–5760. IATED, 2021.

[523] Lee, L., T. Braud, P. Zhou, et al. All one needs to know about metaverse: A complete survey on technological singularity, virtual ecosystem, and research agenda. *CoRR*, abs/2110.05352, 2021.

[524] Inkeles, A., D. H. Smith. *Becoming modern: Individual change in six developing countries*. Harvard University Press, 1974.

[525] Troitzsch, K. G., U. Mueller, G. N. Gilbert, et al., eds. *Social Science Microsimulation [Dagstuhl Seminar, May, 1995]*. Springer, 1996.

[526] Abrams, A. M., A. M. R.-v. der Pütten. I–c–e framework: Concepts for group dynamics research in human-robot interaction: Revisiting theory from social psychology on ingroup identification (i), cohesion (c) and entitativity (e). *International Journal of Social Robotics*, 12:1213–1229, 2020.

[527] Askill, A., Y. Bai, A. Chen, et al. A general language assistant as a laboratory for alignment. *CoRR*, abs/2112.00861, 2021.

[528] Zhang, Z., N. Liu, S. Qi, et al. Heterogeneous value evaluation for large language models. *CoRR*, abs/2305.17147, 2023.

[529] Browning, J. Personhood and ai: Why large language models don’t understand us. *AI & SOCIETY*, pages 1–8, 2023.

[530] Jiang, G., M. Xu, S. Zhu, et al. MPI: evaluating and inducing personality in pre-trained language models. *CoRR*, abs/2206.07550, 2022.

[531] Kosinski, M. Theory of mind may have spontaneously emerged in large language models. *CoRR*, abs/2302.02083, 2023.

[532] Zuckerman, M. *Psychobiology of personality*, vol. 10. Cambridge University Press, 1991.

[533] Han, S. J., K. Ransom, A. Perfors, et al. Inductive reasoning in humans and large language models. *CoRR*, abs/2306.06548, 2023.

[534] Hagendorff, T., S. Fabi, M. Kosinski. Thinking fast and slow in large language models, 2023.

[535] Hagendorff, T., S. Fabi. Human-like intuitive behavior and reasoning biases emerged in language models - and disappeared in GPT-4. *CoRR*, abs/2306.07622, 2023.

[536] Ma, Z., Y. Mei, Z. Su. Understanding the benefits and challenges of using large language model-based conversational agents for mental well-being support. *CoRR*, abs/2307.15810, 2023.

Park, J. S., L. Popowski, C. J. Cai 等. 社会模拟：为社会计算系统创建人口原型。收录于 M. Agrawala, J. O. Wobbrock, E. Adar, V. Setlur 编著，《第35届ACM用户界面软件与技术年会会议论文集》，UIST 2022，地点：美国俄勒冈州本德，2022年10月29日至11月2日，第74:1–74:18页。ACM，2022年。

Gao, C., X. Lan, Z. Lu 等. S³: 具有大语言模型赋能代理的社交网络模拟系统。CoRR, abs/2307.14984, 2023年。

Wang, L., J. Zhang, X. Chen 等. Recagent: 一种新颖的推荐系统模拟范式。CoRR, abs/2306.02552, 2023年。

Williams, R., N. Hosseinichimeh, A. Majumdar 等. 使用生成代理进行流行病建模。CoRR, abs/2307.04986, 2023年。

da Rocha Costa, A. C. 代理社会调控与结构机制的变分基础。Springer, 2019年。

Wimmer, S., A. Pfeiffer, N. Denk. 大流行期间《模拟人生4》中的日常生活：虚拟社会的镜像？收录于 INTED2021会议论文集，第15届国际技术、教育与发展会议，页5754–5760。IATED, 2021年。

Lee, L., T. Braud, P. Zhou 等. 关于元宇宙的全部知识：技术奇点、虚拟生态系统与研究议程的完整综述。CoRR, abs/2110.05352, 2021年。

Inkeles, A., D. H. Smith. 《走向现代化：六个发展中国家的个人变革》。哈佛大学出版社，1974年。

Troitzsch, K. G., U. Mueller, G. N. Gilbert 等，编. 社会科学微观模拟。Dagstuhl研讨会，1996年5月。Springer, 1996年。

Abrams, A. M., A. M. R.-v. der Pütten. I–c–e 框架：人机交互中群体动力学研究的概念：重新审视社会心理学中关于内群体认同（i）、凝聚力（c）和实体性（e）的理论。《国际社会机器人期刊》，12: 1213–1229, 2020年。

Askill, A., Y. Bai, A. Chen 等. 作为对齐实验室的通用语言助手。CoRR, abs/2112.00861, 2021年。

Zhang, Z., N. Liu, S. Qi 等. 大语言模型的异质价值评估。CoRR, abs/2305.17147, 2023年。

Browning, J. 人格与人工智能：为何大语言模型无法理解我们。《人工智能与社会》，第1–8页，2023年。

Jiang, G., M. Xu, S. Zhu 等. MPI: 评估与引导预训练语言模型中的人格。CoRR, abs/2206.07550, 2022年。

Kosinski, M. 心理认知理论可能在大语言模型中自发出现。CoRR, abs/2302.02083, 2023年。

Zuckerman, M. 《人格的心理生物学》，第10卷。剑桥大学出版社，1991年。

Han, S. J., K. Ransom, A. Perfors 等. 人类与大语言模型中的归纳推理。CoRR, abs/2306.06548, 2023年。

Hagendorff, T., S. Fabi, M. Kosinski. 大语言模型中的快速与慢速思维，2023年。

Hagendorff, T., S. Fabi. 类人直觉行为与推理偏差在语言模型中出现——在GPT-4中消失。CoRR, abs/2306.07622, 2023年。

Ma, Z., Y. Mei, Z. Su. 理解基于大语言模型的对话代理在心理健康支持中的优势与挑战。CoRR, abs/2307.15810, 2023年。

[537] Bates, J. The role of emotion in believable agents. *Commun. ACM*, 37(7):122–125, 1994.

[538] Karra, S. R., S. Nguyen, T. Tulabandhula. AI personification: Estimating the personality of language models. *CoRR*, abs/2204.12000, 2022.

[539] Zhang, S., E. Dinan, J. Urbanek, et al. Personalizing dialogue agents: I have a dog, do you have pets too? In I. Gurevych, Y. Miyao, eds., *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, ACL 2018, Melbourne, Australia, July 15-20, 2018, Volume 1: Long Papers*, pages 2204–2213. Association for Computational Linguistics, 2018.

[540] Kwon, D. S., S. Lee, K. H. Kim, et al. What, when, and how to ground: Designing user persona-aware conversational agents for engaging dialogue. In S. Sitaram, B. B. Klebanov, J. D. Williams, eds., *Proceedings of the The 61st Annual Meeting of the Association for Computational Linguistics: Industry Track, ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 707–719. Association for Computational Linguistics, 2023.

[541] Maes, P. Artificial life meets entertainment: Lifelike autonomous agents. *Commun. ACM*, 38(11):108–114, 1995.

[542] Grossmann, I., M. Feinberg, D. C. Parker, et al. Ai and the transformation of social science research. *Science*, 380(6650):1108–1109, 2023.

[543] Wei, J., K. Shuster, A. Szlam, et al. Multi-party chat: Conversational agents in group settings with humans and models. *CoRR*, abs/2304.13835, 2023.

[544] Hollan, J. D., E. L. Hutchins, L. Weitzman. STEAMER: an interactive inspectable simulation-based training system. *AI Mag.*, 5(2):15–27, 1984.

[545] Tambe, M., W. L. Johnson, R. M. Jones, et al. Intelligent agents for interactive simulation environments. *AI Mag.*, 16(1):15–39, 1995.

[546] Vermeulen, P., D. de Jongh. ‘dynamics of growth in a finite world’ – comprehensive sensitivity analysis. *IFAC Proceedings Volumes*, 9(3):133–145, 1976. IFAC Symposium on Large Scale Systems Theory and Applications, Milano, Italy, 16-20 June.

[547] Forrester, J. W. System dynamics and the lessons of 35 years. In *A systems-based approach to policymaking*, pages 199–240. Springer, 1993.

[548] Santé, I., A. M. García, D. Miranda, et al. Cellular automata models for the simulation of real-world urban processes: A review and analysis. *Landscape and urban planning*, 96(2):108–122, 2010.

[549] Dorri, A., S. S. Kanhere, R. Jurdak. Multi-agent systems: A survey. *Ieee Access*, 6:28573–28593, 2018.

[550] Hendrickx, J. M., S. Martin. Open multi-agent systems: Gossiping with random arrivals and departures. In *56th IEEE Annual Conference on Decision and Control, CDC 2017, Melbourne, Australia, December 12-15, 2017*, pages 763–768. IEEE, 2017.

[551] Ziems, C., W. Held, O. Shaikh, et al. Can large language models transform computational social science? *CoRR*, abs/2305.03514, 2023.

[552] Gilbert, N., J. Doran. *Simulating Societies: The Computer Simulation of Social Phenomena*. Routledge Library Editions: Artificial Intelligence. Taylor & Francis, 2018.

[553] Hamilton, J. D. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica: Journal of the econometric society*, pages 357–384, 1989.

[554] Zhang, G. P. Time series forecasting using a hybrid ARIMA and neural network model. *Neurocomputing*, 50:159–175, 2003.

[555] Kirby, S., M. Dowman, T. L. Griffiths. Innateness and culture in the evolution of language. *Proceedings of the National Academy of Sciences*, 104(12):5241–5245, 2007.

贝茨, J. 情感在可信代理中的作用. *Commun.ACM*, 37(7): 122–125, 1994年。 Karra, S. R., S. Nguyen, T. Tulabandhula. 人工智能拟人化: 估算语言模型的个性. *CoRR*, abs/2204.12000, 2022年。 张, S., D. Dinan, J. Urbanek 等. 个性化对话代理: 我有一只狗, 你也有宠物吗? 在 I. Gurevych, Y. Miyao 编, 《第56届计算语言学协会年会论文集, *ACL 2018*, 澳大利亚墨尔本, 2018年7月15-20日, 第1卷: 长篇论文》, 第2204–2213页. 计算语言学协会, 2018年。 权, D. S., S. Lee, K. H. Kim 等. 何时、何地以及如何进行基础设定: 设计面向用户角色的对话代理以实现引人入胜的对话。在 S. Sitaram, B. B. Klebanov, J. D. Williams 编, 《第61届计算语言学协会年会行业专场论文集, *ACL 2023*, 加拿大多伦多, 2023年7月9-14日》, 第707–719页. 计算语言学协会, 2023年。 梅斯, P. 人工生命与娱乐的结合: 逼真的自主代理. *Commun. ACM*, 38(11): 108–114, 1995年。 格罗斯曼, I., M. Feinberg, D. C. Parker 等. 人工智能与社会科学研究的变革. *Science*, 380(6650): 1108–1109, 2023年。 魏, J., K. Shuster, A. Szlam 等. 多方聊天: 群体环境中的对话代理, 包括人类与模型. *CoRR*, abs/2304.13835, 2023年。 霍兰, J. D., E. L. Hutchins, L. Weitzman. S TEAMER: 一种交互式可检视的基于模拟的培训系统. *AI Mag.*, 5(2): 15–27, 1984年。 坦贝, M., W. L. Johnson, R. M. Jones 等. 面向交互式模拟环境的智能代理. *AI Mag.*, 16(1): 15–39, 1995年。 弗米伦, P., D. de Jongh. “有限世界中的增长动力”——全面敏感性分析. *IFAC Proceedings Volumes*, 9(3): 133–145, 1976年。意大利米兰, IFAC 大规模系统理论与应用研讨会, 6月16-20日。 福雷斯特, J. W. 系统动力学与35年经验教训。在基于系统的政策制定方法中, 第199–240页. 施普林格, 1993年。 桑特, I., A. M. García, D. Miranda 等. 用于模拟现实城市过程的元胞自动机模型: 综述与分析。景观与城市规划, 96(2): 108–122, 2010年。 多瑞, A., S. S. Kanhere, R. Jurdak. 多智能体系统: 综述. *IEEE Access*, 6: 28573–28593, 2018年。 亨德里克斯, J. M., S. Martin. 开放式多智能体系统: 带有随机到达与离开的闲聊。在第56届IEEE决策与控制年会, *CDC 2017*, 加拿大墨尔本, 2017年12月12-15日, 第763–768页. IEEE, 2017年。 齐姆斯, C., W. Held, O. Shaikh 等. 大型语言模型能否变革计算社会科学? *CoRR*, abs/2305.03514, 2023年。 吉尔伯特, N., J. Doran. 模拟社会: 社会现象的计算模拟。劳特利奇图书馆系列: 人工智能. 泰勒与弗朗西斯, 2018年。 汉密尔顿, J. D.。非平稳时间序列与商业周期经济分析的新方法. *Econometrica*: 计量经济学会期刊, 第357–384页, 1989年。 张, G. P.。结合ARIMA与神经网络模型的时间序列预测. *Neurocomputing*, 50: 159–175, 2003年。 柯比, S., M. Dowman, T. L. Griffiths. 语言演化中的天生性与文化. 美国国家科学院院刊, 104(12): 5241–5245, 2007年。

[556] Shibata, H., S. Miki, Y. Nakamura. Playing the werewolf game with artificial intelligence for language understanding. *CoRR*, abs/2302.10646, 2023.

[557] Junprung, E. Exploring the intersection of large language models and agent-based modeling via prompt engineering. *CoRR*, abs/2308.07411, 2023.

[558] Phelps, S., Y. I. Russell. Investigating emergent goal-like behaviour in large language models using experimental economics. *CoRR*, abs/2305.07970, 2023.

[559] Bellomo, N., G. A. Marsan, A. Tosin. *Complex systems and society: modeling and simulation*, vol. 2. Springer, 2013.

[560] Moon, Y. B. Simulation modelling for sustainability: a review of the literature. *International Journal of Sustainable Engineering*, 10(1):2–19, 2017.

[561] Helberger, N., N. Diakopoulos. Chatgpt and the AI act. *Internet Policy Rev.*, 12(1), 2023.

[562] Weidinger, L., J. Mellor, M. Rauh, et al. Ethical and social risks of harm from language models. *CoRR*, abs/2112.04359, 2021.

[563] Deshpande, A., V. Murahari, T. Rajpurohit, et al. Toxicity in chatgpt: Analyzing persona-assigned language models. *CoRR*, abs/2304.05335, 2023.

[564] Kirk, H. R., Y. Jun, F. Volpin, et al. Bias out-of-the-box: An empirical analysis of intersectional occupational biases in popular generative language models. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan, eds., *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 2611–2624. 2021.

[565] Nadeem, M., A. Bethke, S. Reddy. Stereoset: Measuring stereotypical bias in pretrained language models. In C. Zong, F. Xia, W. Li, R. Navigli, eds., *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, ACL/IJCNLP 2021, (Volume 1: Long Papers), Virtual Event, August 1-6, 2021*, pages 5356–5371. Association for Computational Linguistics, 2021.

[566] Roberts, T., G. Marchais. Assessing the role of social media and digital technology in violence reporting. *Contemporary Readings in Law & Social Justice*, 10(2), 2018.

[567] Kandpal, N., H. Deng, A. Roberts, et al. Large language models struggle to learn long-tail knowledge. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *Proceedings of the 40th International Conference on Machine Learning*, vol. 202 of *Proceedings of Machine Learning Research*, pages 15696–15707. PMLR, 2023.

[568] Ferrara, E. Should chatgpt be biased? challenges and risks of bias in large language models. *CoRR*, abs/2304.03738, 2023.

[569] Haller, P., A. Aynetdinov, A. Akbik. Opiniongpt: Modelling explicit biases in instruction-tuned llms, 2023.

[570] Salewski, L., S. Alaniz, I. Rio-Torto, et al. In-context impersonation reveals large language models’ strengths and biases. *CoRR*, abs/2305.14930, 2023.

[571] Lin, B., D. Bouneffouf, G. A. Cecchi, et al. Towards healthy AI: large language models need therapists too. *CoRR*, abs/2304.00416, 2023.

[572] Liang, P. P., C. Wu, L. Morency, et al. Towards understanding and mitigating social biases in language models. In M. Meila, T. Zhang, eds., *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, vol. 139 of *Proceedings of Machine Learning Research*, pages 6565–6576. PMLR, 2021.

[573] Henderson, P., K. Sinha, N. Angelard-Gontier, et al. Ethical challenges in data-driven dialogue systems. In J. Furman, G. E. Marchant, H. Price, F. Rossi, eds., *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society, AIES 2018, New Orleans, LA, USA, February 02-03, 2018*, pages 123–129. ACM, 2018.

[556] 柴田浩史，三木慎一，中村康之。利用人工智能进行狼人游戏中的语言理解。*CoRR*, abs/2302.10646, 2023年。[557] Junprung, E. 探索大规模语言模型与基于代理的建模的交集：通过提示工程。*CoRR*, abs/2308.07411, 2023年。[558] Phelps, S., Y. I. Russell。利用实验经济学研究大规模语言模型中的新兴目标导向行为。*CoRR*, abs/2305.07970, 2023年。[559] Bellomo, N., G. A. Marsan, A. Tosin。复杂系统与社会：建模与仿真，第2卷。Springer, 2013年。[560] Moon, Y. B. 可持续性模拟建模：文献综述。国际可持续工程期刊, 10(1): 2–19, 2017年。[561] Helberger, N., N. Diakopoulos。ChatGPT与人工智能法案。互联网政策评论, 12(1), 2023年。[562] Weidinger, L., J. Mellor, M. Rauh 等。语言模型的伦理与社会风险。*CoRR*, abs/2112.04359, 2021年。[563] Deshpande, A., V. Murahari, T. Rajpurohit 等。ChatGPT中的毒性：人格分配语言模型的分析。*CoRR*, abs/2304.05335, 2023年。[564] Kirk, H. R., Y. Jun, F. Volpin 等。偏见一览：对流行生成式语言模型中交叉职业偏见的实证分析。在M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, J. W. Vaughan 编著，神经信息处理系统进展34：2021年神经信息处理系统年度会议，*NeurIPS 2021, 2021年12月6-14日*，虚拟会议，第2611–2624页。2021年。[565] Nadeem, M., A. Bethke, S. Reddy。Stereoset：衡量预训练语言模型中的刻板偏见。在C. Zong, F. Xia, W. Li, R. Navigli 编著，第59届计算语言学协会年会暨第11届自然语言处理国际联合会会议论文集（第1卷：长论文），虚拟会议，2021年8月1-6日，第5356–5371页。计算语言学协会，2021年。[566] Roberts, T., G. Marchais。评估社交媒体和数字技术在暴力报道中的作用。当代法律与社会正义读物, 10(2), 2018年。[567] Kandpal, N., H. Deng, A. Roberts 等。大型语言模型难以学习长尾知识。在A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett 编著，第40届国际机器学习会议论文集，第202卷，机器学习研究论文集，第15696–15707页。PMLR, 2023年。[568] Ferrara, E. ChatGPT应否存在偏见？大型语言模型中的偏见挑战与风险。*CoRR*, abs/2304.03738, 2023年。[569] Haller, P., A. Aynetdinov, A. Akbik。Opiniongpt：建模指令调优大型语言模型中的显式偏见，2023年。[570] Salewski, L., S. Alaniz, I. Rio-Torto 等。在上下文中模仿揭示大型语言模型的优势与偏见。*CoRR*, abs/2305.14930, 2023年。[571] Lin, B., D. Bouneffouf, G. A. Cecchi 等。迈向健康的人工智能：大型语言模型也需要治疗师。*CoRR*, abs/2304.00416, 2023年。[572] Liang, P. P., C. Wu, L. Morency 等。理解与缓解语言模型中的社会偏见。In M. Meila, T. Zhang 编著，第38届国际机器学习会议论文集，*ICML 2021, 2021年7月18-24日*，虚拟会议，第139卷，机器学习研究论文集，第6565–6576页。PMLR, 2021年。[573] Henderson, P., K. Sinha, N. Angelard-Gontier 等。数据驱动对话系统中的伦理挑战。In J. Furman, G. E. Marchant, H. Price, F. Rossi 编著，2018年 AAAI/ACM人工智能、伦理与社会会议论文集，*AIES 2018, 路易斯安那州新奥尔良，美国, 2018年2月2-3日*，第123–129页。ACM, 2018年。

[574] Li, H., Y. Song, L. Fan. You don’t know my favorite color: Preventing dialogue representations from revealing speakers’ private personas. In M. Carpuat, M. de Marneffe, I. V. M. Ruíz, eds., *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2022, Seattle, WA, United States, July 10-15, 2022*, pages 5858–5870. Association for Computational Linguistics, 2022.

[575] Brown, H., K. Lee, F. Miresghallah, et al. What does it mean for a language model to preserve privacy? In *FAccT ’22: 2022 ACM Conference on Fairness, Accountability, and Transparency, Seoul, Republic of Korea, June 21 - 24, 2022*, pages 2280–2292. ACM, 2022.

[576] Sebastian, G. Privacy and data protection in chatgpt and other ai chatbots: Strategies for securing user information. *Available at SSRN 4454761*, 2023.

[577] Reeves, B., C. Nass. *The media equation - how people treat computers, television, and new media like real people and places*. Cambridge University Press, 1996.

[578] Roose, K. A conversation with bing’s chatbot left me deeply unsettled, 2023.

[579] Li, K., A. K. Hopkins, D. Bau, et al. Emergent world representations: Exploring a sequence model trained on a synthetic task. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.

[580] Bai, Y., S. Kadavath, S. Kundu, et al. Constitutional AI: harmlessness from AI feedback. *CoRR*, abs/2212.08073, 2022.

[581] Bai, Y., A. Jones, K. Ndousse, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *CoRR*, abs/2204.05862, 2022.

[582] Liu, X., H. Yu, H. Zhang, et al. Agentbench: Evaluating llms as agents. *CoRR*, abs/2308.03688, 2023.

[583] Aher, G. V., R. I. Arriaga, A. T. Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 337–371. PMLR, 2023.

[584] Liang, Y., L. Zhu, Y. Yang. Tachikuma: Understanding complex interactions with multi-character and novel objects by large language models. *CoRR*, abs/2307.12573, 2023.

[585] Xu, B., X. Liu, H. Shen, et al. Gentopia: A collaborative platform for tool-augmented llms. *CoRR*, abs/2308.04030, 2023.

[586] Kim, S. S., E. A. Watkins, O. Russakovsky, et al. " help me help the ai": Understanding how explainability can support human-ai interaction. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pages 1–17. 2023.

[587] Choi, M., J. Pei, S. Kumar, et al. Do llms understand social knowledge? evaluating the sociability of large language models with socket benchmark. *CoRR*, abs/2305.14938, 2023.

[588] Wilson, A. C., D. V. Bishop. " if you catch my drift...": ability to infer implied meaning is distinct from vocabulary and grammar skills. *Wellcome open research*, 4, 2019.

[589] Shuster, K., J. Urbanek, A. Szlam, et al. Am I me or you? state-of-the-art dialogue models cannot maintain an identity. In M. Carpuat, M. de Marneffe, I. V. M. Ruíz, eds., *Findings of the Association for Computational Linguistics: NAACL 2022, Seattle, WA, United States, July 10-15, 2022*, pages 2367–2387. Association for Computational Linguistics, 2022.

[590] Ganguli, D., L. Lovitt, J. Kernion, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *CoRR*, abs/2209.07858, 2022.

[591] Kadavath, S., T. Conerly, A. Askell, et al. Language models (mostly) know what they know. *CoRR*, abs/2207.05221, 2022.

[574] Li, H., Y. Song, L. Fan. 你不知道我最喜欢的颜色：防止对话表示泄露说话者的私人信息。载于 M. Carpuat, M. de Marneffe, I. V. M. Ruíz 编, 2022 年北美计算语言学协会人类语言技术会议论文集： *NAACL 2022*，西雅图，华盛顿，美国，2022年7月10-15日，第5858-5870页。计算语言学协会，2022年。[575] Brown, H., K. Lee, F. Miresghallah 等。语言模型如何实现隐私保护？载于 *FAccT ’22：2022 年公平、责任与透明度 ACM 会议*，首尔，大韩民国，2022年6月21-24日，第2280-2292页。ACM，2022年。[576] Sebastian, G. 聊天机器人及其他人工智能聊天系统中的隐私与数据保护：保障用户信息的策略。可在 *SSRN 4454761* 获取，2023年。[577] Reeves, B., C. Nass. 媒体等式——人们如何对待计算机、电视和新媒体，就像对待真实的人和场所一样。剑桥大学出版社，1996年。[578] Roose, K. 与 Bing 聊天机器人的对话让我感到非常不安，2023年。[579] Li, K., A. K. Hopkins, D. Bau 等。新兴的世界表征：探索在合成任务上训练的序列模型。载于 第十一届学习表征国际会议（*ICLR 2023*），卢旺达基加利，2023年5月1-5日。OpenReview.net，2023年。[580] Bai, Y., S. Kadavath, S. Kundu 等。宪法人工智能：来自人工智能反馈的无害性。 *CoRR*，abs/2212.08073，2022年。[581] Bai, Y., A. Jones, K. Ndousse 等。利用强化学习从人类反馈中训练有帮助且无害的助手。 *CoRR*，abs/2204.05862，2022年。[582] Liu, X., H. Yu, H. Zhang 等。Agentbench：评估大型语言模型作为代理的能力。 *CoRR*，abs/2308.03688，2023年。[583] Aher, G. V., R. I. Arriaga, A. T. Kalai. 利用大型语言模型模拟多个人类并复制人类受试者研究。载于 A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett 编，机器学习国际会议（*ICML 2023*），2023年7月23-29日，夏威夷檀香山，第 202 卷 机器学习研究论文集，第337-371页。PMLR，2023年。[584] Liang, Y., L. Zhu, Y. Yang. Tachikuma：通过大型语言模型理解复杂交互中的多角色和新颖对象。 *CoRR*，abs/2307.12573，2023年。[585] Xu, B., X. Liu, H. Shen 等。Gentopia：一个用于工具增强型大型语言模型的协作平台。 *CoRR*，abs/2308.04030，2023年。[586] Kim, S. S., E. A. Watkins, O. Russakovsky 等。“帮我帮 AI”：理解可解释性如何支持人机交互。载于 2023 年人机交互系统会议（*CHI*）论文集，第1-17页。2023年。[587] Choi, M., J. Pei, S. Kumar 等。大型语言模型是否理解社会知识？用 socket 基准评估其社交能力。 *CoRR*，abs/2305.14938，2023年。[588] Wilson, A. C., D. V. Bishop. “如果你明白我的意思……”：推断暗示性意义的能力与词汇和语法技能不同。 *Wellcomeopen research*，第4期，2019年。[589] Shuster, K., J. Urbanek, A. Szlam 等。我是谁还是你？最先进的对话模型无法保持身份一致性。载于 M. Carpuat, M. de Marneffe, I. V. M. Ruíz 编，计算语言学协会会议论文集： *NAACL 2022*，西雅图，华盛顿，美国，2022年7月10-15日，第2367-2387页。计算语言学协会，2022年。[590] Ganguli, D., L. Lovitt, J. Kernion 等。通过“红队”方法减少语言模型的危害：方法、扩展行为及经验教训。 *CoRR*，abs/2209.07858，2022年。[591] Kadavath, S., T. Conerly, A. Askell 等。语言模型（大多）知道它们知道什么。 *CoRR*，abs/2207.05221，2022年。

- [592] Colas, C., L. Teodorescu, P. Oudeyer, et al. Augmenting autotelic agents with large language models. *CoRR*, abs/2305.12487, 2023.
- [593] Chaudhry, A., P. K. Dokania, T. Ajanthan, et al. Riemannian walk for incremental learning: Understanding forgetting and intransigence. In *Proceedings of the European conference on computer vision (ECCV)*, pages 532–547. 2018.
- [594] Hou, S., X. Pan, C. C. Loy, et al. Learning a unified classifier incrementally via rebalancing. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 831–839. 2019.
- [595] Colas, C., T. Karch, O. Sigaud, et al. Autotelic agents with intrinsically motivated goal-conditioned reinforcement learning: A short survey. *J. Artif. Intell. Res.*, 74:1159–1199, 2022.
- [596] Szegedy, C., W. Zaremba, I. Sutskever, et al. Intriguing properties of neural networks. In Y. Bengio, Y. LeCun, eds., *2nd International Conference on Learning Representations, ICLR 2014, Banff, AB, Canada, April 14-16, 2014, Conference Track Proceedings*. 2014.
- [597] Goodfellow, I. J., J. Shlens, C. Szegedy. Explaining and harnessing adversarial examples. In Y. Bengio, Y. LeCun, eds., *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*. 2015.
- [598] Madry, A., A. Makelov, L. Schmidt, et al. Towards deep learning models resistant to adversarial attacks. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018.
- [599] Zheng, R., Z. Xi, Q. Liu, et al. Characterizing the impacts of instances on robustness. In A. Rogers, J. L. Boyd-Graber, N. Okazaki, eds., *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 2314–2332. Association for Computational Linguistics, 2023.
- [600] Zhiheng, X., Z. Rui, G. Tao. Safety and ethical concerns of large language models. In *Proceedings of the 22nd Chinese National Conference on Computational Linguistics (Volume 4: Tutorial Abstracts)*, pages 9–16. 2023.
- [601] Akhtar, N., A. Mian, N. Kardan, et al. Threat of adversarial attacks on deep learning in computer vision: Survey II. *CoRR*, abs/2108.00401, 2021.
- [602] Drenkow, N., N. Sani, I. Shpitser, et al. A systematic review of robustness in deep learning for computer vision: Mind the gap? *arXiv preprint arXiv:2112.00639*, 2021.
- [603] Hendrycks, D., T. G. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019.
- [604] Wang, X., H. Wang, D. Yang. Measure and improve robustness in NLP models: A survey. In M. Carpuat, M. de Marneffe, I. V. M. Ruíz, eds., *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2022, Seattle, WA, United States, July 10-15, 2022*, pages 4569–4586. Association for Computational Linguistics, 2022.
- [605] Li, J., S. Ji, T. Du, et al. Textbugger: Generating adversarial text against real-world applications. In *26th Annual Network and Distributed System Security Symposium, NDSS 2019, San Diego, California, USA, February 24-27, 2019*. The Internet Society, 2019.
- [606] Zhu, C., Y. Cheng, Z. Gan, et al. Freedb: Enhanced adversarial training for natural language understanding. In *8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020*. OpenReview.net, 2020.
- [607] Xi, Z., R. Zheng, T. Gui, et al. Efficient adversarial training with robust early-bird tickets. In Y. Goldberg, Z. Kozareva, Y. Zhang, eds., *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, pages 8318–8331. Association for Computational Linguistics, 2022.

- [592] Colas, C., L. Teodorescu, P. Oudeyer, 等. 利用大型语言模型增强自发性代理. *CoRR*, abs/2305.12487, 2023年.
- [593] Chaudhry, A., P. K. Dokania, T. Ajanthan, 等. 测地线行走在增量学习中的应用: 理解遗忘与不妥协. 发表于欧洲计算机视觉会议 (ECCV) 论文集, 第532–547页, 2018年.
- [594] Hou, S., X. Pan, C. C. Loy, 等. 通过再平衡逐步学习统一分类器. 发表于*IEEE/CVF*计算机视觉与模式识别会议论文集, 第831–839页, 2019年.
- [595] Colas, C., T. Karch, O. Sigaud, 等. 具有内在动机目标条件强化学习的自发性代理: 简要综述. 人工智能研究杂志, 74: 1159–1199, 2022年.
- [596] Szegedy, C., W. Zaremba, I. Sutskever, 等. 神经网络的引人入胜的特性. 收录于Y. Bengio, Y. LeCun编辑, 《第二届学习表示国际会议 (ICLR) 2014》, 班夫, AB, 加拿大, 2014年4月14-16日, 会议论文集. 2014年.
- [597] Goodfellow, I. J., J. Shlens, C. Szegedy. 解释与利用对抗样本. 收录于Y. Bengio, Y. LeCun编辑, 《第三届学习表示国际会议 (ICLR) 2015》, 圣地亚哥, CA, 美国, 2015年5月7-9日, 会议论文集. 2015年.
- [598] Madry, A., A. Makelov, L. Schmidt, 等. 迈向对抗攻击具有抵抗力的深度学习模型. 收录于《第六届学习表示国际会议 (ICLR) 2018》, 温哥华, BC, 加拿大, 2018年4月30日至5月3日, 会议论文集. OpenReview.net, 2018年.
- [599] Zheng, R., Z. Xi, Q. Liu, 等. 影响实例对模型鲁棒性的影响特征分析. 收录于A. Rogers, J. L. Boyd-Graber, N. Okazaki编辑, 《计算语言学协会2023年会议论文集 (ACL 2023) 》, 多伦多, 加拿大, 2023年7月9-14日, 第2314–2332页. 计算语言学协会, 2023年.
- [600] Zhiheng, X., Z. Rui, G. Tao. 大型语言模型的安全性与伦理问题. 收录于第22届中国计算语言学全国会议 (第四卷: 教程摘要), 第9–16页, 2023年.
- [601] Akhtar, N., A. Mian, N. Kardan, a bs/2108.00401, 2021年.
- [602] Drenkow, N., N. Sani, I. Shpitser, 等. 深度学习在计算机视觉中的鲁棒性系统性综述: 弥合差距? *arXiv预印本 arXiv:2112.00639*, 2021年.
- [603] Hendrycks, D., T. G. Dietterich. 基准测试神经网络对常见损坏和扰动的鲁棒性. 收录于第七届学习表示国际会议 (*ICLR 2019*), 新奥尔良, LA, 美国, 2019年5月6-9日. OpenReview.net, 2019年.
- [604] Wang, X., H. Wang, D. Yang. 测量与提升NLP模型的鲁棒性: 综述. 收录于M. Carpuat, M. de Marneffe, I. V. M. Ruíz编辑, 《2022年北美计算语言学协会人类语言技术会议论文集 (NAACL 2022) 》, 西雅图, WA, 美国, 2022年7月10-15日, 第4569–4586页. 计算语言学协会, 2022年.
- [605] Li, J., S. Ji, T. Du, 等. Textbugger: 针对实际应用生成对抗文本. 收录于第26届网络与分布式系统安全研讨会 (*NDSS 2019*), 圣地亚哥, 加利福尼亚, 美国, 2019年2月24-27日. 互联网协会, 2019年.
- [606] Zhu, C., Y. Cheng, Z. Gan, 等. Freedb: 增强的自然语言理解对抗训练. 收录于第八届学习表示国际会议 (*ICLR 2020*), 亚的斯亚贝巴, 埃塞俄比亚, 2020年4月26-30日. OpenReview.net, 2020年.
- [607] Xi, Z., R. Zheng, T. Gui, 等. 利用鲁棒的早鸟票进行高效对抗训练. 收录于 Y. Goldberg, Z. Kozareva, Y. Zhang 编辑, 《2022年自然语言处理实证方法会议 (EMNLP 2022) 》, 阿布扎比, 阿联酋, 2022年12月7-11日, 第8318–8331页. 计算语言学协会, 2022年.

- [608] Pinto, L., J. Davidson, R. Sukthankar, et al. Robust adversarial reinforcement learning. In D. Precup, Y. W. Teh, eds., *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, vol. 70 of *Proceedings of Machine Learning Research*, pages 2817–2826. PMLR, 2017.
- [609] Rigter, M., B. Lacerda, N. Hawes. RAMBO-RL: robust adversarial model-based offline reinforcement learning. In *NeurIPS*. 2022.
- [610] Panaganti, K., Z. Xu, D. Kalathil, et al. Robust reinforcement learning using offline data. In *NeurIPS*. 2022.
- [611] Lab, T. K. S. Experimental security research of tesla autopilot. *Tencent Keen Security Lab*, 2019.
- [612] Xu, K., G. Zhang, S. Liu, et al. Adversarial t-shirt! evading person detectors in a physical world. In A. Vedaldi, H. Bischof, T. Brox, J. Frahm, eds., *Computer Vision - ECCV 2020 - 16th European Conference, Glasgow, UK, August 23-28, 2020, Proceedings, Part V*, vol. 12350 of *Lecture Notes in Computer Science*, pages 665–681. Springer, 2020.
- [613] Sharif, M., S. Bhagavatula, L. Bauer, et al. Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition. In E. R. Weippl, S. Katzenbeisser, C. Kruegel, A. C. Myers, S. Halevi, eds., *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, Vienna, Austria, October 24-28, 2016*, pages 1528–1540. ACM, 2016.
- [614] Jin, D., Z. Jin, J. T. Zhou, et al. Is BERT really robust? A strong baseline for natural language attack on text classification and entailment. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence, AAAI 2020, The Thirty-Second Innovative Applications of Artificial Intelligence Conference, IAAI 2020, The Tenth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2020, New York, NY, USA, February 7-12, 2020*, pages 8018–8025. AAAI Press, 2020.
- [615] Ren, S., Y. Deng, K. He, et al. Generating natural language adversarial examples through probability weighted word saliency. In A. Korhonen, D. R. Traum, L. Màrquez, eds., *Proceedings of the 57th Conference of the Association for Computational Linguistics, ACL 2019, Florence, Italy, July 28- August 2, 2019, Volume 1: Long Papers*, pages 1085–1097. Association for Computational Linguistics, 2019.
- [616] Zhu, K., J. Wang, J. Zhou, et al. Promptbench: Towards evaluating the robustness of large language models on adversarial prompts. *CoRR*, abs/2306.04528, 2023.
- [617] Chen, X., J. Ye, C. Zu, et al. How robust is GPT-3.5 to predecessors? A comprehensive study on language understanding tasks. *CoRR*, abs/2303.00293, 2023.
- [618] Gu, T., B. Dolan-Gavitt, S. Garg. Badnets: Identifying vulnerabilities in the machine learning model supply chain. *CoRR*, abs/1708.06733, 2017.
- [619] Chen, X., A. Salem, D. Chen, et al. Badnl: Backdoor attacks against NLP models with semantic-preserving improvements. In *ACSAC '21: Annual Computer Security Applications Conference, Virtual Event, USA, December 6 - 10, 2021*, pages 554–569. ACM, 2021.
- [620] Li, Z., D. Mekala, C. Dong, et al. Bfclass: A backdoor-free text classification framework. In M. Moens, X. Huang, L. Specia, S. W. Yih, eds., *Findings of the Association for Computational Linguistics: EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 16-20 November, 2021*, pages 444–453. Association for Computational Linguistics, 2021.
- [621] Shi, Y., P. Li, C. Yin, et al. Promptattack: Prompt-based attack for language models via gradient search. In W. Lu, S. Huang, Y. Hong, X. Zhou, eds., *Natural Language Processing and Chinese Computing - 11th CCF International Conference, NLPCC 2022, Guilin, China, September 24-25, 2022, Proceedings, Part I*, vol. 13551 of *Lecture Notes in Computer Science*, pages 682–693. Springer, 2022.
- [622] Perez, F., I. Ribeiro. Ignore previous prompt: Attack techniques for language models. *CoRR*, abs/2211.09527, 2022.

Pinto, L., J. Davidson, R. Sukthankar 等。鲁棒对抗强化学习。收录于 D. Precup, Y. W. Teh 编著的第34届国际机器学习会议论文集, *ICML 2017*, 澳大利亚悉尼, *NSW*, 2017年8月6-11日, 第70卷机器学习研究会议论文集, 第2817–2826页。PMLR, 2017年。

Rigter, M., B. Lacerda, N. Hawes. RAMBO-RL: 鲁棒对抗模型基础的离线强化学习。发表于*NeurIPS*, 2022年。 Panaganti, K., Z. Xu, D. Kalathil 等。利用离线数据的鲁棒强化学习。发表于*NeurIPS*, 2022年。 Lab, T. K. S. 特斯拉自动驾驶安全性实验研究。腾讯安全实验室, 2019年。 Xu, K., G. Zhang, S. Liu 等。对抗T恤! 在实体世界中规避人员检测器。收录于 A. Vedaldi, H. Bischof, T. Brox, J. Frahm 编著的计算机视觉——*ECCV 2020*——第16届欧洲会议, 格拉斯哥, 英国, 2020年8月23-28日, 论文集, 第V部分, 第12350卷计算机科学讲义, 第665–681页。Springer, 2020年。 Sharif, M., S. Bhagavatula, L. Bauer 等。装饰成犯罪: 对最先进面部识别的真实且隐秘的攻击。收录于 E. R. Weippl, S. Katzenbeisser, C. Kruegel, A. C. Myers, S. Halevi 编著的2016年 *ACM SIGSAC* 计算机与通信安全会议论文集, 维也纳, 奥地利, 2016年10月24-28日, 第1528–1540页。ACM, 2016年。 Jin, D., Z. Jin, J. T. Zhou 等。BERT真的具有鲁棒性吗? 针对文本分类和蕴含任务的强基线自然语言攻击。收录于第34届AAAI人工智能会议, *AAAI 2020*, 第32届人工智能创新应用会议, *IAAI 2020*, 第10届AAAI教育进展研讨会, *EAAI 2020*, 纽约, 美国, 2020年2月7-12日, 第8018–8025页。AAAI出版社, 2020年。 Ren, S., Y. Deng, K. He 等。通过概率加权词重要性生成自然语言对抗样本。收录于 A. Korhonen, D. R. Traum, L. Màrquez 编著的第57届计算语言学协会会议论文集, *ACL 2019*, 意大利佛罗伦萨, 2019年7月28日-8月2日, 第1卷: 长论文, 第1085–1097页。计算语言学协会, 2019年。 Zhu, K., J. Wang, J. Zhou 等。Promptbench: 评估大型语言模型在对抗性提示下的鲁棒性。*CoRR*, abs/2306.04528, 2023年。 Chen, X., J. Ye, C. Zu 等。GPT-3.5对前辈的鲁棒性如何? 关于语言理解任务的全面研究。*CoRR*, abs/2303.00293, 2023年。 Gu, T., B. Dolan-Gavitt, S. Garg. Badnets: 识别机器学习模型供应链中的漏洞。*CoRR*, abs/1708.06733, 2017年。 Chen, X., A. Salem, D. Chen 等。Badnl: 针对NLP模型的后门攻击及语义保持改进。收录于 *ACSAC '21: 年度计算机安全应用会议, 虚拟会议, 美国, 2021年12月6-10日*, 第554–569页。ACM, 2021年。 Li, Z., D. Mekala, C. Dong 等。Bfclass: 一种无后门文本分类框架。收录于 M. Moens, X. Huang, L. Specia, S. W. Yih 编著的计算语言学协会会议论文集: *EMNLP 2021*, 虚拟会议/多米尼加共和国蓬塔卡纳, 2021年11月16-20日, 第444–453页。计算语言学协会, 2021年。 Shi, Y., P. Li, C. Yin 等。Promptattack: 基于梯度搜索的提示攻击方法。收录于 W. Lu, S. Huang, Y. Hong, X. Zhou 编著的自然语言处理与中文计算——第11届CCF国际会议, *NLPCC 2022*, 广西桂林, 2022年9月24-25日, 论文集, 第I部分, 第13551卷计算机科学讲义, 第682–693页。Springer, 2022年。 Perez, F., I. Ribeiro。忽略前述提示: 语言模型的攻击技术。*CoRR*, abs/2211.09527, 2022年。

- [623] Liang, P., R. Bommasani, T. Lee, et al. Holistic evaluation of language models. *CoRR*, abs/2211.09110, 2022.
- [624] Gururangan, S., D. Card, S. K. Dreier, et al. Whose language counts as high quality? measuring language ideologies in text data selection. In Y. Goldberg, Z. Kozareva, Y. Zhang, eds., *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022, Abu Dhabi, United Arab Emirates, December 7-11, 2022*, pages 2562–2580. Association for Computational Linguistics, 2022.
- [625] Liu, Y., G. Deng, Y. Li, et al. Prompt injection attack against llm-integrated applications. *CoRR*, abs/2306.05499, 2023.
- [626] Carlini, N., D. A. Wagner. Audio adversarial examples: Targeted attacks on speech-to-text. In *2018 IEEE Security and Privacy Workshops, SP Workshops 2018, San Francisco, CA, USA, May 24, 2018*, pages 1–7. IEEE Computer Society, 2018.
- [627] Morris, J. X., E. Lifland, J. Y. Yoo, et al. Textattack: A framework for adversarial attacks, data augmentation, and adversarial training in NLP. In Q. Liu, D. Schlangen, eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, EMNLP 2020 - Demos, Online, November 16-20, 2020*, pages 119–126. Association for Computational Linguistics, 2020.
- [628] Si, C., Z. Zhang, F. Qi, et al. Better robustness by more coverage: Adversarial and mixup data augmentation for robust finetuning. In C. Zong, F. Xia, W. Li, R. Navigli, eds., *Findings of the Association for Computational Linguistics: ACL/IJCNLP 2021, Online Event, August 1-6, 2021*, vol. ACL/IJCNLP 2021 of *Findings of ACL*, pages 1569–1576. Association for Computational Linguistics, 2021.
- [629] Yoo, K., J. Kim, J. Jang, et al. Detection of adversarial examples in text classification: Benchmark and baseline via robust density estimation. In S. Muresan, P. Nakov, A. Villavicencio, eds., *Findings of the Association for Computational Linguistics: ACL 2022, Dublin, Ireland, May 22-27, 2022*, pages 3656–3672. Association for Computational Linguistics, 2022.
- [630] Le, T., N. Park, D. Lee. A sweet rabbit hole by DARCY: using honeypots to detect universal trigger’s adversarial attacks. In C. Zong, F. Xia, W. Li, R. Navigli, eds., *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, ACL/IJCNLP 2021, (Volume 1: Long Papers), Virtual Event, August 1-6, 2021*, pages 3831–3844. Association for Computational Linguistics, 2021.
- [631] Tsipras, D., S. Santurkar, L. Engstrom, et al. Robustness may be at odds with accuracy. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019.
- [632] Zhang, H., Y. Yu, J. Jiao, et al. Theoretically principled trade-off between robustness and accuracy. In K. Chaudhuri, R. Salakhutdinov, eds., *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, vol. 97 of *Proceedings of Machine Learning Research*, pages 7472–7482. PMLR, 2019.
- [633] Wong, A., X. Y. Wang, A. Hryniowski. How much can we really trust you? towards simple, interpretable trust quantification metrics for deep neural networks. *CoRR*, abs/2009.05835, 2020.
- [634] Huang, X., D. Kroening, W. Ruan, et al. A survey of safety and trustworthiness of deep neural networks: Verification, testing, adversarial attack and defence, and interpretability. *Comput. Sci. Rev.*, 37:100270, 2020.
- [635] Huang, X., W. Ruan, W. Huang, et al. A survey of safety and trustworthiness of large language models through the lens of verification and validation. *CoRR*, abs/2305.11391, 2023.
- [636] Raffel, C., N. Shazeer, A. Roberts, et al. Exploring the limits of transfer learning with a unified text-to-text transformer. *J. Mach. Learn. Res.*, 21:140:1–140:67, 2020.

梁, P., R. Bommasani, T. Lee 等。语言模型的整体评估。 *CoRR*, abs/2211.09110, 2022年。 Gururangan, S., D. Card, S. K. Dreier 等。谁的语言被认为是高质量的? 衡量文本数据选择中的语言意识形态。收录于Y. Goldberg, Z. Kozareva, Y. Zhang 编, 《2022年自然语言处理实证方法会议论文集, *EMNLP 2022*, 阿布扎比, 阿联酋, 2022年12月7-11日》, 第2562–2580页。计算语言学协会, 2022年。 刘, Y., G. Deng, Y. Li 等。针对大语言模型集成应用的提示注入攻击。 *CoRR*, abs/2306.05499, 2023年。 Carlini, N., D. A. Wagner. 音频对抗样本: 针对语音转文本的定向攻击。收录于 *2018年IEEE安全与隐私研讨会, SP Workshops 2018*, 旧金山, 加利福尼亚, 美国, 2018年5月24日, 第1–7页。IEEE计算机学会, 2018年。 Morris, J. X., E. Lifland, J. Y. Yoo 等。Textattack: 一个用于对抗攻击、数据增强和对抗训练的自然语言处理框架。收录于Q. Liu, D. Schlangen 编, 《2020年自然语言处理实证方法会议: 系统演示, *EMNLP 2020* - 演示, 线上, 2020年11月16-20日》, 第119–126页。计算语言学协会, 2020年。 Si, C., Z. Zhang, F. Qi 等。通过更全面的覆盖实现更强的鲁棒性: 对抗性和混合数据增强以实现稳健微调。收录于C. Zong, F. Xia, W. Li, R. Navigli 编, 《计算语言学协会研究成果: *ACL/IJCNLP 2021*, 线上活动, 2021年8月1-6日》, 第1569–1576页。计算语言学协会, 2021年。 Yoo, K., J. Kim, J. Jang 等。在文本分类中检测对抗样本: 通过稳健密度估计的基准和基线。收录于S. Muresan, P. Nakov, A. Villavicencio 编, 《计算语言学协会研究成果: *ACL 2022*, 爱尔兰都柏林, 2022年5月22-27日》, 第3656–3672页。计算语言学协会, 2022年。 Le, T., N. Park, D. Lee. DARCY的“甜蜜兔子洞”: 利用蜜罐检测通用触发器的对抗攻击。收录于C. Zong, F. Xia, W. Li, R. Navigli 编, 《第59届计算语言学协会年会暨第11届自然语言处理国际联合会议论文集(第1卷: 长论文), 虚拟会议, 2021年8月1-6日》, 第3831–3844页。计算语言学协会, 2021年。 Tsipras, D., S. Santurkar, L. Engstrom 等。鲁棒性可能与准确性相冲突。收录于第7届学习表示国际会议 (*ICLR 2019*), 新奥尔良, 路易斯安那, 美国, 2019年5月6-9日。OpenReview.net, 2019年。 Zhang, H., Y. Yu, J. Jiao 等。理论上原则性的鲁棒性与准确性之间的权衡。收录于K. Chaudhuri, R. Salakhutdinov 编, 《第36届国际机器学习会议 (*ICML 2019*), 2019年6月9-15日, 长滩, 加利福尼亚, 美国》, 第97卷《机器学习研究论文集》, 第7472–7482页。PMLR, 2019年。 Wong, A., X. Y. Wang, A. Hryniowski. 我们到底能多信任你? 面向深度神经网络的简单、可解释的信任量化指标。 *CoRR*, abs/2009.05835, 2020年。 Huang, X., D. Kroening, W. Ruan 等。深度神经网络的安全性与可信性综述: 验证、测试、对抗攻击与防御, 以及可解释性。 *Comput. Sci. Rev.*, 37: 100270, 2020年。 Huang, X., W. Ruan, W. Huang 等。通过验证与确认视角对大语言模型的安全性与可信性进行综述。 *CoRR*, abs/2305.11391, 2023年。 Raffel, C., N. Shazeer, A. Roberts 等。探索统一文本到文本转换器在迁移学习中的极限。 *J. Mach. Learn. Res.*, 21: 140:1–140:67, 2020年。

[637] Chen, Y., L. Yuan, G. Cui, et al. A close look into the calibration of pre-trained language models. In A. Rogers, J. L. Boyd-Graber, N. Okazaki, eds., *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 1343–1367. Association for Computational Linguistics, 2023.

[638] Blodgett, S. L., S. Barocas, H. D. III, et al. Language (technology) is power: A critical survey of "bias" in NLP. In D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault, eds., *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020*, pages 5454–5476. Association for Computational Linguistics, 2020.

[639] Guo, W., A. Caliskan. Detecting emergent intersectional biases: Contextualized word embeddings contain a distribution of human-like biases. In M. Fourcade, B. Kuipers, S. Lazar, D. K. Mulligan, eds., *AIES '21: AAAI/ACM Conference on AI, Ethics, and Society, Virtual Event, USA, May 19-21, 2021*, pages 122–133. ACM, 2021.

[640] Bolukbasi, T., K. Chang, J. Y. Zou, et al. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. In D. D. Lee, M. Sugiyama, U. von Luxburg, I. Guyon, R. Garnett, eds., *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, pages 4349–4357. 2016.

[641] Caliskan, A., J. J. Bryson, A. Narayanan. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017.

[642] Ji, Z., N. Lee, R. Frieske, et al. Survey of hallucination in natural language generation. *ACM Comput. Surv.*, 55(12):248:1–248:38, 2023.

[643] Mündler, N., J. He, S. Jenko, et al. Self-contradictory hallucinations of large language models: Evaluation, detection and mitigation. *CoRR*, abs/2305.15852, 2023.

[644] Maynez, J., S. Narayan, B. Bohnet, et al. On faithfulness and factuality in abstractive summarization. In D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault, eds., *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020*, pages 1906–1919. Association for Computational Linguistics, 2020.

[645] Varshney, N., W. Yao, H. Zhang, et al. A stitch in time saves nine: Detecting and mitigating hallucinations of llms by validating low-confidence generation. *CoRR*, abs/2307.03987, 2023.

[646] Lightman, H., V. Kosaraju, Y. Burda, et al. Let’s verify step by step. *CoRR*, abs/2305.20050, 2023.

[647] Guo, Y., Y. Yang, A. Abbasi. Auto-debias: Debiasing masked language models with automated biased prompts. In S. Muresan, P. Nakov, A. Villavicencio, eds., *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2022, Dublin, Ireland, May 22-27, 2022*, pages 1012–1023. Association for Computational Linguistics, 2022.

[648] Du, M., F. He, N. Zou, et al. Shortcut learning of large language models in natural language understanding: A survey. *CoRR*, abs/2208.11857, 2022.

[649] Brundage, M., S. Avin, J. Clark, et al. The malicious use of artificial intelligence: Forecasting, prevention, and mitigation. *CoRR*, abs/1802.07228, 2018.

[650] Bommasani, R., D. A. Hudson, E. Adeli, et al. On the opportunities and risks of foundation models. *CoRR*, abs/2108.07258, 2021.

[651] Charan, P. V. S., H. Chunduri, P. M. Anand, et al. From text to MITRE techniques: Exploring the malicious use of large language models for generating cyber attack payloads. *CoRR*, abs/2305.15336, 2023.

[652] Wang, Z. J., D. Choi, S. Xu, et al. Putting humans in the natural language processing loop: A survey. *CoRR*, abs/2103.04044, 2021.

[637] Chen, Y., L. Yuan, G. Cui 等. 深入探讨预训练语言模型的校准问题. 收录于 A. Rogers, J. L. Boyd-Graber, N. Okazaki 编., 第61届计算语言学协会年会论文集（第1卷：长论文），*ACL 2023*，加拿大多伦多，2023年7月9-14日，第1343–1367页。计算语言学协会，2023年。

[638] Blodgett, S. L., S. Barocas, H. D. III 等. 语言（技术）即权力：对自然语言处理中的“偏见”批判性综述. 收录于 D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault 编., 第58届计算语言学协会年会论文集，*ACL 2020*，线上会议，2020年7月5-10日，第5454–5476页。计算语言学协会，2020年。

[639] Guo, W., A. Caliskan. 识别新兴的交叉偏见：上下文化词嵌入包含人类偏见的分布. 收录于 M. Fourcade, B. Kuipers, S. Lazar, D. K. Mulligan 编., *AIES '21: AAAI/ACM人工智能、伦理与社会会议*，虚拟会议，美国，2021年5月19-21日，第122–133页。ACM，2021年。

[640] Bolukbasi, T., K. Chang, J. Y. Zou 等. 男人是程序员，女人是家庭主妇？词嵌入的去偏差方法. 收录于 D. D. Lee, M. Sugiyama, U. von Luxburg, I. Guyon, R. Garnett 编., 神经信息处理系统进展29：神经信息处理系统年度会议，2016年12月5-10日，西班牙巴塞罗那，第4349–4357页。2016年。

[641] Caliskan, A., J. J. Bryson, A. Narayanan. 自动从语料库中提取的语义包含人类偏见. *Science*, 356(6334):183–186, 2017年。

[642] Ji, Z., N. Lee, R. Frieske 等. 自然语言生成中的幻觉现象综述. *ACMComput. Surv.*, 55(12):248:1–248:38, 2023年。

[643] Mündler, N., J. He, S. Jenko 等. 大型语言模型的自我矛盾幻觉：评估、检测与缓解. *CoRR*, abs/2305.15852, 2023年。

[644] Maynez, J., S. Narayan, B. Bohnet 等. 摘要生成中的忠实性与事实性问题研究. 收录于 D. Jurafsky, J. Chai, N. Schluter, J. R. Tetreault 编., 第58届计算语言学协会年会论文集，*ACL 2020*，线上会议，2020年7月5-10日，第1906–1919页。计算语言学协会，2020年。

[645] Varshney, N., W. Yao, H. Zhang 等. 一针及时：通过验证低置信度生成内容检测与缓解大型语言模型的幻觉. *CoRR*, abs/2307.03987, 2023年。

[646] Lightman, H., V. Kosaraju, Y. Burda 等. 让我们逐步验证. *CoRR*, abs/2305.20050, 2023年。

[647] Guo, Y., Y. Yang, A. Abbasi. 自动去偏：利用自动偏见提示对掩码语言模型进行去偏. 收录于 S. Muresan, P. Nakov, A. Villavicencio 编., 第60届计算语言学协会年会论文集（第1卷：长论文），*ACL 2022*，爱尔兰都柏林，2022年5月22-27日，第1012–1023页。计算语言学协会，2022年。

[648] Du, M., F. He, N. Zou 等. 大型语言模型在自然语言理解中的捷径学习：综述. *CoRR*, abs/2208.11857, 2022年。

[649] Brundage, M., S. Avin, J. Clark 等. 人工智能的恶意使用：预测、预防与缓解. *CoRR*, abs/1802.07228, 2018年。

[650] Bommasani, R., D. A. Hudson, E. Adeli 等. 关于基础模型的机遇与风险. *CoRR*, abs/2108.07258, 2021年。

[651] Charan, P. V. S., H. Chunduri, P. M. Anand 等. 从文本到MITRE技术：探索大型语言模型在生成网络攻击载荷中的恶意用途. *CoRR*, abs/2305.15336, 2023年。

[652] Wang, Z. J., D. Choi, S. Xu 等. 将人类纳入自然语言处理循环：综述. *CoRR*, abs/2103.04044, 2021年。

[653] Galsworthy, J. *The inn of tranquillity: studies and essays*. W. Heinemann, 1912.

[654] Yao, S., K. Narasimhan. Language agents in the digital world: Opportunities and risks. *princeton-nlp.github.io*, 2023.

[655] Asimov, I. Three laws of robotics. *Asimov, I. Runaround*, 2, 1941.

[656] Elhage, N., N. Nanda, C. Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 1, 2021.

[657] Bai, J., S. Zhang, Z. Chen. Is there any social principle for llm-based agents? *CoRR*, abs/2308.11136, 2023.

[658] Baum, S. A survey of artificial general intelligence projects for ethics, risk, and policy. *Global Catastrophic Risk Institute Working Paper*, pages 17–1, 2017.

[659] Lecun, Y. <https://twitter.com/ylecun/status/1625127902890151943>.

[660] Zhao, S. [Can Large Language Models Lead to Artificial General Intelligence?](#)

[661] Brandes, N. [Language Models are a Potentially Safe Path to Human-Level AGI](#).

[662] Zocca, V. [How far are we from AGI?](#)

[663] Ilya Sutskever, L. F. [Ilya Sutskever: Deep Learning | Lex Fridman Podcast #94](#).

[664] Lecun, Y. <https://twitter.com/ylecun/status/1640063227903213568>.

[665] LeCun, Y. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62, 2022.

[666] Shridhar, M., X. Yuan, M. Côté, et al. AlfworlD: Aligning text and embodied environments for interactive learning. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021.

[667] Chowdhury, J. R., C. Caragea. Monotonic location attention for length generalization. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett, eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, vol. 202 of *Proceedings of Machine Learning Research*, pages 28792–28808. PMLR, 2023.

[668] Duan, Y., G. Fu, N. Zhou, et al. Everything as a service (xaas) on the cloud: Origins, current and future trends. In C. Pu, A. Mohindra, eds., *8th IEEE International Conference on Cloud Computing, CLOUD 2015, New York City, NY, USA, June 27 - July 2, 2015*, pages 621–628. IEEE Computer Society, 2015.

[669] Bhardwaj, S., L. Jain, S. Jain. Cloud computing: A study of infrastructure as a service (iaas). *International Journal of engineering and information Technology*, 2(1):60–63, 2010.

[670] Serrano, N., G. Gallardo, J. Hernantes. Infrastructure as a service and cloud technologies. *IEEE Software*, 32(2):30–36, 2015.

[671] Mell, P., T. Grance, et al. The nist definition of cloud computing, 2011.

[672] Lawton, G. Developing software online with platform-as-a-service technology. *Computer*, 41(6):13–15, 2008.

[673] Sun, W., K. Zhang, S.-K. Chen, et al. Software as a service: An integration perspective. In *Service-Oriented Computing—ICSOC 2007: Fifth International Conference, Vienna, Austria, September 17-20, 2007. Proceedings 5*, pages 558–569. Springer, 2007.

[674] Dubey, A., D. Wagle. Delivering software as a service. *The McKinsey Quarterly*, 6(2007):2007, 2007.

[675] Sun, T., Y. Shao, H. Qian, et al. Black-box tuning for language-model-as-a-service. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato, eds., *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, vol. 162 of *Proceedings of Machine Learning Research*, pages 20841–20855. PMLR, 2022.

高尔斯沃斯, J. 《宁静客栈: 研究与随笔》. W. Heinemann, 1912.[654] 姚, S., Narasimhan, K. 数字世界中的语言代理: 机遇与风险. *princeton-nlp.github.io*, 2023.[655] 艾萨莫夫, I. 《机器人三定律》. 艾萨莫夫, I. 《Runaround》, 2, 1941.[656] Elhage, N., N. Nanda, C. Olsson 等. 转换器电路的数学框架. *Transformer Circuits Thread*, 1, 2021.[657] 白, J., 张, S., 陈, Z. 大型语言模型代理是否存在社会原则? . *CoRR*, abs/2308.11136, 2023.[658] 鲍姆, S. 关于人工通用智能项目的伦理、风险与政策的综述. 全球灾难风险研究所工作论文, 页码17–1, 2017.[659] LeCun, Y. <https://twitter.com/ylecun/status/1625127902890151943>. [660] 赵, S. 大型语言模型能否引领人工通用智能? [661] 布兰德斯, N. 语言模型是通向人类水平AGI的潜在安全路径. [662] 佐卡, V. 我们距离AGI还有多远? [663] 伊利亚·苏茨克弗, L. F. 伊利亚·苏茨克弗: 深度学习 | Lex Fridman播客#94.[664] LeCun, Y. <https://twitter.com/ylecun/status/1640063227903213568>. [665] LeCun, Y. 通向自主机器智能的路径版本0.9. 2022-06-27. 开放评审, 62, 2022.[666] Shridhar, M., X. Yuan, M. Côté 等. AlfworlD: 对齐文本与具身环境以实现交互式学习. 在第九届国际学习表征会议 (ICLR 2021), 虚拟会议, 奥地利, 2021年5月3-7日. OpenReview.net, 2021.[667] Chowdhury, J. R., C. Caragea. 用于长度泛化的单调位置注意力. 收录于A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, J. Scarlett 编., 国际机器学习会议 (ICML 2023), 2023年7月23-29日, 夏威夷檀香山, 美国, 第202卷, 页码 28792–28808. PMLR, 2023.[668] 段, Y., G. Fu, N. Zhou 等. 云端一切即服务 (XaaS): 起源、现状与未来趋势. 收录于C. Pu, A. Mohindra 编., 第八届IEEE云计算国际会议 (CLOUD 2015), 纽约市, NY, 美国, 2015年6月27日-7月2日, 页码621–628. IEEE计算机学会, 2015.[669] Bhardwaj, S., L. Jain, S. Jain. 云计算: 基础设施即服务 (IaaS) 研究. 《工程与信息技术国际期刊》, 2(1):60–63, 2010.[670] 塞拉诺, N., G. Gallardo, J. Hernantes. 基础设施即服务与云技术. 《IEEE软件》, 32(2):30–36, 2015.[671] 梅尔, P., T. Grance 等. NIST对云计算的定义, 2011.[672] Lawton, G. 利用平台即服务技术在线开发软件. 《计算机》, 41(6):13–15, 2008.[673] 孙, W., Z. Zhang, S.-K. Chen 等. 软件即服务: 一种集成视角. 在服务导向计算——ICSOC 2007: 第五届国际会议, 奥地利维也纳, 2007年9月17-20日. 会议录第5卷, 页码558–569. Springer, 2007.[674] Dubey, A., D. Wagle. 软件即服务的交付. 《麦肯锡季刊》, 6(2007):2007, 2007.[675] 孙, T., S. Shao, H. Qian 等. 面向语言模型即服务的黑箱调优. 在K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, S. Sabato 编., 第六届国际机器学习会议 (ICML 2022), 2022年7月17-23日, 马里兰州巴尔的摩, 美国, 第162卷, 页码20841–20855. PMLR, 2022.